Speech during exercise

A study of respiratory, phonatory, and articulatory adaptation

Heather Weston

Studies in Laboratory Phonology 17

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ISSN: 2363-5576

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Heather Weston. 2026. *Speech during exercise: A study of respiratory, phonatory, and articulatory adaptation* (Studies in Laboratory Phonology 17). Berlin: Language Science Press.

This title can be downloaded at:
http://langsci-press.org/catalog/book/590

ISBN: 978-3-96110-584-7 (Digital)
978-3-98554-201-7 (Hardcover)

ISSN: 2363-5576
DOI: 10.5281/zenodo.20607629
Source code available from www.github.com/langsci/590
Errata: paperhive.org/documents/remote?type=langsci&id=590

Cover and concept of design: Ulrike Harbort
Typesetting: Heather Weston
Proofreading: Elliott Pearl, Gerald Delahunty, Jean Nitzke, Mary Ann Walter, Nicoletta Romeo
Fonts: Libertinus, Arimo, DejaVu Sans Mono
Typesetting software: XƎLATEX

Language Science Press
Scharnweberstraße 10
10247 Berlin, Germany
http://langsci-press.org
support@langsci-press.org

Storage and cataloguing done by FU Berlin

FREIE
UNIVERSITÄT
BERLIN

Contents

Acknowledgments

This book is based on a doctoral dissertation accepted by the Faculty of Language, Literature and Humanities at Humboldt-Universität zu Berlin in 2023. My thanks go to Susanne Fuchs for introducing me to the fascinating topic of speech breathing and for giving me the freedom to pursue it in my own way. I am also indebted to Tine Mooshammer and Oliver Niebuhr, whose questions and insights shaped this study, as did many opportunities for *fachlicher Austausch* with fellow phoneticians at HU and SDU. I also benefited from the advice of Amélie Rochet-Capellan and Laura Koenig, who generously shared their expertise on respiratory analyses and glottal source measures, respectively.

Much of this research took place at ZAS, where I had the privilege of meeting all kinds of linguists. I thank Marzena Żygis, Stef Jannedy, and Frank Seifart for helpful discussions, and Ola, Oksana, Nasim, Ludger, Zoka, and Jocelyn for their camaraderie. Theres Weißgerber deserves special mention for her tireless assistance with the SALAMMBO project. Some of this work was conducted at the Cambridge Phonetics Lab, and I am grateful to Brechtje Post, Kirsty McDougall, and Francis Nolan for welcoming me into such a vibrant community and for the many conversations that enriched my time there.

My sincere thanks go to the participants, who volunteered for a rather demanding experiment during the pandemic. I also gratefully acknowledge funding provided by the DFG/ANR, Erasmus, and the Wikimedia Foundation.

For the preparation of this book, I would like to thank Martine Grice and Aviad Albert for their guidance and patience throughout the process. ChatGPT (GPT-5.5; OpenAI, 2026) assisted with manuscript-related tasks: formatting LaTeX tables, transforming pencil sketches into line drawings, refining formulations, double-checking translations, and hunting for inconsistencies.

On a personal note, I thank my parents for giving me so many opportunities to learn throughout my life, and my brother for discussing the academic nitty-gritty. I am grateful to my German family for their kindness and unfailing good humor over the years. Special thanks go to Maxim and Ari, whose friendship has been one of the great constants of the past two decades. Finally, I thank my husband. He has made possible far more than I ever imagined, and for that I will always be grateful.

1 Introduction

1.1 Why study speech during physical activity?

Much of human communication occurs while people are moving, exerting themselves, or otherwise not at rest. We speak while walking down a crowded street, carrying objects, or performing physical work, but also during emotionally charged or cognitively demanding situations. Yet the majority of speech production research is based on recordings from seated speakers in quiet, isolated environments. This emphasis on tightly controlled laboratory conditions has been essential for establishing foundational models of speech production. However, this focus also means that comparatively little is known about how speech is produced during physical activity. This gap raises a straightforward question: Do everyday levels of physical exertion influence speech production? If so, what kinds of adjustments emerge?

1.1.1 Speech production as a physical system

Speech production entails highly complex motor coordination across multiple body systems. At its core, speech production is a sequence of physical events triggered by the intention to speak: muscles in the chest and abdomen regulate airflow from the lungs; laryngeal muscles adjust the vocal folds to enable voicing; and additional muscles position the vocal tract to shape that signal into distinct speech sounds. The anatomical structures involved are not used exclusively for speech but serve primary biological functions, most notably respiration, which must be maintained under all conditions. Changes in bodily state can therefore be expected to affect speech production. Physical activity, beginning with commonplace behaviors such as walking, is one such state: its onset triggers an immediate, whole-body response that places new demands on respiration and motor activation, creating conditions under which speech production must adapt.

However, speech production adaptations need not occur uniformly across speakers. Prior work has shown that speakers diverge in various production habits. Breathing patterns, for example, vary widely across individuals in both read and spontaneous speech (Winkworth et al. 1994, 1995), contributing to differences in habitual utterance duration. Speakers are also known to differ in habitual laryngeal "settings," such as preferred vocal fold configurations (Laver 1980), as well as in articulatory habits, including vowel realization and constriction postures (e.g., Johnson et al. 1993), giving rise to different voice qualities. Comparable variability is observed in physiological responses to exercise. For example, some individuals experience elevated mood during exercise (often termed a "runner's high"), while others do not (Dietrich & McDaniel 2004), leading to different emotional states that may influence vocal output. These examples illustrate that variability is inherent to both speech and human physiology. Rather than treating individual differences as noise to be averaged away, examining speaker differences can help reveal structure: Where do speakers converge? Where are common limits? Where is there the most freedom? Reframing variability in this way can thus provide insight into the flexibility, constraints, and adaptive capacity of the speech production system.

Exercise is a useful paradigm for examining the flexibility of speech production because it induces coordinated, whole-body physiological change. Physically active states are common in everyday life, so speakers have substantial experience producing speech under such conditions. Examining speech during exercise therefore allows us to observe how the production system is managed under demands that are both routine and physiologically meaningful. At the same time, exercise affects multiple levels of the speech production mechanism simultaneously, making it possible to observe how adaptations may unfold across the system as a whole as physiological demands shift. Because exercise can be graded, it provides a scalable, continuous perturbation, allowing examination of progressive change as well as potential thresholds in adaptation.

Viewed at an abstract level, studying speech during physical activity offers more than a catalog of acoustic or articulatory changes. Exercise functions as a controlled means to move the body away from a resting equilibrium, placing sustained demands on respiration and motor coordination while speakers continue to communicate. Observing how speech adapts under these conditions provides a window into how a complex production system reorganizes in real time. Points at which speakers converge in their responses suggest shared constraints or limits of the system, whereas points of divergence reveal flexibility and alternative strategies for maintaining speech. In this sense, speech during exercise can reveal not only *what* changes under increased physical demands, but how such adaptations might be linked across interacting subsystems. This perspective shifts the focus from level-specific effects to the underlying organization of speech production as a dynamic, adaptive system.

This view of exercise as a perturbation that pushes the body out of equilibrium can be extended to other everyday situations that affect respiration and motor activation. This idea lies at the core of Hans Selye's theory of the physiological stress response. In today's non-linguistic usage, 'stress' is typically understood as strain or tension associated with particular situations, such as deadlines or interpersonal conflict. In other words, stress is often regarded as something external to the individual. Selye, however, conceptualized stress as a *response* of the body. In his early work, he described a general adaptation syndrome, observing that diverse adverse conditions elicited a similar pattern of physiological change in laboratory animals (Selye 1936). He later defined stress more broadly as the "nonspecific response of the body to any demand." From this perspective, stress is defined less by the external stressor than by the internal pattern of bodily activation recruited to restore homeostasis.

This nonspecificity is particularly relevant for understanding why different situations affect speech production. Because the same physiological machinery is recruited across different stressors, research on vocal change can be meaningfully compared across diverse events, from physical stressors such as exercise or high altitude to cognitive or emotional challenges. Selye's notion of stress therefore provides a useful conceptual framework for explaining why different contexts can give rise to similar vocal effects. Additionally, a body-centered account is especially appropriate for speech production, which relies on the respiratory and motor systems – both central to bodily regulation and thus among the first to be affected by a stress response.

At the same time, Selye's framework explicitly allows for substantial individual variation. Although the physiological stress response is general, the perception of what constitutes a stressor is highly individual. What one speaker experiences as demanding or uncomfortable may be neutral or even positive for another. Stress responses thus emerge from the interaction between an individual and a situational demand, rather than from the situation alone. Importantly, this accommodation of individuality aligns with observed speaker differences in speech production. As a result, the stress framework provides a unifying way to examine situational changes in the voice across different stressors and individuals. Within this context, physical exercise offers a particularly useful entry point for examining how the stress response shapes speech output: it places controlled demands on bodily systems relevant for speech while avoiding the ethical challenges associated with inducing cognitive or emotional stress.

1.1.2 Speech under stress: An emerging research field

Speech produced under stress has attracted interest across multiple disciplines, in part because changes in bodily and cognitive state are known to affect spoken communication. Such observations are not a modern insight: authors from Aristotle[1] to Shakespeare[2] to Darwin[3] provide us with texts describing how exertion, emotion, or agitation shape how speech sounds. Such observations underscore that these effects are salient, perceptible, and deeply embedded in human communicative experience. For modern researchers, however, the problem remains vexing. As Klaus Scherer, a pioneer in the study of emotion in the voice, noted in 1986, speech under stress is often impressionistically identifiable, yet difficult to characterize systematically: "Whereas judges seem to be rather accurate in decoding emotional meaning from vocal cues, researchers in psychoacoustics and psychophonetics have so far been unable to identify a set of vocal indicators that reliably differentiate a number of discrete emotions" (Scherer 1986: 143–144). Thus while it is clear that speech is a sensitive, externally observable output of an individual's internal state, fundamental questions remain about why different emotions "sound different" and which production-level settings give rise to them. Understanding situational changes in the voice is of clear value for disciplines concerned with human performance and behavioral state, such as psychology, as well as for applied domains including occupational health monitoring and stress detection. As such, several reviews have synthesized findings on speech under stress, approaching the topic from different disciplinary perspectives.

Two early reviews of speech under stress focused primarily on psychological and emotional stressors. Reviews by Kirchhübel et al. (2011) and Giddens et al. (2013) were motivated largely by applied goals, such as understanding what the voice conveys about emotional state, stress, or deception in forensic and

[1] Why do those who have taken exercise and weak persons speak in a shrill voice? Is it because the weak move but little air, and a little air travels more quickly than a larger quantity. But those who have taken exercise cause a violent movement in the air and violently moved air travels more quickly. And rapid movement in voice implies shrillness.

Problemata XI, 21. ca. 3 BC–AD 5

[2] Juliet to her nurse

How art thou out of breath when thou hast breath
To say to me that thou art out of breath?

Romeo and Juliet, ca. 1595

[3] That the pitch of the voice bears some relation to certain states of feeling is tolerably clear. A person gently complaining of ill-treatment, or slightly suffering, almost always speaks in a high-pitched voice.

The Expression of the Emotions in Man and Animal, 1872 performance-related contexts. Despite some differences in disciplinary orientation, these reviews showed overlap in their conceptual framing and empirical focus. Stress was defined within Selye's (1936) framework, emphasizing that identical stressors can elicit different responses across individuals. Empirically, the studies reviewed included laboratory paradigms and real-life situations, focusing on a narrow set of measures – most prominently f0, along with intensity, jitter, shimmer, and speech rate. While increased f0 was identified as the most consistent effect of stress, both reviews concluded that there is little evidence for universal acoustic markers of stressed speech. Both reviews also acknowledge substantial individual variability, which may account for mixed findings. At the same time, the sources of this variability remain unclear, potentially arising from differences in stressor type, methodology, or inherently speaker-specific responses. This degree of variability is further recognized as a central challenge for developing an overarching theoretical account of speech under stress.

The most recent review (Van Puyvelde et al. 2018) broadened this perspective by shifting attention toward the bodily effects of stress. While again grounding its framework in Selye's (1936) concept of stress, the authors emphasize both the nonspecific nature of the stress response across stressor types and its impact on physiological systems directly relevant to speech production. In particular, respiration is highlighted as a central link between stress and speech, given its role in both metabolic regulation and voice production. This broader definition of stress encompasses cognitive, emotional, and physical workload, and the review is explicitly organized by levels of the speech production mechanism. In doing so, it connects physiological processes to acoustic outcomes while opening the door to system-wide accounts of speech adaptation. Methodologically, the review also departs from earlier work by explicitly treating individual differences as a meaningful signal rather than noise. Although the authors again note the "miscellany of approaches" in the literature (Van Puyvelde et al. 2018: 14) and the prominence of f0 effects, they propose a conceptual account in which stress induces bottom-up bodily arousal – reflected, for example, in increased f0 – that may be modulated by top-down cognitive regulation. Variation in stress responses is thus implicit in their framework. Accordingly, the authors call for future studies to examine multiple voice parameters simultaneously in order to identify multivariate patterns of stress response.

Considered together, these reviews converge on several core conclusions while also underscoring persistent gaps in the literature. Across perspectives, there is broad agreement that speech does change under stress, yet the nature and direction of these changes are inconsistent and highly variable. Reported effects depend on the type of stressor examined, the methodological choices made, and the individual speaker, with substantial individual variability repeatedly noted. At the same time, these shared observations point to unresolved issues. Physical stress – including everyday activities like walking – remains markedly underrepresented relative to cognitive and emotional stressors. Respiration is identified as central to understanding stress–speech interactions, yet it is rarely investigated directly. Most studies focus on isolated acoustic measures or single levels of speech production, limiting insight into system-wide adaptation. Finally, although individual variability is widely acknowledged, it is rarely explained or incorporated into theoretical accounts. Thus, while the literature agrees that stress affects speech, it offers limited insight into the precise nature of changes and speaker variability.

1.2 Aims and overview of the present study

1.2.1 Research aims

Despite how commonplace speech during physical activity is, remarkably little is known about how it is produced. Previous work has focused on the vocal effects of cognitive and emotional stress, calling for further study of diverse voice quality measures. Work on speech during physically active states highlights increased respiration as a potential driver of changes across the speech production mechanism, calling for further study of respiratory patterns. Additionally, the substantial individual variability observed is seen as a factor that must be taken into account when examining speech under stress. Against this backdrop, the present study takes a multi-level, production-oriented approach to investigating speech during physical activity, integrating respiratory, phonatory, articulatory, and experiential data. The goal is to capture both dominant patterns of adaptation and individual variability across different levels in order to better understand how speech production adapts to physical activity. Specific research questions are derived from the detailed literature review in Chapter 2.

1.2.2 How this book is structured

The structure of this book zooms in and out between research background and the present results by design. Early chapters establish shared motivation and methodology for the three studies, while each empirical chapter includes specific background and methods that lay the foundation for the study of that individual speech production level. Thus, while literature and methodology sections reappear throughout the book, they do so at progressively finer levels of resolution, moving from a general physiological perspective to subsystem-specific mechanisms rather than revisiting the same material. Each study likewise includes an in-depth discussion of the findings at that level, making each empirical chapter self-contained. In this sense, readers may approach the empirical chapters independently, although the full contribution of the book emerges from their cumulative interpretation in the final chapter.

Chapter 2 provides an overview of the relevant physiology together with a systematic review of the literature. Because this research area is interdisciplinary and extends beyond traditional linguistic work, the aim is to broadly orient the reader to the kinds of questions, methods, and findings that characterize research on speech under physical and other forms of stress. Rather than detailing individual studies, the review emphasizes general patterns and clarifies the physiological rationale for why exercise – and stress more broadly – should affect speech production at multiple levels. In this way, the systematic review is used to derive the level-specific research questions that are addressed in the subsequent chapters of the book.

Chapter 3 describes the methods used to *collect* the speech and respiratory data. The chapter covers participant selection, speech materials, experimental design, laboratory setup, recording procedures, post-processing, and transcription. This level of detail is intended to lay the foundation for understanding the later analyses and offer detailed information for researchers interested in conducting similar studies. Further methods used to *analyze* the speech and respiratory data are presented in the relevant empirical chapters.

The next three chapters present the empirical studies that form the core of the book. Chapter 4 investigates speech breathing, assessing multiple respiratory parameters to determine how speakers modify breath timing and breath depth to speak during physical activity. The chapter begins with an overview of methods for assessing respiratory activity, followed by a chronological review of prior work. The methodological section provides detailed descriptions of preparing and analyzing respiratory data captured via chest kinematics, offering a reference point for future studies. Individual speaker responses are visualized using line plots and heatmaps to assess systematic variability. Additionally, post-task questionnaires are analyzed to capture speakers' subjective experiences of speaking during exercise, providing interpretive context for the quantitative findings.

Chapter 5 focuses on the voice source, first outlining commonly used glottal source measures before briefly reviewing the inconsistent findings reported in prior studies. This chapter explores new territory by assessing a range of acoustic measures to capture the multidimensional nature of glottal behavior, examining both group-level trends and individual patterns across multiple vowels.

Chapter 6 addresses supralaryngeal effects by analyzing changes in formant frequency to infer changes in articulatory targets. Given the scarcity of such work on speech during physical activity, the chapter reviews articulatory findings from the related domains of speech under other types of stress and loud speech. The methodology provides an in-depth description of speech sample selection and formant estimation procedures. As in the other experimental chapters, the results include both statistical findings and visual explorations of individual differences.

Finally, Chapter 7 provides a general discussion that synthesizes the findings across all levels of speech production. This chapter integrates results across workload conditions, speech tasks, and vowel identities to identify general patterns of adaptation as well as points of divergence. Individual differences observed across levels are examined, and a conceptual framework is proposed to account for the coexistence of robust group-level trends and substantial speaker-specific variability. The chapter concludes by addressing methodological considerations and outlining directions for future research.

2 Background and literature review

To lay the foundation for the present study, this chapter outlines the physiological basis of speech production and presents a systematic review of the literature on speech during physical activity. The key anatomical structures and processes used to produce speech sounds are introduced for each stage of the speech production mechanism – respiration, phonation, and articulation. Then, an overview of the body's response to exercise describes how physical activity can influence the delicate balance of these speech mechanisms. The second section maps the research landscape to establish what is currently known about speech during physical activity and to identify the most significant gaps in the field. The chapter concludes with the research objectives emerging from the review.

2.1 The speech production mechanism: Anatomy and physiology

In its essence, the speech production mechanism is a sequence of physical events triggered by the intention to speak. The brain encodes a thought as a speech motor plan, nerves tell respiratory muscles to move air into the lungs, the vocal folds move closer together to generate a source signal, and further muscles position the vocal tract to modulate the signal into the intended speech sounds. The physical act of producing speech can thus be broadly divided into three levels: respiration, phonation, and articulation. The following sections describe the relevant anatomical structures and functions for each level, based on details given in Schneiderman & Potter (2002), Sataloff et al. (2011), and Gick et al. (2013) unless otherwise noted.

2.1.1 Respiration

In the context of speech production, the main role ascribed to the lungs is to generate the airflow needed to make the vocal folds vibrate. However, in the context of the current study, it is important to remember that the primary – and vital – function of the respiratory system is to facilitate *blood gas exchange*, that is, to bring oxygen into the body and remove carbon dioxide.

The lungs (Figure 2.1) are masses of spongy tissue supported by the skeletal system, including the vertebral column and ribs, and indirectly manipulated by muscles of the abdomen, chest, and back. The lungs are covered by a lubricated membrane called the *visceral pleura*, while the chest wall is lined with the *parietal pleura*. A thin fluid-filled space between these membranes maintains negative pressure, mechanically coupling the lungs to the chest wall. As a result, expansion and contraction of the chest wall causes the lungs to expand and contract.

Air moves into and out of the lungs via the *trachea*, an elastic tube whose upper end connects to the larynx and whose lower end branches into two *bronchi*. The bronchi subdivide into ever-smaller branches across the lungs, leading to millions of tiny air sacs, or *alveoli*, which are responsible for blood gas exchange. When oxygen-rich air reaches the alveoli, oxygen diffuses across the thin alveolar lining into the surrounding blood vessels. At the same time, carbon dioxide diffuses from the circulating blood into the alveoli and is expelled during exhalation. This process of oxygen uptake and carbon dioxide removal is known as *respiration*.

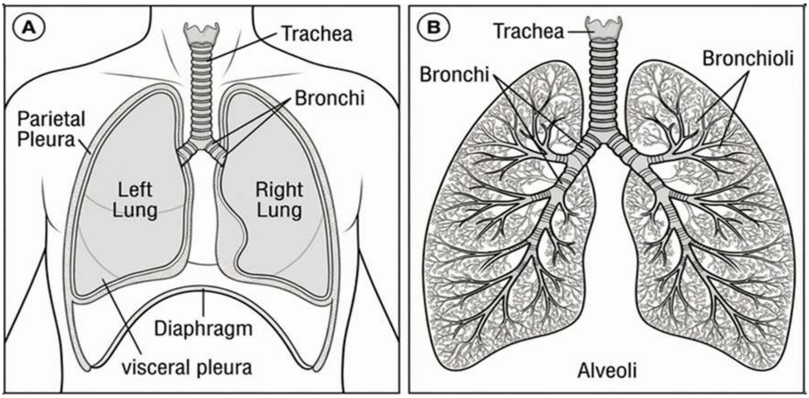

Figure 2.1: Overview of the respiratory system

The act of moving air into and out of the lungs – or breathing – is referred to as *ventilation*. It is a mechanical process that relies on muscle activity, air pressure, and mechanical forces to move air. Ventilation consists of two phases: breathing in (*inspiration*) and breathing out (*expiration*). Inspiration is an active phase during which various muscles contract. Important muscles for inspiration are the diaphragm and the intercostals.[1] The *diaphragm* is a dome-shaped muscle located beneath the lungs, and it separates the chest and abdominal cavities.

[1]For details on all muscles involved in respiration, see Schneiderman & Potter (2002: 25–35).

The intercostal muscles run between the ribs in different directions. The *external intercostals* lie on the outside surface of the ribs and extend from the vertebral column. The parts of the *internal intercostals* that contract during inspiration extend from the sternum, running along the inner surface of the rib cage between the cartilaginous parts of the top eight pairs of ribs. When the inspiratory muscles contract, they enlarge the chest cavity in different ways. The diaphragm pulls its dome shape downwards, which enlarges the chest cavity by lowering its floor and thus displacing the contents of the abdominal cavity – much like pinching the middle of a balloon. The external intercostals angle the ribs up and outward, while the interchondral parts of the internal intercostals pull the lower cartilages up and away from the body's center, with both actions enlarging the chest cavity. As the chest cavity expands, it draws with it the elastic tissue of the lungs, increasing the volume of the lungs. Because air pressure and volume are inversely related (Boyle's Law), the increase in lung volume lowers the pressure inside the lungs below the atmospheric pressure outside, causing air to rush into the lungs to equalize the pressure.

Expiration, in contrast, is generally passive: when the inspiratory muscles stop contracting, the bones and elastic soft tissues that have been pulled outward are allowed to recoil – to return to their rest positions thanks to a combination of gravity, the elastic nature of soft tissue, and the force associated with the rotation of the ribs (torque). When this happens, the chest cavity – and with it, the volume of the lungs – contracts, thereby increasing the pressure in the lungs above atmospheric pressure and causing air to rush out to equalize pressure.

This basic description of ventilation applies to breathing at rest, known as quiet breathing or tidal breathing. But ventilation can change considerably depending on what an individual is doing. For the present study, it is crucial to understand that breathing differs between resting, exercising, speaking, and doing combinations of these activities. For example, during exercise, greater force is required to expel larger volumes of air. In these cases, expiration becomes active, and additional muscles are recruited to actively drive expiration. Figure 2.2 shows muscles involved in inspiration and active expiration. The key muscles are the main sections of the internal intercostals – which run perpendicular to the external intercostals and thus pull the ribs downward when contracted – and the abdominal muscles. The *rectus abdominis* is a deep muscle running from the groin to the cartilages that links ribs 5–7 to the sternum; contraction pulls the ribs down and inward. The *external obliques* run along the sides of the body; contraction pushes on the abdominal organs, which, in turn, push the diaphragm upward into the chest cavity. The *internal obliques* and the *transversus abdominis* do the same thing, but the former run perpendicular to the external obliques, and the latter is positioned deeper in the body and runs horizontally around the front of the thorax. Additional "accessory muscles" (e.g., the *latissimus dorsi*) may also be recruited as needed to squeeze out as much air as possible.

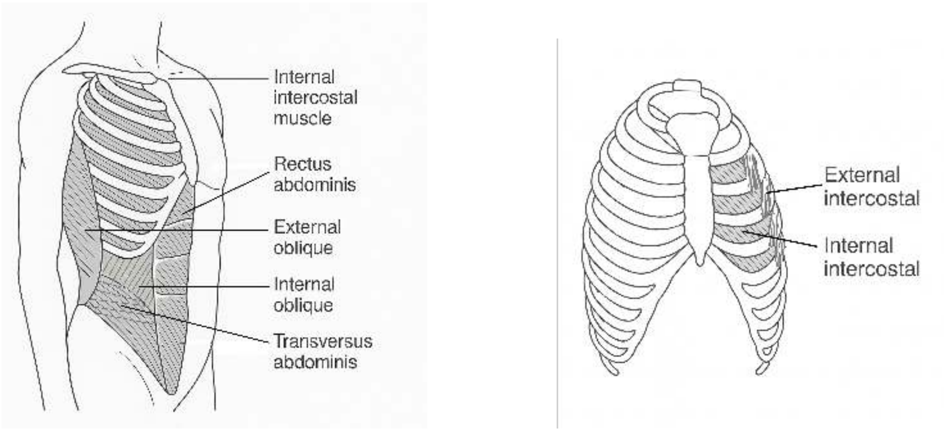

Figure 2.2: Muscles involved in inspiration and active expiration

Another breathing mode that is relevant for the present study is speech breathing, which differs in several fundamental ways from tidal breathing. For one, tidal breathing is automatic, while speech breathing is voluntarily controlled. For another, breath depth and timing differ. Tidal breathing is characterized by relatively shallow, regular inspirations followed by slightly longer expirations. In speech breathing, the depth and duration of each breath is different, largely dictated by what a speaker intends to say – the volume of air inspired and the duration of a given breath cycle reflect the length of the following utterance (Huber 2008, Sperry & Klich 1992, Winkworth et al. 1995). For example, a speaker would typically take a short, lower-volume breath prior to a short utterance, like "It's over there," and a longer, deeper breath prior to a complex utterance, such as a series of directions to walk to a destination. This means that speech breathing requires varying and often greater volumes of air to be controlled differently over time, which is accomplished by a different pattern of muscle activation than used for tidal breathing. Inspiration is quick, which may require accessory muscles to be recruited to inspire large volumes of air. Expiration, on the other hand, is long and slow, precisely controlled by keeping the inspiratory muscles active. Abdominal muscles are also contracted to control the breath stream. Because the amount of air in the lungs is constantly changing, the muscles must constantly work to keep subglottal pressure constant across different lung volumes.

2.1.2 Phonation

Phonation is the process of producing vocal sounds. It occurs when the airstream exiting the lungs and throat passes up through the vocal folds – two multilayered bands of muscle and tissue that, when brought together, begin to vibrate, creating the basic source sound of human speech. The vocal folds are located in the larynx (Figure 2.3), a complex cartilaginous system that extends up from the trachea and is suspended from the hyoid bone above. The main cartilages are the thyroid, the largest cartilage; the cricoid, which is attached to the trachea; and the two arytenoids, which are the rear attachment points for the vocal folds.

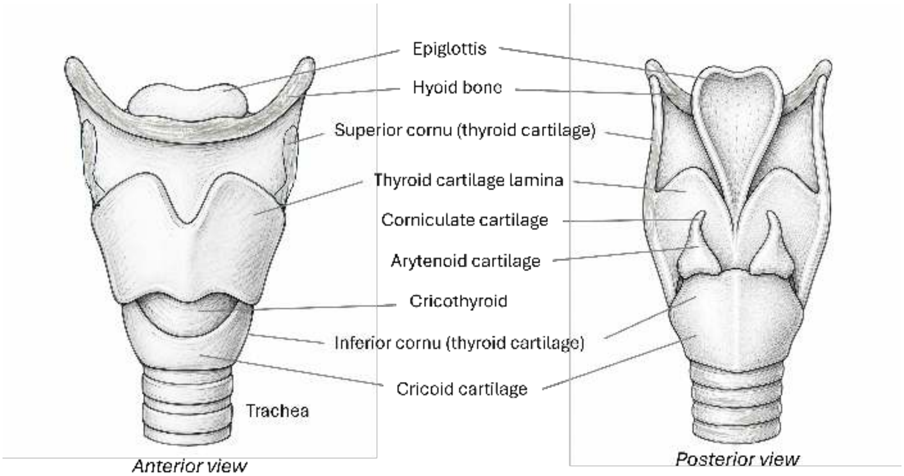

Figure 2.3: Cartilages and structures of the larynx shown from the front (*left*) and the back (*right*)

The cartilages are connected to one another (and to the hyoid bone and trachea) by different membranes and muscles. *Extrinsic muscles* support the larynx and originate from or attach to a structure outside of it; *intrinsic muscles*, depicted in Figure 2.4, control the position and tension of the vocal folds. One intrinsic muscle, the *posterior cricoarytenoid*, separates (abducts) the vocal folds and thus widens the *glottis* – the space between the vocal folds. Four intrinsic muscles close (adduct) them: the *oblique arytenoids*, the *transverse arytenoid*, the *lateral cricoarytenoids*, and the *thyroarytenoids*. The lateral parts of the thyroarytenoids form the vocal folds (these folds of muscle are also referred to as the vocalis muscles). Finally, the *cricothyroid* muscle increases the tension on the vocal folds by stretching them out in length – when the cricothyroid contracts, the thyroid cartilage is pulled down and the distance between the arytenoids is increased.

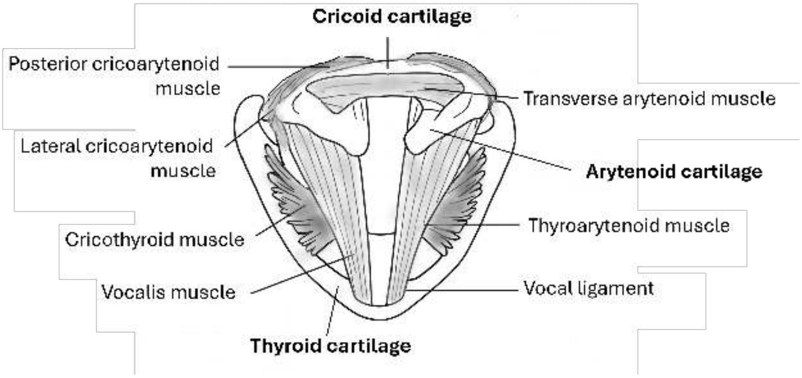

Figure 2.4: Intrinsic muscles of the larynx: The cricothyroid pivots the thyroid cartilage to move it further from the arytenoids; the posterior cricoarytenoids rotate the arytenoids to draw the vocal process outward; the lateral cricoarytenoids pivot the arytenoids to draw the vocal process inward; the transverse arytenoid connects the back of the arytenoid cartilages and slides them toward each other; the thyroarytenoids draw the arytenoids forward.

Phonation – the generation of voiced sounds – occurs when the vocal folds are brought together while air flows through the glottis. The glottal cycle starts when the vocal folds are adducted, and enough subglottal pressure builds up to force them apart from the bottom up. As the glottis widens, airflow increases, and pressure decreases along their edges, which pulls the vocal folds back together (Bernoulli's principle) in tandem with the muscle's elastic force. The process then begins anew. In phonetics, this account of voice production is known as the myoelastic-aerodynamic theory of phonation (van den Berg 1958).

It should be emphasized that vocal fold movement is complex. The vocal folds consist of two membranes layered over the vocal ligament and vocalis muscle, and the outer membranes are less elastic than the muscle that comprises the main substance of the vocal fold. The entire structure can remain somewhat thick during voicing, which means that the vocal folds do not open all at once but rather in a wavelike manner from the bottom up and from back to front. These mechanical properties of the vocal folds, along with air pressure, affect the number of opening and closing cycles per second – frequency, or its perceptual correlate, pitch.

The anatomical complexity of the laryngeal system also permits a range of phonatory qualities through different configurations of its cartilages, muscles, and soft tissues. Changes in the angles and distances between the cartilages alter the shape and tension of the tissues suspended between them, thereby affecting the resulting acoustic signal.

2.1.3 Articulation/resonance

After the source signal is generated at the vocal folds, it is propagated up through the *supraglottal* vocal tract, through the tube-shaped *pharynx* (in general terms: the throat), and toward the tongue, teeth, palate, oral and nasal cavities, and movable articulators (tongue, lips, jaw, velum). The characteristics of these structures filter the source signal, creating formant frequencies that give vowels their distinctive acoustic properties, while the configuration of the articulators determines the identity of specific speech sounds (articulation). Whereas phonation produces a basic source signal, this stage of the speech production mechanism shapes that signal in a multitude of ways to create what we recognize as distinctive speech sounds. The following descriptions focus on the pharynx/oronasal cavities and the movable articulators because they are most relevant for the study in Chapter 5, which investigates vowel production.

The pharynx and the oral and nasal cavities, shown in Figure 2.5, form a series of interconnected *resonators* – systems that naturally oscillate with greater amplitude at certain frequencies. This means that the systems inherently boost some frequencies and dampen others. As a result, particular frequencies are radiated in the acoustic signal with greater relative intensity. In speech, these concentrations of energy in the vocal tract are referred to as formants. Formants specify vowel identity and will be further discussed in Chapter 6. The pharynx consists of three sections and three pairs of overlapping constrictor muscles that can increase tension during speech (but whose primary function is to move food down the esophagus). The oral and nasal cavities may be joined by lowering the velum (soft palate), which allows air to flow through the nose and produces nasal sounds. Changing the size and shape of the resonators, whether by increasing tension on the pharyngeal muscles or lowering the velum, affects the resonance properties of the entire system.

In addition to the velum, the tongue, lips, and jaw are configured in different ways to produce different phonemes. The tongue is mainly composed of muscular tissue and is extremely flexible – the contractions of four intrinsic muscles enable it to bulge upward (*superior longitudinal muscles*), shorten (*inferior longitudinal muscles*), narrow (*transverse muscles*), and flatten (*vertical muscles*). Three extrinsic muscles alter the tongue's shape and position. The *genioglossus muscles* move the tongue tip forward and lower the whole tongue body in the mouth. The *hyoglossus muscles*, which extend from the hyoid bone, pull the tongue down and back. The *styloglossus muscles* raise and retract the tongue. The tongue is the main actor in producing vowel sounds. Most vowels can be identified by the first two formant frequencies, which reflect the tongue's height and backness, respectively.

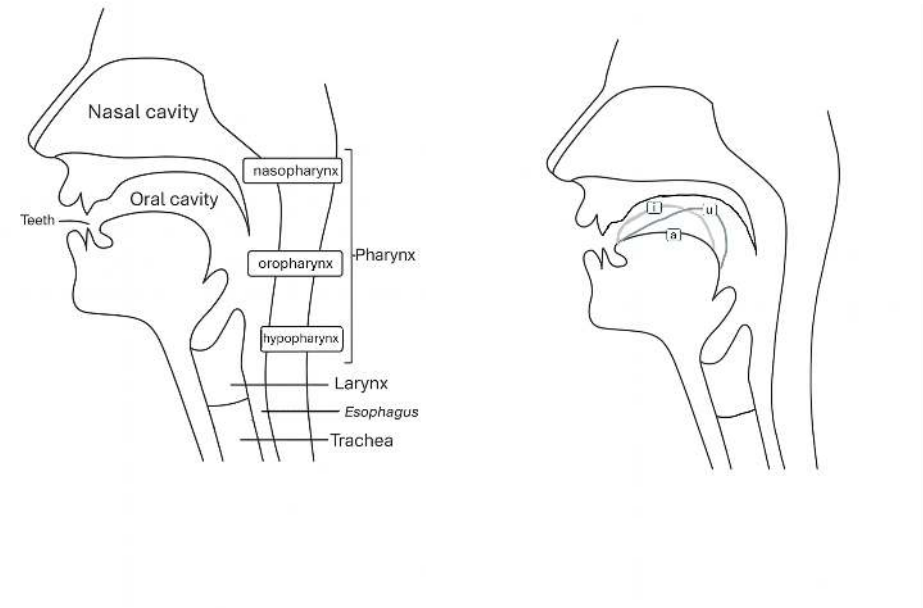

Figure 2.5: The structures of resonation in relation to the larynx (*left*), and different tongue positions for different vowels (*right*)

2.2 Physical activity and the body's physiological response

This section begins by defining physical activity and introducing related terminology. It then draws on the extensive overviews given in Travers et al. (2022), Haverkamp et al. (2005), and Sapolsky (2004) to describe the body's acute response to a bout of physical activity. The goal is to lay out how such exercise affects the anatomical structures involved in speech production processes. Potential interactions that may arise between the respiratory, laryngeal, and supralaryngeal levels will also be considered.

2.2.1 Definitions of key terms

One challenge in investigating speech during physical activity is navigating the varied terminology found in the literature. Discipline-specific traditions and frameworks give rise to different terms, some of which may be used interchangeably, while others may refer to different concepts. For the present study, it was thus important to select and define key terms that would apply across diverse fields. An appropriate reference was found in the *Physical Activity Guidelines Advisory Committee Report*, published by the U.S. Department of Health and Human Services. Compiled by a committee of researchers in medicine and public health, the 600-page report reviewed recent scientific literature with the goal of defining physical activity recommendations for the public. The authors explicitly "attempted to use definitions that have been generally accepted in the scientific literature and in major reports and recommendations for physical activity and public health" (Physical Activity Guidelines Advisory Committee 2008: C-1). The following terminology and descriptions are based on this report unless otherwise noted.

Across the literature, human movement is predominantly referred to using two terms: physical activity and exercise. *Physical activity* refers to "any bodily movement produced by the contraction of skeletal muscle that increases energy expenditure above basal level" (Physical Activity Guidelines Advisory Committee 2008: C-1). *Exercise* refers to a specific type of physical activity carried out for the purpose of improving physical fitness. One general definition of *physical fitness* cited in the report comes from the World Health Organization: "the ability to perform muscular work satisfactorily" (WHO Meeting on Exercise Tests in Relation to Cardiovascular Function & World Health Organization 1968: 6).

Physical activity can be further specified according to its purpose (e.g., occupational, recreational), its *mode*, or type of activity (e.g., aerobic exercise, muscle-strength activities), and its level of *intensity*, which refers to the amount of work being performed. Intensity may be expressed in *absolute measures*, such as kilometers per hour, or in *relative measures*, which account for a person's specific capacities. Common relative measures to express the rate of work are the percentage of maximal oxygen uptake (commonly referred to as VO_2 max) and the percentage of maximum heart rate (HR max). Perceptual measures of a person's own effort are also used to assess intensity from the exercising person's point of view (e.g., the Borg Scale of perceived exertion).

A taxonomy of key terms related to physical activity is given in Figure 2.6. The most common terms are given as headings, and alternative terms found in the literature are given beneath them. In some cases, the alternative terms appear to be specific to a researcher's conceptual framework (e.g., "physical task stress," used in John Hansen's work) or a particular field. For example, in the phonetics literature, physical activity is sometimes referred to as "physical effort" (e.g., Primov-Fever et al. 2014). This terminology may be in analogy to "vocal effort." However, this term itself has been shown to be inconsistently defined and at times conflated with related constructs such as "vocal load" and "vocal fatigue" (Hunter et al. 2020). The use of "physical effort" therefore risks similar conceptual ambiguity.

The present study uses the following terminology: "physical activity" will refer to human movement of all types and "exercise" will refer specifically to endurance-type activities "that are repetitive and produce dynamic contractions of large muscle groups for an extended period of time (e.g., walking, running, cycling, swimming)" (Physical Activity Guidelines Advisory Committee 2008: C-2). Note that these terms are used interchangeably to refer to the experimental conditions because the physical activity was in fact endurance exercise. To maintain the distinction between objective and subjective measures of exercise intensity, the amount of work being performed will be referred to as "workload" in the context of the experimental conditions and as "(perceived) effort" in the context of the participants' subjective assessments of the work being performed.

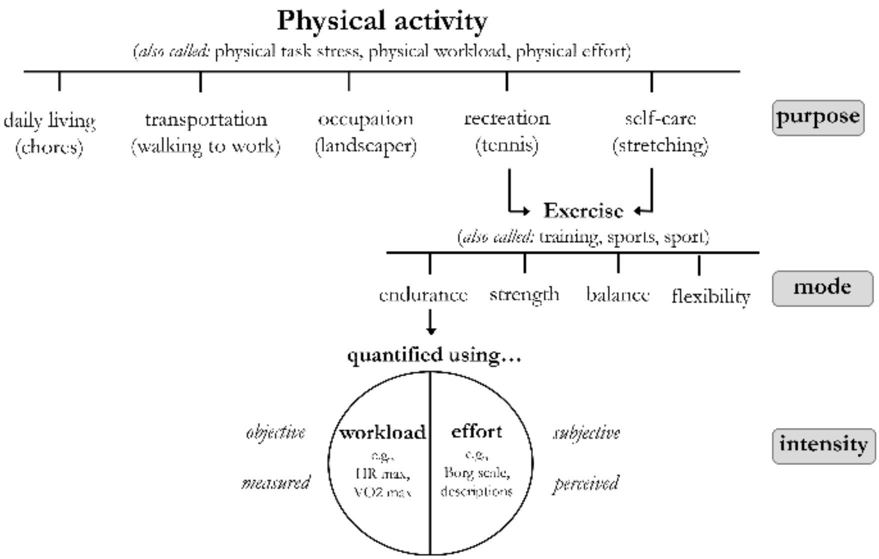

Figure 2.6: Taxonomy of key terms used to describe physical activity

2.2.2 Physiological effects of exercise on speech production

The onset of physical activity triggers an immediate, whole-body response that has deep evolutionary roots. As biologist Robert Sapolsky explains using an example of animals on the African savanna: whether you are a zebra running from a predator or a lion chasing its prey, "the core of the stress-response is built around the fact that your muscles are going to work like crazy" (Sapolsky 2004: 11). This response entails a cascade of events. The force and power needed for locomotion is generated by muscle contractions – when muscles start to work, they require fuel, so energy substrates stored in the body must be mobilized. The sympathetic nervous system is activated while the parasympathetic nervous system is depressed to coordinate the ventilatory and cardiovascular responses needed to deliver fuel to the working muscles – the lungs increase their work to exchange more air, and the heart pumps faster and harder to circulate nutrients and remove waste. Blood flow is thus increased to the working muscles but decreases in some organs, like the liver, pancreas, and gut, whose processes may also be affected by sympathetic stimulation (Heinonen et al. 2014). As a result, when an individual initiates physical activity, they are not simply moving their limbs but are fundamentally changing conditions across their entire body.

The biggest changes, however, occur in the cardiorespiratory system. Faster, deeper breathing – *hyperpnea* – is "the first line of defense" when adapting to an exercising state (Haverkamp et al. 2005: 535). Ventilation increases immediately at exercise onset and then gradually reaches a steady state that precisely meets the body's needs for oxygen supply and carbon dioxide removal. At low and moderate exercise levels, ventilation is proportional to metabolic demands, so an increase in exercise intensity will be accompanied by an increase in ventilation. At high exercise intensities, however, this relationship breaks down. Breathing becomes excessively deep and rapid, beyond what is required for blood-gas exchange, producing a state known as *hyperventilation*. Although carbon dioxide levels continue to fall, respiratory drive remains high, often leading to gasping. Consequently, exercise leading to hyperventilation would be expected to hinder an individual's ability to speak.

Less obvious changes in other parts of the respiratory system may also affect speech production. Ventilation is increased by increasing the mechanical work of the diaphragm and the supporting respiratory muscles. To make this work of breathing more efficient, the upper and lower airways widen to reduce airflow resistance. In the upper airway, this is accomplished through four means. First, the predominantly nasal breathing route seen during rest changes to oronasal breathing when ventilation reaches around 30 l/min (Haverkamp et al. 2005: 528). Based on a table of gas exchange levels on the preceding page (*ibid.*, p. 527), oronasal breathing appears to become predominant in healthy young adults at around 40% VO_2 max, which is a low-to-moderate level of exercise. Second, vasculature in the nasal passages constricts to decrease airflow resistance in the nose. Third, resistance is further decreased via contraction of the nasal dilator muscles immediately prior to activation of the inspiratory muscles. Haverkamp et al. (2005: 528) presume that the pharyngeal and laryngeal muscles also contract in this manner to further aid inspiration. Fourth, mean glottic aperture is widened. Taken together, these adaptations reflect considerable changes to airflow and muscle activation in parts of the upper airway that are used for speech.

The physiological adaptations to exercise may thus be predicted to impact speech in various ways. To start with, speech breathing is directly affected. In the temporal domain, faster breathing means shorter breaths and thus less time to speak between breath pauses. In terms of lung volumes, deeper breaths increase the amount of air coming into and out of the lungs, with the airstream flowing up through the glottis becoming more forceful. At the same time, airway dilation decreases airway resistance, which could increase airflow rate and make the airstream unpredictable or more difficult to control. This could lead to difficulties with airflow management during speech. For example, if speakers misjudge flow rates, they may run out of air mid-utterance.

Respiratory adjustments may also affect phonation. For example, increased subglottal pressure requires tighter vocal fold adduction to maintain phonation. In addition, exercise triggers acute laryngeal adjustments that can alter vocal fold positioning. The larynx acts as a bottleneck in the airway (Røksund et al. 2015), because the laryngeal passage is narrower than the passages above and below it. To accommodate the increased ventilatory demands of exercise, the upper part of the larynx above the vocal folds – the *epilaryngeal tube* – opens more fully. This structure consists of the epiglottis (front), the aryepiglottic folds (sides) and the arytenoid cartilages (back). As exercise intensity increases, the larynx progressively dilates to reduce airflow resistance: The epiglottis rotates forward toward the base of the tongue, stretching the aryepiglottic folds (Bates et al. 2015), while the posterior parts of the vocal folds are lifted upward, and air increasingly flows through the posterior glottis (Røksund et al. 2015). These laryngeal adjustments alter the angle and aperture of the glottis and thus influence the shape and positioning of the vocal folds.

Glottal behavior during exercise also highlights a fundamental tension between breathing and speaking. During quiet breathing, glottal aperture increases during inspiration and decreases during expiration, moving the vocal folds further apart and closer together, respectively (Brancatisano et al. 1983, Chen et al. 2013). During exercise, however, the vocal folds tend to remain separated during both inspiration and expiration, maintaining a wider glottal aperture (Beaty et al. 1999, England & Bartlett Jr. 1982). This respiratory posture is directly at odds with the adducted position of the vocal folds required for phonation. Consequently, speakers may need to make larger or faster laryngeal adjustments to initiate and maintain voicing. If the vocal folds must move a greater distance to get from "breath pause position" to "phonation position," speech timing may be affected. Indeed, VOT has been shown to increase at high lung volumes, where the flattening of the diaphragm can pull down the trachea and larynx, which generally abducts the vocal folds (Hoit et al. 1993).

Phonation may also be affected by changes in muscle activity. For instance, extrinsic muscles that support the larynx, such as the four infrahyoid muscles ("strap muscles") that run along the front of the neck, are also activated during deep inspirations (Gick et al. 2013: 61). These muscles support the hyoid bone, so contracting them can change the vertical position of the larynx. This modifies the position of the cartilages relative to one another and affects the resting length of the intrinsic muscles (Sataloff et al. 2011). A combination of neck tension and altered intrinsic muscle function may make it more difficult for the laryngeal musculature to maintain smooth, even vocal quality in the face of continuous changes in airflow and subglottal pressure. As a result, voicing in connected speech may present voicing breaks, tremulousness, or periods of non-modal phonation.

There are also non-respiratory-related effects of exercise that could affect phonation. One example is heart rate. Work conducted by Robert Orlikoff has shown that variation in fundamental frequency (jitter) and amplitude during sustained phonation is partially accounted for by variation in heart rate in both sexes (Orlikoff 1990b, Orlikoff & Baken 1989), with the effect becoming more pronounced with age, possibly as a result of age-related changes in vasculature and soft tissues (Orlikoff 1990a). Another way in which exercise can affect simultaneous speech is through the production of different neurotransmitters, or chemical messengers, in the brain. Of particular relevance are endocannabinoids, which are responsible for the "runner's high," or elevated mood, some people experience during or after exercise. The endocannabinoid system helps to regulate the body's metabolism and stress response and is involved with controlling motor activity, cognitive functions, and emotions. During exercise, circulating endocannabinoid levels significantly increase, and the individual experiences an improved mental state compared to before exercise (Brellenthin et al. 2017). While the mechanisms are not fully clear, the endocannabinoid system may activate to suppress pain (Dietrich & McDaniel 2004). Although many open questions remain, and the magnitude of effects reported in the literature varies greatly (Desai et al. 2022, Matei et al. 2023), there is consensus that acute exercise can strongly affect emotional state. This is interesting because the effects of emotion on voice production are well documented. In particular, the work of Klaus Scherer has pursued the hypothesis that different emotional states are reflected by different phonatory and articulatory correlates (Scherer 1986) and demonstrated that voice quality reflects not only the type but the intensity of emotion (Banse & Scherer

1996). This means that in addition to physical changes in the position and shape of the vocal folds, mental states may also affect the outcome of phonatory processes, adding another layer of complexity to exercise–voice interactions.

Further exercise effects on speech could emerge at the supralaryngeal level. For example, several physiological changes could alter articulator position. High autonomic arousal has been found to affect lip aperture variability (Kleinow & Smith 2006). At the same time, mouth aperture could generally increase, both to facilitate airflow and due to the breathing route changing from nasal at rest to oronasal during exercise. Tongue position may change as well. During deep inspirations, the tongue tends to move forward to keep the airway open (Mitchinson & Yoffey 1947), which could lead to a more fronted position at utterance onset. A crucial point is that the articulators are interdependent – so changes in the position of one may affect changes in the positions of other articulators to maintain intended resonance properties and acoustic output.

Endurance-type exercise may also affect processes of muscle activation and coordination, potentially affecting the articulators. For example, genioglossus motor units are coactivated during inspiration in non-exercising states (Bailey 2011). But during exercise, motor units are also activated during expiration (Walls et al. 2013), with activation of the genioglossus muscle progressively increasing as ventilation increases (Shi et al. 1998). While muscle activity in the upper airway is complex and still incompletely understood, a review on the topic concluded that "the bulk of the evidence suggests that the entire assembly of tongue and laryngeal muscles operate together but differently during breathing and swallowing, like a ballet rather than a solo performance" (Fregosi & Ludlow 2014: 291). One possible consequence is that when an individual speaks during exercise, activation of the genioglossus during the expiratory phase could tense other muscles involved in articulation and in turn affect articulatory precision. For example, greater tension in the articulators could generate more force, leading to target overshoot, or alternatively manifest as stiffness and target undershoot. Acoustically, this could result in an expanded or centralized vowel space, respectively.

Further laryngeal–supralaryngeal interactions are conceivable, although very little is currently known about how laryngeal adjustments affect pharyngeal motor behavior. One recent study – building on some of the work and findings above – hypothesized that laryngeal airway aperture may be linked to genioglossus activity to dilate the airway and facilitate expiration (LaCross et al. 2017). To test this, genioglossus activity was recorded during production of different sustained vowels in whispered, voiced, and electro-larynx conditions; the first two conditions inherently changed laryngeal aperture creating a high-airflow and a low-airflow condition, respectively, and the third condition used a battery-powered sound source in the airway to generate sound with no airflow. During speech, genioglossus motor unit firing rate increased from low to high expiratory airflow (phonation to whisper) and was lowest for the electro-larynx rate. These results demonstrate that the aperture of the larynx – which itself is responding to changes at respiratory level – impacts muscle activity in the pharynx. In other words, exercise-induced breathing could plausibly affect the entire speech production chain.

A more recent study of vocal effort further illustrates potential interactions between respiratory, laryngeal, and articulatory processes during speech (Abur et al. 2022). Vocal effort refers to a speaker's perceived difficulty speaking and is associated with increased laryngeal muscle tension and subglottal pressure. Participants read a passage under different levels of vocal effort – but without changing speech intensity – while respiratory and articulatory data were recorded. Maximum vocal effort was associated with increased lung volume at speech initiation and decreased lip aperture, whereas tongue kinematics remained unchanged. Although the mechanisms underlying these articulatory adjustments remain unclear (e.g., reduced lip aperture may have helped speakers maintain vocal intensity), the findings suggest that changes in respiratory and laryngeal conditions may be accompanied by articulatory adjustments. It is therefore plausible that similar adjustments may arise under the increased demands of speaking during exercise.

This section has established some anatomical and physiological reasons why physical activity may affect various levels of the speech production chain, and where potential interactions between levels may arise. The survey of empirical studies on the body's response to exercise provided evidence of numerous adjustments across the body and that both physiological and psychological factors can shape speech during exercise. Given the extent of changes in the parts of the body used to produce speech, we may expect a similarly extensive range of changes in speech production. The following section presents a systematic review of reported changes in speech breathing, voice quality, and articulation during exercise.

2.3 Literature review

2.3.1 Method

A systematic review was conducted to identify literature. Four databases were searched: PubMed, Scopus, Web of Science, and Google Scholar. For the first three databases, searches were conducted in June 2021 using the MeSH[2] keywords "physical fitness," "exercise," "sports," and "sport" in combination with the following MeSH keywords pertaining to speech and language: "phonetics," "speech," "speech production measurement," "speech perception," "speech intelligibility," "speech-language pathology," "language disorders," "speech disorders," "language development disorders," "specific language disorder," "speech sound disorder," "apraxia," "rehabilitation of speech and language disorders," "language therapy," "speech therapy," "language development," and "child language." These terms were selected to sample across a range of linguistic research areas indexed in the MeSH system. This search returned 1,366 non-duplicate studies. Of these, 215 were identified as potentially relevant based on the titles. The abstracts of these studies were then checked against two criteria: (1) the study involved physical activity, and (2) the variable of interest was a linguistic parameter (e.g., f0, pause type, etc.). This second screening yielded a shortlist of 72 studies pertaining to speech during physical activity.

Because some relevant phonetic terms were not among the MeSH keywords, Google Scholar was searched in March 2022 using synonyms for physical activity – "physical exercise," "physical fitness," "physical stress," "physical task," "physical workload," "sports," and "sport" – in combination with the following terms related to speech production: "linguistic," "speech," "speaking," "phonetic," "articulation/articulatory," "glottal," "phonation/phonatory," "vocal," and "voice/voicing." Because Google Scholar offers limited support for Boolean operators, each activity term was paired with each speech term separately (e.g., "physical exercise AND linguistic," "physical exercise AND speech," etc.). This search returned around 40 further non-duplicate references. Most of these were not published via the traditional journal system, but rather as conference proceedings, chapters in edited collections and textbooks, master's theses, doctoral dissertations, or government reports.

In total, 112 references were identified as potentially relevant based on title and abstract. These references were given a final screening for inclusion in the review. The criterion was whether the study answered the question: *How does physical activity affect simultaneous speech?* A total of 40 studies were rejected because they did not investigate a linguistic parameter or address the present research question; 15 studies were rejected because they investigated the talk test (an informal method to gauge exercise intensity); a group of 11 studies and a group of 9 studies were not reviewed because they focused on laryngeal effects of

[2] MeSH stands for Medical Subject Headings – the terms used by the National Library of Medicine to index articles for PubMed.

exercise and factors that affect general stress reactivity, respectively. Though not included in the systematic review, some of these references are cited elsewhere in the book.

A total of 37 studies were thus identified for review. The references fall into two broad groups based on the type of linguistic parameter investigated: studies that assessed effects on speech planning and studies that assessed effects on the acoustic signal. In the traditional psycholinguistic hierarchy of linguistic representations (Levelt 1989), these would be respectively categorized as higher-level (semantic and syntactic) and lower-level (phonological and phonetic) processes. The studies concerned with higher-level effects comprise the smaller group and will be discussed together, while the studies looking at different acoustic effects constitute the bulk of the literature and will be discussed in subgroups according to the level of the speech production mechanism:

- 13 studies investigating higher-level processes
- 24 studies investigating the speech production mechanism
  - 8 studies on **respiration** (5 via Google Scholar search)
  - 13 studies on **phonation** (6 via Google Scholar search)
  - 3 studies on **articulation** (3 via Google Scholar search)

Because several studies investigated multiple aspects of speech production, these categories are not mutually exclusive. In particular, four studies classified under respiration also reported laryngeal measures, bringing the total number of studies reporting on phonation to 17.

This review aims to paint a picture in broad strokes – to establish what is currently known about speech during physical activity and to identify the most pressing open questions. The following sections therefore focus on the central findings across all levels of speech production rather than on methodological detail. Fuller discussions of methods and results are provided in the background sections of Chapters 4–6, each of which is devoted to a specific level of speech production.

2.3.2 Higher-level speech processes during physical activity

The studies in this group were published by researchers in diverse fields, including psychology, cognitive science, stroke rehabilitation, and speech-language pathology.[3] Most studies used a dual-task paradigm – the simultaneous execution of two tasks – in this case, walking while talking (WWT). This paradigm stems from a seminal study of older adults that found that individuals who stopped walking when starting a conversation were significantly more likely to suffer a fall (Lundin-Olsson et al. 1997). The authors attributed this to the cognitive load of doing two things at once – when attention is divided between two tasks, performance is compromised in one or both tasks. Although speech is generally viewed as the secondary task (i.e., the research question is: *How does the cognitive load of talking influence gait?*), some studies also reported on linguistic measures, ranging from different metrics of fluency to grammatical and semantic complexity to discourse coherence. Unless otherwise noted, the speech task was to speak freely for several minutes on familiar topics.

In general, measures of speech fluency showed a decline in performance during concurrent walking. However, a rather complex picture emerges, with results depending on both the nature of the walking task and speaker characteristics. For example, in a study of 16 healthy young adults, silent pause duration was unaffected by the easy task of unobstructed walking, but increased when participants had to avoid obstacles or avoid obstacles while carrying a tray (Raffegeau et al. 2018). Such fluency effects may also be compounded by other factors, such as age. In another study, younger adults were found to use more lexical fillers during unobstructed walking, while older adults spoke more slowly (Kemper et al. 2005). But increasing walking difficulty by climbing stairs or carrying groceries led to greater fluency effects for older adults – in addition to speaking more slowly, they spoke in shorter utterances with more lexical fillers. However, a different study of older adults with and without stroke found no such fluency effects – speech rate, utterance length, and percentage of grammatical utterances did not change for either group from talking only to talking while walking on an unobstructed elliptical path (Pohl et al. 2011). In addition, most participants used no lexical fillers in any condition. The lack of fillers could reflect an effect of speech

[3] Only studies that assessed speech during physical activity are reviewed here, but two types of longer-term speech–activity interactions found in the systematic search are mentioned here for the interested reader. One topic is how speech reflects the cognitive effects of extreme environments: during expeditions to mountain summits, speakers showed an increase in comprehension time for simple sentences (Lieberman et al. 1995) and were observed to use discursive strategies similar to those of individuals with depression (Blanchet et al. 1997). Another topic concerns how exercise may facilitate language study. Chinese learners of English were found to improve vocabulary recall speed and accuracy and sentence-level semantic judgement speed after learning vocabulary while cycling compared to sitting still (Liu et al. 2017). Similarly, vocabulary recall was better for German learners of Polish when words were learned during light treadmill walking compared to sitting (Schmidt-Kassow et al. 2014).

topic, with speakers having more to say about some topics than others. Another reason for these differences may be task difficulty – Kemper et al. (2003: 181) instructed participants to walk at a "brisk but comfortable" pace while participants self-selected pace in Pohl et al. (2011)'s study.

Effects on grammatical and semantic complexity are similarly complex. Comparisons of younger and older adults show that walking changed speech differently for the two age groups (Kemper et al. 2003, 2005). During physical tasks, young adults simplified their speech, producing fewer clauses per utterance and fewer modifying clauses while decreasing propositional density. This speech style was closer to that of the older adults at baseline. The older adults mainly reduced speech rate during physical tasks. The same pattern was reported for whole-body physical activities of different complexities (walking alone and while carrying groceries, carrying groceries up stairs) (Kemper et al. 2005) and for fine motor tasks (complex finger-tapping) (Kemper et al. 2003), which underscores the role of divided attention over physical effort. These findings are consistent with comparisons of older adults with and without stroke. Grammatical complexity and semantic content decreased during walking only for participants with stroke, if at all. Adults with stroke used significantly fewer clauses per utterance and modifying clauses in speech during walking and showed subtle declines in propositional density and type–token ratio, but no change was observed in age-matched healthy participants (Pohl et al. 2011). Different indicators of sentence complexity and repetitiveness – the number of verbs per utterance and the prevalence of new information, respectively – were not affected by concurrent walking for either group (Plummer et al. 2020). However, the number of verbs used per sentence varied considerably across participants in the dual-task compared to the single-task condition, with some participants performing better and some worse during walking.

Less is currently known about effects on higher-level processes of discourse structuring. The systematic search did not return any studies on healthy individuals, but a study of adults with stroke found no effect of concurrent walking on discourse coherence, either in terms of general topic maintenance or the linking of utterances (Rogalski et al. 2010). The authors suggest that the dual-task effect could have been obscured by task-order or speech-topic effects but also point out that discourse coherence is generally lower in older adults. Thus, it remains to be seen whether younger (or healthy older) adults show a change in discourse management while walking.

Two final observations are worth noting. The first is that while most WWT studies consider speech effects to arise from divided attention – that is, at the cognitive level – several studies have found evidence of interactions between speech and other motor programs. One study found that executing speech and speech-like oral movements while walking increased temporal but not spatial (stride length) gait parameters (Davie et al. 2012). Because silent articulation and nonce words had the same effect on gait as voiced real words, the authors concluded that dual-task interference with gait may stem from oral motor components rather than speech itself. While the authors do not speculate as to whether walking would also affect speech motor programs, other studies have reported temporal coupling between speech and walking speeds. In a study of speech read at normal, slow, and fast paces while individuals walked at normal, slow, and fast paces, significant correlations were found between both speech rate and walking pace, as well as utterance duration and step length, in both healthy older adults and adults with Parkinson's disease (Cantiniaux et al. 2010). The authors propose that those different rhythms may be regulated by the same mechanism.

The second observation is that dual-task costs may be higher for individuals with certain personality traits or physical characteristics. In a study of 295 older adults, individuals with high neuroticism and low extraversion scores were found to walk more slowly and make more mistakes on a serial 7's subtraction task when doing both tasks at once than those with lower neuroticism/high extraversion (LeMonda et al. 2015). The authors conclude that these personality traits can affect how speakers allocate their attention in dual-task situations. Although it is not clear how these findings would generalize to speech (e.g., in terms of reduced fluency or grammatical complexity) or to younger adults, it seems plausible to hypothesize that stable personality traits could also affect higher-level speech processes in different situations. A study looking at speech effects on gait speed in older and younger healthy adults found that while speech was found to slow gait in both groups, older adults that walked slower showed more dual-task interference than fast walkers (Plummer-D'Amato et al. 2011). In contrast, the older, faster-walking speakers showed a similar dual-task cost compared to the younger walkers. Again, it is not clear how these results might relate to speech, but the finding suggests that age alone may not predict dual-task performance. These observations thus hint at the breadth of factors that may influence speech–gait interactions.

Taken together, these studies demonstrate that physical activity also imposes a *cognitive* load on individuals, which can impact higher-level planning processes. Although walking is a common and well-practiced task, it still requires cognitive resources, which decreases the resources available for planning speech. WWT paradigms hence generally view effects on speech performance as arising at the cognitive level. This is a different domain of speech than that investigated in this book. However, it brings an important perspective to the study of speech during physical activity by highlighting the fact that speaking imposes a cognitive load, and that simultaneously executing the two tasks can result in complex interactions shaped by individual physical *and* mental characteristics, including age, fitness, and personality traits. An important conclusion from this body of work is that regardless of the physiological effects of exercise, individuals may adopt different attentional strategies when speaking in a dual-task situation – focusing on speech, focusing on the motor task, dividing attention between the two tasks, or alternating attention between them. This variation could thus give rise to different patterns of speech change.

2.3.3 Respiration during exercise

Studies of speech breathing specifically examined the effects of exercise rather than those of lower-intensity physical activities such as walking. In the following studies, exercise was generated in lab contexts on stationary equipment (ergometer, treadmill). When exercise intensity was increased, increases were observed in respiratory rate (Doust & Patrick 1981, Fuchs et al. 2015, Meckel et al. 2002, Trouvain & Truong 2015), the volume of inspired air per breath (tidal volume) (Doust & Patrick 1981, Meckel et al. 2002), and the volume of air inspired per minute (Baker et al. 2008, Doust & Patrick 1981, Meckel et al. 2002, Patil et al. 2010, Ziegler et al. 2019). Speech during exercise also showed increased subglottal pressure (Ziegler et al. 2019) and expiratory flow rates (Doust & Patrick 1981, Ziegler et al. 2019).

One topic of interest in this body of work is how the faster breathing required by exercise affects utterance duration and pause placement. An increase in respiratory rate implies a decrease in breath cycle duration and thus a shorter expiratory phase in which to speak. Support comes from empirical studies showing that speech breaths during exercise are on average shorter than those at rest, with both inspiratory and expiratory time decreasing (Meckel et al. 2002), and that utterances contain fewer syllables (Baker et al. 2008) and overall pause time increases (Fuchs et al. 2015, Trouvain & Truong 2015).

There is noteworthy overlap here with some of the dual-task studies reviewed above, which also assessed utterance duration and pause time but interpreted them as measures of fluency rather than as consequences of increased respiratory demands. This illustrates that separate underlying mechanisms can explain the same behavior: in the dual-task literature, changes in breathing were attributed to a strain on cognitive resources, while the phonetic studies put forth an underlying physiological mechanism. But these different interpretations also suggest that multiple mechanisms may be at play in speech during exercise. For example, Kemper et al.'s (2005) dual-task study found that older adults produced shorter utterances when climbing stairs, but Pohl et al. (2011) found no such effects for participants walking on a flat track. Climbing stairs could certainly increase respiration – so it is possible that the shorter utterances were due in part to faster breathing. In the same vein, the number of ungrammatical pauses – which also relates to fluency – was found to increase in speech during exercise (Baker et al. 2008). The authors interpreted this as an effect of unintended breath pauses, but it is equally plausible that the dual task of reading while cycling strained cognitive resources. A different theoretical perspective thus adds some ambiguity to the results in both cases.

Another measure, articulation rate, was also examined in both groups of studies. The dual-task studies found either no effect or that walking slowed speech, while the linguistic studies consistently found faster speech during exercise compared to speech at rest (Baker et al. 2008, Fuchs et al. 2015, Trouvain & Truong 2015). It is not currently clear what would account for the differences, but the discrepancy in findings highlights the need to assess multiple speech and respiratory parameters to obtain a comprehensive picture of speech during physical activity.

Lastly, two studies on speech breathing investigated perceived speaking difficulty during exercise. One study found that discomfort due to breathlessness progressively grew as exercise intensity increased (Baker et al. 2008), but the other study reported that the correlation between exercise intensity and perceived difficulty was speaker-specific (Rotstein et al. 2004). Nevertheless, participants also experienced more discomfort when speaking during exercise than while exercising alone at the same intensity. Together, these results demonstrate that speaking during exercise is perceived as uncomfortable *and* that the level of discomfort varies between individuals. Since emotions, like discomfort, are known to impact different aspects of speech, the perceived speaking difficulty during exercise is a critical area to assess.

2.3.4 Phonation during exercise

At the laryngeal level, the most widely obtained finding by far is an increase in mean fundamental frequency (f0) (Dallaston & Rumbach 2016, Doust & Patrick 1981, Evdokimova & Zakharchenko 2019, Fuchs et al. 2015, Johannes et al. 2007, Koblick 2004, Otis & Clark 1968, Primov-Fever et al. 2014, Trouvain & Truong 2015). There are reports, however, of considerable individual variability. In a study of 51 speakers producing sentences while exercising on an elliptical machine, Godin & Hansen (2008) reported a statistically significant increase in f0 in 60.8% of the participants; of the remainder, 25.5% showed no change, and 13.6% showed a statistically significant decrease compared to speech at rest. Individual means were also reported by (Evdokimova & Zakharchenko 2019), with eight speakers consistently showing increases in f0 during exercise. In this study, speakers produced connected speech immediately *after* (not during) an exercise protocol, so this could account for the difference in result. Conflicting results were also reported for the effect of exercise intensity. While fundamental frequency was generally found to increase linearly with increasing exercise intensity (Fuchs et al. 2015, Primov-Fever et al. 2014, Ziegler et al. 2019), Johannes et al. (2007) found that f0 increased when cycling activity commenced but did not change with incrementally increasing intensity until the pre-exhaustion stage was reached. The authors concluded that f0 is not affected when physical activity is easily tolerated – the 11 participants were Special Forces soldiers and thus had high fitness levels. Interestingly, a similar non-linear effect was reported in a study on speech intelligibility at high atmospheric pressures published by the U. S. Naval Submarine Medical Research Laboratory (Murry et al. 1972). Cycling exercise resulted in a significant decrease in perceptual intelligibility at surface level and at high pressures in a hyperbaric chamber, but no linear decrease in intelligibility was observed as exercise intensity was incrementally increased. Again, the participants had high levels of physical fitness as well as experience in physically demanding conditions. The authors speculated that perhaps it was the act of moving itself rather than the intensity of activity that diminished intelligibility, or that once the diver had adjusted to the task the increasing physical workload did not pose an additional burden.

Vocal intensity was also found to increase during exercise (Dallaston & Rumbach 2016, Doust & Patrick 1981, Fuchs et al. 2015, McCaig 2013, Otis & Clark 1968, Patil et al. 2010, Trouvain & Truong 2015, Ziegler et al. 2019). This is unsurprising, since vocal intensity is correlated with f0, and louder speech would also be expected due to the increased airflow and subglottal pressure arising from increased ventilation. In one of the few studies that investigated a population other than healthy young adults, McCaig (2013) found that intensity also increased during walking for healthy older speakers (58–80 years old) and age-matched speakers with Parkinson's disease. These findings demonstrate that exercise can affect speech in diverse populations and also that low-intensity activities can already have an effect on speech.

Measures of frequency and amplitude perturbation have also been investigated, although the results are mixed. In her study of fitness instructors during class, Koblick (2004) found no significant effects on jitter or shimmer, although the absolute values increased. Dallaston & Rumbach (2016) also reported no significant changes in jitter or shimmer, although they recorded fitness instructors immediately after teaching a 60-minute class, so it is possible that if perturbation effects are found, they are short lived. Evdokimova & Zakharchenko (2019) also recorded speech immediately post-exercise, and the descriptive statistics for mean jitter for their eight participants show effects in different directions. Although no group-level statistical analysis was reported, this degree of individual variation may plausibly have yielded a non-significant result. An alternative explanation could lie with the speech material. Primov-Fever et al. (2014) investigated the sustained vowels /i, a/ produced by 14 speakers at rest and while cycling at 60%, 75% and 90% heart rate reserve (moderate, moderate-heavy, and heavy workloads), finding a significant increase in jitter when workload changed from the 75% to 90% condition and an increase in shimmer between the 60% and 90% conditions. Hence it is possible that changes in some parameters may be conditioned by speech material.

Several further measures of voice quality returned mainly non-significant results. Noise-to-harmonics ratio was not found to change significantly from rest to exercise (Dallaston & Rumbach 2016, Koblick 2004, Primov-Fever et al. 2014). Normalized amplitude quotient and harmonic richness factor only showed changes in a subset of speakers, who showed a breathier quality during exercise (Godin & Hansen 2015).

Finally, one study investigated physiological parameters of voice production (Sandage et al. 2013). After an 8-minute bout of cycling, 18 participants repeated the syllable /pi/ and read a passage. Phonation threshold pressure was found to significantly increase during /pi/ as did perceived phonatory effort during reading. Additionally, pharyngeal temperature decreased by 1.9°C. The authors concluded that these results are consistent with the physiologically driven hypothesis that phonation during physical activity requires greater laryngeal effort and force to adduct the vocal folds.

2.3.5 Articulation during exercise

Only three studies – each presented as a conference paper – were found that examined the articulatory domain of speech during exercise. This limited number may reflect the methodological challenges involved: many methods of collecting articulatory data, such as ultrasound, electromagnetic articulography (EMA), and magnetic resonance imaging (MRI), require participants to remain relatively still, making such methods difficult to apply in physically active settings. To circumvent this limitation, articulatory changes can be inferred from the acoustic signal, as was the case for the studies reviewed here.

One of the papers was co-authored by John Hansen, who has conducted a number of studies on speech during exercise, referred to as “physical task stress.” Godin & Hansen (2011b) assessed changes in the first two formants of the vowels /i, ɑ/ produced by four speakers during constant elliptical exercise at 10 mph. Formant frequencies were estimated with the WaveSurfer algorithm. No significant change in F1 frequency was seen during exercise, suggesting no change in tongue height. A significant change was observed in F2, however, implying a shift in tongue backness (though no details on direction were given). The reported F1 changes were in the range of 7–20 Hz, with participants varying in direction and magnitude of effect.

Marquard et al. (2017) investigated the vowels /i, e, a, o, u/ produced by four speakers during cycle exercise. Formants were estimated with the Praat algorithm. The results were vowel specific: F1 decreased for /a/ and increased for /u/, while F2 increased for /i, a/ and decreased for /e, o/. Together, these changes suggest a fronting, centralizing tendency in speech during exercise.

Williamson et al. (2018) investigated the cumulative effects of exercise, heat, and altitude on articulatory precision. Thirteen participants produced speech samples after cycling for 45 minutes in a hypobaric chamber at altitude (3,000 m) and then at altitude plus heat. The authors assessed articulatory precision using eigenspectra of correlation matrices constructed from time series data of the first three formants. Eigenvalues increased during physical activity, which the authors interpreted as evidence of *improved* articulatory coordination. However, they noted that the observed changes could also reflect greater articulatory variability (i.e., less precision).

All in all, very little empirical data is currently available on supralaryngeal-level changes in speech during exercise. These three acoustic studies provide some tentative evidence of articulatory effects. Both Godin & Hansen (2011b) and Marquard et al. (2017) report changes in F2 (the front–back dimension) across vowels. Fronting of the tongue is consistent with the data obtained on genioglossus activation during exercise (Walls et al. 2013), but a systematic investigation is needed to establish the direction and magnitude of effect. The results obtained by Williamson et al. (2018) support two contrary hypotheses – increased or diminished articulatory precision. Given the other studies’ suggestion of a centralizing tendency, the latter hypothesis may be the more appropriate interpretation. Regardless of the interpretation of these results, the study suggests that exercise can affect articulatory targets.

2.3.6 State of the field

This systematic review of literature on speech during physical activity aimed to chart the current research landscape, identifying explored topics and knowledge gaps. Two observations about the review process are noteworthy. First, published literature was scarce, perhaps due to the difficulty of research requiring specialized equipment and interdisciplinary methodologies. Many studies were published as conference proceedings, theses, dissertations, or government reports rather than journal articles. This diverse publication format suggests both the field's exploratory nature and the substantial resources required for such research. Consequently, relevant studies are often not indexed in major academic databases, complicating the literature search process.

The second observation concerns the wide variety of keywords used in studies of speech during physical activity. This diversity stems from both the interdisciplinary nature of the subject and the use of different terms within single disciplines (e.g., "physical task stress" and "effort" to refer to "exercise"). This variability can hinder researchers from finding relevant studies outside their immediate field. Moreover, searches require combining terms related to both "speech" and "exercise," resulting in numerous possible search pairs when options for Boolean operators are limited. To identify relevant studies, researchers must look carefully and creatively beyond their own field, considering various publication formats and sources beyond academic journals.

The present systematic review, using this search strategy, identified two broad linguistic topics. A smaller group of studies examined effects on higher-level speech planning, finding that various fluency and complexity measures were often negatively affected by concurrent exercise. The larger group investigated effects on physical speech production, focusing on respiration, phonation, or articulation.

Most of these studies concentrate on laryngeal-level effects. Interestingly, relatively few parameters were investigated. Approximately half of the studies assessed fundamental frequency, consistently finding an increase during exercise, although some also reported notable individual variability. The few studies reporting on jitter, shimmer, and noise-to-harmonics ratio yielded mixed or non-significant results. Consequently, these findings provide only a rudimentary understanding of laryngeal-level effects. The core issue to address is: *Which glottal source effects characterize speech during exercise?*

Turning to respiration, the review revealed a remarkably consistent body of research. Most studies investigated multiple parameters, with results showing increases in respiratory rate, tidal volume, minute ventilation, subglottal pressure, and expiratory flow rates. Participants reported discomfort while talking during exercise that typically increased with exercise intensity. However, a critical gap emerged in the understanding of changes in breath volume. Most studies assessed a combined temporal-volumetric measure, minute ventilation, leading to disagreement about how speakers modify breathing during exercise. The key question is: *Do speakers increase breath rate, breath depth, or both?*

Articulation remains the least explored area in research on speech during exercise. Only three studies were found, each using different methods to infer articulatory changes from the acoustic signal. While all reported formant-frequency changes, the effects were small, vowel- and speaker-specific, and open to multiple interpretations. At this stage, the most pressing question seems to be: *Does exercise affect articulation at all?* Physiological adaptations that facilitate airflow in the pharynx could potentially affect articulator position during inspiration and expiration, but it remains unknown whether articulatory targets are affected.

Beyond the specific topics identified, the systematic review uncovered some general methodological trends in the research landscape. Figure 2.7 charts four notable research patterns, providing further insights into the state of knowledge in the field. First, research skews heavily towards laryngeal-level and higher-level cognitive changes, as previously discussed. This focus highlights a gap in our understanding of physical speech production processes. Second, researchers employed diverse speech tasks to elicit linguistic material. This heterogeneity complicates cross-study comparability, because the extent to which different speech tasks influence results remains unclear. Third, studies generally used small sample sizes, raising concerns about statistical power and result generalizability, especially given reported individual variability. Lastly, while earlier studies predominantly recorded male speakers, recent research tends to include both biological sexes, moving towards overall gender balance. However, physiological differences between sexes make cross-study comparisons of results challenging. These methodological trends highlight both progress and persistent challenges in the field, underscoring the need to address specific knowledge gaps and refine research methodologies.

2.4 Research objectives

Current research on speech during physical activity has examined different linguistic parameters, providing a foundation for further inquiry. Studies have established consistent changes in respiratory parameters, increases in fundamental frequency and vocal intensity, and potential articulatory adjustments during

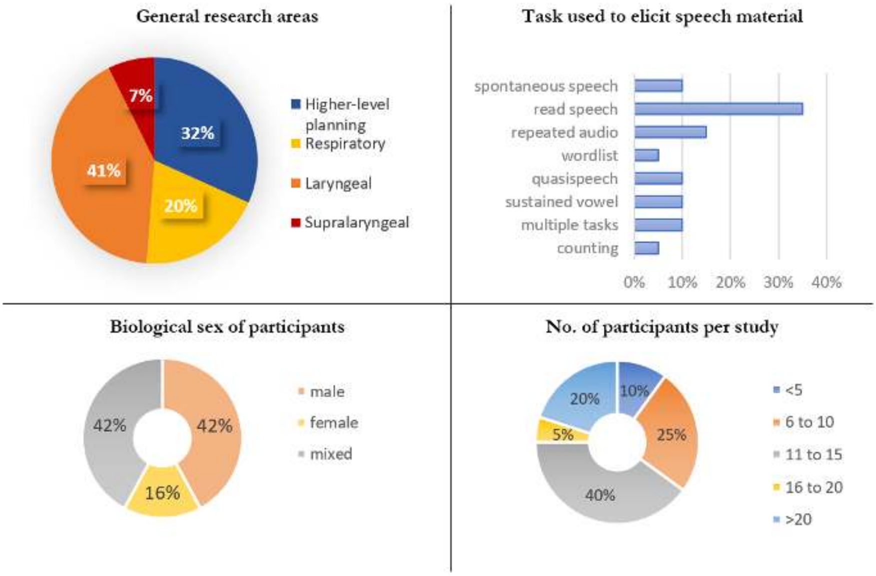

Figure 2.7: Overview of research topics and methodological choices in studies of speech during physical activity

physical activity. However, significant knowledge gaps remain, particularly regarding the specific mechanisms of respiratory adaptation that enable speech during exercise. Conflicting findings on laryngeal and supralaryngeal adjustments highlight the need for further research to clarify the underlying mechanisms. Moreover, heterogeneous methodologies have left key questions unresolved, particularly concerning the effects of different exercise levels and speech tasks, as well as the extent of speaker differences. The present study aims to provide a more comprehensive understanding of speech during exercise by assessing changes in breath support, voice quality, and articulatory precision. The methodological approach will thus incorporate different speech tasks and exercise intensities, combining empirical analysis of respiratory and acoustic data with qualitative explorations to examine speech during exercise from different perspectives.

Emerging from the identified gaps in the literature, the following research questions will be addressed:

(1) How does exercise intensity influence respiratory patterns during simultaneous speech? Specifically, do speakers increase breath rate, breath depth, or both?

(2) What are the effects of exercise on laryngeal function, specifically on measures of fundamental frequency, intensity, and voice quality? Which glottal source effects characterize speech during exercise?

(3) How does exercise impact articulatory precision and acoustic vowel space?

3 Methodology: Data collection

A database of speech, respiratory and motion-capture recordings was created to investigate how physical activity influences concurrent speech. Data were collected in a controlled laboratory setting using a detailed experimental protocol and participant questionnaires. The database consists of signal files, annotation files, and metadata. This chapter describes the data collection methods, covering participants, speech materials, the laboratory setup, the experimental procedure, and the guidelines for transcribing and segmenting speech recordings.

3.1 Participants and sample size

3.1.1 Participants

Data were collected from 48 healthy, non-smoking female native speakers of standard German, aged 18 to 35. All participants gave informed consent prior to the experiment. No participants reported any speech, hearing, or breathing pathologies, nor any physical disabilities that would prevent them from riding a bicycle during the experiment. Participants were compensated €25 for taking part in the 2-hour experiment.

Participants were recruited using the LingEx study database.[1] Invitations to take part in a study on speech during exercise were emailed to all individuals meeting four criteria: (1) female,[2] (2) aged 18–35, (3) non-smoker or occasional smoker (≤3 cigarettes/day), and (4) native speaker of standard German.

The decision to include only female participants was based on practical and scientific considerations. Significant differences in respiratory physiology (LoMauro & Aliverti 2018) and breathing during exercise (Horiuchi et al. 2019, Sheel et al. 2004) exist between males and females. The potential impact of these differences on the measured variables would necessitate separate analyses for each sex.

[1]Managed by Leibniz-Zentrum Allgemeine Sprachwissenschaft (Berlin, Germany), the database had a pool of around 1,700 active users when the study was conducted. Of these, 701 users met criteria for inclusion and were sent a weekly recruitment email with a registration link during August and September 2020.

[2]While it is acknowledged that sex is not binary, this wording refers to that used in the recruitment text (“weiblich”) and the way in which the respondents self-identified.

Given the resources available for the study, recording participants of both sexes would have resulted in smaller sample sizes for each group, reducing statistical power. To maintain adequate statistical power, a larger number of participants from a single sex was recruited. Female participants were selected for this study to align with the participant demographics used in recent research on speech during physical activity and facilitate comparison of results. Age was restricted to adults under 35 to control for age-related changes in lung volume and function that typically begin after this age (Zeleznik 2003). Occasional smokers (≤3 cigarettes/day) were accepted to widen the pool of respondents. Of the 48 participants, 40 reported being nonsmokers and 8 reported being occasional smokers. Participants self-selected as speakers of standard German.

3.1.2 Sample size

The current study adopted a pragmatic approach to determining sample size because existing literature did not report effect sizes, precluding the traditional *a priori* power analysis described by Cohen (1988) and implemented in software such as G*Power (Faul et al. 2009). Following Brysbaert's (2019) proposal, which drew on recent replication and metastudies in psychology, the current study took an effect size of d = .4 as a reasonable estimate for detecting a non-negligible effect. This aligns with Plonsky & Oswald's (2014) recommendation that d = .4 be considered a small effect in L2 research, a field adjacent to the present one.

For a comparison of two within-participants conditions with 80% power and an effect size of d = .4, a minimum of 50 participants is required. This was set as the target of the current study, and multiple observations per condition per participant were planned to further increase power. To accommodate counterbalancing of speech task order, 60 participants were initially registered. Due to cancellations (12 due to illness and 4 no-shows), an additional 4 participants were recruited through convenience sampling. Data collection was halted in early October due to rising COVID-19 cases in Berlin, resulting in a final sample of 48 participants.

3.1.3 Data protection

To ensure data protection, participant data were anonymized. Each participant was assigned a random three-letter code before recording, and all associated data (including questionnaires and recordings) were stored under this code. Participants' names were documented only on the signed informed consent forms, which were stored separately from the experimental data and were not included in any published materials. This procedure ensured that participants' identities could not be linked to their recordings. Participants were also asked to sign a second consent form to permit the publication of their data. In total, 43 participants consented to the publication of their speech, respiratory, and motion-capture recordings.

3.2 Speech materials

Three types of speech data were collected: (1) sustained vowels, (2) a read passage, and (3) unscripted monologues. Each type is described below. The inclusion of multiple speech tasks served two methodological purposes. First, different tasks enable multiple lines of analysis. For example, sustained vowels are ideal for assessing changes in voice quality, while monologues provide opportunities to assess disfluencies during exercise. Second, the same measures can be compared across different types of speech material, which helps determine whether findings are conditioned by speech task. Such methodological insights are important for interpreting the present study's results as well as designing future studies.

3.2.1 Sustained vowels

A *sustained vowel* refers to the continuous phonation of an isolated vowel, during which the vocal tract configuration remains relatively fixed and the vocal folds maintain a relatively steady vibration (Van Soom & de Boer 2020). Sustained vowels are often elicited in clinical contexts to obtain isolated vowel samples free from the coarticulatory and prosodic influences of connected speech. In this study, sustained vowels were recorded to provide a highly controlled context for investigating voice quality and vowel space during physical activity.

The corner vowels /i a u/ were chosen to enable investigations of both high and low vowels and the overall vowel space. Each vowel was elicited using the carrier phrase *sie sagt* /V/ 'she says /V/' to create a more speech-like context – inspired by the "sentence frame" approach described by Iwarsson et al. (2020) – and to produce additional tokens of the voiced fricative /z/ for separate analysis. Before the experiment, participants were trained to sustain the vowels at a consistent pitch for 2–3 seconds prior to the experiment. Immediately prior to recording, participants were instructed: "Read the phrase and draw out the vowel until the presentation screen goes blank." Three repetitions of each vowel were elicited in random order in each condition, resulting in a total of 27 vowel tokens per speaker (3 repetitions × 3 vowels × 3 conditions).

3.2.2 Read passage

A read passage was included to elicit controlled connected speech. The passage was designed to approximate spoken language, making it suitable for comparison with the unscripted monologues in discourse-level analyses (e.g., pause placement). The passage was excerpted from unscripted speech as follows: a female German speaker (24 years old), recruited via convenience sampling, was prompted to spontaneously respond to the question *Worauf muss man achten, um ein perfektes Fest zu organisieren?* ('What does one need to organize a perfect party?') The speaker recorded her response as a WhatsApp voice message. The recording was transcribed, disfluencies were removed, and punctuation was added to reflect the spoken text. The first section, a 126-word passage, was selected for use. The passage is shown below as it appeared on the presentation screen (33-point font, 20-inch monitor; Times New Roman; single-spaced):

> Um ein perfektes Fest zu organisieren, braucht man eigentlich gar nicht so viel. Vor allem braucht man eigentlich gar nicht so viel Zeit, man kann das eigentlich auch relativ spontan machen, so lange Läden noch offen haben. Wichtig ist einfach, dass man gute Laune hat, dass auf dem Fest gute Stimmung herrscht, deshalb ist die Gästeliste etwas sehr Wichtiges. Die Leute, die man einlädt, sollten keinen Streit untereinander haben, sie sollten sich alle untereinander ganz gut verstehen, vielleicht auch teilweise schon kennen, jeder sollte irgendeine Person haben, die er als Ansprechpartner haben kann und sich nicht einsam fühlt. Und außerdem sollte natürlich für das leibliche Wohl gesorgt sein, also sollte man sich vorher Gedanken machen was man zu Essen anbietet, das vorher einkaufen und zubereiten.

The English translation (by the author) is as follows:

> 'To organize a perfect party you[3] actually don't need all that much. And you actually don't even need that much time, you can actually do it pretty spontaneously as long as the shops are still open. The important thing is just being in a good mood, having a good atmosphere at the party, which is why the guest list is very important. The people you invite shouldn't have any issues with one another, they should all get along well, maybe some already know each other, everyone should have a person they can talk to and not feel alone. And of course food and drinks should be provided, so you should think about what to serve beforehand, buy it, and prepare it.'

Participants were familiarized with the text prior to the experiment. The experimenter explained that the passage had originally been spoken, making it different from typical written texts. Participants then read the passage aloud once to become comfortable with the text. They were instructed: "You may read the passage as if you were speaking or in any way that feels natural. If you deviate from the text, you can reread the section if you like, but it's not necessary. We are interested in how people talk in real life, so feel free to read the passage in your own way." The passage was subsequently read three times in each condition, yielding a total of 10 repetitions per speaker – 1 practice reading and 9 recordings for analysis (3 repetitions × 3 conditions) – totaling roughly 5.5 hours of read speech.

3.2.3 Unscripted monologues

Unscripted monologues refer to single-speaker narratives that are produced without any notes or preparation. Unscripted speech was elicited to obtain speech samples that are closer to each participant's natural speaking style. This approach also captures idiosyncratic speech features that might not emerge in more structured tasks, such as map tasks, which constrain the lexicon or linguistic structures. Monologues were chosen over dialogues to avoid the influence of turn-taking, which impacts breath timing in conversation (Rochet-Capellan & Fuchs 2014, Włodarczak & Heldner 2020). At the same time, findings from monologues were expected to generalize well to interactional speech, because longer speaking turns often appear in conversation and because breath volumes do not significantly differ between spontaneous monologues and dialogues (Winkworth

[3] Note that the impersonal pronoun *man* 'one', which is common in spoken German, has been translated as 'you', which is more idiomatic in informal spoken English.

et al. 1995). The unscripted monologues were elicited using nine stimulus questions about general preferences (*What is your favorite season and why?*) and opinions (*How do you feel about virtual lectures/meetings?*). The questions were selected based on an online pre-study, conducted as follows: Thirty female native German speakers (aged 18–35 years), recruited through an online platform (clickworker.com), were asked to rate a set of questions in terms of emotional neutrality and ease of discussion. They were compensated €2 for their participation (estimated time: 8 minutes). These participants did not take part in the main experiment. The survey aimed to identify topics that would (a) be easy to talk about to generate around 2 minutes of speech and (b) not evoke strong emotions, which can alter both respiratory behavior (e.g., Boiten 1998, Homma & Masaoka 2008, Jerath & Beveridge 2020) and phonatory parameters such as fundamental frequency and voice quality measures (e.g., Banse & Scherer 1996).

The questions for the pre-study were developed collaboratively with native German speakers to ensure suitability for the target demographic, because the author is neither a native speaker of German nor within the age range of the target group. A male native German speaker (37 years old) helped brainstorm initial ideas, and the first 10 entries from a Google search on "small talk questions" were used to create a list of 50 questions. A female native German speaker (25 years old) then selected 23 questions based on their perceived emotional neutrality and ease of discussion. Seven emotionally charged or controversial questions (e.g., *What is the greatest disappointment of your life?*) were included as fillers to reduce monotony and counteract survey fatigue, resulting in a final list of 30 questions for the pre-study. The survey thus consisted of 75% neutral "small talk" questions that participants could address with little mental or emotional engagement and 25% filler items.

The study was conducted online using a German online survey platform (soscisurvey.de). Following the design in Mitchell et al. (1996: 96), participants were asked to rate each question on a Likert scale of 1 to 5, with 1 indicating "Not at all emotional" or "Not easy to talk about" and 5 indicating "Extremely emotional" or "Very easy to talk about." The questions were presented in randomized order to each participant.

Responses from 26 participants (mean age: 26.3; range: 18–32 years) were analyzed. One participant was excluded due to missing data, and three others were excluded for providing dubious responses (e.g., giving the same rating to all questions). Questions with a mean rating exceeding 3 for the emotional criterion were excluded. The remaining items were ranked according to their mean ease-of-discussion ratings. This yielded a shortlist of 15 topics, of which four were excluded – one because it was a filler question and three because they required responses in the conditional tense. The remaining 11 questions were piloted with two female participants, leading to the exclusion of one topic and the rephrasing of two others for clarity. The final set of nine questions used in the study is shown in Table 3.1. The topic for the reading task, shown in the last row, was selected based on emotional neutrality.

Table 3.1: Questions with mean ratings for discussion ease and emotional impact (scale: 1=low, 5=high)

Questions	*Easy to talk about*	*Emotional*
What kind of vacations do you like? How does your idea of a nice vacation differ from what your friends, family, etc. prefer?	4.0	3.3
How has your daily routine changed since the pandemic?	3.7	3.0
Which phone app do you use the most and why?	4.0	2.5
What is your favorite season and why?	4.3	3.0
In your experience, does technology make life easier or more complicated?	3.7	2.6
How do you feel about virtual lectures/meetings?	3.7	2.6
What do you do to relax after a strenuous event or project?	3.5	3.0
Describe what you like about Berlin. What's missing in Berlin?	4.3	3.5
What is your idea of a nice weekend?	4.2	3.5
What does one need to organize a perfect party?*	3.2	2.7

*This question was used to generate read speech, while all others generated spontaneous speech.

For the main experiment, question order was randomized per participant. To familiarize participants with the task, instructions emphasized that there was no "correct" way to respond:

> "Answer the question in any way you like. There is no right or wrong way to respond, and your answers will not be judged. There is also no right or wrong way to speak. We are interested in how people talk in real life, so

> anything you say is fine. The recording will last for two minutes. You don't need to talk continuously. Feel free to pause when you wish. When the screen goes blank, finish your thought and stop speaking."

Each participant responded to three questions per condition, resulting in 9 monologues of approximately 2 minutes each (3 questions × 3 conditions) per speaker. In total, roughly 14 hours of unscripted speech was recorded.

3.3 Laboratory setup and equipment

3.3.1 Laboratory setting

The data were collected at Leibniz-Zentrum Allgemeine Sprachwissenschaft (ZAS) in Berlin, Germany, between 18 August and 1 October 2020. This was during the COVID-19 pandemic, and the laboratory setup was adapted to prevent transmission of the virus.[4] To comply with COVID-19 safety measures, participants were alone in the recording room, while the experimenters monitored the recordings from the hallway. Participants were informed that the experimenters were listening via headphones and were asked to speak in a normal voice. A camera was set up in the recording room to allow the experimenters to monitor participants' physical well-being remotely.

The exercise task was performed on a low-noise ergometer (daum electronic, Fürth, Germany) under thermoneutral conditions (22 °C, 40% humidity). Temperature and humidity were recorded with a digital hygrometer. The speech materials were presented on a 20-inch monitor at eye level. The camera was positioned in a corner of the room, outside of the participants' line of sight. Figure 3.1 shows the laboratory setting.

3.3.2 Speech recording equipment

Speech was recorded at a sampling rate of 22,050 Hz using a head-mounted condenser cardioid microphone (OPUS 54.16/3, beyerdynamic, Heilbronn, Germany) equipped with a mini foam windscreen. The microphone was placed 4 cm from the corner of the mouth at a 45° angle. For a subset of 12 speakers,[5] vocal intensity was recorded with an omnidirectional measurement microphone (MiniSPL, NTi Audio, Schaan, Liechtenstein). The microphone was positioned 30 cm from

[4] No cases of COVID-19 were reported by participants or experimenters.

[5] The calibrated microphone was acquired late in the experiment, so only the last 12 speakers could be recorded.

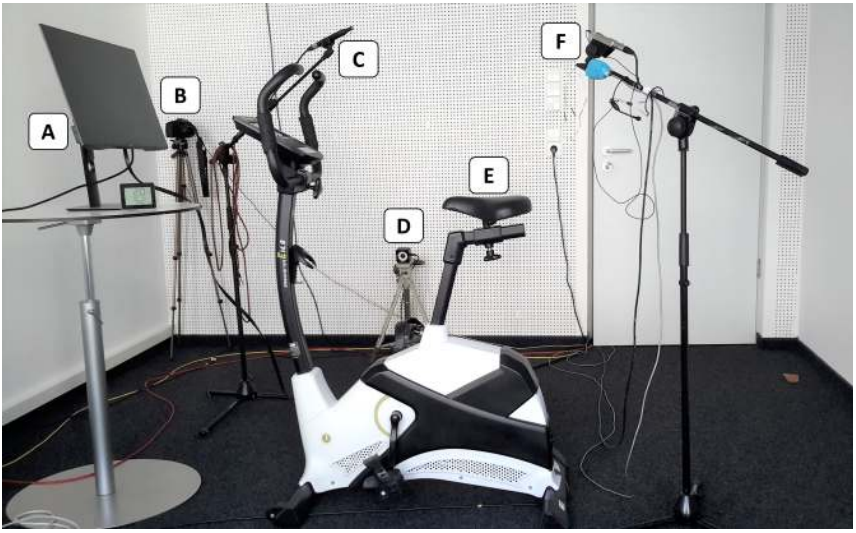

Figure 3.1: Laboratory setting with equipment: (A) presentation screen; (B) live-feed camera; (C) calibrated microphone; (D) motion-capture camera; (E) low-noise ergometer; (F) head-mounted microphone with battery pack on mic stand

the participant's mouth at the participant's preferred cycling posture, which was maintained across recordings. Sound pressure levels (SPL) were measured using an AL1 Acoustilyzer (NTi Audio, Schaan, Liechtenstein) as equivalent continuous sound level in dB (Leq, A-weighted[6]).

3.3.3 Respiratory data recording equipment

Respiratory data were recorded at a sampling rate of 22,050 Hz[7] using respiratory inductance plethysmography (RIP), a highly sensitive, non-invasive method that captures the movements of the chest and abdomen during breathing. The expansion and contraction of the chest and abdomen directly reflects the volume of each inhalation and exhalation. The RIP data were recorded with the Inductotrace system (Ambulatory Monitoring, Ardsley, NY, USA). Shown in Figure 3.2, the system comprises (a) elasticated belts encircling the rib cage and abdomen that contain an insulated wire; (b) an oscillator module; and (c) a calibrator box.

[6] A-weighting adjusts instrument-measured sound levels to approximate human perception of sound level.

[7] Note that respiratory data are generally recorded at much lower sampling rates; this rate was determined by the recording setup, which simultaneously recorded speech data. See Section 4.2.2 for details on downsampling and filtering.

The rib cage belt was positioned around the chest just under the armpits, and the abdomen belt was positioned over the belly button but below the last rib. If the participant was wearing a loose or slippery top (e.g., spandex sportswear), double-sided tape or safety pins were used as described in Sperry & Klich (1992: 1248) to counteract slippage of the belt without interfering with its expansion and contraction. The belts were connected to the oscillator module at the participant's back to be less cumbersome, and the oscillator was affixed to the arm of a microphone stand.

Figure 3.2: Participant fitted with Inductotrace belts: front view (*left*) and rear view showing the connection to the oscillator module (*right*)

RIP captures the movements of the chest and abdomen as a series of changing voltage values over time. The technique relies on *inductance* – the property of a wire to oppose a change in the current flowing through it. The wire is attached to the belts in a zigzag shape to allow it to expand and contract with each breath. An alternating current is sent through the wires, generating a magnetic field that is specific to the area enclosed by the belts (i.e., the circumference of the chest and abdomen). When that area expands or contracts during breathing, the shape of the magnetic field continually changes with it, which in turn generates a series of opposing currents. When the current enters the oscillator module, it changes the frequency of oscillation of the oscillator circuit. The oscillator module outputs this change in frequency to the calibration box, where demodulation circuitry produces an electric voltage that is proportional to the circumference of the chest and abdomen. When plotted over time, the changes in voltage values result in a series of peaks and troughs that reflect the duration and depth of each breath. The resulting voltage signals can then be analyzed directly or calibrated and normalized using different chest-wall maneuvers.

3.3.4 Motion-capture data recording equipment

An OptiTrack motion-capture system (Prime 13 cameras; Corvallis, OR, USA) was used to record the pace of pedaling. A motion-capture marker was affixed to one bike pedal, and a motion-capture camera was positioned 2 meters from the ergometer, approximately level with the pedal. Further motion-capture cameras were positioned around the perimeter of the room at ceiling height. The motion-capture data are not analyzed in this study.

3.3.5 Signal path and file synchronization

Audio, respiratory, and motion-capture signals were recorded using two computers and a multi-channel data acquisition system (Zodiac Aerospace DIC6B, ADM Meßtechnik; Gelnhausen, Germany). The signal path is shown in Figure 3.3. Motion-capture data were recorded and displayed on Computer 1 using OptiTrack's Motive software. When recording started, Motive sent a synchronization impulse to channel 1 of the data acquisition system. Channels 2 and 3 received two respiratory signal inputs – from the rib cage and abdomen, respectively – from the RIP calibrator box. Channel 4 received the audio signal via a mixer (Behringer Ultragain MIC2000; Willich, Germany). To monitor the signals for disturbances, these four channels were displayed in real time on Computer 2 using the EWinView and EdWin software packages (versions 1.94 and 5.15, respectively; MH, Erftstadt, Germany). During the experiment, recording was started for the respiratory and audio signals first and then for the motion-capture signal. The synchronization impulse generated by the motion-capture software was then used to align the start and end of the audio and respiratory recordings to the motion-capture recording duration before saving the files.

3.4 Experimental design and conditions

3.4.1 Experimental design

This study employed a repeated-measures design – each participant was tested under each experimental condition. The three physical-activity conditions (described below) were presented in fixed order: rest, low workload, and moderate workload. Because the physiological response to exercise persists for some time following activity (Romero et al. 2017), and responses may vary greatly between individuals (Travers et al. 2022), randomizing condition order would have made it difficult to ensure that rest captured an individual's speech and breathing at baseline, that is, with no physical workload.

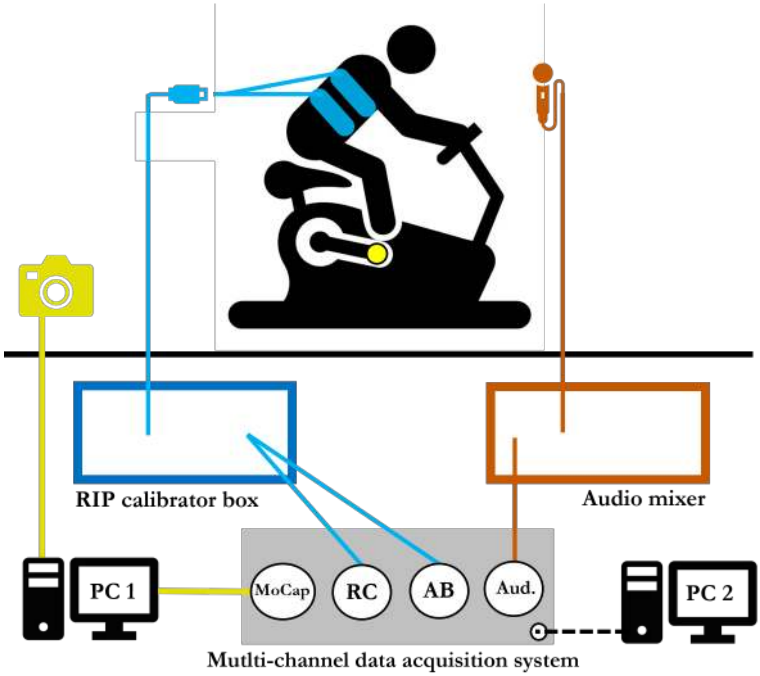

Figure 3.3: Schematic drawing of the signal path: motion-capture path shown in yellow, RIP path in blue, audio in orange, and data acquisition system in gray

The order of the three speech tasks – producing sustained vowels, reading a passage, and producing unscripted monologues – was alternated between participants to minimize order effects such as boredom or fatigue in later trials. Each participant was randomly assigned one of the six possible task orders (vowels–reading–monologues, monologues–reading–vowels, etc.). Each task order had an equal number of participants (eight).

3.4.2 Definition of experimental conditions

Exercise was performed using a low-noise cycle ergometer (daum electronic, Fürth, Germany). Cycling was chosen for the experiment because it is a weight-supported aerobic activity that stabilizes the upper body, thereby minimizing the potential for non-breathing movements to contaminate the respiratory recordings. Two levels of exercise intensity – low and moderate – were chosen to approximate the physical workload required by daily activities (e.g., walking) and some occupations (e.g., emergency services). The workload level for each condition was defined using a relative measure of exercise intensity,[8] percentage of heart rate reserve (HRR), calculated as the difference between maximum heart rate (HR_{max}) and resting heart rate (HR_{rest}). Low workload was defined as 35% of HRR and moderate workload as 65% of HRR, based on public health guidelines (e.g., Centers for Disease Control and Prevention 2022). These relative levels were converted into participant-specific target heart rates (beats per minute, or bpm) using a standard method in sports science, the Karvonen formula:

$$\text{target HR} = [(HR_{\text{max}} - HR_{\text{rest}}) \times \%\text{intensity}] + HR_{\text{rest}}$$

Different versions of the formula are in use. The version used here makes two adjustments for greater accuracy in estimating heart rate for each exercise level. First, maximum heart rate (HR_{max}) was estimated following Tanaka et al.'s (2001) equation: $208 - (0.7 \times \text{age})$. Second, resting pulse (HR_{rest}) was approximated using a 10-minute resting period (see Section 3.5.1) rather than predicted by age. This approach ensured that the physical effort required to exercise at the target exercise intensity was comparable across participants, regardless of their age or level of fitness.

To validate participants' perceptions of the low and moderate workloads, participants were asked to rate their level of physical effort for cycling only. This was done using the Borg Rating of Perceived Exertion (Borg 1982), a 15-point scale ranging from 6 to 20. The average rating for low workload was 9.7 ("very light") and for moderate workload 13.8 ("somewhat hard").

3.5 Recording procedure

Participants were in the lab for approximately 2 hours. Preparation for the experiment took approximately 1 hour, and the experimental data were recorded in the second hour. Figure 3.4 shows the approximate timeline of the experiment. Details of each phase are given below.

3.5.1 Pre-recording (preparation) phase

The first 50 minutes were used for preparation, comprising four stages: welcome and informed consent; collection of metadata and familiarization with speech

[8] Some previous studies defined physical conditions using absolute measures (e.g., miles per hour, power output in watts). However, absolute measures do not account for individual exercise capacities, and thus the physical effort required for a given condition will differ between participants, adding uncontrolled variation.

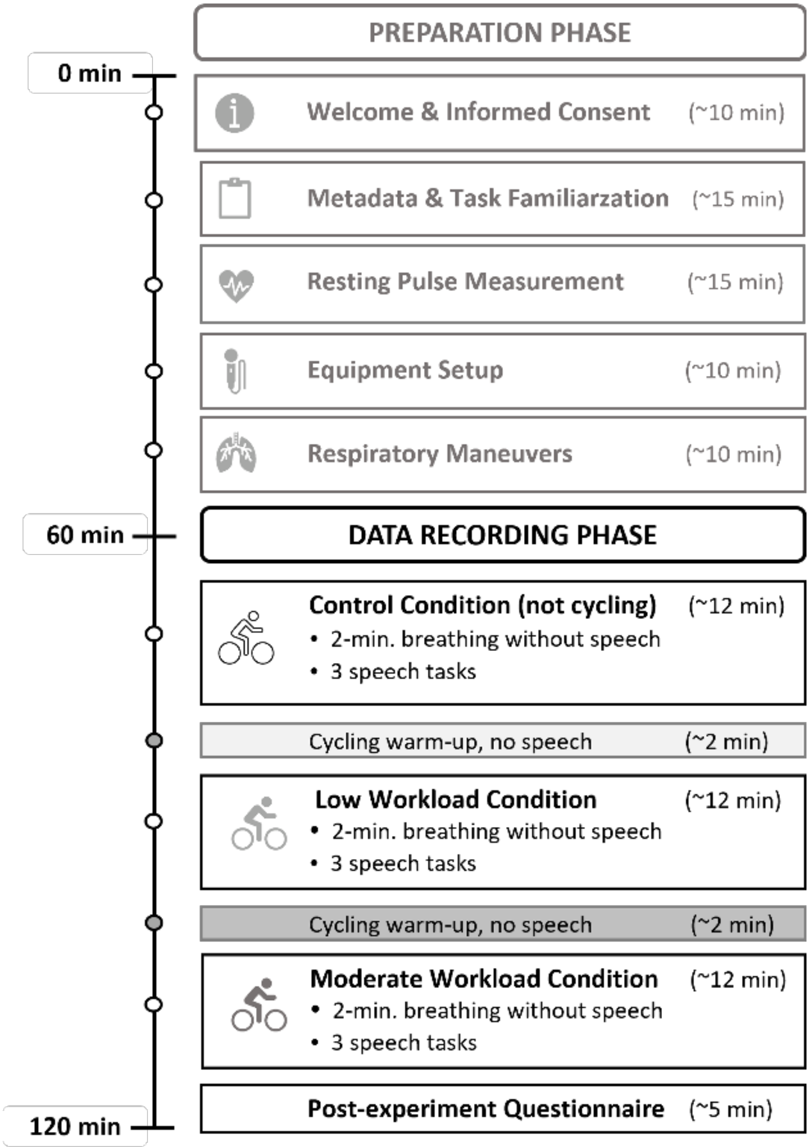

Figure 3.4: Timeline of the experiment, showing the different stages of the preparation and data recording phases

tasks; measurement of resting pulse; and participant setup with ergometer and recording equipment. All activities took place in the recording room.

Stage 1 was intended to acclimate participants to the laboratory setting and to inform them about the experiment. Care was taken to establish rapport with the participant and to encourage questions so that they felt "at home" in the experimental setting. Participants were welcomed, offered water, and asked if they preferred to be addressed with the formal or informal German 'you'. COVID-19 safety precautions were explained, and participants were given written information about the experiment and invited to ask questions. Informed consent was obtained separately for participation and data publication.

Stage 2 was used to collect metadata and introduce the speech tasks. Participants completed four written questionnaires covering various aspects relevant to the study:

(1) **Anthropometric data** were collected because some physiological characteristics are known to be related to respiratory behavior. For example, individuals with higher body mass have been found to have higher respiratory rates and lower tidal volumes (Littleton 2012). Additionally, height is associated with measures of lung capacity, at least in males (Bhatti et al. 2014). Participants self-reported[9] weight (in kilograms) and height (in centimeters) to calculate body mass index (BMI).

(2) **Participants' fitness level** was assessed using the German short form of the International Physical Activity Questionnaire (IPAQ). Fitness level affects how efficiently an individual meets increased respiratory needs during exercise (e.g., Cheng et al. 2003).

(3) **Relevant lifestyle information** was collected about experience with activities that would affect breath control (e.g., singing, playing a wind instrument), cycling performance, and reading performance (e.g., acting, reading aloud). Participants also specified whether they smoked or spoke other languages. This information was intended to identify participants who might be particularly skilled at the experimental tasks.

[9]Participants self-reported height and weight. Self-reporting was chosen because body metrics can be a sensitive issue for young women (Brunson et al. 2014) and may trigger emotional distress. Self-reporting was also deemed accurate enough for the purposes of the present research, because the anthropometric data were collected to calculate BMI values to categorize participants in terms of standard body-mass categories. The potential negative impact of weighing someone would have been greater than the value of the weight measurement.

(4) **Participant mood** was assessed before and during the experiment using a validated German version (Maibach et al. 2020) of the Feeling Scale (Hardy & Rejeski 1989), because emotional state can affect both vocal and respiratory behavior (Banse & Scherer 1996, Jerath & Beveridge 2020). The mood data were intended as additional information to contextualize individual results.

After completing the questionnaires, participants were familiarized with the three speech tasks using printed versions of the screen presentation format. They were shown the speech prompt for the reading task and asked to read the passage aloud once to ensure familiarity. The experimenter also demonstrated sustained phonation for the vowel stimuli.

Stage 3 was used to calculate individual heart rate ranges for each condition. Participants were fitted with a wrist-worn heart rate monitor and reclined on a cushioned mat in a supine position for 10 minutes to approximate their resting pulse. To maximize the potential for genuine relaxation, the experimenter left the room, dimmed the lights, and monitored heart rate remotely via a tablet device. The average heart rate during minute 11 was considered the resting pulse (HR_{rest}). These data and the participant's age were used to calculate the target heart rates for each participant for the low workload (30–40% of HRR) and moderate workload (60–70% of HRR) conditions. Ranges were calculated to accommodate short-term fluctuations in heart rate during exercise.

In *Stage 4*, participants were prepared for recording. A low-noise ergometer (daum electronic, Fürth, Germany) was adjusted to each participant's preferred configuration. A head-mounted condenser microphone (OPUS 54.16/3, beyerdynamic, Heilbronn, Germany) was placed 4 cm from the corner of the participant's mouth at a 45-degree angle away from the mouth. The RIP belts were placed around the abdomen (roughly at the belly button but below the last rib) and rib cage (underneath the armpits). The belts were fastened together at the participant's back and connected to the oscillator module mounted on a microphone stand for an unobtrusive setup.

3.5.2 Data recording

Following the preparation phase, the data recording process began with a series of chest wall maneuvers designed to calibrate and normalize the rib cage (RC) and abdomen (AB) signals. These maneuvers were performed while participants were seated on the ergometer.

The first maneuver, known as the isovolume maneuver, was used to determine the relative contributions of the rib cage and abdomen. This procedure, based on the work of Konno & Mead (1967), required participants to hold their breath and shift the inspired air between their chest and abdominal compartments. The experimenter demonstrated the technique and provided instructions in German, adapted from McKenna & Huber (2019: 4): "I want you to hold your breath, but do not take a breath in, just stop breathing. Then, while holding your breath, I will ask you to suck your belly in and let it flop out." Participants repeated this maneuver three times while wearing a nose clip to prevent air from escaping through the nose.

To normalize lung volumes across participants, vital capacity (VC) and inspiratory capacity (IC) maneuvers were also performed. These standard measures of static lung capacity estimate the volume of air an individual can move in and out of their lungs. For the VC maneuver, participants took their deepest breath possible and then exhaled as much air as possible. For the IC maneuver, participants exhaled to their typical resting level and then took their deepest breath possible. Each maneuver was performed three times, with the attempt showing the greatest RIP signal excursion selected as the participant's VC or IC value. Breaths can then be normalized across participants as a percentage of either VC or IC.

After completing the chest wall maneuvers, tidal breathing was recorded for 2 minutes to establish a resting baseline. Participants were then briefed on the upcoming speech task and encouraged to drink water as needed.

The speech materials were presented using PowerPoint, allowing the experimenter to adjust the pace of presentation for each participant. After the first trial of each task, participants were given the same positive feedback ("That's perfect, great job") to minimize self-consciousness. The only exception was the sustained vowel task. Eight participants did not initially produce a vowel with a steady pitch. In this case, the experiment was halted, and the experimenter repeated the training instructions and demonstrated a sustained vowel. If a participant did not produce vowels with a steady pitch after several attempts, recording proceeded (data not meeting criteria for analysis were excluded in a later step).

The control condition was followed by a cycling warm-up of approximately 4 minutes, following Baker et al. (2008), to transition from rest to low workload. After participants reached the lower end of the target heart rate range (after approximately 1–2 minutes), breathing without speech was recorded for 2 minutes. The speech tasks were presented in the same order as the control condition. To ensure participants stayed within their target heart rate range, the experimenter monitored heart rate using a wrist-worn heart monitor (Scosche Rhythm 1.9, Scosche Industries, Oxnard, California, USA) and a tablet (Galaxy SM-T800, Samsung, Seoul, South Korea).[10] If participants moved out of their target heart rate range, the resistance on the ergometer was decreased or increased as needed between trials. If the lab temperature rose above 24 °C, or at the participant's request, the windows were opened between task blocks and fans were used to cool the room.

After each speech task was recorded, participants rated their mood on the Feeling Scale (Hardy & Rejeski 1989). The warm-up procedure was repeated to reach the moderate-workload heart rate. After the speech tasks were recorded, participants completed a second Feeling Scale form, after which they were disconnected from the recording equipment and given the opportunity to freshen up. Throughout the recording process, care was taken to maintain consistent procedures while adapting to individual participant needs, ensuring high-quality data collection for subsequent analysis.

3.5.3 Post-recording

After completing the experiment, participants rated their perceived physical exertion levels for each exercise condition using the Borg Scale (Borg 1982). Participants also completed a written qualitative questionnaire, providing free-form answers about their subjective experience of speaking while cycling. Participants were asked questions such as: "How difficult was it for you to say the vowels/read the text/speak freely while cycling in the first round? In the second round?" Additional questions explored whether they felt that cycling affected their speech and whether they experienced any discomfort or breathlessness. Participants were then given the opportunity to ask questions and offered an optional oral debriefing on the purpose of the experiment. After each participant left, all equipment was disinfected, and the lab was ventilated for at least 1.5 hours to minimize the risk of COVID-19 transmission.

3.6 Data processing

The recording setup yielded two types of raw data that were transformed for further analysis. The audio signal, two RIP signals (one rib cage, one abdomen), and the motion-capture synchronization signal were recorded together and saved as a single binary file in Inductotrace's proprietary file format for each trial. These

[10] A wrist-worn monitor was chosen for hygienic reasons. Wrist-worn heart rate monitors are comparable to chest-strap monitors in terms of accuracy and have a level of error that is clinically acceptable (Sartor et al. 2018: 8).

files were imported into MATLAB and converted into three separate WAV files for analysis in Praat: one audio recording, one rib cage respiratory recording, and one abdomen respiratory recording. The synchronization signal was used to trim the WAV files to match the duration of the motion-capture recordings. The motion-capture data were recorded in the Motive software's TAK format and exported in C3D format.

3.7 Transcription and segmentation of speech material

All speech recordings were transcribed and segmented into stretches of speech and pauses in Praat to prepare the data for further analysis. This section describes the semi-automatic transcription pipeline, transcription conventions, and criteria used to segment pauses.

3.7.1 Transcription and segmentation pipelines

Each type of speech task was transcribed using distinct processes. Sustained vowels were labeled and segmented manually at the onset and offset of periodicity in the waveform. The connected speech tasks – reading and monologues – were transcribed using a combination of automatic processes for speed and human input for accuracy. Two annotators were involved: the author (Annotator 1) and a native German-speaking research assistant (Annotator 2). Each transcription was seen by both annotators in a serial review process, schematized in Figure 3.5.

First, a Praat TextGrid was created for each file using the "Annotate to Silence" function, which segments the audio signal into stretches of speech and pauses. The annotators then added the transcriptions manually. For the recordings of the read passage, one annotator manually copied the relevant parts of the text into the appropriate TextGrid intervals. Any deviations from the text – repetitions, false starts, etc. – were transcribed by hand. The second annotator then reviewed the transcription and manually adjusted the pause boundaries. Each annotator completed the first step for half of the speakers; for the second stage, the annotators swapped speakers, ensuring that each annotator reviewed every speaker's transcription. Any disagreements were resolved through discussion.

For the monologues, an additional step was introduced: each trial was automatically transcribed using the Fraunhofer ASR service – the FhG IAIS ASR engine – accessed via the BAS WebService, provided by the Bavarian Archive for Speech Signals and hosted by the Institute of Phonetics and Speech Processing at Ludwig-Maximilians-Universität, Munich, Germany. The ASR transcriptions were output as text files. Annotator 2 checked the transcriptions for errors, and then Annotator 1 reviewed the transcriptions by listening to the recordings. The corrected text files were subsequently aligned with the prepared TextGrids using a Praat script. The pause boundaries were adjusted manually.

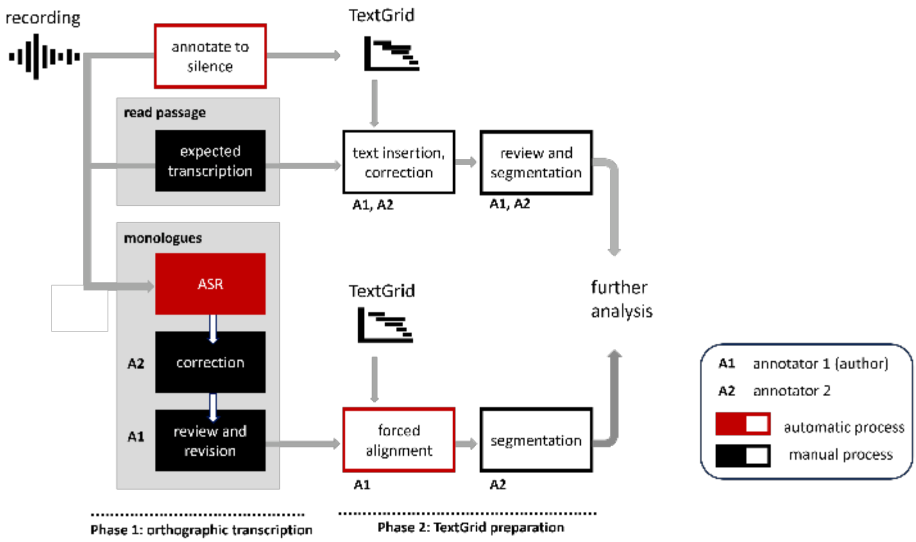

Figure 3.5: Schematization of TextGrid creation, transcription, and acoustic segmentation (adapted from Rodd et al. 2021: 746)

3.7.2 Transcription conventions

All speech was transcribed orthographically to make the material accessible for diverse research interests. Because orthographic transcription does not impose any analysis on the speech material, it provides a better basis for comprehensive searches compared to other transcription approaches, which may unintentionally exclude relevant phenomena (Deppermann 2010). To capture the speech as accurately as possible, disfluent speech was transcribed in detail. This included interrupted words (e.g., *be-*) and sublexical errors (e.g., *Tagesaflauf* instead of *Tagesablauf*). Table 3.2 provides an overview of the transcription conventions.

3.7.3 Segmentation of pauses

Connected speech was segmented into stretches of speech and acoustic pauses. Pauses were defined as lasting more than 100 ms or more than 130 ms before a

Table 3.2: Overview of transcription decisions

Category	Decision
Capitalization	Nouns only ***Note:*** Per German orthographic rules
Disfluencies	Interrupted words: Indicated with a hyphen (e.g., *ich ge- ich ging*) Sublexical errors: Transcribed as heard (e.g., *Tagesaflauf*) Intraword pauses: Indicated with double hyphens (e.g., *Frühlings- -gefühle*) ***Note:*** Annotated if pause exceeded 130 ms
Filled pauses	Standard forms: *äh, ähm, hm, mm* Other forms: rendered with 2–3 letters (e.g., *ff, aa, pff*)
Laughter	Transcribed as <laugh> when produced separately from speech (e.g., *nun* <laugh> *also*) ***Note:*** Laughter overlapping with speech was not transcribed
Lexicon	Informal register: Transcribed if listed in *Duden*, the standard German dictionary (e.g., *sone, gerne, irgendwas*) Regional usage: Transcribed (e.g., *ebend, jetzte*)
Non-speech sounds	Placed in angled brackets (e.g., <sigh>, <cough>)
Punctuation	None
Unintelligible speech	Placed in double angled brackets: «unintelligible»
Metaspeech	Enclosed by square brackets: [metaspeech.start] … [metaspeech.end] ***Note:*** Speech excluded from analysis, e.g., when participants addressed the experimenter

plosive to account for stop closure duration (Hieke et al. 1983). The acoustic onsets and offsets of each stretch of speech were determined following the guidelines in Keating et al. (1994). Additional guidance on handling specific onset and offset phenomena, such as palatal noise, was drawn from Machač & Skarnitzl (2009), while Conrad (2019) was consulted for illustrated examples of German phonemes.

Vowels, sonorants, and plosives were generally segmented based on the waveform. For voiced segments, the acoustic onset was defined as the onset of periodicity in the waveform, with the boundary placed at the zero-crossing of the first upward-going element. The same criterion was applied for offsets, with the boundary placed at the zero-crossing of the last downward-going element. For plosives, which can be released in different ways, from no detectable burst to one or multiple bursts, segmentation followed these rules: If no burst was visible, the boundary was placed at the onset of voicing. If multiple bursts were visible, the boundary was placed at the zero-crossing of the first transient (Conrad 2019: 31–32). Fricatives and aspirated segments were segmented using the spectrogram. Boundaries were placed based on visual inspection. The dynamic range was adjusted as needed to identify noise.

In some cases, creak or breath noise was present at the edges of speech stretches, complicating the identification of precise acoustic onsets or offsets. Creak – characterized by isolated or irregular glottal pulses – primarily occurred at speech onset, although some speakers also produced creaky offsets. Glottal pulses were included as part of the speech stretch if they immediately preceded or followed the onset or offset proper. Isolated glottal pulses (i.e., occurring in the middle of an otherwise silent stretch) were transcribed as <creak> if they numbered three or more.

Audible exhalations sometimes occurred phrase-finally during the physical workload conditions. When a stretch of speech ended with an aspirated plosive or fricative, the aspiration or frication was often seamlessly followed by exhalation noise. In many cases, however, it was still possible to infer the release of the articulatory configuration – the opening of the vocal tract – from a sharp change in the formant structure and amplitude of the waveform. The boundary was placed at this point to demarcate the end of the speech stretch.

4 Respiratory level: Speech breathing during exercise

4.1 Introduction

As established in the survey of the research landscape given in Chapter 2, speech and exercise place opposing demands on the respiratory system. Speech requires extended exhalation phases to support utterances, whereas exercise requires rapid breathing to meet increased metabolic needs. When exercise intensity increases, speakers must reconcile these conflicting requirements, balancing faster breathing with the temporal demands of speech.

While previous research has documented general physiological responses to combined speech and exercise – including increased respiratory rate, deeper breaths, and higher subglottal pressure – it remains unclear exactly how speakers adjust their breathing patterns, since breath volume and frequency can be modulated independently. Speakers could increase breath volumes to meet increased gas exchange demands without altering breath frequency, or maintain breath volume while breathing faster to achieve the same metabolic goal.

This raises a critical question: how do individuals coordinate adjustments in breath depth and frequency to speak during exercise? Specifically, when faced with greater metabolic demands, do speakers adjust breath rate, breath volume, or both? Moreover, are these strategies used consistently across individuals, or do they vary? Understanding these adaptive patterns is essential for characterizing how speech production functions under the variable physiological conditions speakers routinely encounter.

The following sections examine the current state of knowledge on speech breathing during exercise. Section 4.1.1 describes the methods and measures used to assess respiratory patterns, providing essential background for readers unfamiliar with respiratory studies. Section 4.1.2 reviews what measures have been assessed in previous studies and their key findings, leading to the present study's objectives in Section 4.1.3. The remainder of the chapter presents the empirical study, beginning with the methodological approach to analyzing respiratory data (Section 4.2). The results and visual explorations of individual differences are then reported (Section 4.3), followed by a qualitative analysis of speaker interviews on perceived difficulty of speaking during exercise (Section 4.4). The chapter concludes with a discussion of the implications of these findings for understanding respiratory adaptations during speech (Section 4.5).

4.1.1 Investigating speech breathing: Methods and measures

In mammals, breathing occurs as an automatic rhythmic cycle of inspiration and expiration, during which air is taken into the lungs to supply oxygen to the body and is then expelled to remove carbon dioxide. Various parameters of each breath, including duration, air volume, airflow rate, and gas composition, can be quantified using several established methods. Three primary methods have been employed in previous studies of speech breathing during physical activity, depicted in Figure 4.1. The first two approaches – mass airflow sensors positioned at the mouth (in Meckel et al. 2002 and Rotstein et al. 2004) and oronasal masks (in Baker et al. 2008 and Ziegler et al. 2019) – are coupled with metabolic carts for comprehensive breath analysis. While these methods provide direct measurement of multiple respiratory parameters, they present significant methodological challenges. Both devices cover the mouth, potentially creating unusual sensory feedback for speakers and affecting natural speech behavior. Additionally, mass airflow sensors may not fully capture nasal respirations, while oronasal masks necessitate external microphone placement for speech recording.

The third method, respiratory inductance plethysmography (RIP), used by Doust & Patrick (1981) and Fuchs et al. (2015), offers an alternative approach by measuring chest wall kinematics to capture air displacement, a technique that leaves the face unobstructed. The RIP signal shows the amplitude of chest wall movements (reflecting changes in breath volume), and the duration and symmetry of each breath. Studies have shown that airflow calculations derived from RIP measurements are comparable to those obtained using direct measurement methods such as pneumotachographs (Fiamma et al. 2007). While RIP may register non-respiratory chest movements during vigorous activity, this limitation becomes negligible when participants maintain relatively stable positions, as during cycling.

Various types of measures are assessed in speech breathing research. A key volumetric measure is *tidal volume*, the volume of air that moves into or out of the lungs with each breath; it changes according to metabolic needs. *Minute ventilation* represents the total volume of air delivered over one minute. It is the product of tidal volume and *respiratory rate* (breaths per minute) and thus provides more information on respiratory behavior than either measure alone.

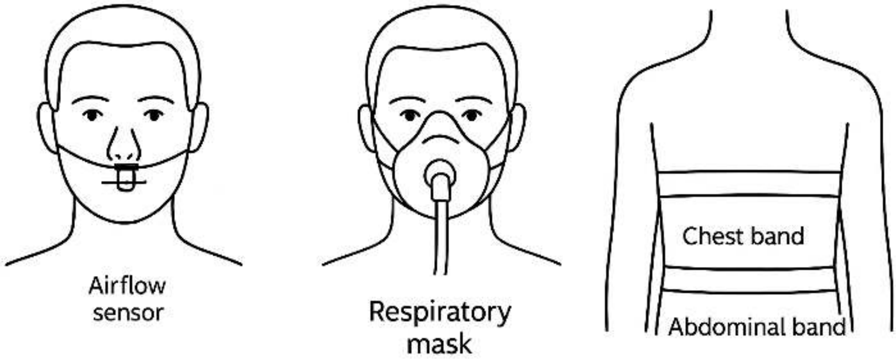

Figure 4.1: Common methods to obtain respiratory data (*from left*): mass airflow sensor at the mouth, oronasal mask, and respiratory plethysmography belts

Temporal measures provide further insights: *breath cycle duration* reflects the total time of inspiration and expiration, while *inspiratory duration* and *expiratory duration* reflect the length of each phase separately. The inspiratory fraction (Lenneberg 1967) quantifies cycle symmetry by expressing inspiratory duration as a proportion of total cycle duration (some studies refer to this as the duty cycle).

Different activities produce characteristic breathing patterns that can be distinguished through these measures. Resting breathing typically exhibits smooth, regular cycles with similar inspiration and expiration amplitudes and balanced phase durations, though some natural breath-to-breath variation occurs. In contrast, speech breathing demonstrates larger, more variable amplitudes with notably shorter inspirations and extended expirations, creating asymmetric cycles. Exercise breathing combines elements of both patterns: it maintains the regular timing and symmetric cycles characteristic of resting breathing while displaying the increased amplitude typical of speech breathing. In other words, exercise breathing is like resting breathing in terms of timing, and like speech breathing in terms of volume.

This understanding of respiratory patterns across different activities provides crucial context for investigating speech during physical exertion. Figure 4.2 illustrates how these patterns relate to the RIP record. The figure shows roughly 30 seconds of breathing in three contexts: resting breathing (top trace), speech breathing (bottom trace in black), and exercise breathing (bottom trace in gray). All three traces are shown approximately to scale. The y-axis displays the amplitude of the chest wall movements with inspiration plotted upward. The x-axis displays the duration of each breath as time progresses.

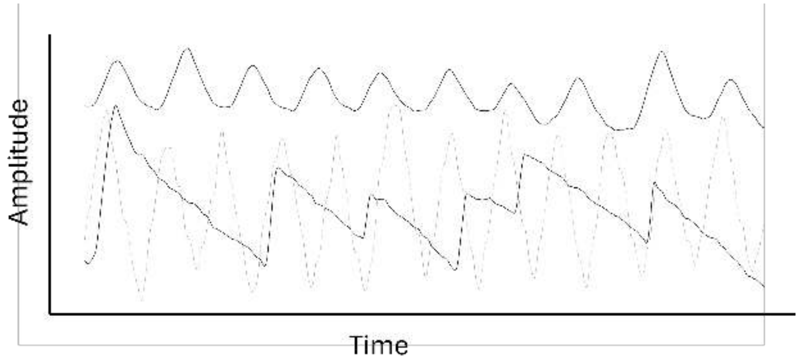

Figure 4.2: Example RIP trace signals showing resting breathing (*top*), speech breathing (*bottom*) and moderate-exercise breathing (*bottom, in gray*)

4.1.2 Speech breathing during exercise: Research history

Research examining interactions between speech breathing and physical activity has evolved significantly since the late 1960s. Early work set out to understand how speech constrains respiratory function during exercise, while subsequent studies explored the inverse relationship – how exercise influences speech breathing patterns. This section traces the chronological development of this research, synthesizing key findings and their theoretical interpretations.

4.1.2.1 Early work (1968–1981)

The foundational studies in this field were conducted by physiologists, beginning with Otis & Clark (1968). Their research was motivated by a physiologically grounded hypothesis outlining the potential for "competition" between the primary function of breathing (gas exchange) and the secondary (but important) function of speech production. As they noted: "Physiologists seem to have given relatively little attention to the interaction between these two functions of breathing, but it does seem that the control mechanism regulating airflow for phonation and that regulating the gas tensions of the body may at times be somewhat competitive and that situations may arise in which one may temporarily override the other" (*ibid.*, p. 122).

To investigate these interactions, Otis and Clark studied six participants during treadmill walking. Breathing patterns were recorded with an unspecified method before and after 5 minutes of exercise while participants performed various speech tasks (counting, reciting the alphabet, reading prose or poetry). The authors found that as minute ventilation increased, the amount of speech material per breath decreased. At the same time, total duration to complete the speech task remained constant. Taken together, these findings suggest that speakers maintained their overall speech rate by taking more frequent breaths and producing shorter utterances. The authors also noted an apparent increase in vocal intensity[1] during exercise, though measurement reliability issues warrant caution in interpreting this finding.

Doust & Patrick (1981) expanded upon this work by investigating the potential "conflict between the gas-exchange and phonatory functions of breathing" (*ibid.*, p. 138). Their study examined six male participants reading a paragraph after 5 minutes of treadmill exercise at various heart rates (85, 95, 125, 150, and 165 bpm). Minute ventilation, respiratory rate, tidal volume, and expiratory flow were assessed using the authors' own version of a respiratory inductance plethysmograph; all parameters were found to decrease during exercise with talking compared to exercise alone.

However, although the authors focused on how speech affected exercise breathing, the effect of exercise on speech breathing can also be seen in the tables and figures. As participants transitioned from walking (85 bpm) to running (165 bpm) while speaking, they exhibited substantial increases across all measures: minute ventilation (125%), tidal volume (54%), respiratory rate (67%), and expiratory flow (60%). Non-phonated expirations – mentioned in passing by Otis & Clark (1968: 128) – were also observed, but only during the last three workload conditions (equivalent to brisk walking, jogging, and running). Doust & Patrick (1981: 146) consider these expirations a major finding of the study and interpret them as evidence that speech breathing can temporarily take priority over gas exchange requirements. They proposed that ventilation is suppressed during speech and that "the transient overshoot in ventilation" (i.e., the non-phonated expiration) then restores blood CO_2 levels immediately afterward. This interpretation represented one of the first theoretical frameworks for understanding the interaction between speech production and respiratory demands during physical activity.

[1]While the authors used a "Brüel and Kjaer apparatus" (Otis & Clark 1968: 127) to measure vocal intensity, the reported mean group increase of 16.3 dB during exercise casts doubt on the accuracy of measurement.

4.1.2.2 Renewed research interest (2002–2008)

In the 2000s physiologists again took up the question of how speaking affects ventilation during exercise. A significant contribution came from a series of sports medicine studies, beginning with two papers analyzing the same dataset. Meckel et al. (2002) evaluated the impact of speech on exercise breathing using a wider range of respiratory measures than previous studies, while Rotstein et al. (2004) sought to relate participants' physiological responses to exercise to their perceived difficulty of speech production.

In the experimental protocol, 14 males ran on a treadmill for 6 minutes at exercise intensities of 65%, 75%, and 85% of their individual maximal oxygen consumption[2] (VO_2 max) with and without talking. Participants read passages at consistent loudness and metronome-guided pace while respiratory data were collected via a mass flow sensor. As in other studies of this kind, the statistical analysis of these data compared exercise breathing with and without speech, but the results tables also show how speech breathing changed with exercise intensity. From moderate (65% VO_2 max) to vigorous exercise (85% VO_2 max), minute ventilation increased by 56%, tidal volume by 17%, and respiratory rate by 18%. Temporal measures showed corresponding decreases in total breath cycle duration (15%), inspiratory duration (13%), and expiratory duration (18%) with increased workload.

Meckel and colleagues interpreted their findings through the lens of competition between respiratory functions, noting that "although speech during exercise is possible by voluntary changes in breathing pattern, it is accomplished at the expense of the physiological needs of efficient gas-exchange and energy production" (Meckel et al. 2002: 1341). A possible application of the findings was considered in Rotstein et al. (2004), which evaluated to what extent speaker perceptions of speech difficulty correlated with physiological measures of exercise intensity. Their findings revealed significant correlations between perceived speech difficulty and various measures of exercise intensity (percentages of maximal oxygen consumption, heart rate, and minute ventilation). However, considerable between-participant differences in difficulty ratings were observed, leading the authors to suggest that the perceived effort of talking during exercise was not aligned with the level of exercise intensity for all speakers. The extent of variability was not quantified, but the concluding paragraph suggests it was considerable, since the authors cautioned against using the talk test for exercise prescription because it "may result in an under- or over-estimation of the optimal training

[2]This measure refers to an individual's ability to extract oxygen from inspired air. VO_2 max is used as an indicator of physical fitness because it improves with increased exercise.

intensity for *a relatively large number of individuals*" [emphasis mine] (Rotstein et al. 2004: 436).

Speaker perceptions and physiological responses to speech during exercise were further investigated by Baker et al. (2008), who advanced this line of research in two important ways. First, they systematically examined exercise effects on specific speech parameters (pause placement, articulation rate, and syllables per phrase). Second, they extended the exercise protocol to 18 minutes to test whether modifications to speech and exercise breathing differed for longer periods of exertion. Their study involved 12 participants (6 females) cycling on an ergometer at 50% and 75% of VO_2 max while reading texts before and during exercise at three-minute intervals. Respiratory measurements using a facemask-connected metabolic cart showed that minute ventilation increased by 32% over 18 minutes at the lower work rate. At the higher work rate, minute ventilation increased compared to the lower rate, but there was only a slight increase over time. This may be because only a 6-minute period was analyzed, because not all speakers completed the full exercise protocol.

Taken together, the results show that the harder exercise is, and the longer it lasts, the deeper and/or more frequent breaths become (recall that minute volume is the product of tidal volume and respiratory rate). Baker and colleagues suggested that increased respiratory rate, rather than tidal volume, was the primary adaptation, because the number of syllables per phrase decreased significantly at both exercise rates, suggesting more frequent pausing and, presumably, inspirations. They proposed a physiological explanation for this pattern: larger volumes of air would make airflow rate more difficult to control, which would in turn require continuous contraction of the inspiratory muscles – known as inspiratory checking/braking – and potentially result in faster muscle fatigue.

Although Baker et al. (2008: 1211) cited Doust & Patrick (1981) and Meckel et al. (2002) as evidence "demonstrat[ing] that tidal volume does not increase during speech in high respiratory drive conditions," this interpretation referenced only the comparisons between exercise with and without speech. The original data actually showed that tidal volume increased with work rate when only the exercise-with-speech conditions were compared across different work rates. Nevertheless, Baker et al.'s (2008) primary conclusions remained valid. The authors also observed an increase in dyspnea over time, which they hypothesize arose from increased use of the respiratory muscles, which would then suggest an increase in tidal volume. Given the uncertainty about which respiratory parameters are affected, the authors recommend that future studies assess both tidal volume and respiratory rate.

Notably, while Baker and collaborators maintain the conceptualization of speech during exercise as a competition between two tasks, they conclude that exercise breathing and linguistic phrasing are both modified rather than being alternately prioritized. This perspective represents an evolution in understanding the complex relationship between speech production and respiratory requirements during physical activity.

4.1.2.3 Shift in perspective (2010–2019)

The field entered a new phase as linguistic researchers began examining speech during exercise as a primary phenomenon rather than an experimental manipulation, a transition anticipated by Baker et al. (2008). Patil et al.'s (2010) conference paper initiated this shift with an application-oriented study developing methods to distinguish between neutral speech and speech under physical stress (i.e., during exercise) by relating speech breathing patterns to prosodic changes in the speech signal. Their experiment involved nine female participants repeating 35 sentences played through headphones before and during stair-stepper exercise at a fixed pace of 10 mph. They innovatively collected data using both a close-talking microphone and a physiological microphone placed on the skin to capture breathing-induced tracheal vibrations that propagate to the surface of the body.

The study analyzed connected speech, decomposed into individual consonant-vowel sequences, for changes in fundamental frequency (f0), vocal intensity, and duration. Respiratory changes were assessed during exhalations using two measures of the energy patterns, total energy and average energy. An energy value was recorded for each frame of an inter-utterance breath. The total energy measure is the sum of energy in all frames and is "assumed to be proportional to the tidal volume of air;" the average energy measure is the mean value across all frames, "assumed to be proportional to minute volume" (Patil et al. 2010: 5147). Total energy was found to *decrease* (-28%) during exercise, while there was no change in average energy (minute volume). At the same time, f0 and vocal intensity were found to increase (+10%) during exercise, which the authors interpreted as "an attempt by speakers to re-direct constrained exhalation resources to speaking under physical stress" (Patil et al. 2010: 5148). In other words, when minute volume increases during exercise, the additional expiratory energy is distributed over both phonated and non-phonated stretches of the expiratory phase. Although it is not clear how to relate the energy measures to the liter-based units in other work, this study notably contributed a novel methodological approach by separately analyzing respiratory parameters during phonated and non-phonated portions of expiration.

Subsequent conference papers explored how exercise breathing patterns influence speech characteristics through temporal parameters. Fuchs et al. (2015) examined 11 participants (10 females) during stationary cycling at fixed power outputs of 70 W and 140 W. Semi-spontaneous speech was elicited using a list-choice task before exercise and at each workload, and respiratory data were recorded using RIP. The results showed significant increases in respiratory rate across all workload comparisons, though no effect size is given. Both pause time and articulation rate increased, leading the authors to conclude that speakers actively restructure utterances in response to elevated respiratory demands.

Trouvain & Truong (2015) complemented these findings in their study of 23 participants (15 females) reading text before and after running until volitional exhaustion on a treadmill. Like Patil et al. (2010), the authors report on breath noise, innovatively deriving respiratory information from the acoustic signal. The results showed a 25% increase in respiratory rate (although this was not statistically significant), significantly increased audible in-breath intensity (suggesting increased tidal volume), and a 43% increase in inspiratory duration – notably contradicting Meckel et al. (2002)'s findings. Analyses of speech measures revealed increased articulation rate alongside longer pauses – a "contradictory strategy regarding speeding up and slowing down" (Trouvain & Truong 2015: 3703) that the authors interpreted, drawing on Baker et al. (2008), as arising from "the competition between metabolic requirements and linguistic phrasing." These findings suggest that speakers may adapt both speech and breathing behavior rather than consistently prioritizing one function over the other.

The culmination of this research phase appeared in Ziegler et al. (2019), based on a doctoral dissertation (Ziegler 2014), which refined the competition concept to focus specifically on respiratory-laryngeal configuration conflicts – closed for phonation versus open for respiration. The authors interpret previous findings on speech during exercise as showing that the voice *reacts* to increased respiratory drive, which raises the question: how flexible are speakers to *actively* modify glottal settings in conditions of high respiratory drive? To pursue this question, the authors recorded loud and normal speech from 32 participants at rest and during two levels of physical activity: treadmill walking at 50% and 70% of age-predicted maximal heart rate.[3] Speech was elicited as sets of seven syllables of /pa:/, first at normal vocal intensity and then in a loud voice (+10 dB). This controlled speech task was chosen specifically to assess glottal valving by calculating

[3]Along with VO_2 max, maximal heart rate is a common index of exercise intensity. At submaximal intensities, oxygen consumption and heart rate show an almost linear relationship (Åstrand et al. 2003), so the workload conditions here and in Baker et al. (2008) are comparable.

the ratio of subglottal pressure to airflow ("laryngeal airway resistance"). Minute ventilation and end-tidal carbon dioxide levels were also recorded to assess respiratory perturbation. The authors found that minute ventilation significantly increased when workload changed from rest to low or high for both loud and normal speech, and when speech changed from normal to loud during both exercise conditions, demonstrating respiratory modifications to simultaneously meet exercise and vocal goals. But participants also modified glottal configuration to produce loud speech at low workload: when instructed to change from their spontaneous vocal intensity to loud vocal intensity participants were observed to decrease airflow to produce loud speech. The authors concluded that these findings demonstrate greater physiological flexibility to meet vocal goals than previously thought – even under conditions of high respiratory drive, speakers can selectively adjust airflow to produce different types of speech.

This study marked a significant theoretical advance by introducing action theory (Reed 1982) as an explanatory framework, proposing that functional goals drive physiological responses to activity. This theoretical integration offers a more comprehensive understanding of speech breathing during exercise by foregrounding the agency of the speaker, highlighting both the voluntary control of breathing and the pursuit of various "vocal goals," from speaking loudly to adopting different speech styles.

4.1.2.4 Synthesis of findings

Research from 1968 to 2019 has revealed remarkably consistent patterns in how exercise affects speech breathing (see Table 4.1), despite varying methodological approaches. During exercise, speakers exhibit increased respiratory rate and breath volume, decreased respiratory cycle duration, elevated vocal intensity and pitch, and greater perceived production difficulty. Theoretical understanding has evolved from viewing speech and exercise breathing as competing functions to recognizing their mutual adaptation, culminating in action theory's proposal that functional goals drive physiological responses.

However, significant knowledge gaps remain in our understanding of how exercise affects speech breathing. Notably, there are limited comparisons between speech during exercise and speech at rest for key parameters like tidal volume, airflow rate, and cycle duration. Effect size comparisons are further complicated by inconsistent reporting and methodological variations (Table 4.2). For instance, defining exercise intensity with absolute measures (e.g., mph) introduces uncontrolled variability in workload due to individual fitness differences, while traditional lab tasks like repeating isolated syllables require breath support that dif-

Table 4.1: Overview of findings on speech during exercise: + indicates a significant increase during exercise; - indicates a significant decrease relative to baseline (rest)

Study	VE	VT	RR	T_{tot}	T_{insp}	T_{exp}	F	P_s	CO_2	VO_2	f0	int.	diff.
Otis & Clark (1968)											+	+	
Doust & Patrick (1981)	+	+	+				+					+	+
Meckel et al. (2002)	+	+	+	–	–	–			+	+			
Rotstein et al. (2004)													+
Baker et al. (2008)	+									+			+
Patil et al. (2010)	+											+	
Fuchs et al. (2015)			+									+	+
Trouvain & Truong (2015)			+		+							+	
Ziegler et al. (2019)	+						+	+	+				+

VE = minute ventilation; VT = tidal volume; RR = respiratory rate; T_{tot} = breath cycle duration; T_{insp} = inspiratory duration; T_{exp} = expiratory duration; F = expiratory flow rate; P_s = subglottal pressure; CO_2 = end-tidal CO_2; VO_2 = Oxygen uptake; f0 = fundamental frequency; int. = vocal intensity; diff. = difficulty/discomfort

Table 4.2: Overview of methodological choices in studies of speech during exercise

Study	Lang.	N	Exercise task	Speech task	Resp. data
Otis & Clark (1968)	English*	6 [?]	treadmill: 3.5 mph, 4.5 mph	read text	unknown
Doust & Patrick (1981)	English	6 ♂	treadmill: 85, 95, 125, 150, 165 HR	read text	RIP
Meckel et al. (2002)	Hebrew*	14 ♂	treadmill: 65%, 75%, 85% VO_2max	read text	facemask
Rotstein et al. (2004)	Hebrew*	14 ♂	treadmill: 65%, 75%, 85% VO_2max	read text	facemask
Baker et al. (2008)	English	12 (6 ♀)	ergometer: 50%, 75% VO_2max	read text	facemask
Patil et al. (2010)	English	9 ♀	stair-stepper: 10 mph	heard text	acoustic
Fuchs et al. (2015)	German	11 (10 ♀)	ergometer: 70 W, 140 W	speech	RIP
Trouvain & Truong (2015)	Dutch	23 (15 ♀)	treadmill: volitional exhaustion	read text	acoustic
Ziegler et al. (2019)	English	32 ♀	treadmill: 50%, 70% HRR max	7× /pa/	facemask

* Language inferred from the country in which research was conducted

fers substantially from connected speech. Temporal aspects of speech breathing, particularly respiratory cycle duration and variability, also remain understudied. While respiratory rate captures global temporal changes, it reflects the frequency of respiratory events (e.g., inspiratory peaks) rather than the durations of respiratory intervals (e.g., inspiration), potentially missing important aspects of speech during exercise. Conflicting findings on individual variation add complexity, with some studies reporting high uniformity across speakers (Doust & Patrick 1981, Meckel et al. 2002) and others suggesting distinct response patterns (Patil et al. 2010). In sum, while we know that people are able to speak during exercise by taking deeper and more frequent breaths, many aspects of how speakers coordinate these respiratory adjustments remain to be explored.

4.1.3 Objectives of the present respiratory study

To address the limitations identified in the literature, the present study systematically examines volumetric and temporal respiratory measures while exploring between-speaker variation in respiratory modifications. Following Baker et al.'s (2008) recommendation to assess these parameters independently, the study combines statistical analyses with descriptive statistics and qualitative insights to investigate two research questions:

(1) **How do speakers modify respiratory patterns in speech during exercise compared to rest? Do they adjust breath volume, breath timing, or both?**

To address this question, the study analyzes multiple volumetric and temporal measures to capture different aspects of respiratory patterns.

(2) **How do respiratory patterns vary across individuals in speech during exercise?**

To address this question, the study uses data visualization techniques and qualitative analyses of speaker experience to examine individual patterns.

To capture a range of respiratory behaviors, the study examines both sustained vowels and connected speech.

4.2 Methodology

This section describes the methods used to analyze the data from the sustained vowel and reading tasks, the collection of which was described in Chapter 3.

A custom MATLAB–Praat routine was used to prepare the respiratory data for analysis. The two RIP signals – capturing movements of the chest and abdomen – were first downsampled, filtered, summed, and labeled at the highest point of each inspiration (peak) and the lowest point of each expiration (trough). The labels were checked manually and then used to derive respiratory measures of amplitude and duration. Statistical analyses were conducted in R using linear mixed models.

4.2.1 Data and participants

The data were obtained via respiratory inductance plethysmography (RIP) as described in Section 3.3.3. The current study analyzes RIP data from three tasks: (1) a maximal-displacement maneuver (to calibrate the raw RIP signal); (2) a reading task, and (3) a sustained vowel task. The speech tasks were performed in each of three physical workload conditions: no workload (rest), low workload (35% of age-predicted maximal heart rate), and moderate workload (65% of age-predicted maximal heart rate). Each recording produced two respiratory signal files: one that captured excursions of the rib cage (RC) and one that captured excursions of the abdomen (AB). The excursions were recorded as changes in raw voltage.

A total of 48 female participants took part in the experiment. Some relevant physiological information is given in Table 4.3. Generally, the participants were relatively young and recreationally active. The results of the standardized physical activity questionnaire showed that the weekly level of physical activity was high for 58% (n = 28) of the participants, moderate for 38% (n = 18), and low for 4% (n = 2).

Table 4.3: Participant characteristics

Characteristic	M (*SD*)
Age (years)	23.6 (3.8)
Body weight (kg)	61.8 (11.8)
Body height (cm)	168.3 (6.8)
BMI	22.0 (3.5)
Resting pulse	61.3 (9.3)

4.2.2 Data processing

The recording setup captured the respiratory signal at the same sampling rate used for speech (22,050 Hz), which resulted in noise due to the slow movements of the chest and abdominal wall. This noise was smoothed in two steps using a MATLAB script (Rochet-Capellan & Serré 2021). First, the RC and AB signal files were downsampled from 22,050 Hz to 100 Hz; then a band-pass filter (passband: 1–10 Hz) was applied to all files.[4] Figure 4.3 shows the effect of downsampling and filter application: the shape and amplitude of the RIP record are preserved but the peaks and troughs are clearer.

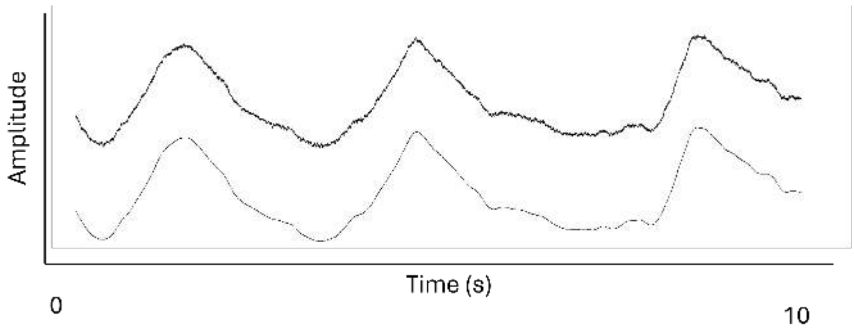

Figure 4.3: An RIP trace of the rib cage signal showing the raw signal (*top*) and the downsampled and filtered signal (*bottom*) (participant ZMQ, reading in the no-workload condition)

To estimate overall lung volume, the rib cage and abdominal traces were combined. This is necessary because the chest and abdomen are both displaced during breathing but function largely independently and therefore contribute differently to changes in lung volume (Konno & Mead 1967). Example rib cage (RC) and abdominal (AB) kinematic traces are shown roughly to scale in Figure 4.4. The traces were combined using a standard 2:1 correction (Banzett et al. 1995), in which the RC signal was multiplied by two before being combined with the AB signal. A study by McKenna & Huber (2019) found that this correction yields lung volume estimates comparable to those obtained using Konno & Mead (1967)'s isovolume maneuver, which requires speakers to take and hold a deep breath

[4]Published studies analyzing RIP data report diverse sampling rates, ranging from 50 Hz (Robles-Rubio et al. 2020) to 1,000 Hz (Silva et al. 2022) to 11,025 Hz (Petrone et al. 2017). Because the choice of appropriate downsampling and filtering values depends on both the sampling rate and the research question, no gold-standard approach can be gleaned from the literature. The method used in this book was recommended by A. Rochet-Capellan (personal communication, 4 August 2021) based on her experience with similar data.

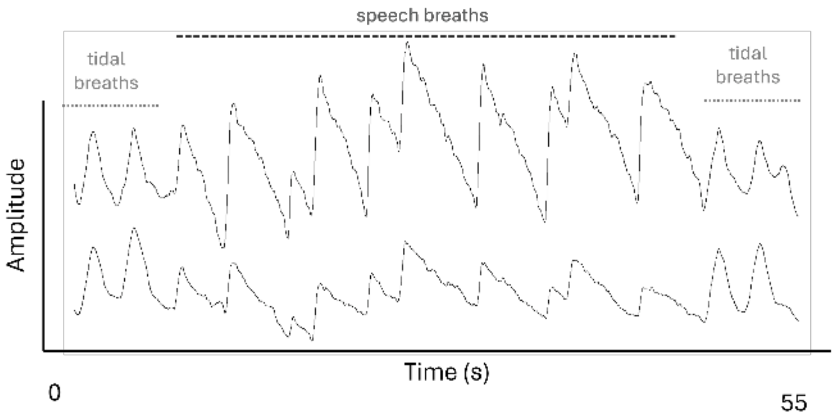

Figure 4.4: RIP traces of rib cage (*top*) and abdomen (*bottom*) signals showing tidal and speech breathing in the no-workload condition (participant ZMQ); signal amplitude reflects changes in rib cage and abdomen circumference.

and then shift air between the chest and abdomen to reveal the displacement of each compartment. Although this maneuver was performed in the present study, some speakers struggled to perform it reliably during recording, and inspection of the data revealed irregularities suggesting that air had not been fully displaced between the two compartments in some cases.

A third calibration method was initially considered but abandoned following the emergence of COVID-19. The least squares method (Chadha et al. 1982) has been found to reduce errors in lung volume estimates compared with both the isovolume maneuver and the standard ratio (McKenna & Huber 2019). However, this method requires prior measurement of lung volumes using a spirometer – an apparatus that measures ventilation when a person inhales fully and then exhales forcefully into a mouthpiece. Because the data were collected during the pandemic (in August and September 2020), the use of a spirometer was considered to pose an unknown risk of spreading the virus.

4.2.3 Detection and labeling of respiratory events

Only the summed RIP traces were labeled for analysis. The traces were first automatically labeled using MATLAB's findpeaks function (The MathWorks, Inc. 2007), which identifies the locations of *local peaks* ("maxima") in a dataset – observations that are larger than both adjacent observations. In the respiratory data, a peak is defined by the local maximum voltage value of the RIP signal.

Greater voltage values reflect greater thoracoabdominal excursions (deflection of the respiratory trace from baseline), so a peak represents the end of an in-breath. The findpeaks function was called in a Praat script (Rochet-Capellan 2021b) that also identified local minima – the end of an out-breath. This point simultaneously marked the start of the next inspiration. The MATLAB visual interface is shown in Figure 4.5. The RIP trace signal appears in blue, and the peaks and troughs are marked with orange asterisks. For each peak and trough, the signal's raw voltage values and timestamps were exported as point labels in TextGrid files for manual validation in Praat.

A Praat script (Rochet-Capellan & Serré 2019) opened the respiratory signal and the speech signal in Praat's stereo view with a TextGrid labeled with breath cycle maxima and minima on a point tier (Figure 4.5, right panel). The temporal locations of the automatically labeled maxima and minima in the TextGrid were cross-checked with the inspiratory peaks and expiratory troughs visible in the respiratory signal and with breath noise and speech sounds visible in the spectrogram and waveform. Note that MATLAB uses a fixed amplitude scale (y-axis) but the findpeaks function identifies the value and location of each peak and trough and stores them in a vector. These vector values are shown in Praat in voltage and milliseconds, respectively. The dashed horizontal line is the zero line, indicating negative voltage (smallest thoracoabdominal excursion).

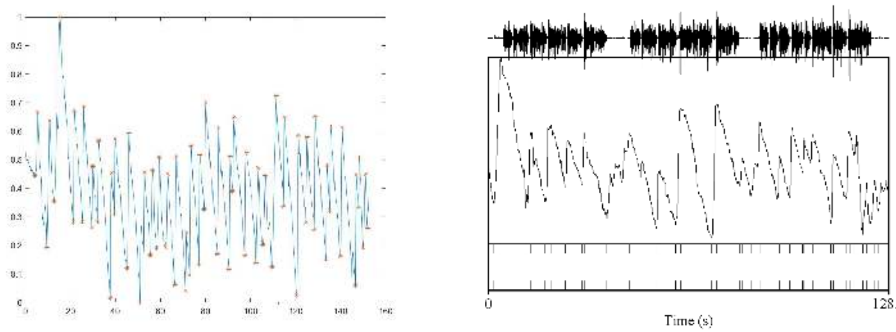

Figure 4.5: RIP traces for reading in the no-workload condition shown in MATLAB (*left*), with local maxima and minima (asterisks) identified using the findpeaks function, and in Praat (*right*), where these points were exported to a point tier (bottom bar).

In Praat, the automatically detected maxima and minima were checked for three possible errors: (1) a respiratory event was not detected; (2) a spurious respiratory event was detected; (3) the location of the respiratory event in time was not plausible (e.g., inhalation during speech; ingressive speech is acoustically distinct from egressive speech and was not observed in the data analyzed here). The findpeaks function identifies peaks based on the distance between adjacent peaks and the prominence of each peak. Consequently, if a peak is not prominent – for example, if little air is inspired or the expiratory phase is relatively flat – it may be missed by the algorithm. Figure 4.6 shows an example of two missed respiratory events in the low workload condition.[5] Note that the signal shows some motion contamination, seen as "wavy" lines. After the in-breath (marked +), the speaker produced the utterance with a relatively small change in thoracoabdominal circumference, resulting in a rather flat signal, which makes the end of the respiratory cycle (marked −) difficult for findpeaks to identify. Inspection of the spectrogram and waveform, however, reveals inhalation noise consistent with two in-breaths, so a peak and a trough were labeled manually at + and −, respectively.

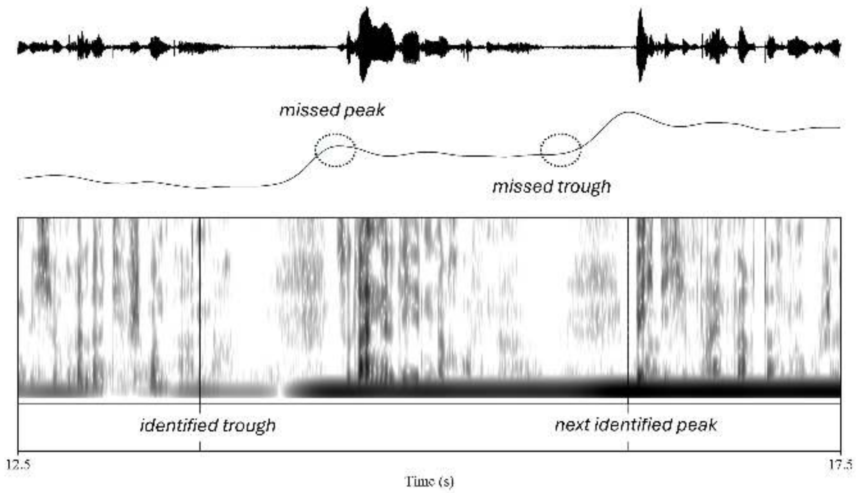

Figure 4.6: Example of missed respiratory events: The findpeaks function did not recognize a local peak or the relative trough at the end of that breath cycle (*dashed circles*) during reading in the low-workload condition (participant HQN).

Spurious peaks may also be identified, especially when the signal contains movement contamination. Figure 4.7 shows a speech sample produced during low workload. What appear to be numerous local maxima and minima are visible during the expiratory phase, and two of them have been identified as a local

[5] For visualization in Praat's stereo display, the respiratory traces (100 Hz) and audio files (22,050 Hz) were resampled to a common sampling rate. The resulting spectrogram artefacts reflect this visualization step only; acoustic analyses were conducted on the original recordings.

trough and peak (marked with *). However, the simultaneous speech signal and lack of inhalation noise in the spectrogram and waveform indicate that these labels are erroneous, and they were removed during manual correction. The motion contamination likely reflects the speaker's pedaling, with rhythmic peaks approximately 500 ms apart – matching the leg strokes at a pedaling rate of about one revolution per second, as confirmed by motion capture data.

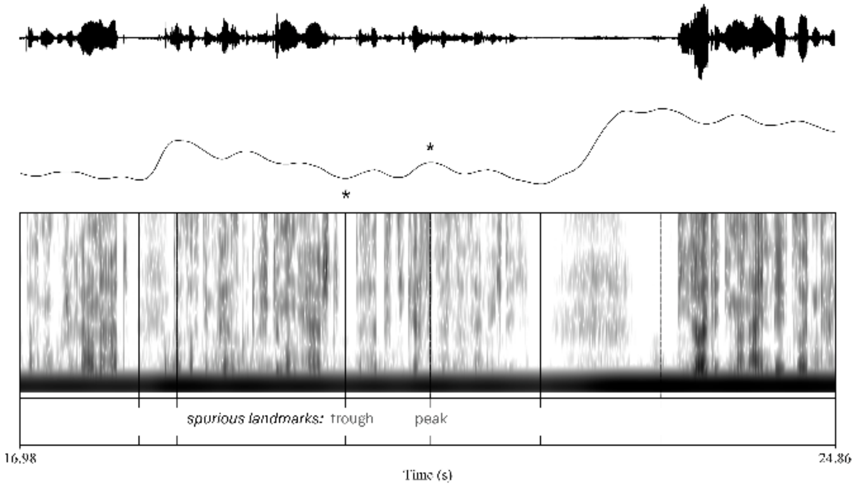

Figure 4.7: Example of spurious respiratory events (*marked* *) in an RIP trace with motion contamination (low workload): The findpeaks function correctly identified a trough and a peak reflecting an in-breath (*at left*), but in the subsequent expiratory phase, movement artifacts were misidentified as a trough and peak.

In addition to creating spurious local peaks, motion contamination can also "shift" the timing of the respiratory event. The rhythmicity of the pedaling motion creates what appear to be numerous local peaks. Because the findpeaks function uses prominence (highest amplitude values) to detect peaks, it selects whichever value is highest. In the current dataset, small differences in the amplitude of the rhythmic noise components therefore resulted in the spurious location of an inspiratory peak during egressive speech. Figure 4.8 illustrates such a case, where the peak immediately prior to speech onset was slightly lower in amplitude than the following (motion-induced) "peak" in the signal.

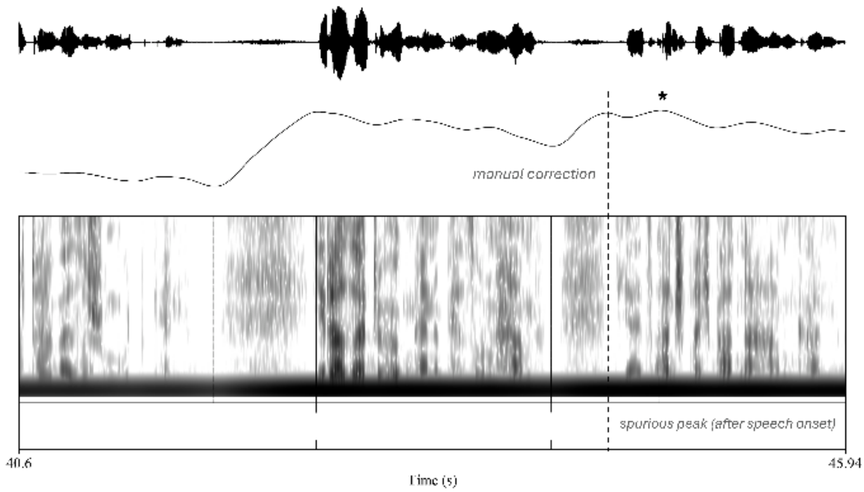

Figure 4.8: Example of implausible timing of a respiratory event: Motion contamination (moderate workload) created a local respiratory peak (*marked* *) after speech onset. The label was manually shifted to the adjacent peak (*dashed line*).

4.2.4 Calculation and extraction of respiratory data for analysis

The amplitude and timestamp of all maxima and minima were extracted from the corrected TextGrids using a MATLAB function (Rochet-Capellan 2021c) that calculated further parameters as follows. The function regarded the first labeled respiratory event as the onset of inspiration (1), the second labeled event as the inspiratory peak (2), and the third labeled event as the expiratory offset (3). This offset was then taken as the onset of inspiration for the following respiratory cycle. These respiratory events were used to calculate the amplitude and duration of each inspiration and each expiration, the duration of each breath cycle, and the ratio of inspiration to expiration, as shown in Table 4.4 and in Figure 4.9. These parameters, and the number of respiratory cycles per trial, were exported to CSV format using a MATLAB script (Rochet-Capellan 2021a).

Respiratory rate during speech was calculated in breaths per minute. In clinical contexts, nursing textbooks (e.g., Ball et al. 2021, Wilson & Giddens 2022) generally recommend counting breaths for a fixed period of time – ranging from 15 to 60 seconds (Hill et al. 2018) – and multiplying as needed to obtain breaths per minute. This approach was used in an adapted form in the current study because most trials were less than 1 minute in duration (roughly 45–50 seconds for the reading task and 45–60 seconds for the sustained vowel task). A trial was

Table 4.4: Respiratory measures derived from the RIP signal: Each RIP landmark yields displacement measures and temporal measures; additional measures are calculated from these landmarks.

Measure	Source/calculation
Displacement (volumetric) measures	
Inspiratory onset value	RIP landmark: (1)
Inspiratory peak value	RIP landmark: (2)
Expiratory offset value	RIP landmark: (3)
Inspiratory amplitude	Calculated as (2) − (1)
Expiratory amplitude	Calculated as (3) − (2)
Temporal measures	
Inspiratory onset time	RIP landmark: (1)
Inspiratory peak time	RIP landmark: (2)
Expiratory offset time	RIP landmark: (3)
Inspiratory duration	Calculated as (2) − (1)
Expiratory duration	Calculated as (3) − (2)
Cycle duration	Calculated as (3) − (1)
Inspiratory fraction	Calculated as: Insp. duration / Cycle duration
Respiratory rate	Inspirations per minute

deemed to start at the inspiratory onset of the first breath prior to speech onset and to end with the expiratory offset of the last breath prior to the final utterance. For trials of less than 60 seconds, the number of breaths was divided by trial duration and multiplied by 60 to obtain breaths/minute. For trials of 60 seconds, breaths were simply counted.

4.2.5 Expressing RIP signals as percentage of maximal displacement

Respiratory inductance plethysmography captures thoracoabdominal kinematics as changes in raw voltage. To directly compare results between speakers, RIP data must be calibrated to a more physiologically functional unit of measure, typically liters. Another option is to express the data as a percentage of vital capacity. Vital capacity is defined as the "largest volume measured on complete exhalation after full inspiration" (Gold & Koth 2016: 406) – that is, the lung volume actively used by a given individual. This approach was chosen in the current study, but because the kinematic data reflect the maximal expansion and compression of the

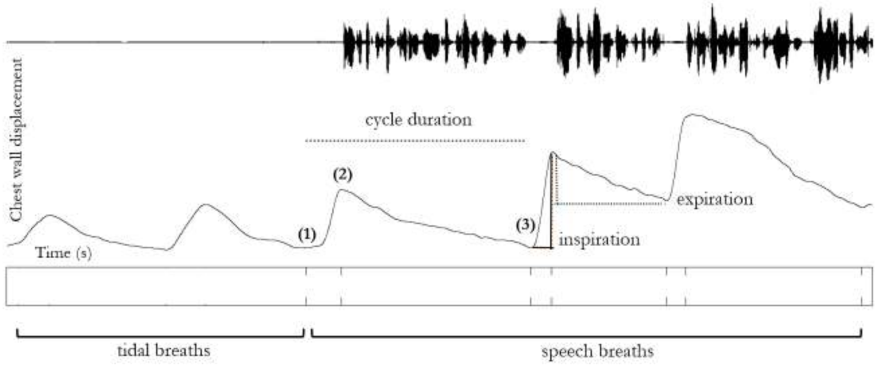

Figure 4.9: RIP trace and TextGrid point tier showing a labeled breath cycle: (1) inspiratory onset; (2) inspiratory peak; and (3) expiratory offset, which is the inspiratory onset of the next breath.

thoracic and abdominal cavities, the measure will be referred to as the *percentage of maximal displacement*.

To obtain maximal displacement values, a vital capacity maneuver was conducted prior to the experiment. Figure 4.10 shows an example kinematic trace: participants were instructed to inhale as deeply as possible and to exhale as fully as possible at least 3 times at their preferred pace (here, the trials are separated by several tidal breaths). The trial with the greatest thoracoabdominal excursion (distance from peak to trough) was taken as that participant's maximal displacement. To express each breath phase as a percentage of maximal displacement, the RIP signal amplitude of each inspiration and expiration was divided by the participant's maximal displacement amplitude.

4.2.6 Outliers and exclusion of observations

No *a priori* criteria for excluding outliers were defined. During labeling, speech breaths were excluded that contained swallowing/burping (n = 4) or laughter (n = 2) or that showed signal malfunctions (n = 5) or clipped signals (n = 5). Data for one participant (PHP) were excluded due to apparent belt slippage that rendered the displacement data unreliable. The moderate workload trial for one participant (OET) was excluded due to motion contamination that prevented reliable identification of maxima and minima. Subsequent boxplots of the data generally showed positive skew with far outliers, defined as values farther than 3 IQRs from the quartiles (De Veaux et al. 2009: 91). The far outliers were spot-checked in the

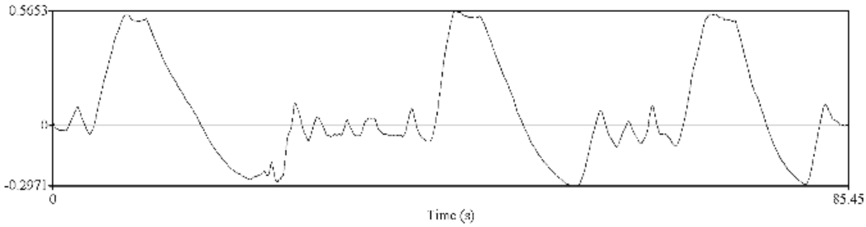

Figure 4.10: Example of maximal displacement maneuver in the RIP trace: The three peaks reflect the participant's attempts at their fullest inspiration, and the troughs reflect attempts at greatest expiration; the trials are separated by tidal breaths (participant YBH).

RIP traces and found to be true observations. At the same time, many appeared to be related to the lab-speech tasks, particularly for inspiratory duration – inspection of the data showed some participants slowly drawing in air while awaiting presentation of the next stimulus, be it the read text or the next sustained vowel. Because extreme values can unduly influence means and affect statistical models, the far outliers (Tables 4.6 and 4.7) were removed prior to analysis. Table 4.5 lists the total number of observations analyzed for each speech task and condition.

Table 4.5: Total number of observations (breaths) for analysis

Workload	Read passage	Sustained vowels
Rest	1518	425
Low workload	1537	431
Moderate workload	1808	425

4.2.7 Statistical analyses

Statistical analyses were conducted using linear mixed-effects models[6] to account for multiple observations from the same participants across workload levels. Separate models were fitted for each speech task (reading and sustained vowels), with physical workload (rest, low, moderate) as the primary predictor and

[6]Analyses were implemented in R (Version 4.2.2) using the following packages: lme4 (Bates et al. 2015), lmerTest (Kuznetsova et al. 2017), and emmeans (Lenth et al. 2023). The BOBYQA optimizer (Powell 2009), with a set maximum of 200,000 iterations, was used to maximize the likelihood function.

Table 4.6: Far outliers (more than 3 IQRs from the quartiles) excluded from analysis in the reading task

Measure	rest	low	moderate
Inspiratory duration	57	31	12
Expiratory duration	4	2	2
Cycle duration	4	1	3
Inspiratory fraction	8	8	6
Inspiratory amplitude	7	2	–
Expiratory amplitude	10	–	–
Respiratory rate	–	–	–
Total	90	44	23

Table 4.7: Far outliers (more than 3 IQRs from the quartiles) excluded from analysis in the sustained vowel task

Measure	rest	low	moderate
Inspiratory duration	–	–	–
Expiratory duration	5	–	–
Cycle duration	4	1	–
Inspiratory fraction	3	6	1
Inspiratory amplitude	4	1	–
Expiratory amplitude	2	–	1
Respiratory rate	–	–	–
Total	18	8	2

respiratory measures as outcome variables. This approach reflects the study's focus on the effects of exercise on speech breathing, rather than differences between speech tasks themselves. Sustained vowels were included to align with analyses of glottal source measures in the same dataset, where this speech task is used to minimize coarticulatory influences. Modeling the tasks separately provided a parsimonious way to assess workload effects within each speech context without introducing additional interaction terms. At the outset of the study, no empirical studies were found that would have motivated modeling task differences in speech breathing.

To capture between-speaker variability, two random-effects structures were compared for each outcome variable within each speech task. The more complex structure allowed both participants' baseline levels (random intercepts) and their sensitivity to workload (random slopes) to vary across individuals, reflecting the possibility that participants differ not only in their resting respiratory behavior but also in how strongly they respond to exercise. The simpler structure allowed only baseline differences between participants, assuming a common workload effect. The best-fitting model was determined based on three criteria: lower Akaike information criterion (AIC) to indicate goodness of fit, lower Bayesian information criterion (BIC) to balance goodness of fit with model simplicity, and absence of singularity.

The selected random-effects structure for each measure is reported in Table 4.8. For most measure–task combinations, models with both random intercepts and random slopes provided the best fit. However, for inspiratory fraction in both tasks, singularity indicated that individual differences in workload sensitivity could not be reliably estimated, and the simpler structure was therefore selected. The simpler random structure was also selected for respiratory rate in the sustained vowel task because the AIC and BIC values were smaller. For expiratory duration and cycle duration in the sustained vowel task, the AIC and BIC values showed opposite effects, indicating an unclear effect of the random-effects structure, so the simplified random structure was selected as the best-fitting model.

Model fit was evaluated using diagnostic plots generated with the performance package (Lüdecke et al. 2021), including assessments of linearity, homogeneity of variance, residual normality, and influential observations. Most respiratory measures showed right-skewed distributions that violated model assumptions. After removal of extreme outliers, upper skew remained, so data were log-transformed and the models refitted. The transformed models showed substantially improved fit and met statistical assumptions; Table 4.8 indicates which analyses were conducted on log-transformed data. Respiratory rate met distributional assumptions

Table 4.8: Random structure and log-transformation choices for each respiratory measure and speech task

Variable	Speech task	Random structure	Log?
Respiratory rate	reading	(1 + workload | participant)	no
	vowels	(1 | participant)	no
Cycle duration	reading	(1 + workload | participant)	yes
	vowels	(1 | participant)	no
Inspiratory duration	reading	(1 + workload | participant)	yes
	vowels	(1 + workload | participant)	yes
Expiratory duration	reading	(1 + workload | participant)	yes
	vowels	(1 | participant)	no
Inspiratory fraction	reading	(1 | participant)	yes
	vowels	(1 | participant)	yes
Inspiratory amplitude	reading	(1 + workload | participant)	yes
	vowels	(1 + workload | participant)	yes
Expiratory amplitude	reading	(1 + workload | participant)	yes
	vowels	(1 + workload | participant)	yes

without transformation, though diagnostic plots revealed several potentially influential observations.[7] Influence analyses confirmed that these observations did not unduly affect model estimates, so all data points were retained.

Estimated marginal means (EMMs) were used to assess differences between workload conditions. EMMs represent model-based means for each outcome variable at each workload level. Because the number of observations (breaths) varied substantially across workload conditions, speech tasks, and individuals, EMMs were preferred over raw means as they appropriately account for this imbalance. Pairwise differences between workload levels were evaluated using contrast analysis. The degrees of freedom needed to compute *p*-values were calculated using the Kenward-Roger method (Kenward & Roger 1997), which provides conservative inference in mixed-effects models.

[7] A few participants took relatively many or few breaths compared to others. For example, participant ILU read the passage in the rest condition using just 3–4 breaths compared to 10–16 for other participants.

4.3 Results

4.3.1 Descriptive statistics: Respiratory measures

To provide context for the statistical analyses, violin plots overlaid with boxplots display the distribution of raw data across workloads for each speech task. Violin plots show how data points are distributed across the measurement range, while boxplots summarize the median and interquartile range. This presentation is particularly useful for physiological data, which often show skewed or uneven distributions due to biological constraints and individual differences. In addition to illustrating central tendency and spread, the violin plots make it possible to identify multimodality, skewness, and changes in distributional shape across conditions. Direct comparison of these distributional shapes across tasks and workload conditions can thus reveal patterns not captured by summary statistics alone. Means and standard deviations from the observed data are given in Appendix A.

Figure 4.11 displays inspiratory and expiratory displacement during reading and sustained vowels, reflecting the volume of air inhaled and exhaled per breath. Across both tasks, breath volumes increase progressively with workload. However, rather than simply shifting toward larger values, the distributions also become more dispersed across the measurement range, indicating that increasing workload was associated not only with larger breath volumes but also with greater variability in breath volume across observations. Expiratory displacement largely mirrors inspiratory patterns, although sustained vowels show a more compact shape and the reading task displays greater variability, including a bimodal distribution at low workload. Lower outliers appear only for expiratory displacement in the vowel task, indicating unusually small expirations.

In contrast, respiratory durations show stark task differences. Inspiratory and expiratory durations are plotted on a common scale in Figure 4.12 to illustrate how both phases relate within each task: reading shows short, consistent inspiratory phases paired with longer, more variable expiratory phases, whereas sustained vowels show inspiratory and expiratory phases closer in duration. Inspiratory durations show distributions that are compact and symmetrical across workloads in the reading task. Sustained vowels exhibit a wider range of durations, skewed toward longer values. This pattern suggests greater variability in the timing of inspiration during sustained vowel production than during reading. With increasing workload, the sustained-vowel distribution becomes more compact and symmetrical.

For expiratory duration, the distributions in the reading task are asymmetrical and bimodal across workloads, indicating that observations are unevenly

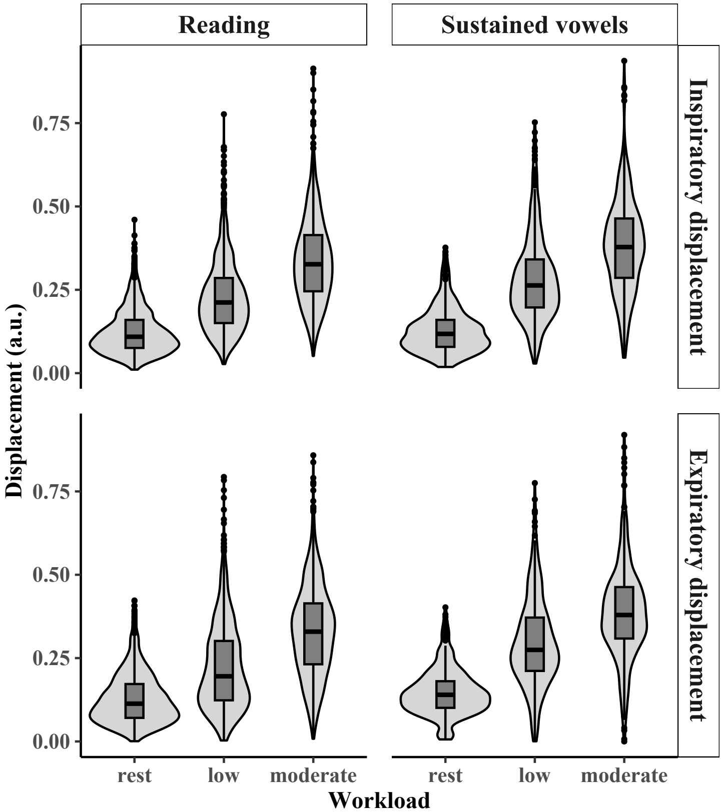

Figure 4.11: Inspiratory and expiratory displacement expressed in arbitrary units (a.u.), shown as violin plots overlaid with boxplots

distributed and cluster into two ranges. Sustained vowels show more symmetrical and compact distributions. In both tasks, increasing workload compresses the upper range, showing fewer very long expirations compared to speech at rest.

The relationship between breath phases is captured by inspiratory fraction (Figure 4.13), the proportion of inspiration within the total breath cycle. In reading, distributions are compact and symmetrical across all workloads. In contrast, sustained vowels show broader distributions at rest and low workload, reflecting greater variability in how speakers balanced inspiratory and expiratory duration. At moderate workload, the distribution narrows, suggesting more uniform relative timing.

Unlike the previous breath-by-breath measures, respiratory rate (Figure 4.14) is an aggregate measure capturing breaths per minute across each trial. Both

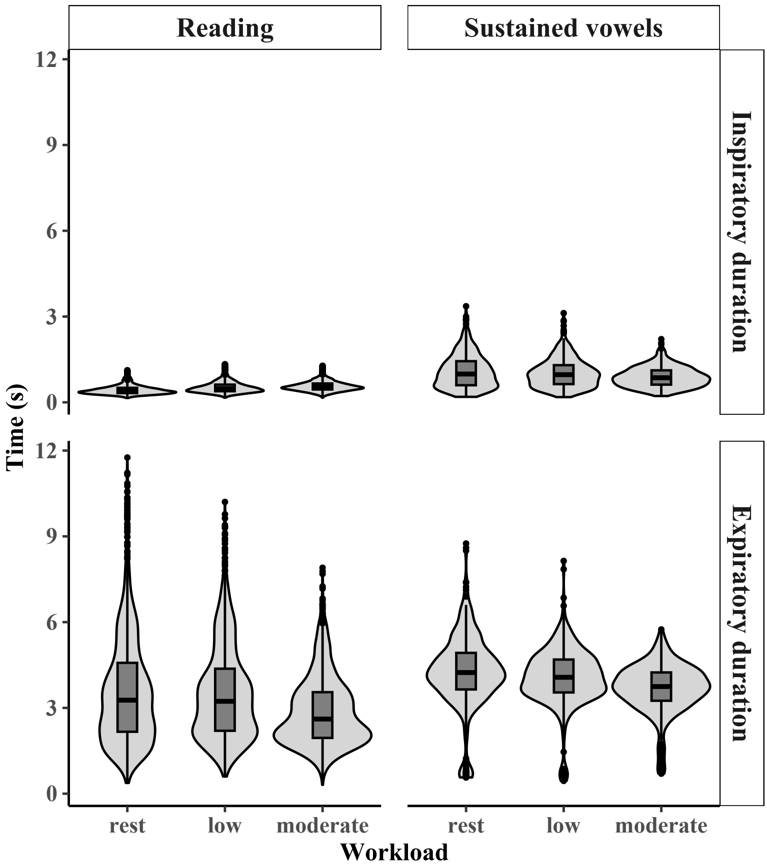

Figure 4.12: Inspiratory and expiratory duration, shown as violin plots overlaid with boxplots

tasks show relatively similar ranges yet variable distributional shapes across workloads, with no clear systematic pattern. Unlike the preceding measures, respiratory rate summarizes overall breathing behavior across a trial and is therefore influenced by multiple factors, including speech task, phrase structure, and breath volume.

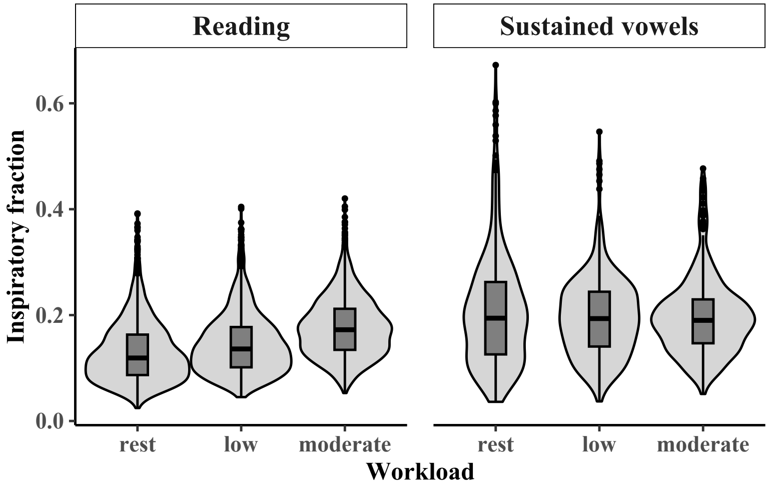

Figure 4.13: Inspiratory fraction by workload and task, shown as violin plots overlaid with boxplots

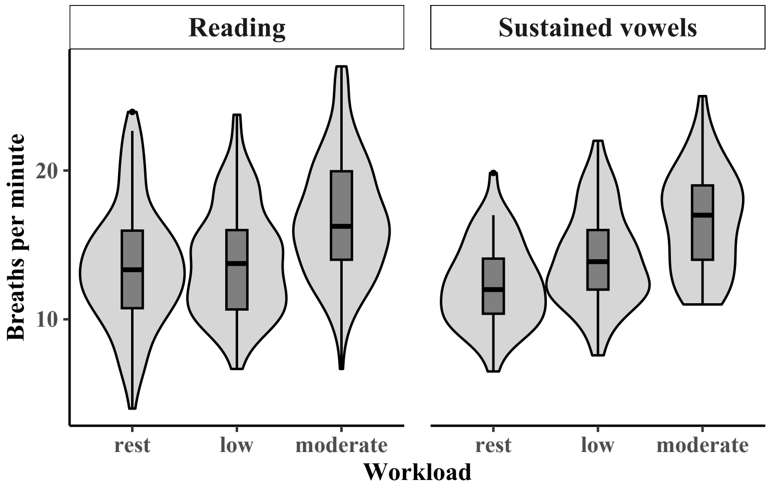

Figure 4.14: Respiratory rate by workload and task, shown as violin plots overlaid with boxplots

4.3.2 Inferential statistics: Respiratory measures

To evaluate workload effects on speech breathing, separate models were fit for each speech task, and pairwise comparisons among rest, low, and moderate conditions were conducted using estimated marginal means (EMMs). Results are presented for displacement measures first, followed by duration and other timing measures. Both original and log-transformed model predictions are shown for comparison. Where predictions diverge, the log-transformed model is used for inference, as it better accommodates the distributional properties of the data. Pairwise comparisons and model statistics are given in Appendix A.

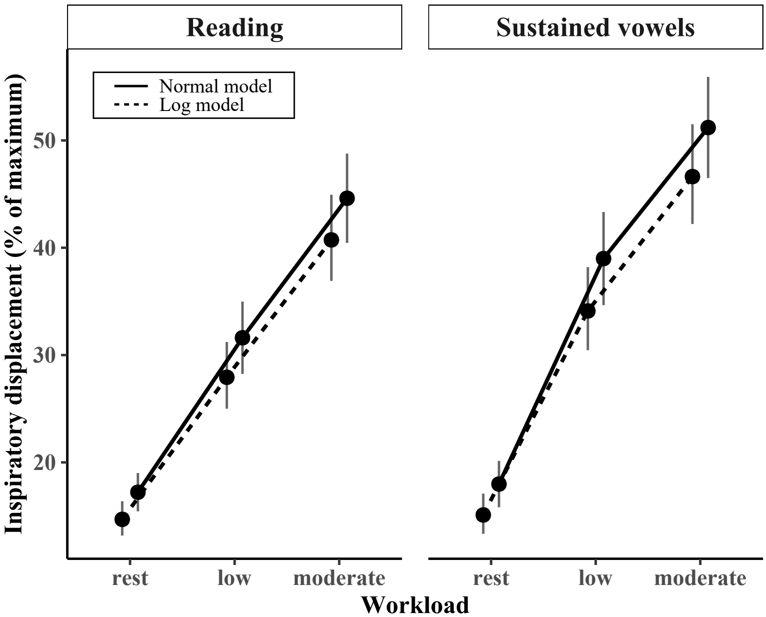

Figure 4.15: EMMs for inspiratory displacement by workload and task; error bars represent 95% confidence intervals.

Figure 4.15 shows inspiratory displacement across workloads for reading and sustained vowels. Although statistical models were fit to the original arbitrary-unit values, EMMs are expressed here as percentages of maximal displacement (approximating vital capacity; see Section 4.2.5) to provide a physiologically meaningful scale. In both tasks, inspiratory displacement increased significantly across all workload comparisons (all $p < .001$). The overall pattern indicates progressively deeper breaths with rising physical demand, though the effect was somewhat larger for sustained vowels.

Expiratory displacement (Figure 4.16) showed a similar pattern, with significant increases across all workload levels (all $p < .001$). Such similarity is not surprising, given that inhaled air must ultimately be exhaled. However, it is important to consider timescales. These results reflect averages over time, but the underlying RIP data showed that individual breaths were not always balanced in terms of the volume of air inspired and expired.

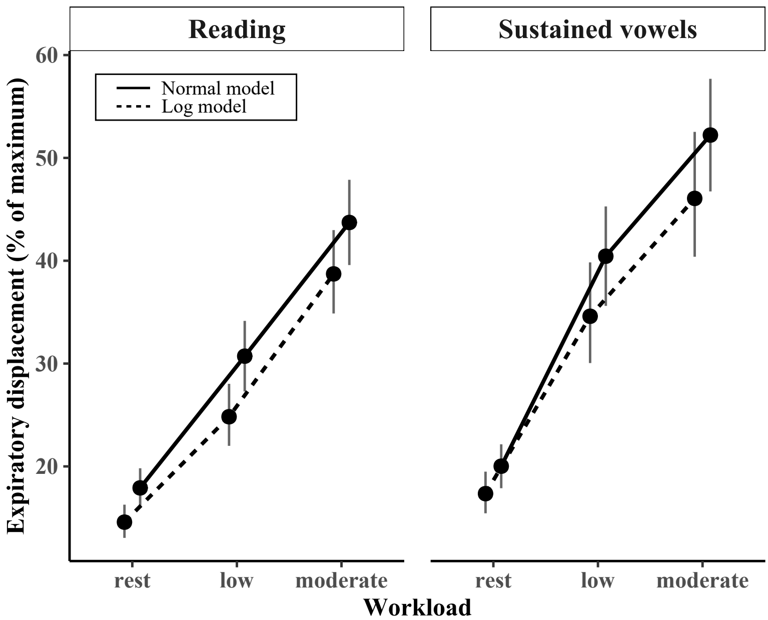

Figure 4.16: EMMs for expiratory displacement by workload and task; error bars represent 95% confidence intervals.

Durational measures revealed clear differences between tasks and between inspiratory and expiratory phases. At rest, inspirations for sustained vowels were nearly twice as long as those in the reading task, highlighting a substantial task difference even before workload increased. Workload effects also differed between tasks. During the reading task, inspiratory duration (Figure 4.17) increased significantly for all workload comparisons (all $p < .001$). In contrast, sustained vowels showed no significant change from rest to low workload ($p > .160$ in both models), but showed significant decreases from low to moderate workload and from rest to moderate workload (all $p < .012$). Inspiratory durations during sustained vowels were also more variable, particularly at rest and low workload.

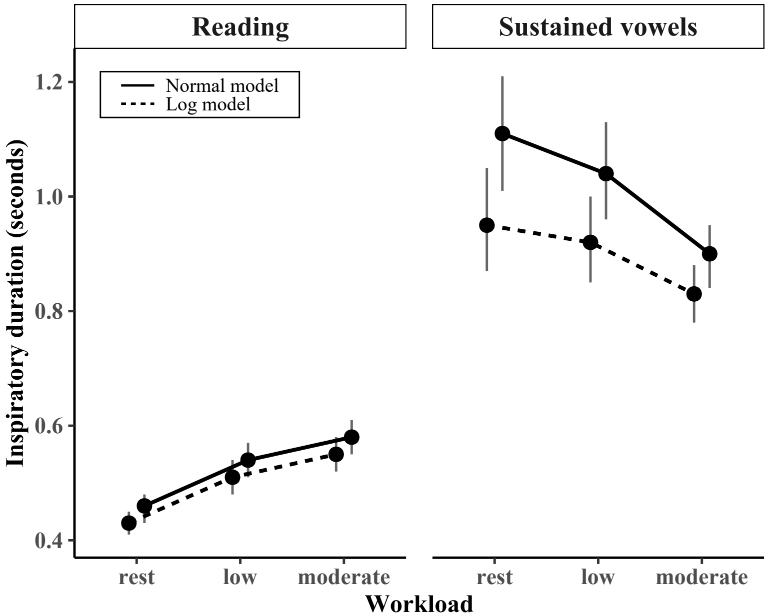

Figure 4.17: EMMs for inspiratory duration by workload and task; error bars represent 95% confidence intervals.

Figure 4.18 displays expiratory duration; total breath cycle duration closely mirrored these patterns and is discussed here but not plotted separately. Overall, both measures decreased as workload intensified, though the pattern differed by task. During reading, neither measure changed significantly from rest to low workload ($p > .151$), but both decreased significantly from low to moderate and from rest to moderate (all $p < .001$). In sustained vowels, both measures showed significant decreases across all workload comparisons (all $p < .001$).

Inspiratory fraction (Figure 4.19) increased significantly across all workload comparisons in the reading task (all $p < .001$), indicating that inspiration occupied progressively more of each breath cycle. In sustained vowels, no significant changes emerged for any workload comparison (all $p > .121$).

Respiratory rate (Figure 4.20) increased with workload in both tasks. During reading, the increase was significant from low to moderate and from rest to moderate ($p < .001$), with no change from rest to low ($p = .300$). In sustained vowels, respiratory rate increased across all workload comparisons (all $p < .001$).

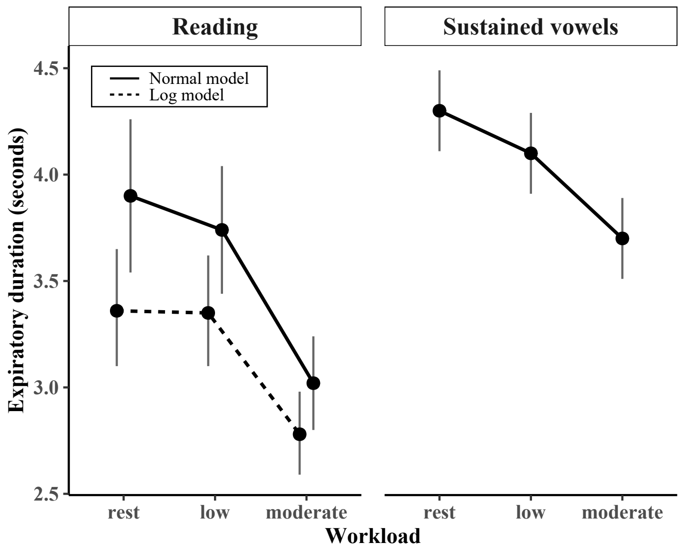

Figure 4.18: EMMs for expiratory duration by workload and task; error bars represent 95% confidence intervals.

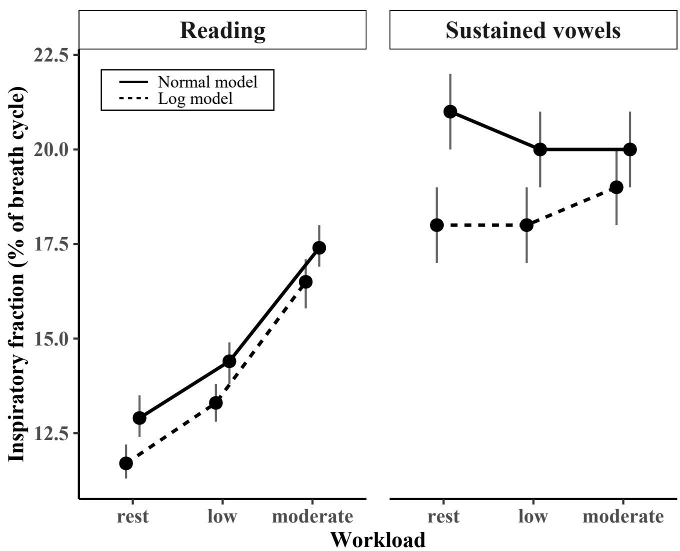

Figure 4.19: EMMs for inspiratory fraction by workload and task; error bars represent 95% confidence intervals.

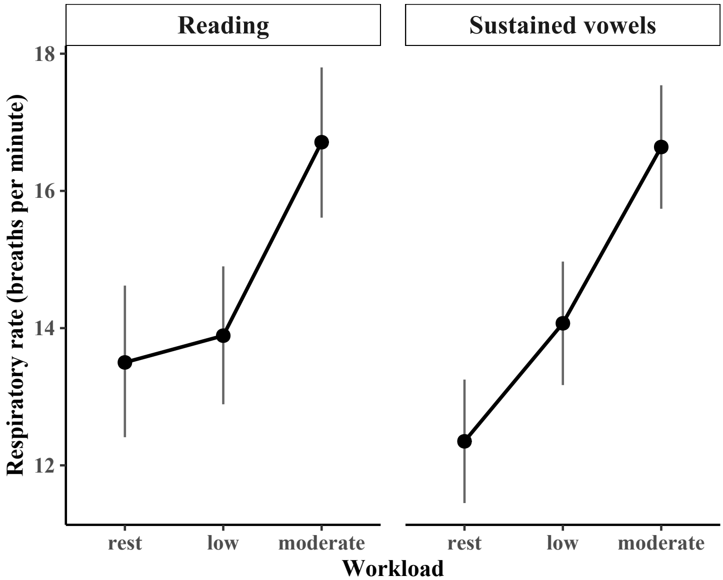

Figure 4.20: EMMs for respiratory rate by workload and task; error bars represent 95% confidence intervals.

4.3.3 Individual differences by respiratory measure: Line plots

Between-speaker differences were explored using line plots to display each speaker's response pattern alongside the group mean. This enables visual assessment of how individual patterns align with the group trend and displays the range of variation across exercise intensities.

As in the group-level models, inspiratory and expiratory displacement patterns were similar across workloads and tasks, including for participants who appeared as outliers. To avoid redundancy, only inspiratory displacement is shown. Figure 4.21 shows broadly consistent patterns across tasks: participants generally followed the group trend, with steady increases in breath volume across workloads that sometimes leveled off at higher intensities. The upper range was somewhat higher for vowels, driven at low workload by a few participants, while in reading, three participants took notably deeper breaths at moderate workload.

In contrast, inspiratory duration (Figure 4.22) showed clear task differences in both workload effects and individual variability. In reading, the range was smaller, with most speakers showing modest increases as workload rose. The form of change varied: some exhibited steady, incremental rises, while others

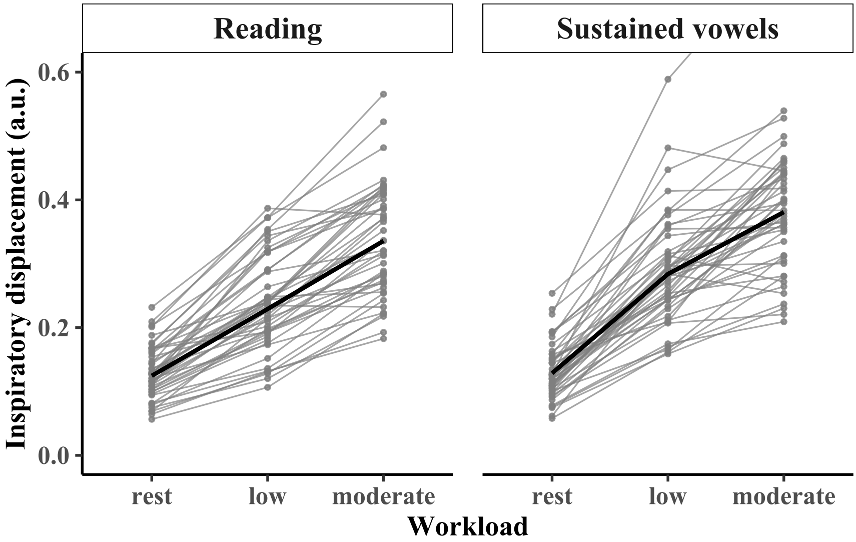

Figure 4.21: Inspiratory displacement by workload and task; participant means are shown in gray and the group mean in black. The y-axis is truncated to improve readability (one participant: max = 0.8 a.u.).

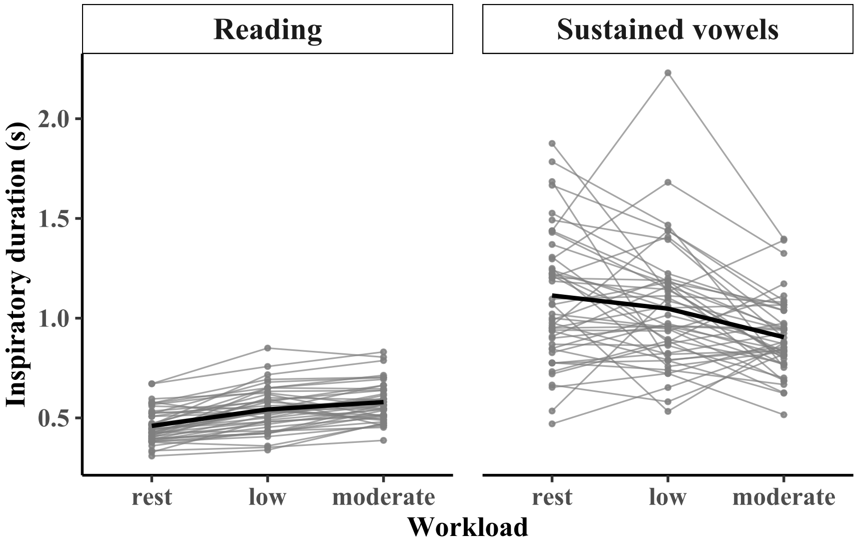

Figure 4.22: Inspiratory duration by workload and task; participant means are shown in gray and the group mean in black.

showed an early increase followed by a plateau, or little change until moderate workload. The vowel task displayed greater diversity. While many speakers demonstrated progressive decreases in inspiratory duration with rising workload, others showed "mountain" or "valley" patterns – initial increases or decreases at low workload followed by a return toward baseline levels at moderate workload. Overall, the effect of workload on inspiratory duration during sustained vowels was highly variable, with several speakers even showing incremental increases contrary to the group trend.

Expiratory duration (Figure 4.23) showed broadly similar patterns across tasks, with both exhibiting shorter expirations as workload increased. However, some task-specific differences emerged: the range was larger for reading at baseline, indicating wide variation in typical expiratory durations, whereas the vowel task showed less variability around a higher median. In both tasks, low workload elicited divergent responses – some speakers shortened expirations, while others lengthened them – yielding the "mountain" and "valley" patterns also seen for inspiratory duration. The line plots further suggest that the bimodality observed in Figure 4.12 reflects clusters of speakers at baseline. As workload increased, speakers tended to shorten their expirations while maintaining these relative groupings.

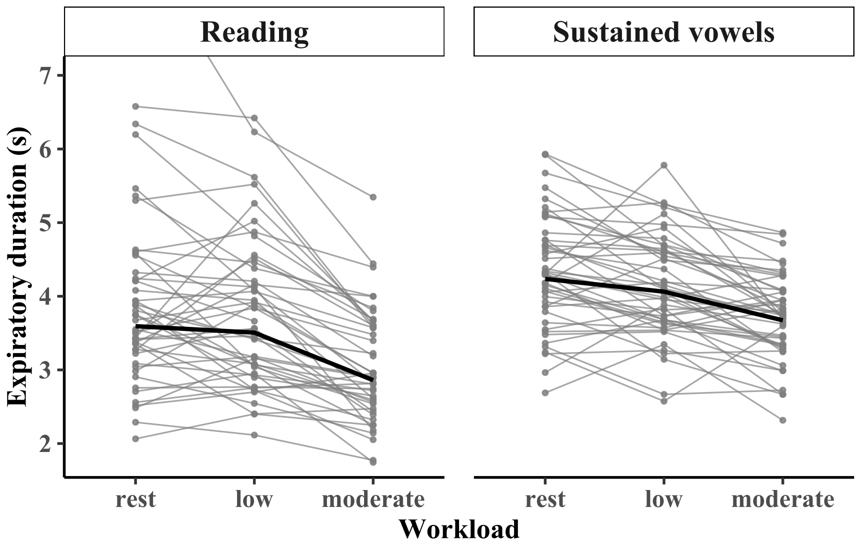

Figure 4.23: Expiratory duration by workload and task; participant means are shown in gray and the group mean in black. The y-axis is truncated to improve readability (one participant: max = 9 s).

Patterns for inspiratory fraction differed across tasks (Figure 4.24). In reading, the range was smaller and the increase with workload more uniform, indicating that while inspiration occupied a slightly larger portion of the breathing cycle, the change was limited. In contrast, the vowel task showed much greater variability: participants displayed widely differing relations between inspiratory and expiratory duration, with some upper outliers at baseline and "mountain" or "valley" trajectories across workload. Overall, the effect of workload on the timing balance between inspiration and expiration was far less consistent during sustained vowels than during reading.

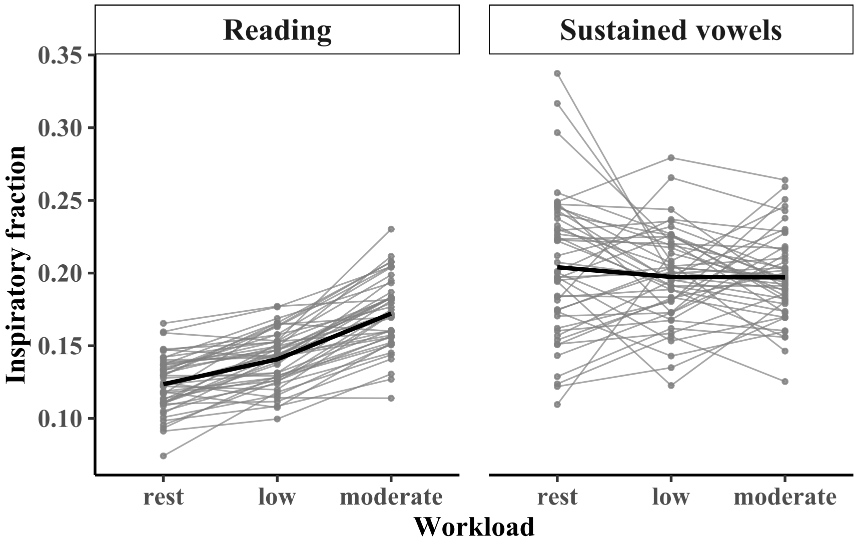

Figure 4.24: Inspiratory fraction by workload and task; participant means are shown in gray with the group mean overlaid in black.

Respiratory rate (Figure 4.25) showed greater variability during reading than sustained vowels, mirroring the spread observed in expiratory duration. Neither task showed consistent patterns of change with workload, illustrating substantial individual variability. Participants differed particularly in their responses to low workload.

These line plots demonstrate that while group means capture broad workload effects, individual patterns for timing measures differ markedly. The relative uniformity across speakers in displacement patterns versus variability in timing highlights that these measures exhibit different degrees of individual variation. Relationships among the respiratory measures may become more apparent

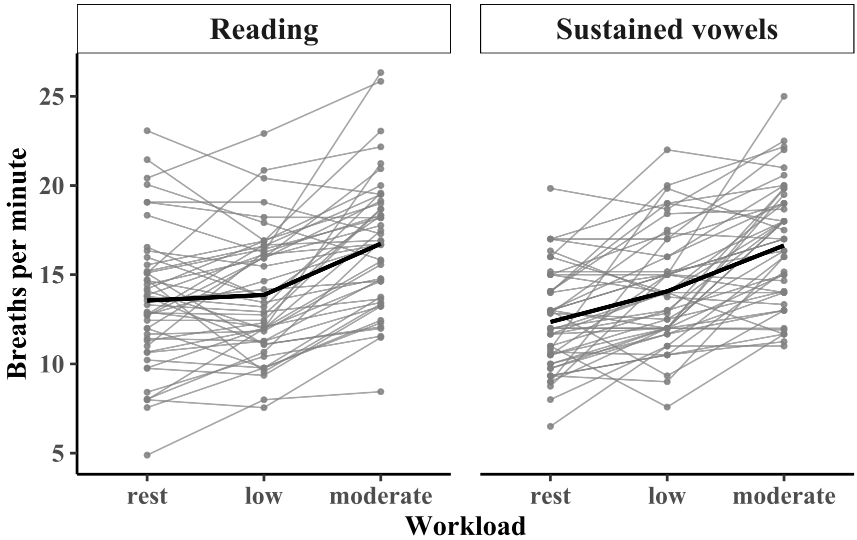

Figure 4.25: Respiratory rate by workload and task; participant means are shown in gray with the group mean overlaid in black.

when temporal and volumetric dimensions are viewed together. The following heatmap analysis takes this multivariate perspective, examining all respiratory measures simultaneously.

4.3.4 Speaker groups across respiratory measures: Heatmaps

Heatmaps display discrete categories as a matrix of cells. In the present study, the respiratory measures form columns and individual participants – denoted by lower-case three-letter codes (e.g., lwa) – form rows. Each cell's color indicates the percentage change in that measure from rest to exercise (in 10% increments), with red showing increases and blue showing decreases. The matrix applies hierarchical clustering to group similar columns and rows, revealing respiratory measures that change in parallel as well as participants with similar response patterns. The accompanying *dendrogram* shows these relationships through tree-like branches (*clades*): shorter clades indicate greater similarity, while taller clades show larger differences between groups.

The heatmap analysis was conducted only for the reading task because connected speech better represents natural speech-breathing patterns than sustained vowels. Figure 4.26 shows changes from rest to low workload. At the top of the heatmap, the dendrogram distinguishes two main clusters of measures: amplitude (volume) measures on the left and timing measures to the right. The cells in the inspiratory and expiratory amplitude columns range from orange to deep red, indicating moderate to substantial increases from rest to low workload. The timing measures show smaller increases (yellow) or decreases (blue). Whereas the line plots highlighted differences in speaker response for a single measure, the heatmaps point to groups of speakers that behave in similar ways.

Hierarchical clustering reveals two main speaker groups: a small group in the upper left, visible as a block of deep red cells, that shows considerable increases in breath volume (often exceeding 100%) and a larger group showing a wider range of increases. The larger group further separates into two subgroups. The smaller subgroup, in the upper third of the heatmap, shows relatively small volume increases but decreases in expiratory and cycle duration. The larger subgroup at the bottom of the heatmap shows moderate volume increases together with more variable changes in expiratory duration. These clusters suggest different patterns of adaptation during exercise, with some speakers relying primarily on greater breath volume, others combining more modest volume increases with shorter expiratory phases, and many showing intermediate adjustments.

The heatmap showing percentage change from rest to moderate workload (Figure 4.27) shows that increases in volume become more pronounced and widespread, with some speakers exhibiting increases of up to 300%. Hierarchical clustering again separates speakers into groups characterized by large volume increases with relatively small timing changes and groups showing more moderate volume increases accompanied by greater decreases in expiratory duration. Comparing participants across workloads suggests that individuals maintain similar response patterns as exercise intensity increases. For example, many of the speakers clustering in the volume-dominant group in the top-left corner showed a similar pattern at low workload. While several speakers appeared to shift toward this volume-dominant group at moderate workload, nearly all speakers who showed timing-focused patterns at low workload showed a combined pattern characterized by more moderate adjustments to both volume and timing at moderate workload.

4.4 Examining individual differences: Qualitative analysis

The qualitative analysis aimed to uncover potential reasons for individual differences by examining participants' experiences. Both planned questions about perceived difficulty and insights emerging from the visual analyses were considered, including: Why did speakers respond so differently at low workload? What

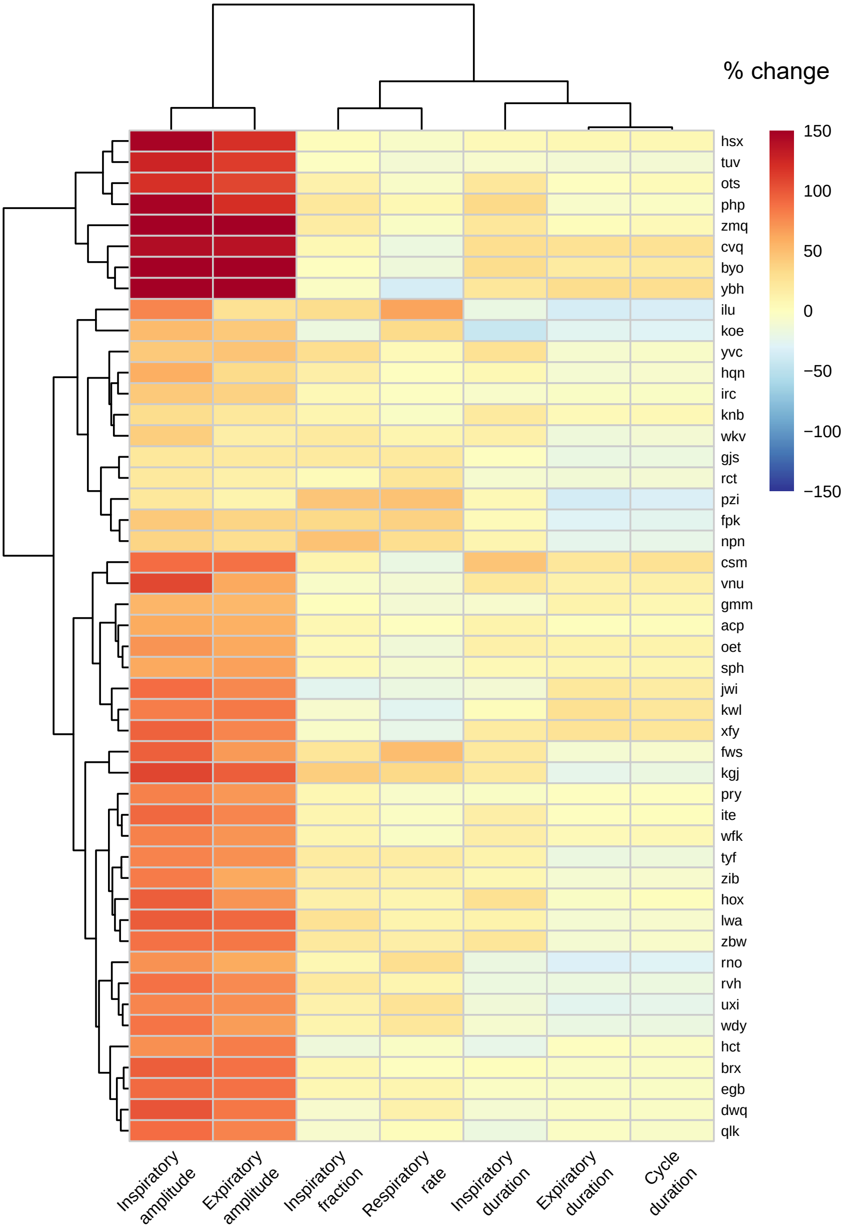

Figure 4.26: Heatmap showing percentage change from rest to low workload for all respiratory measures; hierarchical clustering is shown by the dendrograms.

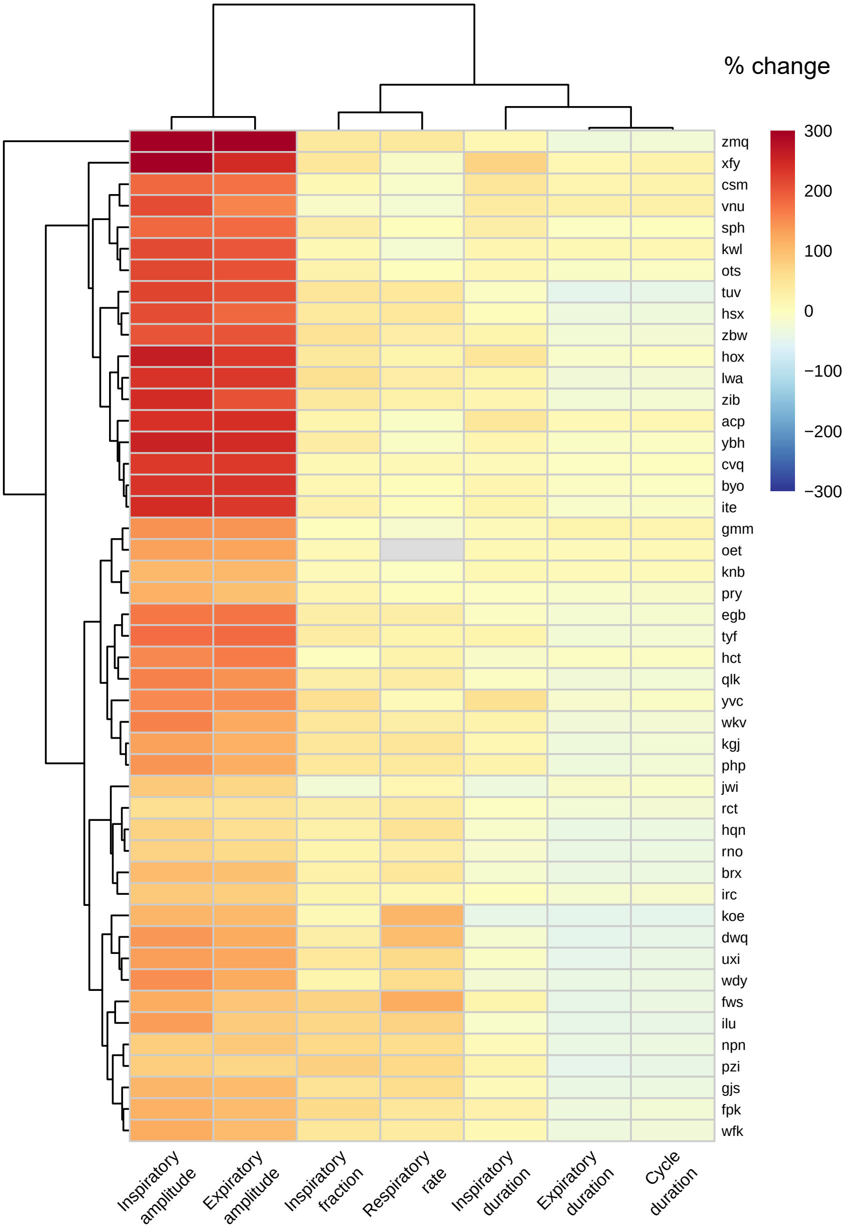

Figure 4.27: Heatmap showing percentage change from rest to moderate workload for all respiratory measures; hierarchical clustering is shown by the dendrograms.

factors influence different speech-breathing strategies? This section presents participants' subjective experiences of producing speech during exercise, beginning with the methodology and analytical approach, followed by a summary of perceived task difficulty across workloads. The analysis then focuses on three main sources of difficulty – physical, cognitive, and emotional – before exploring how these challenges varied across task contexts.

4.4.1 Methodology and analytical approach

Research on human performance often draws on self-perceptual assessments to explore connections between observable behavior and inner experience. This approach has also proven valuable in areas of linguistics concerned with real-world communication, including clinical linguistics, where qualitative interviews reveal how neurological conditions affect speakers' sense of effort, control, and communicative success (e.g., Damico & Simmons-Mackie 2003, Guendouzi & Müller 2006). A small body of work has applied this perspective to speech during physical activity. While participants typically perceive speech during exercise as difficult or uncomfortable to produce (Baker et al. 2008, Rotstein et al. 2004), these studies provide little insight into what participants actually found difficult or why — leaving the underlying sources of perceived effort largely unexplored. To address this gap, a semi-structured questionnaire was developed to capture both task evaluation and psychological dimensions of speaking during exercise.

The questionnaire captured participants' subjective experiences through free-form written responses administered immediately after the experiment. As shown in Figure 4.28, participants rated the difficulty of each speech task across workload conditions and described how cycling affected their speech and breathing comfort. The open-ended format allowed participants to express their experiences in their own words, providing rich qualitative data about the challenges and strategies involved in speaking during exercise.

The qualitative data were interpreted using the classic approach to qualitative analysis summarized in Müller (2013). This iterative method involves multiple readings of the data to compile potentially relevant topics. Each topic is then "tested" for relevance through critical read-throughs to identify the most important themes by considering

- whether statements by many participants could be categorized under the topic or if mention of the topic appears confined to a few speakers;
- whether the topic is connected with certain contexts; and
- how participants relate it to other identified topics.

(1) Did you find any topic hard to talk about?		
(2) Did you find yourself getting annoyed or excited while talking about any topic?		
	Biking, first round	**Biking, second round**
How difficult was it for you to respond to the questions?		
How difficult was it for you to read the paragraph?		
How difficult was it for you to say "She says aaaaaa"?		
(3) How much did cycling affect your speech?		
(4) Did you feel discomfort due to breathlessness?		

Figure 4.28: Post-experiment questionnaire (condensed format). *Note:* "Biking, first round" indicates low workload, while "Biking, second round" refers to moderate workload.

For example, when multiple participants mentioned negative effects on concentration, subsequent readings identified related phrases (e.g., "lost my train of thought"), leading to the identification of "cognitive effects" as an important aspect of perceived difficulty. In this way, the qualitative analysis remains data-driven while being actively shaped by the researcher's focus on certain themes and questions.

4.4.2 Overview of task and workload difficulty

Participants' difficulty ratings provide important context for understanding individual variation in speech breathing during exercise. Figure 4.29 summarizes participants' difficulty ratings as percentages, while Tables 4.9 and 4.10 provide the corresponding participant counts. Together, these results reveal clear differences in perceived task difficulty across workloads. Most participants found reading more difficult than sustained vowels, with this gap widening at higher workloads (63% at low, 83% at moderate workload). Only a small minority (9% at low workload) rated vowels as more difficult, and none did so at moderate workload. Workload effects varied dramatically by task. While 91% of participants found reading more difficult at moderate compared to low workload, responses for the vowel task were far more variable: 41% found it harder at moderate workload, 29% slightly harder, 24% saw no difference, and a few even rated low workload as more challenging. These results highlight substantial individual differences in participants' experiences of the tasks, particularly for sustained vowels.

Table 4.9: Relative difficulty of reading vs. vowel task by workload level; values indicate the number of participants per response category.

Difficulty	Low workload	Moderate workload
Reading more difficult than vowels	22	29
Reading and vowels equally difficult	10	6
Vowels more difficult than reading	3	0
No response	13	13

Table 4.10: Relative difficulty of low vs. moderate workload by speech task; values indicate the number of participants per response category.

Difficulty	Reading	Vowels
Moderate more difficult than low	41	17
Moderate slightly more difficult than low	0	12
Moderate and low equally difficult	2	10
Moderate less difficult than low	2	2
No response	3	7

4.4.3 Perceived difficulties and conscious adaptations

All participants reported that cycling affected their speech, typically attributing these effects to changes in breathing, though their experiences varied considerably. Many participants described both the difficulties they encountered and the strategies they used to manage them. The specific challenges fell into three broad categories: physical difficulties, cognitive impact, and emotional response. In the results that follow, participants are quoted directly using three-letter codes to maintain anonymity, with comments referring to moderate workload unless otherwise noted.

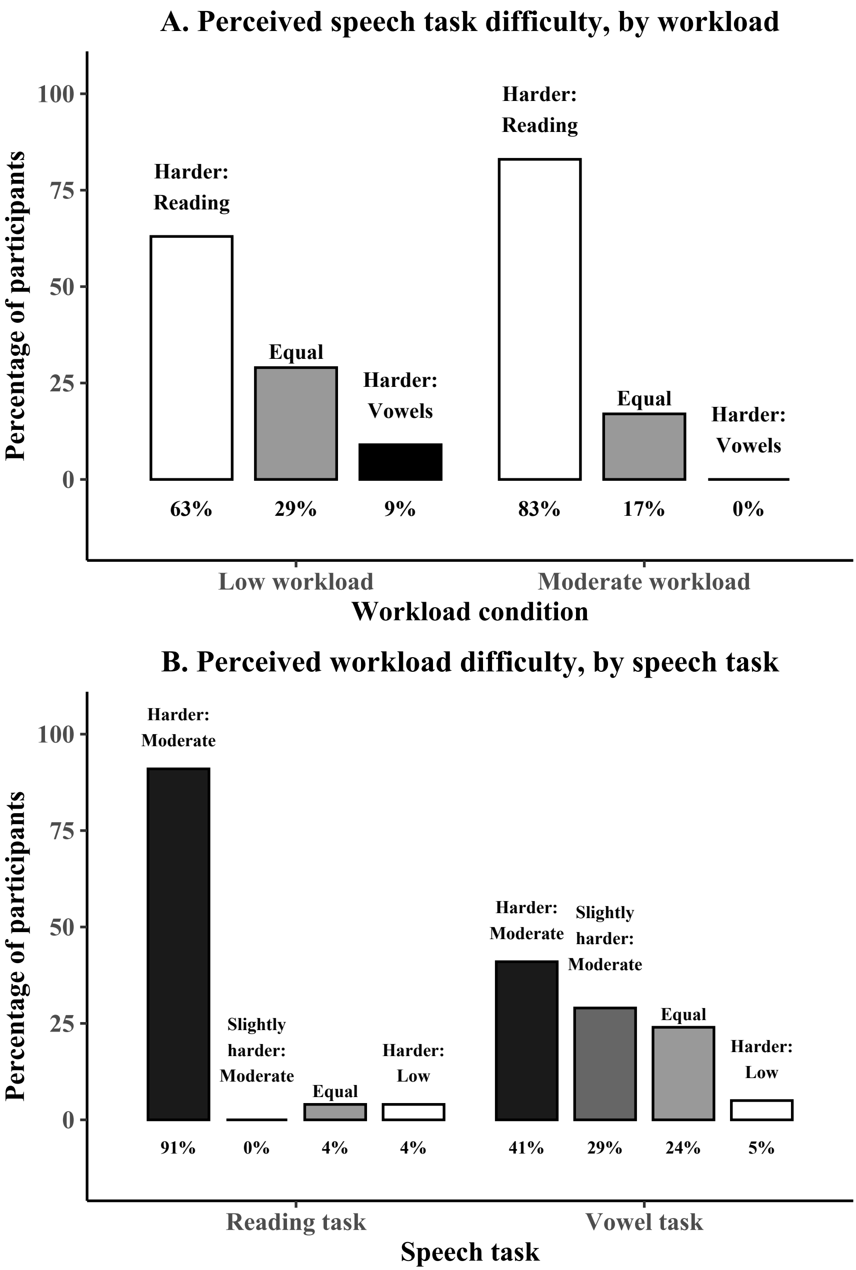

Figure 4.29: Perceived difficulty across speech tasks and workloads

4.4.3.1 Physical challenges

Two specific breathing difficulties dominated participants' physical challenges: breath control issues and disrupted phrasing. Numerous participants reported difficulty managing their breath during exercise, with many describing running out of breath before finishing their intended phrasing. These interruptions were highly salient, often experienced as a breakdown in normal fluency. As one participant explained: "I had to pause for breath right in the middle of sentences sometimes, which felt really unnatural" (TUV).[8] Others described a similar disruption of breath placement: "I increasingly had to take breaths at places where I didn't want to or where it just didn't feel right" (WFK).[9]

In contrast, several participants described the opposite pattern: rather than being forced to breathe mid-utterance, they consciously delayed inspiration to preserve their intended phrasing. As participant TYF explained, the difficulty lay in "trying to pause only after meaningful units."[10] Although participants reported being able to suppress the impulse to breathe, this strategy produced discomfort and an increasingly urgent need for air. One participant noted that "reading in particular took a lot of energy (you quickly felt 'short of breath')" (ZIB).[11] Another described this overextension of the expiratory phase even more vividly: "Sometimes I tried to finish my sentence even though I was almost completely out of breath, and then I had to let out this uncomfortably large amount of air … which made it even more difficult" (KWL).[12]

Yet this same breathing challenge that proved demanding for some participants proved manageable for others through adaptive strategies. Fourteen participants reported no breath-related discomfort at either workload, with about half describing specific modifications. For example, one reported "breathing in more deeply" (FWS)[13] while another explained: "I gave myself a quick break from speaking and then continued" (RNO).[14] Another participant described a more general shift toward active breath management, making a point to "sometimes con-

[8] „Atempausen waren manchmal mitten im Satz, das war unnatürlich."

[9] „Ich musste öfter an Stellen Luft holen, an denen ich nicht wollte oder ich es nicht passend fand."

[10] „merklich schwerer [as low workload], beim Versuch, erst nach Sinnabschnitten zu pausieren"

[11] „Vor allem das Vorlesen … hat viel Kraft gekostet (man kam schnell in ‚Atemnot')."

[12] „Weil ich manchmal versucht habe den Satz noch zu Ende zu bringen obwohl ich schon fast keinen Atem mehr hatte, und dann (ähnlich wie in der letzten der Atemübungen vor dem Experiment) unangenehm/ungewöhnlich viel ausgeatmet habe, was es noch ein bisschen anstrengender gemacht hat."

[13] „kräftiger einatmen"

[14] „Ich habe mir eine kurze Redepause gegönnt und dann weiter gemacht."

sciously take a breath before starting to speak" (SPH).[15] Collectively, these self-reports suggest that participants sometimes made active, individualized adaptations – either to prioritize their intended phrasing or to minimize discomfort. Such choices may partially account for the different patterns seen in the heatmaps that point to volume-based versus timing-based adaptations.

Beyond breathing mechanics, many participants reported various changes to their voice quality. These effects were often explicitly linked to altered breathing patterns. One participant noted: "Speaking & reading [became difficult] when I had to take deeper breaths, because then I couldn't talk as smoothly anymore" (GJS).[16] Several participants perceived their voice as shaky or cracking, often noting emotional connotations. For example, one participant reported: "My voice sounded so shaky – like when you're about to cry" (HCT).[17] Other participants noted difficulties with prosodic control, remarking, for example: "While reading, it was harder to give the words the right emphasis" (HOX).[18] Still others noted an impact on vocal intensity, with one observing that their "voice got quieter toward the end of sentences" (PZI).[19] In addition to these moment-to-moment effects, some participants reported changes operating over longer timescales. For example, when asked how exercise affected speech, participant FPK noted she "spoke more slowly due to the physical effort."[20] These self-reported changes suggest that exercise-induced breathing alterations have perceptually salient effects on multiple dimensions of speech production.

4.4.3.2 Cognitive impact

Although the experimental manipulation was physical, participants also reported cognitive challenges during speech production. Many speakers found reading more difficult than unscripted speech: "While speaking freely, [exercise] affected me less than when reading aloud, because when I'm speaking I don't think about the 'flow'" (RVH).[21] Some participants elaborated on their strategies, describing conscious attempts to pause only after "meaningful units" (TYF)[22] or "to keep the

[15] „manchmal vorher noch bewusst Luft holen bevor Sprechen beginnt“

[16] „beim Sprechen & Lesen – wenn ich tiefer einatmen musste, weil ich dann nicht mehr so flüssig reden konnte“

[17] „weil meine Stimme so brüchig klang. Wie wenn man gleich weint.“

[18] „Beim Vorlesen – die Betonung der Worte war schwer.“

[19] „Stimme wurde zum Satzende hin leiser.“

[20] „in der Sprechgeschwindigkeit langsamer geworden, wegen der Anstrengung“

[21] „beim freien Sprechen weniger als beim Vorlesen, weil ich mir beim Sprechen keine Gedanken um den „Fluss“ mache“

[22] „erst nach Sinnabschnitten zu pausieren“

same pace and flow I had in the first round" (csm).[23] In contrast, other participants appeared less concerned with adhering to earlier phrasing and instead used the text structure opportunistically: "The many subordinate clauses and commas always worked perfectly as breathing pauses" (uxi).[24] These observations offer insight into possible strategies underlying the patterns seen in the quantitative data. Efforts to maintain phrasing may help explain the stable expiratory durations seen at low workload, while different approaches to utterance chunking could underlie the different patterns observed in the heatmaps.

Speakers' comments also highlighted the challenges of divided attention during reading. As participant npn wrote: "Suddenly I had to think about when I pause for breath etc., which I don't have to think about during normal speech."[25] For some, difficulty arose from not knowing how to plan pauses: "Reading the passage was difficult because I didn't really know where a breath pause would be appropriate" (gmm).[26] Others described allocating attention to different aspects of production: "Predictive reading was more difficult because I focused primarily on places in the text that allowed me to pause" (irc).[27] Such uncertainty could help explain some of the observed variability in breathing measures but could also partially explain reports of slowed speech or disrupted intonation. If speakers are focusing on scanning the text to plan breathing, they may have fewer resources to allocate to other aspects of speech production.

Together, these accounts suggest that speech breathing – normally automatic – became less predictable during exercise, adding difficulty to reading. Some of this difficulty appears to stem from adhering to a predetermined structure rather than from speech production itself. To contextualize the cognitive results, a brief discussion of the unscripted monologue data is included. Although these data were not quantitatively analyzed, they provide insight into the challenges speakers are likely to encounter in real-life speech. Moreover, they reveal difficulties that differ in some respects from reading, offering additional perspective on how exercise affects speech production at multiple cognitive and linguistic levels.

Divided attention continued to be a theme: "[Answering the questions] became somewhat harder as exercise became harder, because you had to pay atten-

[23]„da ich versucht habe, das Tempo/ den Redefluss der ersten Runde zu wiederholen"

[24]„Die vielen Nebensätze und Kommata haben immer perfekt für Atempausen gepasst."

[25]„Plötzlich musste ich darüber nachdenken, wann ich Atempausen etc. mache, worüber ich beim normalen Sprechen nicht nachdenken muss."

[26]„Das Lesen des Abschnitts fiel mir schwer, weil ich nicht richtig wusste, wo eine Atempause beim Sprechen angebracht gewesen wäre."

[27]„Das vorausschauende Lesen fiel mir schwerer, da ich vor allem auf Stellen im Text geachtet habe, die mir Sprechpausen ermöglichen."

tion to speaking + thinking + cycling + breathing" (KGJ).[28] However, rather than struggling with breath planning, speakers now reported disruptions to *content* planning. Various participants mentioned difficulty maintaining coherent content. As participant HSX noted, "I felt like I was producing jumbled sentences."[29] Some participants explicitly connected this difficulty to breathing: "Sometimes I ran out of breath before finishing the sentence and then you quickly lose your train of thought" (GJS).[30] However, as with reading, other participants reported that breaks in speech could also be used strategically: "Because you also need time to think, taking breaths worked out fine" (OET).[31]

Participants also noted difficulties at other levels of planning. For example, (YVC) remarked, "The questions were easy to answer, but the [exercise] intensity made it harder to form a complete, coherent sentence."[32] Others described difficulty generating new ideas: "Creativity declines as exertion increases" (HSX).[33] Some emphasized what they saw as the underlying cause, noting that physical exertion limited cognitive resources "because I couldn't express myself the way I normally would and couldn't focus as much on WHAT I was saying" (UXI).[34] In contrast, several participants reported the opposite effect, explaining that speaking could serve as a welcome distraction from physical discomfort. As one participant noted: "With topics I enjoyed, I almost forgot about the cycling and had no problems with (freely) speaking + breathing" (CSM).[35] These observations illustrate that the same speech context could feel burdensome to some and helpful to others, reflecting the broad variability in participants' experiences during exercise.

Finally, the cognitive impact of dual-tasking proved dynamic rather than static, with participants developing varied strategies for allocating attention. Several participants described consciously shifting their focus among cycling, breathing, and speaking over time. For instance, one participant described alternating her

[28] „Die Fragen sind mit zunehmender Anstrengung etwas angestrengter zu beantworten gewesen, da man sich mehr auf das Sprechen + Nachdenken + Radfahren + Atmen konzentrieren musste."

[29] „Ich hatte das Gefühl, zusammenhangslose Sätze zu formulieren."

[30] „Mir fehlte manchmal die Luft, um den Satz zu Ende zu bringen und dann verliert man schnell den Faden."

[31] „Da man auch Zeit zum Nachdenken braucht, ging das mit dem Luft holen."

[32] „Die Fragen waren gut zu beantworten aber durch die Intensität schwieriger einen vollständigen/sinngemäßen Satz zu bilden."

[33] „Kreativität nimmt mit Anstrengung ab."

[34] „Nur, weil ich mich nicht so entspannt ausdrücken konnte wie sonst und nicht so sehr darauf konzentrieren konnte WAS ich sage."

[35] „Bei den für mich schönen Themen habe ich das Radfahren nebenbei fast vergessen und hatte keine Probleme beim (frei) Sprechen + Atmen."

attention between cycling and speaking: “I sometimes let myself ‘ease up’, meaning I concentrated more on cycling than on speaking” (ILU).[36] Others prioritized breathing, as illustrated by one participant’s observation: “I think you have to consciously focus on breathing when you’re exercising like this, and so I couldn’t concentrate as well on formulating sentences” (VNU).[37] These accounts reveal active cognitive resource allocation, with participants making strategic decisions about where to direct their limited attentional capacity.

4.4.3.3 Emotional response

A handful of speakers reported on feelings that arose during the task. Their responses revealed a range of emotional experiences, from self-consciousness to unexpected ease. Several participants (ILU, RVH, CVQ) were bothered by their heavy breathing, with participant ILU feeling particularly self-conscious: “When I had to gasp for air without wanting to, I felt really out of shape. I was embarrassed that I was breathing so loudly by the end.”[38] However, other participants (like EGB) demonstrated a more matter-of-fact attitude, viewing their breathlessness as a normal consequence of physical activity.

Positive psychological effects were also reported by four participants (ITE, QLK, GJS, TYF). Speech during exercise was described as easier (ITE) or freer and less inhibited (GJS, TYF). One participant reported that she felt progressively more carefree as the exercise became increasingly strenuous (QLK). This paradoxical effect was explicitly acknowledged by participant GJS who observed: “I often ran out of breath when I was speaking, but otherwise, the moderate-intensity cycling actually helped me speak in a more relaxed way.”[39]

4.4.4 Perceptions of low workload

Participants varied in how they experienced the shift from rest to low workload. Some reported minimal change from speech at rest, perceiving the task as manageable or even pleasant. For example, participant CVQ noted: “The impact was

[36] „Ich habe mich manchmal ‚absenken‘ lassen. Das heißt, dass ich mich dann mehr auf das Fahren als auf das Sprechen konzentriert habe.“

[37] „Ich glaube, man muss sich schon irgendwie ein bisschen bewusst auf Atmung konzentrieren, wenn man so Sport macht, und deswegen könnte ich mich nicht so gut konzentrieren auf Sätze formulieren.“

[38] „Als ich ungewollt nach Luft schnappen musste, habe ich mich so gefühlt, als sei ich sehr unsportlich. Es war mir peinlich, dass ich gegen Ende hin so laut geatmet habe.“

[39] „Mir fehlte oft die Luft beim flüssigen Sprechen, ansonsten hat mir das moderate Fahren eher geholfen, um locker zu sprechen.“

relatively small. Like talking to someone while taking a walk."[40] Another participant described speaking at low workload as "quite nice, very similar to reading while sitting still" (OET).[41] In contrast, others found the transition surprisingly challenging, describing reading at low workload as "much more difficult than without any activity" (YVC).[42] These different perceptions could help explain the variability observed at low workload – what felt effortless to some proved demanding to others.

It also seems that the nature of the task may have changed for participants as they gained experience speaking during exercise. Several reported developing adaptive strategies over the course of the experiment. For instance, one participant explained, "I could figure out how to breathe best so I could start talking again" (HQN),[43] while another observed, "My body got used to the level of exertion" (RNO).[44] Such responses may explain some of the complex patterns observed in the line plots, where speech measures changed significantly at low workload but plateaued or moved back toward resting levels at moderate workload. More efficient breath coordination could free up cognitive resources for other aspects of the task or reduce dyspnea, making the task feel less effortful. The potential for learning effects to affect task performance adds another layer to understanding why individuals showed such different response patterns across experimental conditions.

The qualitative data suggest that participants did not experience speaking during exercise in the same way. Speakers differed in how they managed breathlessness, allocated attention, and balanced the competing demands of speaking and exercising. Some attempted to preserve their normal phrasing despite discomfort, while others adapted their breathing or speaking patterns more readily. Participants also differed in their emotional responses, with some finding the task stressful or frustrating and others reporting little difficulty or even positive effects. These differences in experience and strategy may help explain why speakers showed divergent patterns of adaptation despite performing the same task under the same physical workload.

[40] „[At low workload] war der Einfluss relativ gering. Vergleichbar, wie wenn man beim Gehen mit Menschen redet."

[41] „Es war recht angenehm und sehr ähnlich zum Lesen beim Sitzen."

[42] „da das Lesen deutlich schwieriger fiel als ohne Aktivität"

[43] „Konnte ich selber einstufen, jetzt wie ich atme am besten, sodass ich wieder sprechen kann."

[44] „Mein Körper hat sich an die Anstrengung gewöhnt."

4.5 Discussion

This study examined (1) how speech breathing changes to accommodate exercise and (2) to what extent such adaptations vary across individuals. The methodology pursued these questions through statistical analyses to identify group-level effects on volume and timing measures, followed by visualizations of individual participant responses to uncover potential variability. Questionnaires about speaker experience and intentions were then analyzed to provide additional context for the quantitative results. The discussion first contextualizes the group findings for each respiratory measure, then turns to individual differences and qualitative insights. These strands of evidence are subsequently integrated to articulate the broader implications of the study, followed by methodological considerations.

4.5.1 Contextualization of results

Both inspiratory and expiratory displacement increased consistently with workload, indicating that speakers took deeper breaths and expelled more air as exercise intensified. This finding is consistent with previous studies on speech breathing showing increases in minute ventilation at higher exercise intensities (Baker et al. 2008, Patil et al. 2010, Ziegler et al. 2019). It also aligns with studies that examined volume and timing separately, though those studies lacked speech-at-rest baselines (Doust & Patrick 1981, Meckel et al. 2002). However, both of these studies found that adding speech to exercise did not further increase breath volume at any intensity examined (low, moderate, vigorous), leading the authors to conclude that breath volume is determined by metabolic demands alone. From a linguistic perspective, this suggests a reversal of the usual control hierarchy: at rest, utterance duration determines breath depth (Huber 2008, Sperry & Klich 1992, Winkworth et al. 1995), but during exercise, physiological needs prevail. This reversal raises questions about the extent of speaker agency in breath control during exercise: Do speakers merely react to changing metabolic demands, or do they make active, strategic choices to sustain speech? This same issue of agency and flexibility is likewise addressed in Ziegler et al. (2019).

To evaluate whether speech adds additional volume requirements to exercise in the present dataset, tidal breathing data recorded before each speech task were analyzed. If breath volume is governed mainly by metabolic factors, comparable increases should be observed in exercise with and without speech. Table 4.11 presents raw means, and Figure 4.30 displays inspiratory displacement for each speech task and workload level. While mean values were similar for speech and no-speech conditions – suggesting that breath volume was indeed primarily driven by metabolic rather than speech demands – the data add two nuances to prior findings.

First, although inspiratory and expiratory displacement increased with workload, the increase at low workload was approximately 20% greater for sustained vowels than for reading and no speech (also seen in the model-predicted means). The difference likely reflects higher airflow during steady-state phonation. During connected speech, airflow is intermittently limited by glottal closures and vocal tract constrictions, whereas sustaining a vowel requires an open glottis and vocal tract, resulting in a steady outflow of air. To compensate, speakers may take in more air prior to phonation. This effect is amplified during exercise, when elevated lung volumes lead to higher driving pressure and airflow. While these patterns suggest that speakers modulate breath volume according to task demands, the considerable variability and uncontrolled prosodic differences limit strong claims from these raw means. Future work should test this directly under controlled loudness and prosodic conditions to determine whether volume adjustments are a practiced strategy.

Second, the large error bars indicate considerable individual variability, particularly at higher workloads. Although variability also increased without speech, it was most pronounced for speech tasks, particularly sustained vowels, indicating a broader range of breath depths. This may reflect individual preferences or flexible adjustment of breath depth to linguistic demands, such as utterance length. When averaged, these adjustments likely canceled out, yielding mean volumes comparable to tidal breathing. Such variability is compatible with an adaptive control perspective: while speakers must take in more air during exercise, they appear to retain flexibility in how that volume is distributed as speech unfolds.

As seen in the EMMs and visualizations, timing results showed substantial variation across tasks and speakers, consistent with active, strategic breath control. The most striking task difference occurred for inspiratory duration: during reading, inspiratory duration increased with workload, whereas for sustained vowels it decreased at moderate intensity. Baselines also differed, with inspiratory durations for sustained vowels roughly twice those for reading (log-model prediction: 430 ms vs. 950 ms). Further, breathing without speech showed average inspiratory durations of about 1.5 s. This continuum suggests that sustained vowels employed an intermediate strategy between tidal and speech breathing. Inspection of the RIP traces confirmed that speakers often maintained tidal breathing until the stimulus appeared, then initiated the vowel material. In some cases, the stimulus even triggered a rapid, speech-like inspiration, demonstrating immediate switching between breathing modes.

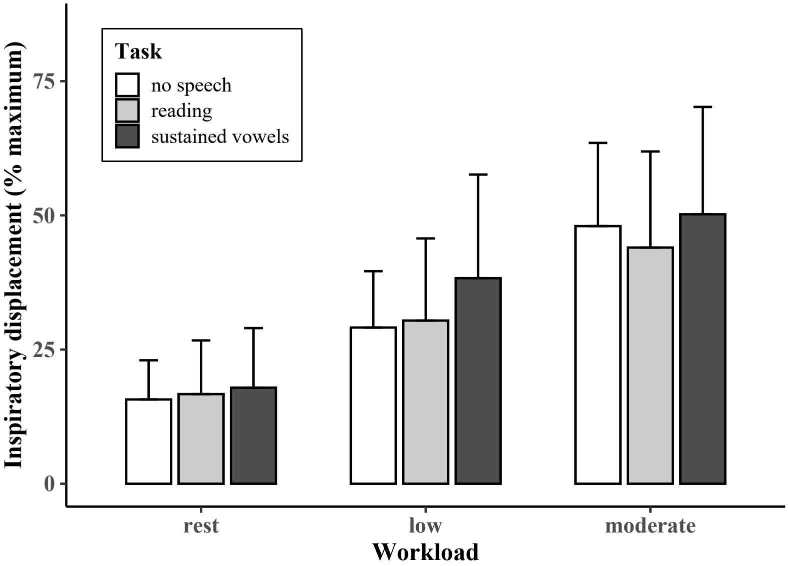

Figure 4.30: Mean inspiratory displacement (% of maximum) during rest, low, and moderate workload conditions for the no-speech, reading, and sustained vowel tasks; error bars represent ±1 SD, illustrating variability across participants.

Table 4.11: Mean (SD) inspiratory and expiratory displacement (% of maximum) across workload and task conditions for the no-speech, reading, and sustained vowel tasks.

Workload	Task	Mean (SD)	
		Inspiratory displacement	Expiratory displacement
Rest	No speech	15.7 (7.3)	15.7 (7.3)
	Reading	16.9 (10.0)	17.5 (11.3)
	Sustained vowels	17.9 (11.1)	20.0 (10.4)
Low	No speech	29.1 (10.5)	29.0 (10.1)
	Reading	30.4 (15.3)	29.5 (18.3)
	Sustained vowels	38.3 (19.3)	39.8 (19.5)
Moderate	No speech	48.0 (15.5)	48.1 (15.3)
	Reading	44.0 (17.9)	43.1 (19.0)
	Sustained vowels	50.2 (19.9)	51.3 (20.8)

The opposite workload effects can be explained by the different constraints of the two tasks. In the sustained vowel task, inspiratory durations were already long at baseline and were not tied to any linguistic unit. Since speakers often appeared to rely on ordinary tidal breathing, it was unproblematic for them to shorten the inspiratory phase when exercise increased respiratory drive. In reading, however, inspiratory duration is linked to utterance planning. As workload increased, speakers needed more air, but they were simultaneously trying to keep inspirations brief so that speech would continue smoothly. Thus, reading appears to have created a linguistic pressure to maintain short inspirations, even as physiology demanded longer ones. Nevertheless, exercise effects were modest in both tasks, indicating stability within each strategy. Earlier studies have reported mixed effects of exercise on inspiratory duration, showing both increases (Trouvain & Truong 2015) and decreases (Meckel et al. 2002) under differing task constraints. The present pattern suggests that such differences likely reflect task-specific adaptation rather than a pure effect of exercise.

Expiratory duration decreased with exercise, but patterns differed across tasks. During reading, expiratory duration remained stable from rest to low workload, then dropped sharply at moderate intensity. Given that expiratory durations are also linked to utterance lengths, a likely explanation for this pattern is that speakers initially preserved utterance length, maintaining expiratory timing close to baseline, before the drive to breathe at moderate workload necessitated shorter expirations. The nearly 1-second reduction directly reduced available speaking time, consistent with some participants' reports of needing to interrupt speech to breathe.

In contrast, expiratory durations for sustained vowels decreased steadily with exercise, suggesting there was no reason to maintain them. As the RIP traces showed, speakers generally incorporated vowel production into an existing breath cycle. Thus, although expiratory duration decreased, there was still ample time to produce a 2–3 second vowel (mean predicted duration at moderate workload: 3.7 s). The overall longer durations for sustained vowels likely reflect mechanical differences. Steady-state phonation allows continuous airflow through an open vocal tract, reducing laryngeal effort and potentially alleviating dyspnea. Reading, however, requires continual glottal valving and vocal fold adjustments against higher subglottal pressures, increasing effort and potentially limiting the ability to maintain long expiratory phases. These mechanical differences were reflected in participants' subjective experiences: around half reported little or no increase in difficulty producing vowels at moderate versus low workload, whereas only 8% reported similar ease for reading, with 85% indicating greater difficulty.

These systematic differences in expiratory duration indicate that speakers modulated breathing according to task demands. During reading, speakers appeared to resist the urge to breathe more frequently, maintaining baseline expiratory duration to satisfy linguistic timing requirements. In contrast, during sustained vowels, they adjusted breathing to accommodate physiological demands, as linguistic constraints were minimal. Comparable evidence is scarce: while Meckel et al. (2002) also reported decreases in expiratory duration with increasing workload, the study lacked resting and low-workload conditions. The current findings therefore suggest greater flexibility in how speakers balance physiological and linguistic demands during the expiratory phase – the portion of the cycle most critical for speech – than previously recognized.

Inspiratory fraction increased with exercise during reading, reflecting the shortening of the expiratory phase. In contrast, sustained vowels showed no systematic change, but displayed greater variability, reflecting the wider range of inspiratory durations. Visualization of the raw data (Figure 4.12) makes this task difference clear, yet overall, inspiratory fraction appears to be a less sensitive indicator of adaptive change than the individual timing measures. No previous studies reviewed here have reported on inspiratory fraction during exercise, leaving little basis for comparison. However, the effect of workload appears modest (0.13–0.18 for reading; 0.20–0.21 for vowels), and values remained far below those typical of tidal breathing (≈ 0.40; Lenneberg 1967) and confirmed in the present data (0.40–0.45). These values suggest that even during increased respiratory drive, speakers preserve a speech-specific organization of breathing, distinct from tidal respiration.

Finally, respiratory rate (breaths per minute) provided a global index of breathing over time. As expected, it was largely the inverse of expiratory and total cycle duration (not discussed here), which are highly collinear measures. Respiratory rate increased with workload in both tasks, mirroring the patterns seen for expiratory duration: during reading, increases appeared only at moderate intensity, whereas sustained vowels showed a steady rise across levels. The magnitude of these changes aligns with previous reports of respiratory rate adjustments during speech under exercise (Doust & Patrick 1981, Trouvain & Truong 2015, Fuchs et al. 2015, Meckel et al. 2002). Beyond confirming this general trend, however, respiratory rate does not reveal additional information about how speakers adjust their breathing patterns.

While the statistical analyses show significant group-level effects, the line plots add nuance by revealing considerable interspeaker variability that did not occur uniformly across measures. Breath volume showed a broadly consistent response to exercise across speakers, whereas timing measures varied widely.

Speakers displayed distinct trajectory shapes, such as rise–plateau, plateau–rise, and mountain/valley. Speaker differences were most pronounced at low workload, suggesting either greater strategic diversity under modest demands or a learning effect in which moderate-workload breathing moved closer to baseline after "practice" at low workload. These patterns complicate previous findings of more uniform behavior. However, those studies may have had methodological constraints – Meckel et al. (2002) required a fixed reading pace, reducing timing flexibility, and Doust & Patrick (1981) had only six participants, likely too few to reveal meaningful variability. Overall, the line plots suggest that individual breathing strategies are far more diverse than group means suggest, providing evidence that breathing adaptations during exercise may be more flexible and individualized than group means suggest.

The heatmap analyses provided a multivariate perspective to complement the statistical and line-plot results, revealing systematic patterns in speaker adaptations that might appear contradictory when individual measures are considered in isolation. They suggest that volume and timing can vary relatively independently as flexibly adjusted dimensions and provided support for individualized breathing patterns, with speakers showing consistent adaptations across workloads. At a conceptual level, adaptation patterns suggested by the heatmaps can be organized along three dimensions: breath depth, frequency, and duration. Speakers appear to fall into broad strategies: (1) volume-dominant, showing substantial increases in volume with minimal timing changes; (2) timing-dominant, showing reduced expiratory duration and increased respiratory rate with smaller volume changes; and (3) an intermediate approach with moderate volume increases with minimal timing adjustments. At moderate workload, balanced strategies remained evident but less distinct, reflecting a trend toward greater reliance on increased breath volume. These patterns highlight the flexibility and individuality of speech breathing during exercise, though further work is needed to confirm and quantify these strategies systematically. The present findings are consistent with an account of speaker-specific breathing strategies during exercise, a possibility also suggested by Patil et al. (2010), who proposed that individuals prioritize different phonatory dimensions.

The qualitative analysis revealed insights into participant experience that help contextualize the quantitative patterns. Participants found reading more difficult than producing vowels, largely because the fixed structure of the text conflicted with their altered and increasingly unpredictable breathing as workload increased. They expressed explicit intentions to maintain their preferred phrasing and described being interrupted mid-phrase by the impulse to breathe. This aligns with expiratory-duration results, suggesting that speakers did indeed attempt to preserve phrasing, and explaining some variability. Participants also described specific breath modifications – such as consciously taking deeper breaths or pausing more frequently – which would be compatible with the different breath patterns displayed in the heatmaps. At the same time, the qualitative data highlighted substantial diversity in experience: the same feature (e.g., punctuation or low workload) could be helpful for one participant and challenging for another. While these contrasts cannot be directly matched to individual trajectories in the line plots, they suggest that individual dispositions and internal states may shape speech during exercise, even under controlled conditions. Finally, several participants reported learning to manage their breathing better as the experiment progressed, suggesting strategic awareness and deliberate adjustment. Future studies might ask participants directly about changes in breath depth versus timing to clarify the basis of these preferences.

4.5.2 Synthesis and broader implications

Together, these strands of evidence addressed the question: How does speech breathing change during exercise – do speakers adjust volume, timing, or both? Prior research suggested that because physical workload primarily drives breath volume (Doust & Patrick 1981, Meckel et al. 2002), speakers mainly adjust timing instead (Baker et al. 2008), resulting in shorter utterances. Yet most studies used combined measures, leaving limited direct evidence on timing-specific or volume-specific changes. On a practical level, understanding how speakers modify breathing during exercise matters because changes such as fragmented utterances could hinder communication. In terms of linguistic theory, systematic shifts in utterance length could reveal something about lookahead or speech planning more broadly.

The methodological choices in this study – particularly the speech tasks and workload levels – revealed nuances in breath adjustments. In terms of volume adjustments, workload was indeed found to be the primary driver – but not the only one. Both speech task and speaker-specific preferences also played clear roles. Group-level patterns showed that speakers inhaled roughly 20% more air for sustained vowels than for reading, suggesting that when required to meet specific speech goals, speakers take in additional air beyond what is needed for metabolic demands. This suggests that speech itself can impose extra volume requirements, which may be relevant for other goals such as loud speech during exercise. The heatmaps further revealed that some individuals consistently took much larger breaths than others, indicating that although volume needs are workload-driven *when averaged over time*, the volume of individual breath can vary in speaker-specific ways. In other words, smaller timescales reveal individual variability in volume at the same workload. Together, these findings suggest that breath volume during exercise is shaped by a combination of task demands, workload, and individual strategy – not workload alone.

The timing results also proved more nuanced than previously assumed. First, timing effects were shown to be workload dependent. When speakers could maintain their baseline timing, they generally did – even when it entailed discomfort. This was evident in the group-level expiratory data, which showed no change in duration from rest to low workload. Such stability had not appeared in earlier work simply because previous studies did not examine low workloads. The results suggest that maintaining baseline timing is feasible up to a certain threshold but becomes difficult at higher intensities, such as the >50% levels tested in Baker et al. (2008). This is also supported by participant observations of attempting to maintain certain phrasing but being interrupted by the drive to breathe at moderate workload. Second, timing was also found to be speaker-dependent. The qualitative data showed that some speakers resisted breaking up their utterances, while others embraced shorter chunks, sometimes even reporting strategic benefits in using syntactic breaks for breath pauses. The heatmaps supported this individual variability, revealing clusters of speakers who made minimal volume adjustments but compensated with higher respiratory rates.

Inspiratory durations revealed other insights into what speakers do to adapt breath timing during exercise. What proved most interesting emerged unexpectedly from the vowel task. Instead of taking short inspirations characteristic of connected speech, speakers produced slow inhalations that fell between tidal and speech breathing. RIP traces showed that many participants showed a tidal-breathing pattern until the stimulus appeared and then initiated the vowel. In some cases, the stimulus even triggered a rapid, speech-like inspiration, showing an instantaneous switch between breathing modes. These patterns suggest that speakers can incorporate short utterances – calling out or backchanneling – into ongoing tidal breathing on a breath-by-breath basis. They may enter "speech mode" only when needed and return to automatic breathing immediately afterward. This moment-to-moment switching resembles other volitional respiratory behaviors, including sniffing, laughter, and sighing, where the default respiratory rhythm can be temporarily overridden to achieve an immediate communicative or physiological goal.

Taken together, the results show that speakers retain flexibility in how they manage their breathing during exercise. They can time breaths economically when speech demands are minimal and can also accommodate sustained utterances when needed. To come back to the main research question: all speakers adjusted both breath volume *and* timing, but some focused on one more than the other. This was especially clear in the heatmaps but was also reported by participants. This evidence of speakers consciously modifying different aspects of their breathing suggests a working hypothesis that breath-adaptation strategies during exercise may be speaker-specific.

This leads to the second research objective of examining how respiratory patterns vary across individuals. This question is not as straightforward to answer. The qualitative data provided insight into participant thinking, showing that speakers were aware of both endeavoring to achieve a linguistic goal and modifying specific aspects of their breathing. However, the factors shaping their intentions appear to be wide-ranging and go beyond workload level (the experimental manipulation). For example, some speakers seemed less concerned with breaking up phrases, which could be rooted in pragmatism (to avoid discomfort) or be more closely aligned with a speaker's habitual phrase length or even reflect a genetic predisposition toward specific breathing patterns (Benchetrit 2000). Some comments hint at underlying personality traits, like conscientiousness (e.g., "I wanted to read the text properly"), which are also known to affect dual-task performance (LeMonda et al. 2015). Previous studies offer little comparison. While Baker et al. (2008) and Meckel et al. (2002) established that participants reported dyspnea, they provide little insight into how speakers experience or respond to that discomfort. The present findings suggest considerable individual variation: while some participants found speaking increasingly uncomfortable, others reported little difficulty or actively adopted breath-management strategies to reduce discomfort. Moreover, while all participants noticed changes in their speech, not all were bothered. Some even reported that speaking felt easier than at baseline despite greater exertion. Physical attributes such as body (lung) size or current fitness level appeared to be less likely explanations in the current study. While data on various potential covariates were collected prior to the experiment, no clear pattern emerged during preliminary analyses of the quantitative results.

4.5.3 Methodological considerations

Three experimental design and analysis choices are particularly relevant in interpreting the findings. First, a fixed order of experimental conditions was used because exercise continues to affect the body after it has ceased, meaning that a rest condition following exercise could compromise the baseline. However, both the "mountain/valley" patterns in individual trajectories and participant comments suggested possible learning or familiarization effects at moderate load. Although not observed for all speakers, such adaptation could have influenced the estimated effect size of moderate workload. At the same time, evidence of learning would be theoretically interesting, because it aligns with findings on rapid adaptation to mechanical speech perturbation (e.g., Savariaux et al. 1995). High respiratory drive may thus be viewed as a perturbation to which speakers learn to adapt – an avenue that could extend current models of speech motor flexibility. Future research could vary workload order (e.g., rest–moderate–light) to test learning effects, or compare individuals who habitually speak during exercise (e.g., fitness instructors) with those who do not. Given the potential for cognitive learning in dual-task performance, such studies should also assess speaker perceptions of attention allocation and conscious breath planning.

Second, the measures necessarily simplified a highly dynamic process. Averaging across trials revealed group-level patterns but obscured the considerable breath-by-breath variability evident in the RIP traces. For example, inspiratory and expiratory volume were often unequal within a single breath cycle, with the volume of air in the lungs increasing as a trial progressed. Such air-trapping is particularly common in speech during exercise because faster breathing shortens expiratory duration and can prevent emptying of the lungs before the next inspiration. Speaking from elevated lung volumes can make subglottal pressure more difficult to regulate efficiently, and glottalized or pressed phonation may further impede airflow, increasing perceived effort and breathlessness – phenomena reported by participants. These analyses therefore capture global effects of physical workload on breathing, while finer-grained patterns that likely shaped moment-to-moment speaking experience remain unexplored. Future work could examine local variability and transient adaptations in spontaneous speech, where speakers manage breathing and phrasing more freely.

Finally, individual traits were not analyzed as explanatory factors for respiratory adaptations. Preliminary visualizations of pre-collected covariates such as fitness level and mood showed no clear alignment with respiratory patterns in the heatmaps. This absence of pattern may reflect both the limitations of the simple measurement instruments and the inherent complexity of factors shaping breathing strategies; post-experiment questionnaires indicated that participants attributed their speech behavior to a diverse mix of mental and physical influences, including prior experience speaking during exercise, bodily discomfort, and engagement with the material. Given this variability and the dynamic nature of some factors (e.g., mood), it seems unlikely that outcomes can be traced to a single physical (and fluctuating) trait such as fitness level. Future research might instead draw on personality measures, as stable traits (e.g., conscientiousness, agreeableness) could offer more insight into how individuals approach speech performance under stress.

4.5.4 Conclusion

Previous studies of speech during exercise have typically focused on single parameters or combined measures, while investigations of both breath depth and timing have examined how speech affects exercise rather than how exercise affects speech. The present study systematically examined both volumetric and temporal dimensions of speech breathing during exercise. The results showed that while breath volume increases systematically with exercise, speakers retain flexibility in breath timing, which varies by speech task and individual. These findings suggest that speech breathing during exercise reflects a balance between obligatory metabolic constraints and speaker-specific adaptation. Different patterns of adaption may reflect differing linguistic goals but may also be shaped by internal factors such as attentional focus or emotional response. Models of speech-respiratory control should therefore account not only for metabolic demands but also for the flexibility speakers exhibit in responding to them.

5 Phonation: Investigation of glottal source measures

5.1 Introduction

This study investigates how speakers achieve phonation during exercise, a form of physiological stress. Already in neutral speech, individuals differ in how they adjust the laryngeal system to achieve phonation, so it is plausible that stress-related changes in phonation may differ across speakers. Much phonetic research on stress-related changes in voice has focused on f0 and loudness, leaving many aspects of phonation unexplored. Additionally, the effect of stress on f0 is not entirely clear. Although most studies report increases and treat elevated f0 as a reliable indicator of stressed speech, some work has reported substantial speaker variability, suggesting that group-level patterns may obscure individual phonatory strategies. One goal of this study is thus to clarify the extent of individual variability in f0 changes during exercise. The other goal is to examine additional acoustic measures that provide insight into vocal fold dynamics, in order to reveal whether speakers make different laryngeal-level changes during exercise. To do this, the study assesses 11 acoustic measures reflecting different aspects of vocal fold dynamics, measured across two speech tasks and three vowel contexts. This chapter begins by outlining relevant acoustic measures and their relation to production conditions, recapping open questions from previous research, and stating the study objectives. The remainder of the chapter presents results from inferential statistical analyses, followed by a visual exploration of individual speaker patterns to highlight variability, and a discussion of all results.

5.1.1 Laryngeal mechanics and acoustic output

Phonation occurs when airflow from the lungs passes through the vocal folds, two multilayered structures in the larynx that can be brought together to support self-sustained vibration and generate the voice source signal. Achieving phonation depends on the coordinated control of different aspects of the vocal folds, including their position (abduction vs. adduction), how firmly they are pressed together, and their longitudinal tension/stiffness. When the folds are sufficiently adducted, rising subglottal pressure forces them apart, and aerodynamic and elastic forces bring them back into contact, producing a repeating oscillatory cycle (myoelastic-aerodynamic theory; van den Berg 1958). Crucially, changes in vocal fold closure, vibratory regularity, and tension have acoustic consequences that can be quantified using different acoustic measures, providing indirect evidence about patterns of vocal fold vibration and glottal configuration.

Such phonatory adjustments are not only measurable acoustically but are also perceptible to listeners as overall impressions or "qualities" of a given voice. Early conceptualizations placed voice quality along a continuum of laryngeal constriction from "breathy" (more open) to "creaky" (more constricted), with modal phonation in the middle (Ladefoged 1971). This view was later extended to account for a wider range of human vocal behaviors. Laver (1980) proposed a multidimensional framework in which voice qualities arise from configurations along multiple parameters of laryngeal muscular tension.

Specifically, Laver (1980) described three fundamental parameters of laryngeal muscular tension, illustrated in Figure 5.1. *Adductive tension* refers to the force that draws the arytenoids together to close the cartilaginous glottis. *Medial compression* refers to the force that draws the vocal processes of the arytenoids together to close the ligamental glottis. *Longitudinal tension* refers to tautness along the length of the vocal folds. Different combinations of these parameters give rise to a range of voice qualities beyond the breathy–creaky continuum and link perceptible acoustic patterns to underlying laryngeal settings.

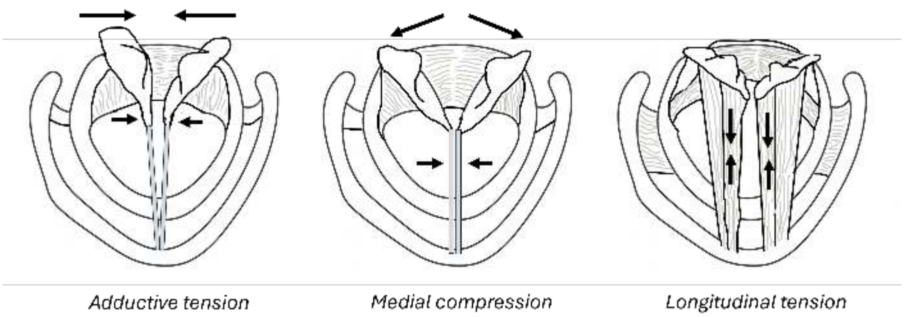

Figure 5.1: Schematic diagram of Laver's (1980) three parameters of laryngeal muscular tension

For the present study, this framework is useful because it connects speakers' physiological adjustments to acoustic changes that give rise to an overall perception that speech under stress is distinct from neutral speech. Importantly, such adjustments need not be intentional – muscular tension may also change as a physiological response to stressors, regardless of whether they are physical, emotional, or cognitive. This perceptual framework helps interpret how production changes may be experienced by listeners, even though the present study is production-focused. The next section outlines the acoustic proxies used in this study, grouped by what aspect of phonation they index, to support interpretation of the results that follow.

5.1.2 Acoustic measures of glottal source dynamics

Phonation is inherently multi-dimensional: Vocal fold behavior unfolds across both time and space, and both dimensions are reflected in the acoustic signal. Temporally, phonation varies in the manner and frequency of vocal fold vibration, whereas spatially it involves changes in the degree of vocal fold approximation. Acoustically, these dimensions can be assessed through measures of periodicity and spectral shape. In the current production-oriented study, these are referred to as "glottal source" measures, whereas perception-oriented studies typically refer to them as "voice quality" measures.

5.1.2.1 Periodicity measures

Fundamental frequency (f0) reflects the repetition rate of the acoustic waveform and is perceived as pitch. This periodicity arises from cyclic opening and closing of the vocal folds. Physiologically, changes in f0 are commonly associated with alterations in vocal fold tension and subglottal pressure, although they do not uniquely specify the underlying mechanism. Cycle-to-cycle irregularity in this repetition rate can be quantified using perturbation measures such as jitter (temporal variability) and shimmer (amplitude variability). However, Hillenbrand (2011) notes that jitter and shimmer show limited reliability and validity. This reflects both their small effect sizes and the difficulty of accurately locating individual cycles in natural speech, something that is already difficult to obtain in modal, healthy voices and further complicated by the phenomena that distinguish non-modal voices, such as may be expected under stress. Consequently, sustained vowels are often used for perturbation analysis because they provide more stable phonation in which cycles can be more reliably identified.

Further periodicity measures include harmonics-to-noise ratio (HNR) and cepstral metrics. HNR (Yumoto et al. 1982) quantifies the relative strength of harmonic (periodic) energy compared to aperiodic energy across frequency bands.

Lower HNR indicates reduced harmonic organization in the signal. While HNR can reflect aspiration noise (Qi & Hillman 1997), the measure is better thought of as an indication of modal vs. non-modal voicing (Garellek & Keating 2011) because aperiodicity can arise from either breathy or glottal phonation. Like jitter and shimmer, HNR requires precise demarcation of each cycle and can be less stable when phonation is highly irregular.

Cepstral peak prominence (CPP; Hillenbrand et al. 1994) provides a more robust estimate of periodic structure in the acoustic signal. Rather than relying on cycle boundary detection, CPP measures the prominence of the harmonic spacing pattern in the cepstrum, allowing periodicity to be estimated even when individual cycles are difficult to identify. Consequently, it is more stable for effortful, pathological, or connected speech. Unlike HNR, which evaluates periodicity within specific frequency bands, CPP reflects harmonic organization across the spectrum.

5.1.2.2 Approximation measures

Spectral measures primarily reflect changes in the distribution of harmonic energy across frequencies, which are associated with differences in glottal configuration and closure characteristics, and are commonly used to examine differences in phonation type (Hillenbrand 2011). Incomplete or gradual glottal closure can manifest as a posterior gap, slowed or non-simultaneous closure along the length and/or width of the vocal folds, or an increased open time in the glottal cycle. Spectral measures have proven valuable in analyzing phonemic contrasts in phonation across languages (for an overview, see Di Napoli 2018: 81–85).

Phonation type can be characterized acoustically through the relative amplitudes of harmonics in the spectrum. H1−H2, the difference between the first two harmonics, is among the most common spectral measures. It indexes the low-frequency spectral slope of the voice source. Larger values indicate a more rapidly decreasing spectrum (a more diffuse voice quality), whereas smaller values indicate a flatter spectrum (a more compact quality). It is associated with open quotient – the proportion of time the glottis remains open per cycle (Holmberg et al. 1995) – and may reflect vocal fold thickness (Zhang 2016), although the measure does not uniquely specify the underlying mechanism.

Additional spectral measures used in studies of phonemic voice quality include H1—A1 and H1—A3, which compare the first harmonic to energy near the first and third formants, respectively. Acoustically, these measures reflect how strongly harmonic structure stands out relative to low- and high-frequency spectral energy. They are frequently interpreted as correlates of glottal configuration, such as incomplete closure or more abrupt closure (Hanson 1997, Hanson et al. 2001), but are also influenced by vowel acoustics.

Collectively, these spectral slope measures are often interpreted as reflecting the degree of vocal fold closure and the abruptness of voicing excitation, while remaining sensitive to vocal-tract filtering. One limitation of harmonic amplitude parameters is their sensitivity to formant frequency and bandwidth. This means that different vowels – and different speakers – cannot be compared directly. To address this, previous work (Hanson 1997, Iseli et al. 2007) developed an algorithm to correct for the effect of formant frequencies and bandwidths. Corrected values are denoted by asterisks (H1*-H2*, H1*-A1*, H1*-A3*).

A final glottal source measure is strength of excitation (SoE). Derived from the algorithm in Murty & Yegnanarayana (2008), SoE estimates the magnitude of the brief excitation impulse that drives each cycle of voicing. In acoustic terms, it reflects how sharply energy is injected into the vocal tract at each cycle. Because the measure targets the excitation event rather than the radiated spectrum, it is less affected by vocal-tract filtering (Chong et al. 2020: EL65). Larger values correspond to a more abrupt excitation, often associated with firmer or more simultaneous vocal fold closure, whereas smaller values indicate a more gradual excitation.

5.1.3 Previous work on laryngeal effects of exercise

Building on the literature review in Chapter 2 describing laryngeal-level effects of exercise, several important research gaps emerge, particularly concerning vocal fold approximation. Further, though several studies have focused on the frequency and manner of vocal fold vibration, two key questions remain unresolved: the nature of fundamental frequency (f0) changes and the effects on harmonics-to-noise ratio (HNR).

Previous research on f0 suggests a complex pattern of results. While several studies examining multiple exercise levels report linear increases in f0 as exercise intensifies (Fuchs et al. 2015, Primov-Fever et al. 2014, Ziegler et al. 2019), other findings suggest more nuanced relationships. Johannes et al. (2007) documented a non-linear relationship in elite soldiers, where f0 showed an initial small increase, maintained a plateau during moderate exercise, and increased sharply only at exhaustion levels. The researchers suggested this pattern reflected the participants' high fitness levels, arguing that because they tolerated the exercise well, their vocal performance remained largely stable until extreme exertion. Further evidence of speaker-specific f0 responses comes from Godin & Hansen (2008), who found considerable individual variation among their 51 participants.

While the majority showed increased f0, approximately 25% showed no change and about 14% showed *decreased* f0. These varying responses raise questions about how f0 is impacted during exercise.

Methodological considerations may partially explain these differences. Godin and Hansen's use of a fixed exercise rate (10 mph on an elliptical) meant that participants experienced different relative workloads based on their individual fitness levels. Understanding the relationship between exercise and f0 carries particular importance given that f0 is both perceptually salient and represents the most frequently measured parameter in research on speech under stress. The documented individual variations suggest a need for more detailed investigation of how physical exertion affects f0 across different speakers and conditions.

Research on HNR during exercise presents similarly mixed results. Several studies found no significant changes during exercise (Dallaston & Rumbach 2016, Koblick 2004, Primov-Fever et al. 2014), which seems surprising given both the presence of high respiratory drive and impressionistic reports of breathier speech in respiratory studies (Doust & Patrick 1981, Otis & Clark 1968). Results pointing toward breathier speech were also obtained by Godin & Hansen (2015) using normalized amplitude quotient (NAQ) and harmonic richness factor (HRF), though some speakers showed opposite trends toward tense or pressed voicing. Further support for exercise effects on HNR comes from Weston et al. (2021), who demonstrated significant decreases in HNR in vowel /a/.

The source of these disparate findings is unclear. Neither Koblick (2004) nor Primov-Fever et al. (2014) discussed their non-significant results, while in other studies methodological factors may play a role. In the case of Dallaston & Rumbach (2016), speech was recorded immediately *after* participants led a 1-hour fitness course, so voicing may have already been returning to baseline. Godin & Hansen (2015)'s study analyzed data from the UT-Scope corpus, where participants exercised at different relative workloads.

Beyond resolving these inconsistent findings, an even more fundamental limitation emerges from the narrow focus of existing research. Analyses of fundamental frequency and harmonics-to-noise ratio capture only the temporal dimension of vocal fold behavior – how the vocal folds vibrate over time. The spatial dimension – specifically, how closely the vocal folds approximate during exercise – remains largely unexplored. This gap in knowledge limits our understanding of glottal source behavior during physical activity. The mixed results in previous research not only highlight areas needing clarification but also underscore how our understanding is constrained by examining just one dimension of vocal fold function.

5.1.4 Objectives of the present phonatory study

This study aims to examine phonatory behavior during physical activity using multiple acoustic measures. Prior research has largely focused on fundamental frequency (f0). However, f0 alone cannot explain how phonation is maintained during exercise and its increased respiratory demands. Additionally, open questions remain about the consistency of f0 effects. The present study therefore sets out to examine glottal source behavior using a multidimensional approach to characterize these phonatory adaptations. Specifically, the study asks:

(1) **How does exercise affect periodicity?**

The periodicity measures analyzed are f0, cepstral peak prominence (CPP), and harmonics-to-noise ratio (HNR) across frequency bands to evaluate changes in the temporal dimension of vocal fold dynamics.

(2) **How does exercise affect glottal closure?**

The measures of vocal fold approximation analyzed are RMS energy, H1*-H2*, H1*-A1*, H1*-A3*, and strength of excitation (SoE) to gather new data on the spatial dimension of vocal fold dynamics.

Finally, the study explores to what extent phonatory adaptations vary across individuals and speech contexts, both to shed light on previous contradictory f0 findings and to better understand variation in speech during exercise.

To capture a range of phonatory behaviors, the study examines both sustained vowels and connected speech.

Table 5.1: Predictions for each glottal source measure during exercise

Measure	**Prediction**	**Basis**
f0	Increases	Higher subglottal pressure
Energy	Increases	Higher subglottal pressure
H1*-H2*	Decreases	Greater vocal fold contact to overcome aerodynamic forces
H1*-A1*	Increases	Greater glottal opening to facilitate airflow
H1*-A3*	Decreases	More abrupt closure from increased pressure and airflow
SoE	Increases	More abrupt closure from increased pressure and airflow
CPP	Decreases	Reduced control of vocal fold tension/length due to muscle activation (e.g., during inspiratory braking)
HNR	Decreases	Reduced control of vocal fold tension/length

5.2 Methodology

5.2.1 Speech materials

Recordings of the German corner vowels /i, a, u/ were obtained from the corpus described in Section 3.2.1. These corner vowels were selected to characterize the acoustic and articulatory working space, as they represent different combinations of height and backness. For analysis, two complementary speech tasks were chosen: 1) reading a passage and 2) producing sustained vowels. Repeated instances of connected speech offer a controlled yet naturalistic production context across speakers and conditions, while sustained vowels allow a focused assessment of a single vocal tract configuration over multiple glottal cycles. This focus can reveal information that is potentially obscured by the dynamic nature of connected speech. Thus, these two speech tasks may provide different insights into phonatory effects in speech during physical activity.

5.2.2 Preparation of vowel samples

The vowel samples were prepared for analysis following the procedure developed in the articulatory-level study. That study was conducted first, as speaker-specific formant settings were required for the method (see Section 6.2), which is summarized here. Vowel samples were created by extracting 200 ms sections from the sustained vowels. In read speech, the entire vowel was extracted from the contexts given in Table 5.2.

Table 5.2: Words used to extract corner vowel tokens, shown in angled brackets, from adjacent sentences in the read passage

Vowel	Word	Sentence
/iː/	*organis<ie>ren*	*Um ein perfektes Fest zu organisieren, braucht man eigentlich gar nicht so viel.* ‘To organize a perfect party, you actually don’t need all that much.’
/aː/	*spont<a>n*	*… man kann das auch relativ spontan machen, so lange Läden noch offen haben.* ‘… you can actually do it pretty spontaneously as long as the shops are still open.’
/uː/	*g<u>te*	*Wichtig ist einfach, dass man gute Laune hat…* ‘The important thing is just being in a good mood…’

Because many of the read-speech tokens were relatively short, a 25 ms buffer was added to both the beginning and end of each token to provide an adequate window for analysis. Acoustic analysis was conducted using VoiceSauce (Shue et al. 2011), implemented in MATLAB. The Praat algorithm was employed to obtain values for the formants and the acoustic measures analyzed. All values were averaged over the duration of the vowel section. A consistent f0 range was applied across all participants (Vogel et al. 2009). After an initial inspection of the data, the pitch floor was set to 130 Hz to avoid pitch-halving errors, while the ceiling was set to 350 Hz to capture the highest observed values.

Formant estimation settings – specifically, the formant ceiling and the number of formants – were determined for each speaker and vowel. This individualized approach has been found to minimize formant-tracking errors (Harrison 2013, Shadle et al. 2016). These settings were identified based on visual inspection of the formant tracks in the spectrogram, ensuring that F1 and F2 remained roughly parallel to each other and centered within their respective formant bands. The values for F1 and F2 were subsequently cross-checked with reference values for relevant German vowels produced by female speakers (for connected speech: Pätzold & Simpson 1997; for sustained vowels: Tolkmitt & Scherer 1986). Vowel tokens with uncertain estimates were excluded from analysis, since poor f0 and formant estimates can lead to unreliable spectral tilt estimates, such as H1–H2, whose values are more challenging to verify. At the time of the study, no reference values for glottal source measures in German were found. The total number of tokens for analysis is given in Table 5.3.

Table 5.3: Number of observations per condition, task, and vowel

	Read passage			**Sustained vowels**		
	/i/	a/	/u/	/i/	/a/	/u/
rest	140	134	137	138	131	137
low	141	133	137	137	133	136
moderate	138	131	135	135	129	133
TOTAL	**419**	**398**	**409**	**410**	**393**	**406**

5.2.3 Acoustic measures analyzed

Several acoustic variables were analyzed to assess the degree of vocal fold approximation and the rate and manner of vocal fold vibration in speech during exercise.

These measures and the methods used to obtain them are given in Table 5.4. Note that although fundamental frequency (f0) and vocal intensity typically correlate, VoiceSauce employs a variable window that normalizes the root mean square (RMS) energy calculation with f0, thereby reducing the correlation between the two. For details, see the VoiceSauce documentation (Shue et al. 2011).

Table 5.4: Acoustic measures analyzed in the study; asterisks indicate corrected harmonic amplitudes.

Measure	Physiological correlate	Method
f0	Glottal pulse frequency	Praat range: 130–350 Hz
Energy	Power source intensity	Calculated per frame over 5-pulse window
H1*-H2*	Open glottis proportion; perhaps vocal fold thickness	(Hanson 1997, Iseli et al. 2007)
H1*-A1*	Incomplete glottal closure	(Hanson 1997, Iseli et al. 2007)
H1*-A3*	Incomplete glottal closure; perhaps closure abruptness	(Hanson 1997, Iseli et al. 2007)
SoE	Vocal fold closure abruptness	(Murty & Yegnanarayana 2008)
CPP	Spectral aperiodicity	(Hillenbrand 2011)
HNR	Frequency-band aperiodicity	(de Krom 1993)

5.2.4 Statistical analyses

Statistical analyses were conducted using linear mixed-effects models[1] to account for the repeated-measures structure of the data, with multiple observations obtained from each participant across workload levels, speech tasks, and vowel contexts. Physical workload (rest, low, moderate) was treated as the primary predictor, with acoustic measures serving as outcome variables. Speech task (read speech vs. sustained vowels) and vowel quality were included as additional predictors, given prior evidence from the respiratory and articulatory studies (Chapters 4 and 6) that both factors influence speech production and may modulate exercise effects.

Interactions between workload, task, and vowel were explicitly tested to assess whether exercise-related changes differed across speech contexts. This was

[1] Analyses were implemented in R (Version 4.2.2) using the following packages: lme4 (Bates et al. 2015), lmerTest (Kuznetsova et al. 2017), and emmeans (Lenth et al. 2023). The BOBYQA optimizer (Powell 2009), with a set maximum of 200,000 iterations, was used to maximize the likelihood function.

particularly important for voice quality measures, which are more likely to be investigated using sustained vowel tasks, whereas respiratory studies typically rely on connected speech. Including task as an interacting factor therefore allowed the analysis to quantify whether effects observed in sustained vowels generalize to more naturalistic speech. Two-way interactions were retained across all models, as they captured meaningful differences in how workload effects manifested across tasks and vowel contexts. No significant three-way interactions were observed.

To capture between-speaker variability, two random-effects structures were compared. The more complex structure allowed both participants' baseline levels (random intercepts) and their sensitivity to workload (random slopes) to vary across individuals, reflecting the possibility that speakers differ not only in their habitual glottal source measures but also in how strongly they respond to exercise. The simpler structure allowed only baseline differences between speakers, assuming a common workload effect across participants. The best-fitting model was determined based on three criteria: lower Akaike information criterion (AIC) to indicate goodness of fit, lower Bayesian information criterion (BIC) to balance goodness of fit with model simplicity, and absence of singularity. For CPP and the spectral tilt measures (H1*-H2*, H1*-A1*, H1*-A3*), the simpler random-effects structure was selected due to singular fits or inconsistent model selection criteria, indicating that individual differences in workload sensitivity could not be reliably estimated for these outcomes. Energy measures showed violations due to upper outliers; after removal of extreme values and log-transformation, model assumptions were met and fit improved substantially. The selected models are summarized in Table 5.5.

5.3 Results

5.3.1 Inferential statistics: Glottal source measures

This section presents results for acoustic measures reflecting various aspects of vocal production during exercise. Periodicity measures, which reflect the temporal dimension, are presented first, followed by spectral shape measures, which reflect the spatial dimension (vocal fold adduction). While the primary focus is on how physical workload affects each glottal source measure, significant interactions are also noted. All data visualizations include error bars to represent statistical uncertainty. Means and standard deviations calculated from the observed data are presented in Appendix B.

Table 5.5: Best-fitting model for each glottal source measure; in all models, y denotes the measure listed in the first column.

Measure	**Model**
f0	y ~ workload * task * vowel + (1 + workload | participant)
Energy	log(y) ~ workload * task * vowel + (1 + workload | participant)
H1*-H2*	y ~ workload * task * vowel + (1 | participant)
H1*-A1*	y ~ workload * task * vowel + (1 | participant)
H1*-A3*	y ~ workload * task * vowel + (1 | participant)
SoE	y ~ workload * task * vowel + (1 + workload | participant)
CPP	y ~ workload * task * vowel + (1 | participant)
HNR <500 Hz	y ~ workload * task * vowel + (1 + workload | participant)
HNR <1500 Hz	y ~ workload * task * vowel + (1 + workload | participant)
HNR <2500 Hz	y ~ workload * task * vowel + (1 + workload | participant)
HNR <3500 Hz	y ~ workload * task * vowel + (1 + workload | participant)

5.3.1.1 Fundamental frequency

Analysis of fundamental frequency (f0; repetition rate of the waveform) (Figure 5.2), revealed significant main effects of workload, task, and vowel, indicating that each factor independently contributed to f0 variation. These effects were further qualified by significant two-way interactions for workload:task and task:vowel (Table 5.6). Pairwise comparisons examining the workload:task interaction showed systematic increases in f0 with increasing physical workload in both tasks, although the magnitude of change varied. In the reading task, the model predicted an increase of 7 Hz from rest to low workload ($p < .001$), and an additional increase of 10 Hz from low to moderate workload ($p < .001$), yielding a total increase of 17 Hz from rest to moderate workload ($p < .001$). In the sustained vowel task, the workload effect was larger: f0 was predicted to rise by 14 Hz from rest to low workload ($p < .001$) and by another 16 Hz from low to moderate workload ($p < .001$), for a total increase of 30 Hz ($p < .001$) – nearly double the change observed in the reading task. Thus, while physical workload consistently increased f0, the magnitude of this effect was more pronounced in the sustained vowel task. Additionally, the task:vowel interaction showed that the model-predicted difference in f0 between the two tasks was influenced by vowel type. According to the model, this difference was smallest for /i/ (5 Hz) and largest for /u/ (28 Hz).

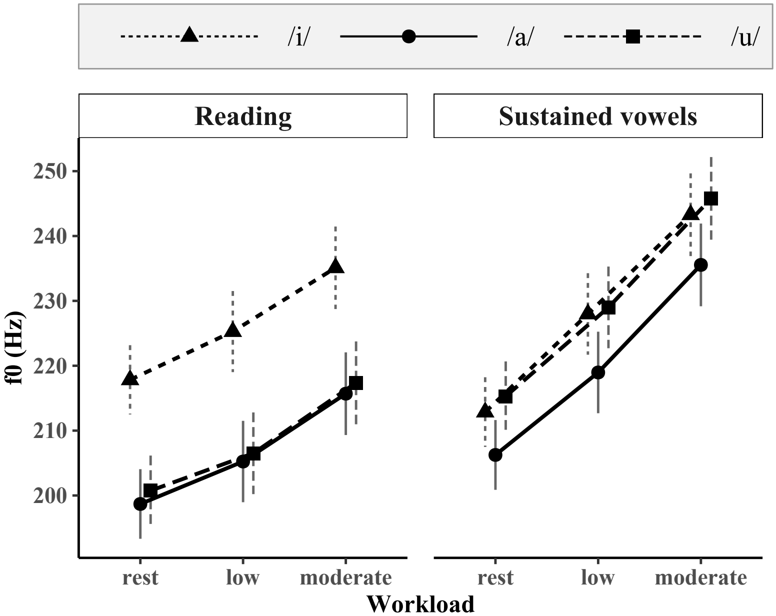

Figure 5.2: EMMs for f0 by workload and speech task; error bars represent 95% confidence intervals.

Table 5.6: Effects and interactions in a linear mixed model of f0 as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
workload	2	44.59	73.42	**<.001**
task	1	2324.82	268.91	**<.001**
vowel	2	2315.39	111.38	**<.001**
workload:task	2	235.11	24.85	**<.001**
workload:vowel	4	2316.57	.19	.945
task:vowel	2	2315.33	58.50	**<.001**
workload:task:vowel	4	2316.55	.05	.994

5.3.1.2 Cepstral peak prominence (CPP)

Cepstral peak prominence captures the strength of periodic structure, with lower values indicating less prominent periodicity (Figure 5.3). The analysis revealed significant main effects of workload, task, and vowel, as well as significant workload:vowel and task:vowel interactions (Table 5.7). Notably, the impact of exercise on CPP was relatively consistent across tasks, as indicated by the non-significant workload:task interaction. The workload:vowel interaction suggests that vowel quality influenced the effect of workload. While the predicted decrease in CPP with increased workload was observed in both speech tasks, pairwise comparisons revealed vowel-specific patterns. For /i/, significant decreases occurred between rest and both workload conditions in both tasks ($p < .001$). In contrast, /u/ showed significant decreases only between rest and moderate workload in the reading task ($p = .021$) and between rest and low workload in the sustained vowel task ($p = .016$). The low vowel /a/ demonstrated no significant workload effects in the reading task, but a significant decrease from rest to low workload was observed in the sustained vowel task ($p = .032$).

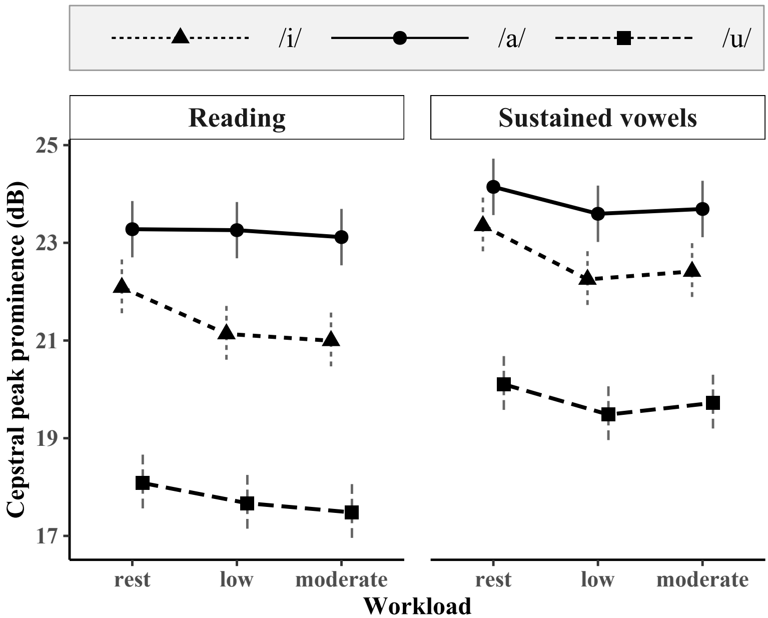

Figure 5.3: EMMs for cepstral peak prominence (CPP) by workload and speech task; error bars represent 95% confidence intervals.

Table 5.7: Effects and interactions in a linear mixed model of CPP as a function of physical workload

Effect	DF	DenDF	statistic	p-value
Workload	2	2411.2	22.2	**<.001**
Task	1	2417.8	223.1	**<.001**
Vowel	2	2411.3	1065.7	**<.001**
Workload:Task	2	2411.2	1.5	.233
Workload:Vowel	4	2411.1	2.8	**.024**
Task:Vowel	2	2411.3	23.1	**<.001**
Workload:Task:Vowel	4	2411.1	0.3	.861

5.3.1.3 Harmonics to noise ratio <500 Hz (HNR05)

The analysis of harmonics-to-noise ratio (HNR; relative periodic vs. aperiodic energy) at the lowest frequency band revealed patterns that generally supported the predicted decrease in periodicity (Figure 5.4), though these patterns varied across speech task and vowel types. Significant main effects of workload, task, and vowel were observed, with all two-way interactions achieving significance as well. The task and vowel effects were particularly pronounced, as indicated by their F-statistics (Table 5.8). Pairwise comparisons examining the workload:task interaction revealed distinct patterns between the sustained vowel and reading tasks. In the sustained vowels, all vowels except /a/ at moderate intensity ($p = .053$) showed significant decreases ($p < .001$) from rest to both exercise conditions. No vowels showed significant differences between low and moderate workload conditions. The reading task yielded more variable results: /i/ decreased significantly only from rest to moderate exercise ($p = .043$), while /a/ showed significant increases during exercise (rest to low: $p = .004$; rest to moderate: $p = .038$). In contrast, /u/ showed no significant changes across workload conditions. Overall, sustained vowels exhibited greater periodicity than read speech, with significantly higher HNR05 values ($p < .001$).

5.3.1.4 Harmonics to noise ratio <1500 Hz (HNR15)

The harmonic patterns below 1500 Hz (Figure 5.5) mirrored those of HNR05, though with elevated values for high vowels. The linear mixed model (Table 5.9) revealed significant main effects of workload, task, and vowel (with task and vowel effects being particularly strong), along with all two-way interactions. Pairwise comparisons of the workload:task interaction showed contrasting patterns between the sustained vowel and reading tasks. In sustained vowels, all vowels exhibited significant decreases from rest to both exercise conditions (p = .001 for all comparisons except /a/ rest to low: p = .003), with no significant differences between low and moderate exercise conditions. The reading task produced more varied results: /i/ showed significant HNR15 decreases across all workload comparisons, while /u/ showed decreased significantly at both moderate (p = .001) and low workload (p = .046) compared to rest. The low vowel /a/ showed only a marginally significant increase at low workload (p = .048).

5.3.1.5 Harmonics to noise ratio <2500 Hz (HNR25)

In the 1500–2500 Hz range, workload effects maintained similar patterns but exhibited more pronounced linear trends and elevated values for high vowels (Figure 5.6). The linear mixed model (Table 5.10) showed task type to have the strongest effect in the model, followed by vowel type, with physical workload showing a smaller but still significant effect. All two-way interactions were significant. Pairwise comparisons revealed that physical workload generally decreased voice periodicity, though its impact varied by both task and vowel. The effect was more pronounced during sustained vowel production, where all vowels showed significant decreases across all workload comparisons for all vowels (p = .037 to $p < .001$). In read speech, high vowels exhibited significant HNR25 decreases across all workload comparisons (p = .017 to $p < .001$), while low vowel /a/ showed a different pattern: a significant increase from rest to low workload (p = .038) followed by a significant decrease from low to moderate workload (p = .036).

5.3.1.6 Harmonics to noise ratio <3500 Hz (HNR35)

The highest analyzed frequency band closely mirrored the HNR25 patterns (Figure 5.7). Significant effects were found for all factors and two-way interactions (Table 5.11), indicating that while each factor independently influenced the response, their effects varied depending on the levels of the other factors. Pairwise comparisons revealed distinct patterns between tasks. In sustained vowels, all three vowels showed significant decreases across all workload comparisons (p = .003 to $p < .001$). In the reading task, the high vowels /i/ and /u/ demonstrated consistent periodicity decreases with increased workload (p = .010 to $p < .001$), while the low vowel /a/ showed a different pattern, with only the low to moderate exercise comparison reaching significance (p = .019). Sustained vowels exhibited consistently higher periodicity than reading across all vowels and workload conditions ($p < .001$), with the effect most pronounced for high vowels /i/ and /u.

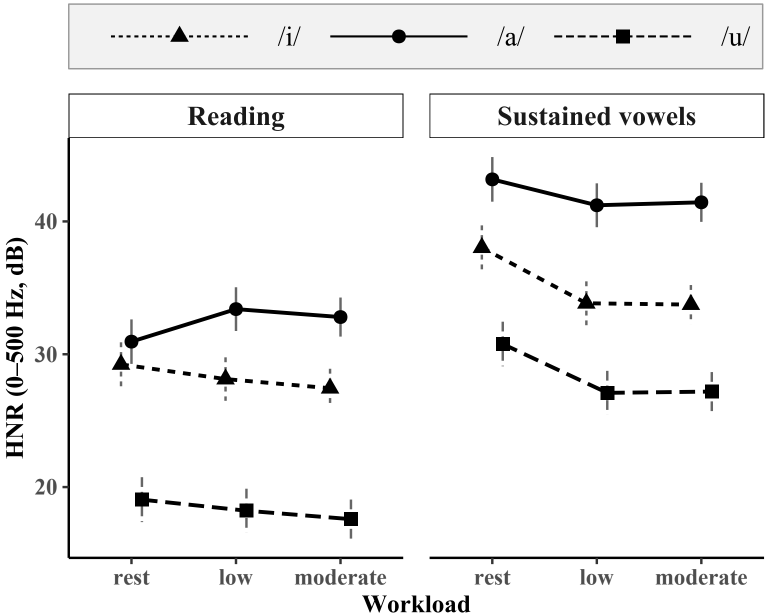

Figure 5.4: EMMs for HNR, 0–500 Hz by workload and speech task; error bars represent 95% confidence intervals.

Table 5.8: Effects and interactions in a linear mixed model of HNR < 500 Hz as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
Workload	2	45.7	6.1	**.005**
Task	1	2330.9	1201.9	**<.001**
Vowel	2	2314.9	1007.8	**<.001**
Workload:Task	2	2353.8	17.0	**<.001**
Workload:Vowel	4	2316.3	6.1	**<.001**
Task:Vowel	2	2314.9	14.9	**<.001**
Workload:Task:Vowel	4	2316.2	0.4	.828

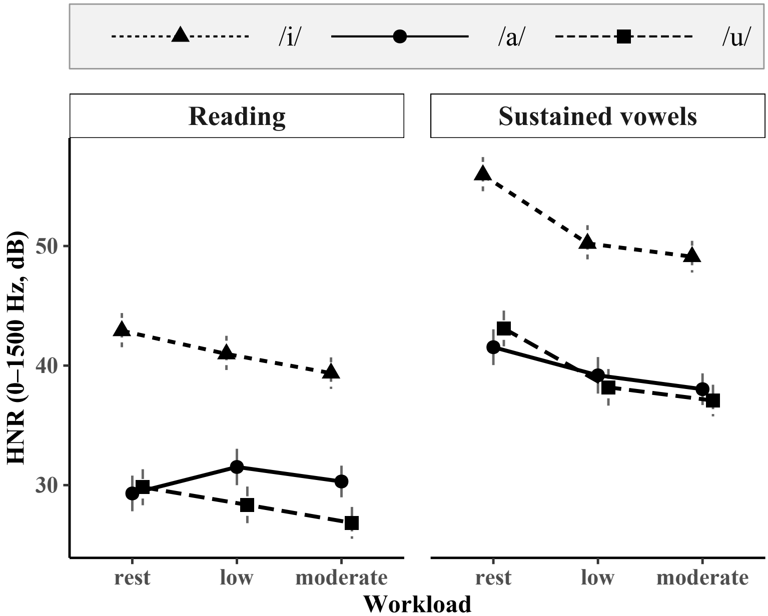

Figure 5.5: EMMs for HNR, 0–1500 Hz, by workload and speech task; error bars represent 95% confidence intervals.

Table 5.9: Effects and interactions in a linear mixed model of HNR < 1500 Hz as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
Workload	2	46.5	24.0	**<.001**
Task	1	2334.1	1952.1	**<.001**
Vowel	2	2317.9	1206.9	**<.001**
Workload:Task	2	2355.2	29.4	**<.001**
Workload:Vowel	4	2319.2	11.8	**<.001**
Task:Vowel	2	2317.9	6.2	**.002**
Workload:Task:Vowel	4	2319.2	0.3	.849

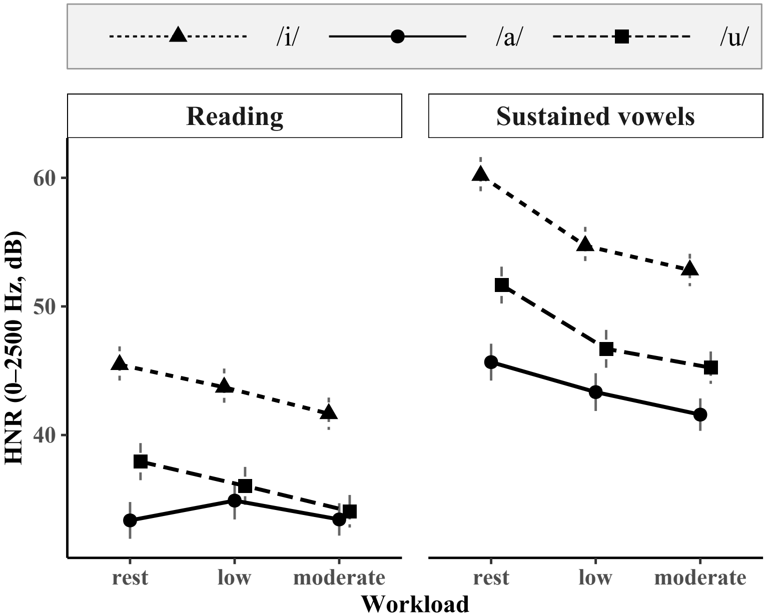

Figure 5.6: EMMs for HNR, 0–2500 Hz, by workload and speech task; error bars represent 95% confidence intervals.

Table 5.10: Effects and interactions in a linear mixed model of HNR < 2500 Hz as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
Workload	2	47.0	35.0	**<.001**
Task	1	2334.8	2641.3	**<.001**
Vowel	2	2319.2	924.0	**<.001**
Workload:Task	2	2355.6	28.3	**<.001**
Workload:Vowel	4	2320.5	11.2	**<.001**
Task:Vowel	2	2319.1	14.5	**<.001**
Workload:Task:Vowel	4	2320.4	0.4	.821

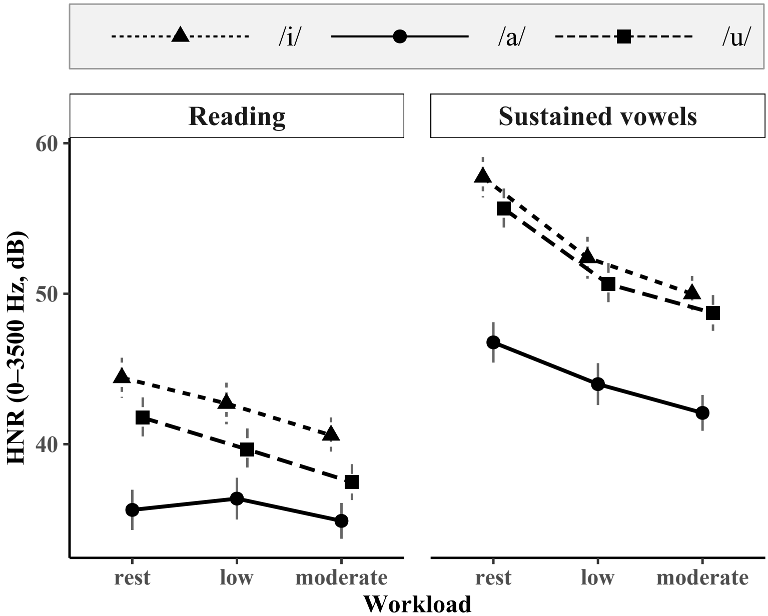

Figure 5.7: EMMs for HNR, 0–3500 Hz, by workload and speech task; error bars represent 95% confidence intervals.

Table 5.11: Effects and interactions in a linear mixed model of HNR < 3500 Hz as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
Workload	2	46.9	49.2	**<.001**
Task	1	2333.1	2844.9	**<.001**
Vowel	2	2318.7	601.6	**<.001**
Workload:Task	2	2354.7	34.1	**<.001**
Workload:Vowel	4	2320.0	10.2	**<.001**
Task:Vowel	2	2318.6	25.9	**<.001**
Workload:Task:Vowel	4	2319.9	0.4	.802

5.3.1.7 Energy

Analysis of acoustic energy levels (signal intensity) showed increases with rising workload (Figure 5.8), with the log-transformed model revealing significant main effects of workload and vowel. Significant two-way interactions were found for workload:task and task:vowel (Table 5.12). Pairwise comparisons examining the workload:task interaction showed that energy increased with greater workload in both tasks, though the increases were slightly larger in the sustained vowel task. However, these changes were modest in both tasks, with model predictions ranging from 1 to 2.5 dB. The task:vowel interaction indicated that task differences in energy were influenced by vowel type, though the effect sizes remained small.

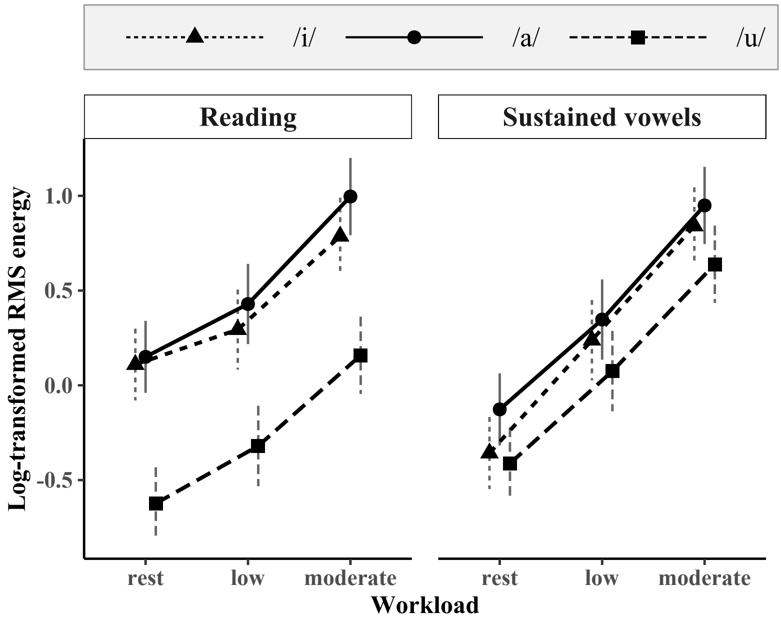

Figure 5.8: EMMs for log-transformed RMS energy by workload and speech task; error bars represent 95% confidence intervals.

5.3.1.8 H1*-H2*

H1*-H2* (voice pulse sharpness) showed complex patterns influenced by both speech task and vowel across workload conditions (Figure 5.9). Significant main

Table 5.12: Effects and interactions in a linear mixed model of RMS energy as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
workload	2	44.95	113.31	**<.001**
task	1	2276.28	3.33	.068
vowel	2	2269.43	238.23	**<.001**
workload:task	2	2300.13	29.49	**<.001**
workload:vowel	4	2271.91	0.21	.932
task:vowel	2	2268.59	73.60	**<.001**
workload:task:vowel	4	2269.55	1.33	.255

effects of workload, task, and vowel were observed, with task showing the strongest effect. These effects were further shaped by significant two-way interactions for workload:task, workload:vowel, and task:vowel (Table 5.13). Notably, the task:vowel interaction was particularly strong, indicating that the task effects on H1*-H2* varied across vowels. While interactions involving workload – the primary variable of interest – were weaker, they remained significant. Pairwise comparisons revealed stronger effects for high vowels, especially in the sustained vowel task. High vowels /i/ and /u/ showed significant decreases in H1*-H2* across all workload levels in the sustained vowel task ($p = .001$ to $p < .001$). In the reading task, these vowels showed significant decreases only at moderate workload compared to both rest and low workload ($p = .008$ to $p < .001$). In contrast, the low vowel /a/ did not show significant changes with workload in either task.

Table 5.13: Effects and interactions in a linear mixed model of H1*-H2* as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
workload	2	2411.8	35.1	**<.001**
task	1	2428.2	302.1	**<.001**
vowel	2	2412.0	56.1	**<.001**
workload:task	2	2411.8	4.3	**.014**
workload:vowel	4	2411.5	11.9	**<.001**
task:vowel	2	2412.0	210.6	**<.001**
workload:task:vowel	4	2411.5	1.2	.307

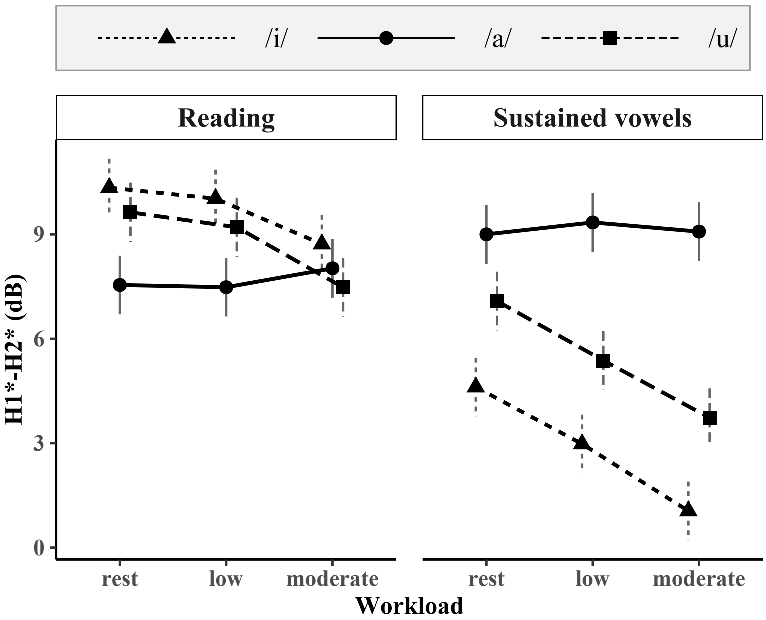

Figure 5.9: EMMs for H1*-H2* by workload and speech task; error bars represent 95% confidence intervals.

5.3.1.9 H1*-A1*

H1*-A1* did not consistently show the predicted increase during physical activity (Figure 5.10). Main effects of workload, task, and vowel were observed, with vowel having a much stronger influence on H1*-A1* than the other factors (Table 5.14). Significant two-way interactions were found for workload:task and task:vowel. Pairwise comparisons revealed distinct patterns between the reading and vowel tasks. In the reading task, no significant differences in H1*-A1* were found across workload conditions for any vowel. However, in the sustained vowel task, high vowels /i/ and /u/ showed significant decreases in H1*-A1* across workload conditions ($p = .001$ or $p < .001$). For the low vowel /a/, significant decreases were found only between rest and moderate load ($p = .001$), with the low-to-moderate comparison at the significance cutoff ($p = .050$). Additionally, a strong task:vowel interaction indicated that H1*-A1* varied significantly between vowels across speech tasks.

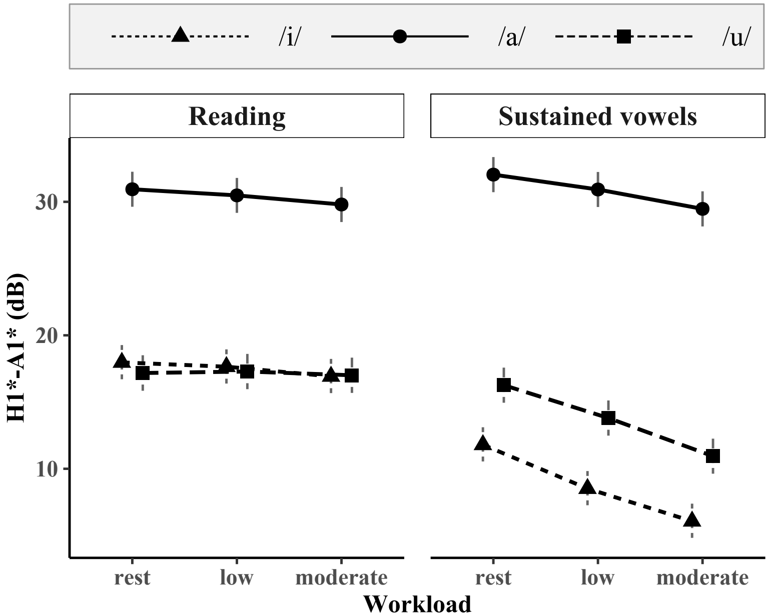

Figure 5.10: EMMs for H1*-A1* by workload and speech task; error bars represent 95% confidence intervals.

Table 5.14: Effects and interactions in a linear mixed model of H1*-A1* as a function of physical workload

Effect	DF	DenDF	statistic	*p*-value
workload	2	2411.7	38.7	**<.001**
task	1	2425.5	249.1	**<.001**
vowel	2	2411.8	1978.2	**<.001**
workload:task	2	2411.7	19.3	**<.001**
workload:vowel	4	2411.5	1.2	.331
task:vowel	2	2411.8	115.6	**<.001**
workload:task:vowel	4	2411.5	2.0	.098

5.3.1.10 H1*-A3*

H1*-A3* was expected to decrease during exercise compared to speech at rest, and the predicted pattern was generally observed (Figure 5.11). Significant main effects of workload, task, and vowel were found, along with significant interactions for workload:task and task:vowel (Table 5.15). Pairwise comparisons of the workload:task interaction indicated that the impact of workload varied by speech task. In the sustained vowel task, workload had a consistent and strong effect across all vowels: high vowels /i/ and /u/ showed significant decreases in H1*-A3* across all workload levels, while /a/ demonstrated significant decreases at both low and moderate workloads compared to rest (p = .013 to p < .001). In the reading task, effects were more selective and smaller. High vowels showed significant decreases only at moderate workload (p = .003 to p = .044), while /a/ showed no significant changes across workload conditions.

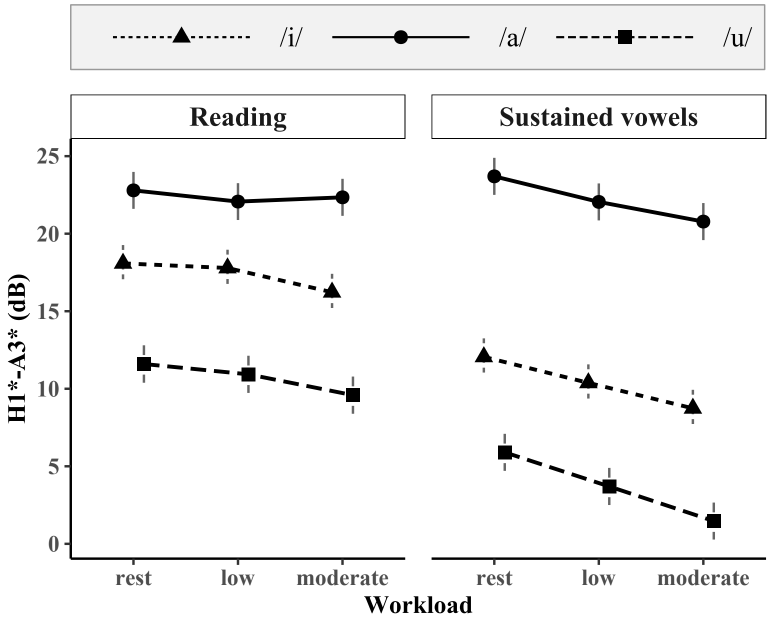

Figure 5.11: EMMs for H1*-A3* by workload and speech task; error bars represent 95% confidence intervals.

Table 5.15: Effects and interactions in a linear mixed model of H1*-A3* as a function of physical workload

Effect	DF	DenDF	statistic	p-value
workload	2	2411.2	42.9	**<.001**
task	1	2424.2	454.8	**<.001**
vowel	2	2411.3	1565.0	**<.001**
workload:task	2	2411.2	7.8	**<.001**
workload:vowel	4	2411.0	1.7	.141
task:vowel	2	2411.3	105.2	**<.001**
workload:task:vowel	4	2411.0	0.4	.830

5.3.1.11 Strength of excitation (SoE)

The strength of excitation (SoE; impact or abruptness of each voice pulse) results were broadly consistent with the H1*-A3* findings but showed more consistent effects of workload (Figure 5.12). Significant main effects of workload, task, and vowel were observed, along with significant workload:task and task:vowel interactions (Table 5.16). Pairwise comparisons of the workload:task interaction showed systematic effects of physical workload on SoE. In the sustained vowel task, physical workload had a strong and consistent impact across all vowels, with significant increases at each workload level. A similar though slightly weaker pattern was observed in the reading task.

Table 5.16: Effects and interactions in a linear mixed model of SoE as a function of physical workload

Effect	DF	DenDF	statistic	p-value
Workload	2	42.5	54.4	**<.001**
Task	1	2324.7	4218.7	**<.001**
Vowel	2	2311.5	136.0	**<.001**
Workload:Task	2	2351.5	7.8	**<.001**
Workload:Vowel	4	2312.9	0.8	.546
Task:Vowel	2	2311.4	104.3	**<.001**
Workload:Task:Vowel	4	2312.9	0.1	.965

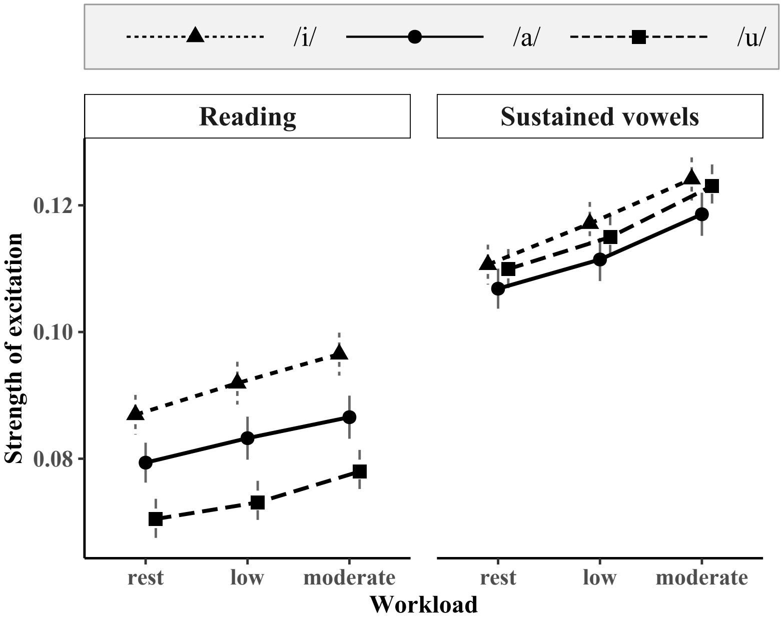

Figure 5.12: EMMs for strength of excitation (SoE) by workload and speech task; error bars represent 95% confidence intervals.

5.3.2 Individual differences in f0: Line plots

Individual speaker trajectories were not visualized for all measures in the statistical analysis, despite being examined in this way in the other studies. The present study examined three vowel contexts across 11 glottal source measures, making comprehensive trajectory visualization difficult. Visualizing individual responses would have required either averaging across vowels, selecting a single vowel for display, or presenting trajectories separately for each vowel and measure. Given that the inferential analyses revealed complex vowel- and task-specific effects, the first two options risked presenting patterns that were either oversimplified or incomplete. Presenting all vowels and measures separately, by contrast, would have produced a large number of heterogeneous line plots that would be difficult to synthesize or interpret. For these reasons, individual trajectory visualization was restricted to mean fundamental frequency (f0), a measure for which exercise-related effects are most consistently reported in the literature and which has typically been analyzed using vowel- or utterance-averaged values in previous studies.

Fundamental frequency (f0) warranted separate analysis based on questions arising from previous research. Although f0 is widely thought to increase with physical activity, most studies reported only group-level analyses. The few studies that have reported speaker-specific patterns found considerable variability, including decreases, minimal change, and plateauing in addition to increases in f0 (Godin & Hansen 2011a, Johannes et al. 2007). To investigate individual response patterns, mean f0 was calculated across all vowels for each speaker at each workload level and speech task.

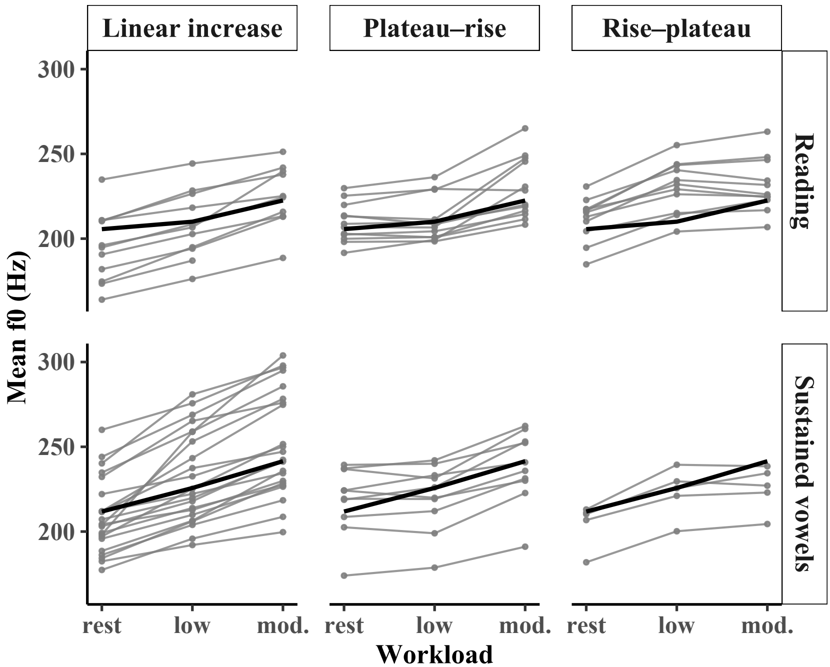

Figure 5.13: Mean f0 change across workloads for individual speakers, grouped by increase pattern, with category means shown in bold for read speech (*top*) and sustained vowels (*bottom*).

The line plot visualization revealed several patterns of f0 change across increasing workload. Different patterns were identified based on the direction and magnitude of f0 change with increasing workload, with >10 Hz serving as an operational threshold for change. In both speech tasks, most participants (≈70%) showed an increase in f0 during exercise, though these increases were not always linear. As shown in Figure 5.13, some participants showed little change until moderate workload ("plateau–rise"), while others showed an initial increase at low workload followed by stabilization ("rise–plateau"). These findings align with observations reported by Godin & Hansen (2008) and Johannes et al. (2007), further underscoring the non-uniform nature of f0 responses to exercise. Notably, a task difference emerged: during sustained vowels, a greater number of participants showed increases in f0 at moderate workload. This difference may reflect task-specific constraints on pitch regulation. In reading, linguistic structure constrains pitch range. In contrast, sustained vowels have no inherent pitch target. Thus, it is possible that vocal fold adjustments to maintain phonation under exercise conditions result in a greater tendency for f0 to increase more freely.

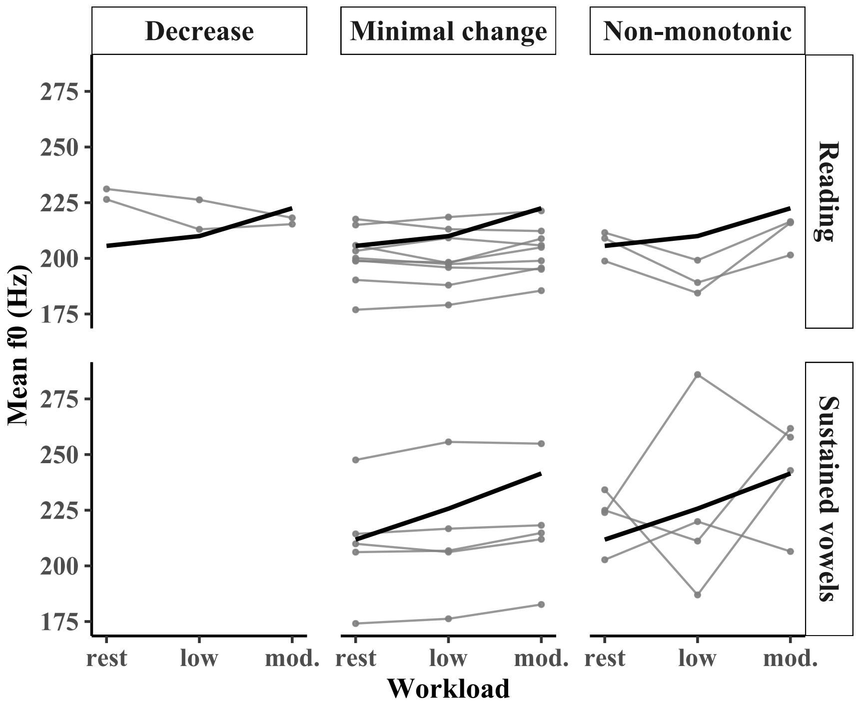

Figure 5.14: Mean f0 change across workloads for individual speakers, grouped by change pattern, with category means shown in bold for read speech (*top*) and sustained vowels (*bottom*).

Importantly, approximately 30% of participants showed no increase in f0 during reading, compared with 20% during sustained vowels (Figure 5.14). In both tasks, the majority of these participants exhibited minimal change across workload levels. Only two participants showed a decrease in f0 during reading, while a small number showed non-monotonic ("mountain/valley") patterns, in which f0 at low workload diverged from both rest and moderate levels. The higher proportion of minimal-change patterns during reading likely reflects the influence of prosodic targets, which may encourage speakers to maintain stable pitch despite increasing physical workload. In contrast, sustained vowels lack such linguistic constraints, making continued increases in f0 more likely as workload increases.

5.3.3 Speaker groups across phonatory measures: Heatmaps

Heatmaps provide an overview of individual response patterns across multiple measures. In the present study, the glottal source measures form columns and individual participants – denoted by lower-case three-letter codes (e.g., lwa) – form rows. Each cell's color indicates the percentage change in that measure from rest to exercise, with red showing increases and blue showing decreases. Reading down a column thus shows how all speakers responded to a given measure, while reading across a row shows how a single speaker responded across all measures. The matrix applies hierarchical clustering to group similar cells, revealing measures that change in parallel as well as participants with similar response patterns. The accompanying dendrogram shows these relationships through tree-like branches (clades): shorter clades indicate greater similarity, while taller clades show larger differences between groups.

Heatmaps were generated only for the reading task because it is closer to natural speech than sustained vowels. Two vowels were chosen: high vowel /i/, which showed the greatest changes at group level, and low vowel /a/, which showed comparatively little change at group level. Energy was not plotted, as it showed larger percentage changes than the other measures, making smaller changes more difficult to see.

Figure 5.15 shows changes from rest to low workload for /i/. Most cells are light in color, indicating modest changes (typically around 10–20%) across speakers and measures. Larger changes are confined to a small number of speakers, particularly for H1*-H2*. However, large changes do not necessarily imply that these speakers differed markedly from others during exercise; they may instead reflect comparatively low baseline values, resulting in larger relative changes. The speaker dendrogram first separates a small group characterized by extreme H1*-H2* responses, often accompanied by marked changes in H1*-A1* and H1*-A3*. The remaining speakers form several subgroups, but these do not reveal obvious response patterns, although some speakers show similar changes across measures, suggesting coordinated phonatory changes. The measure dendrogram likewise appears to be dominated by the extreme H1*-H2* values, with H1*-A1* and

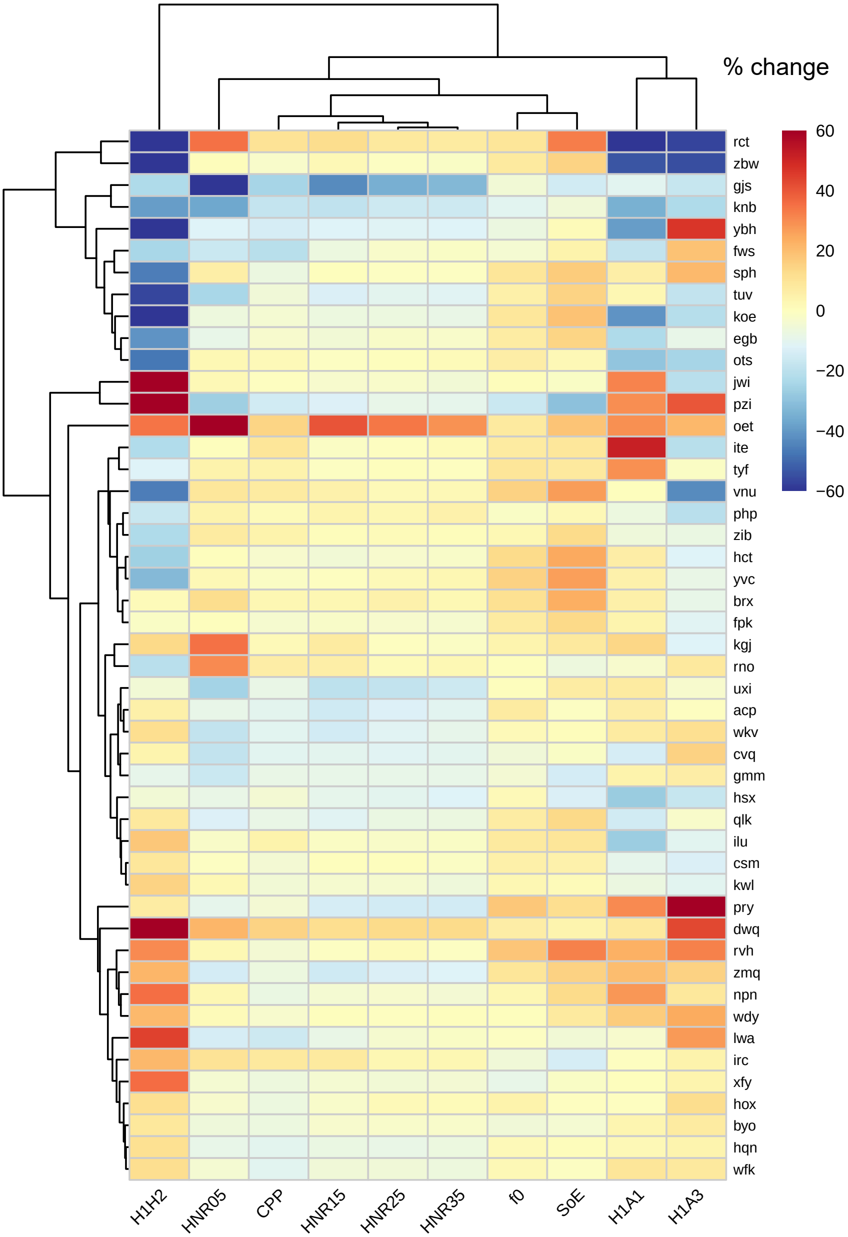

Figure 5.15: Heatmap showing percentage change from rest to low workload for glottal source measures (excluding Energy) for /i/; hierarchical clustering is shown by the dendrograms.

H1*-A3* clustering together, probably reflecting the larger magnitude of change rather than a common response pattern. Overall, the heatmap provides little evidence for a systematic response strategy across speakers.

At moderate workload (Figure 5.16), the hierarchical clustering reveals a somewhat clearer grouping of measures than at low workload. The CPP and higher-band HNR measures cluster together and generally show relatively small changes across speakers, whereas H1*-H2*, H1*-A1*, and H1*-A3* exhibit the largest positive and negative changes. As a result, the measure dendrogram appears to be driven primarily by the magnitude of changes in these harmonic measures rather than by consistent response patterns across participants.

The speaker dendrogram is likewise influenced by a relatively small number of speakers with extreme responses, producing several distinct clusters. However, most speakers show only modest changes across the measures, and no coherent response strategy is evident. Although relative decreases are more prominent for H1*-A1* and H1*-A3* than at low workload, the heatmap continues to display both increases and decreases for most measures. Overall, the results suggest substantial heterogeneity in individual responses rather than a uniform effect of increasing exercise intensity.

Turning to vowel /a/, the change from rest to low workload (Figure 5.17) is characterized predominantly by light-colored cells, indicating that changes were generally small across speakers and measures. A limited number of speakers exhibit pronounced increases or decreases, particularly for the lower-band HNR measures and H1*-H2*, and these extreme responses largely determine the hierarchical clustering. Consequently, the speaker dendrogram does not reveal readily interpretable response groups but rather separates a small number of speakers with unusually large changes from the majority of participants.

The measure dendrogram groups the lower-frequency HNR measures together, while H1*-H2* forms a more distinct branch. However, these groupings appear to be driven primarily by the magnitude of changes rather than by consistent response patterns across speakers. Overall, the heatmap provides little evidence for systematic speaker groupings or response patterns. Instead, most speakers exhibit only modest changes, with the overall clustering largely reflecting a handful of isolated responses.

Turning to the moderate-workload condition for vowel /a/ (Figure 5.18), the overall impression is again one of relatively small changes across speakers and measures, with most cells showing only modest increases or decreases. Although the HNR measures display variation in the direction of change across speakers, with several distinct speaker groups, no consistent response pattern is evident.

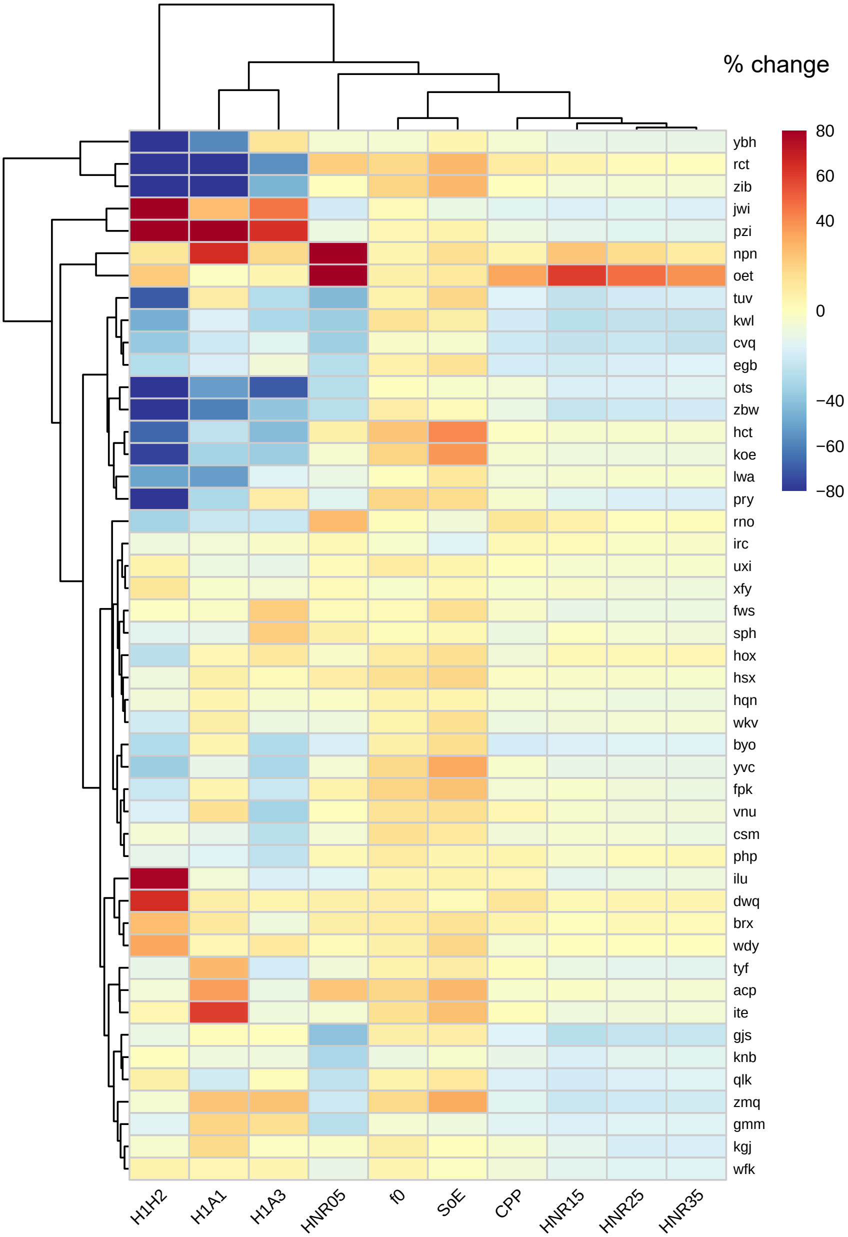

Figure 5.16: Heatmap showing percentage change from rest to moderate workload for glottal source measures (excluding Energy) for /i/; hierarchical clustering is shown by the dendrograms.

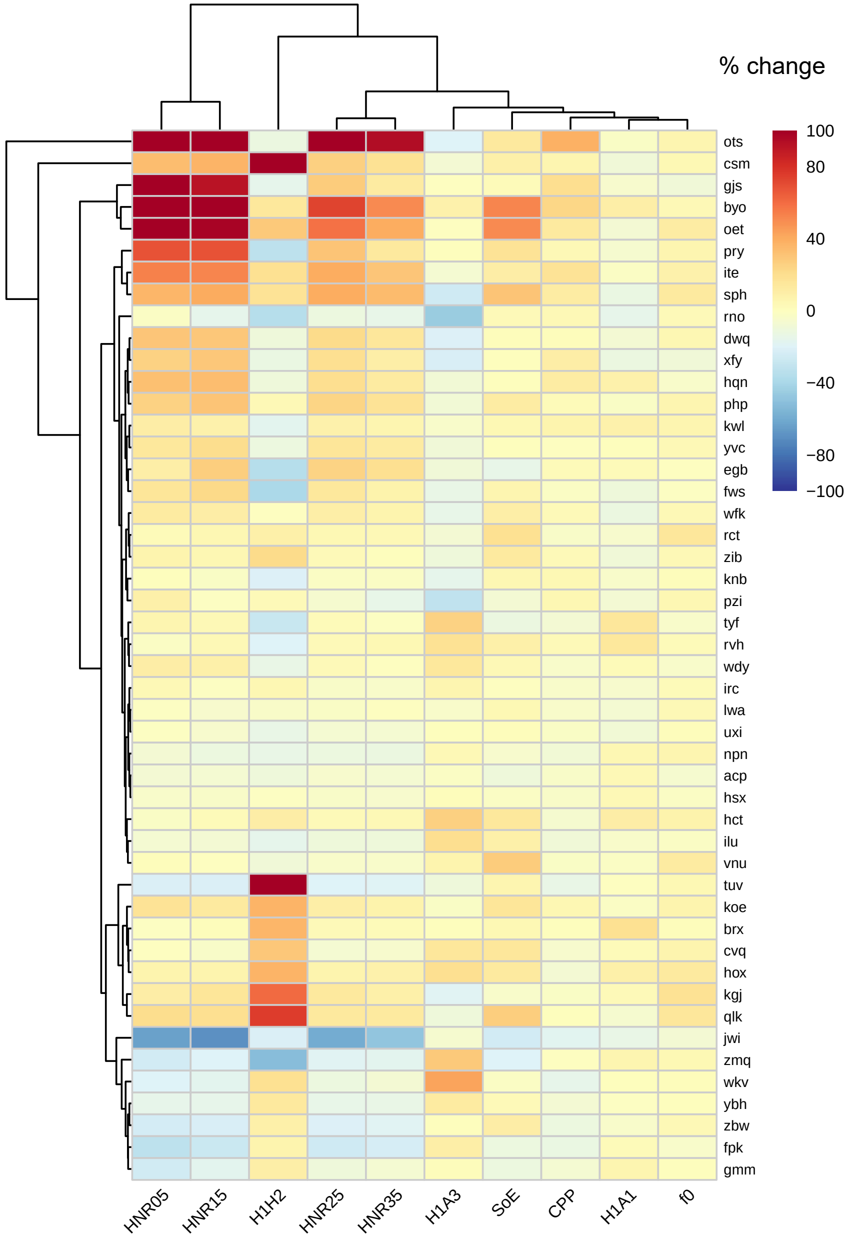

Figure 5.17: Heatmap showing percentage change from rest to low workload for glottal source measures (excluding Energy) for /a/; hierarchical clustering is shown by the dendrograms.

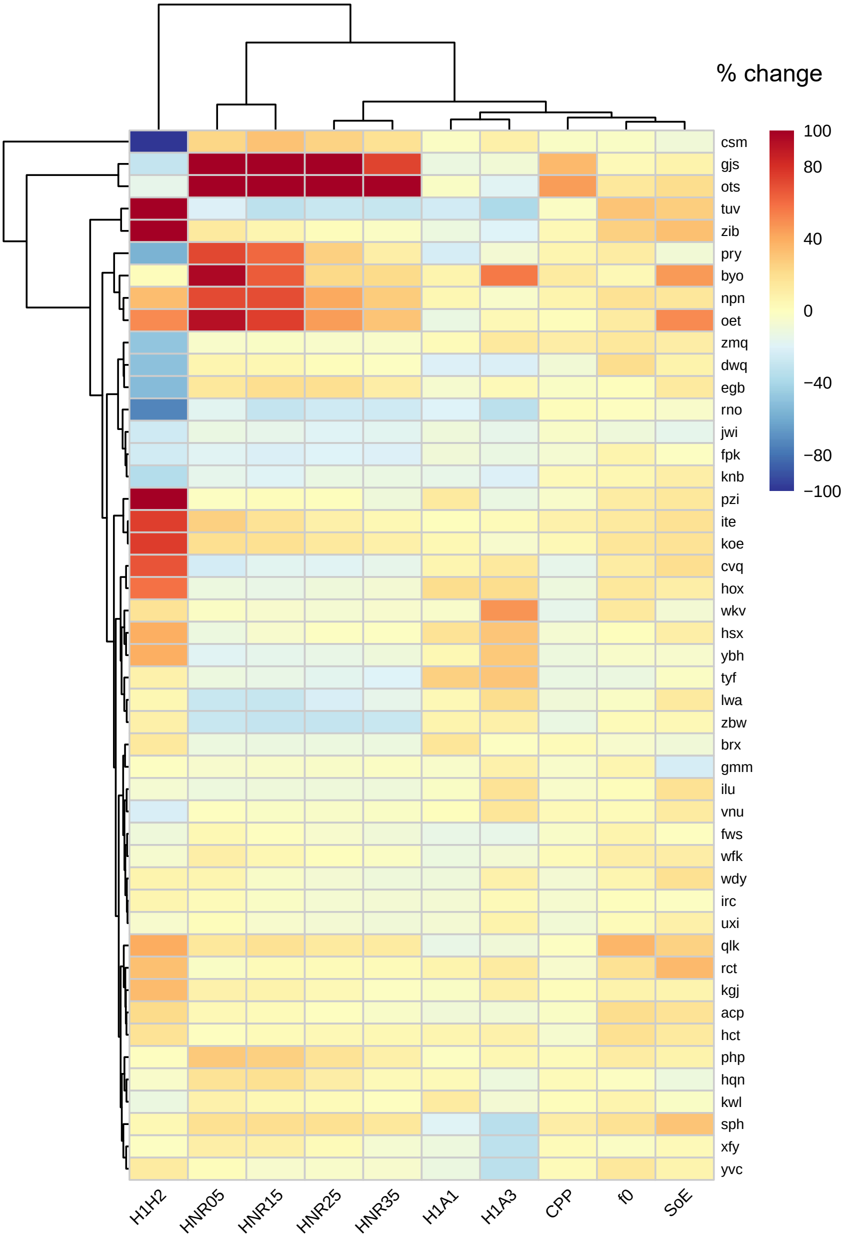

Figure 5.18: Heatmap showing percentage change from rest to moderate workload for glottal source measures (excluding Energy) for /a/; hierarchical clustering is shown by the dendrograms.

Instead, a handful of speakers with extreme values largely determine the hierarchical clustering of both speakers and measures. As in the other heatmap analyses, the visualization provides little evidence for systematic trade-offs among the acoustic measures (e.g., speakers increasing one measure while systematically decreasing another). Rather, it highlights substantial speaker heterogeneity and the absence of a uniform exercise response.

5.4 Discussion

This study investigated various glottal source measures to determine the extent to which physical activity affects glottal behavior. While previous research has predominantly focused on periodicity measures, reporting mixed results on f0 and HNR, vocal fold approximation has remained largely unexplored. Statistical analyses revealed significant group-level effects on both periodicity and vocal fold approximation, while visualizations of individual participant responses established underlying heterogeneity. The discussion first synthesizes results across periodicity and approximation measures, then addresses between-speaker variability. These findings are followed by theoretical and methodological considerations.

5.4.1 Changes in periodicity measures

Across all measures of periodicity, physical workload showed a clear influence on vocal fold vibration. As expected, fundamental frequency increased with physical workload in both speech tasks, indicating faster vocal fold oscillation as breath rate and volume increased. At the same time, the measures quantifying the regularity of these oscillations, CPP and HNR, showed the inverse effect. CPP decreased between rest and exercise conditions, with effects emerging at low workload and stabilizing or plateauing at moderate workload. HNR values decreased across all frequency bands, with greater reductions at higher frequencies. These workload effects were strongest and most consistent in sustained vowels, but they were also present in read speech. Taken together, the periodicity measures suggest that during exercise, vocal fold cycles become faster but less regular.

During exercise, speech is produced under increased breathing demands; increased breath volume and frequency shift the baseline pressure conditions under which phonation occurs. As workload increases, speakers inhale and exhale more forcefully, likely raising the subglottal pressure operating point rather than simply increasing airflow. This produces faster vocal fold vibration, reflected in the rise in fundamental frequency, but also reduces the speaker's ability to finely regulate pressure from moment to moment. Thus, phonation must be maintained while respiratory pressure fluctuates more strongly. Small pressure fluctuations therefore translate into irregular cycle timing, reducing CPP and HNR even though voicing is maintained. The effect is most evident in sustained phonation, which requires continuous regulation, but it also appears in connected speech, where consonant–vowel transitions repeatedly re-coordinate voicing and make these fluctuations less apparent.

These physiological sources of instability were also apparent to participants, who reported on their subjective experience in post-experiment questionnaires (see Section 4.4). For example, participant EGB expressed a sense of overall change: "My voice no longer felt steady."[2] Others explicitly linked their breathing to perceived vocal effects and reported active compensation: "I couldn't control my voice as well because my breathing was so strained. I had to be more conscious of my intonation" (participant HCT).[3] Another participant linked tremulousness with potential communicative consequences: "My voice shook a bit from the effort, which made me sound less sure of myself" (participant UXI)[4]. These experiences suggest that the reduced periodicity indicated by lower CPP and HNR values is not observable only through acoustic analysis but may also correspond to perceived reductions in voice stability.

Overall, these results reveal a general decrease in periodicity with increasing workloads. However, methodological choices influenced how strongly the effect appeared. For example, the difference between CPP and HNR likely reflects what each measure captures. CPP tracks overall harmonic organization, whereas HNR is more sensitive to cycle-to-cycle irregularity and aperiodic energy. Exercise appears to disrupt these fine-grained aspects of vibration without fully altering the broader harmonic pattern, explaining why HNR declined further while CPP plateaued at moderate workload.

The changes in all measures were most consistently observed in sustained vowels. Sustained phonation of a single vowel requires continuous fine regulation over longer durations, and the observed effects during exercise indicate that maintaining this configuration becomes more difficult under increased respiratory demands. In contrast, read speech contains segmental variation that repeat-

[2] „Meine Stimme hat sich nicht mehr sicher angefühlt."

[3] „Meine Stimme konnte ich nicht so kontrolliert verwenden, weil meine Atmung sehr angestrengt war. Ich musste bewusster intonieren."

[4] „Weil meine Stimme etwas gezittert hat durch die Anstrengung und ich dadruch etwas unsicher klang."

edly interrupts and reinitiates voicing, allowing breathing and phonation to be recoordinated. This, together with the shorter duration of the analyzed vowel samples, likely makes small control fluctuations less apparent in connected speech. The lack of effects for /a/ during reading likely reflects a combination of such factors: the preceding voiceless stop in *spontan* 'spontaneously' interrupts voicing, and the open vowel /a/ remains stable across a wider range of pressure conditions. Similar contextual factors could explain why neither Koblick (2004) nor Primov-Fever et al. (2014) found HNR effects for /a/, though the latter study also found no effect for /i/, which, in the current study, shows the most prominent decrease. These differences in speech materials illustrate some factors that make aperiodicity more or less detectable, and they should be considered when generalizing results or designing future studies.

5.4.2 Changes in approximation measures

Among the measures of vocal fold approximation, several consistent changes emerged with increasing workload. Acoustic energy showed small but reliable increases, indicating greater vocal intensity, and SoE increased as well, reflecting a stronger push of energy driving the voicing signal at each glottal cycle. Effects on the remaining measures were shaped by speech task and vowel context. H1*-H2* decreased, but only for high vowels and more strongly in sustained vowels, indicating a redistribution of energy in the lowest harmonics under those conditions. H1*-A1* showed no workload effects in read speech, while sustained vowels showed decreases, indicating relatively greater energy in the first formant region during exercise. Finally, decreases in H1*-A3* indicated a shift toward greater relative energy at higher frequencies, aligning with the observed increases in SoE. These effects were again clearest for sustained vowels and high vowels. Taken together, the results observed for the approximation measures suggest systematic changes in how energy is distributed during phonation as workload rises, with the magnitude of these effects varying across tasks and vowels.

These observed changes can be understood in relation to aerodynamic changes induced by exercise. As shown in the respiratory study (Chapter 4), breath volume increased linearly with workload, indicating progressively greater ventilation. Increased airflow places greater demands on phonatory control and is associated with reduced periodicity, as reflected in the observed decreases in CPP and HNR. At the same time, the increases in acoustic energy and SoE suggest that speakers increased the driving strength of voicing to sustain phonation under exercise conditions. If increased airflow were allowed to pass through the glottis with minimal regulation, acoustic energy would be expected to accumulate at low frequencies. Instead, the observed decreases in H1*–H2*, H1*–A1*, and H1*–A3* indicate a redistribution of energy toward higher frequencies, suggesting that as breathing becomes heavier, speakers modify how airflow is converted into acoustic energy by adjusting the timing and degree of glottal opening.

This general shift toward a more constrained glottal configuration can plausibly be understood in relation to the demands of initiating and sustaining phonation during exercise. To facilitate tidal breathing, the larynx adopts a more open configuration that reduces airway resistance (Beaty et al. 1999, Røksund et al. 2015). This means that speech must then be initiated from a more open, high-flow respiratory state than at rest, requiring larger and more rapid adjustments to achieve effective adduction. These adjustments may involve increased medial compression at phonation onset or more complete vocal fold contact. Once voicing is established, continued high airflow may further require tighter regulation of glottal opening during each cycle, for example through shorter open phases or increased longitudinal tension. Rapid alternation between breathing-oriented and phonation-oriented laryngeal states may contribute to increased perceived vocal effort. This account aligns with Sandage et al. (2013)'s findings of elevated phonation threshold pressure (PTP) and greater perceived effort during exercise. Increased PTP is required when the folds begin farther apart, as during exercise tidal breathing, or are functionally stiffer or less hydrated – patterns that correspond to the higher f0 and the exercise duration of the present study. Such changes provide a plausible link between increased ventilation and the emergence of more effortful or tightly regulated phonation during exercise.

Breathing-related differences may also explain why vocal intensity increased with workload. Higher subglottal pressure produces larger and more forceful airflow pulses through the glottis, raising sound pressure level. Unlike the spectral-tilt measures, which reflect how acoustic energy is shaped during phonation, intensity is a more direct aerodynamic consequence of elevated respiratory drive. This pattern aligns with prior studies showing greater vocal intensity during exercise (Dallaston & Rumbach 2016, Doust & Patrick 1981, Fuchs et al. 2015, McCaig 2013, Otis & Clark 1968, Patil et al. 2010, Trouvain & Truong 2015, Ziegler 2014). It is also consistent with participants' reports that their voices "got quieter" at utterance ends: even when average intensity increases, speakers may run out of usable pressure during speech, resulting in a noticeable loss of loudness. Other participants' reports of difficulty with intonation also fit this account, since fine-grained pitch modulation requires tight and stable control of subglottal pressure – something that becomes more difficult under elevated respiratory load.

Although the results consistently indicate that laryngeal settings are altered during exercise, the magnitude of these effects varied by task and vowel. Sustained vowels showed stronger and more reliable changes across all approximation measures, consistent with the demands of continuous phonation under elevated subglottal pressure. In this static, open laryngeal configuration, speakers must maintain voicing during uninterrupted airflow. Read speech, in contrast, continuously intersperses phonation with vocal tract closures, which naturally interrupts airflow and may attenuate exercise-related effects. Vowel quality further modulated these patterns. High vowels (/i, u/) showed the strongest effects. Their articulatory configurations involve a greater supraglottal constriction, which increases supraglottal pressure and makes phonatory control more difficult. The low vowel /a/, in contrast, has a more open vocal tract that allows airflow to pass more freely and maintains a larger transglottal pressure drop, potentially reducing the need for additional laryngeal adjustment during exercise.

5.4.3 Implications of exercise-related phonatory changes

The present findings extend knowledge about acoustic effects of physical activity in several ways. First, they corroborate prior findings of increased f0 and vocal intensity (Dallaston & Rumbach 2016, Doust & Patrick 1981, Evdokimova & Zakharchenko 2019, Fuchs et al. 2015, Johannes et al. 2007, Koblick 2004, McCaig 2013, Otis & Clark 1968, Patil et al. 2010, Primov-Fever et al. 2014, Trouvain & Truong 2015, Ziegler 2014) and also show that such effects can emerge already at low workload, meaning that everyday activities, such as brisk walking, can induce acoustic changes. Second, the present study demonstrates between-speaker variability in both the magnitude and direction of f0 changes. This aligns with prior work reporting speaker differences (Godin & Hansen 2008, Johannes et al. 2007) and provides further evidence that while increased f0 is common, it should not be understood as a universal acoustic correlate of physical stress. Third, the study gathered new data on further glottal source measures during exercise, again revealing differing responses across speakers.

Similarly diverse individual responses have been reported in studies of speech produced under other types of stress. Much of this literature has focused on increased cognitive or emotional load, and a recurring finding is substantial variability across speakers, which has long been recognized as a central challenge for characterizing stress-related speech changes (e.g., Scherer 1986, Giddens et al. 2013, Kirchhübel et al. 2011, Van Puyvelde et al. 2018). A key open question, therefore, is why stress elicits such heterogeneous vocal outcomes. Many interpretations draw on Selye (1936)'s concept of stress, which emphasizes that a stressor – anything that disrupts the body's internal state of balance – is not itself the cause of change; rather, it is the individual's *response* to that stressor. The implication of this conceptual framing is that identical stressors can elicit different physiological and behavioral responses across individuals. Variation is therefore expected in this framework. Applied to speech, this implies that acoustic outcomes are not determined by the situation per se (e.g., exercise, taking an oral exam, or having an argument), but by the changes an individual makes relative to their own production baseline. This account is consistent with the individual trajectories observed in the present study, including cases of opposite-direction acoustic effects, suggesting that speakers modulated laryngeal measures in different ways under the same physical workload.

One plausible source of this variability is that speakers who appear to be in the same stress condition may in fact be operating under quite different speech production conditions. The results from the respiratory study presented in Chapter 4 showed that participants altered the volume and timing of breaths during exercise in distinct ways – for example, fewer, deeper breaths versus more frequent, shallower breaths. Such differences imply variation in lung volume at speech initiation, and, when combined with expiratory drive, can create different aerodynamic starting points for phonation. From the perspective of myoelastic–aerodynamic theory (van den Berg 1958), such differences are expected to influence phonation. Aerodynamic forces are a primary driver of vocal fold vibration, shaping how the folds move over time and space. The vocal folds are complex, multi-layered structures that allow dynamic variations in effective stiffness, mass, and elasticity. As a result, changes in airflow and pressure can alter vibratory behavior in multiple ways across the glottal cycle, including where separation begins, how rapidly and evenly the folds open, how long they remain open, and how quickly and in what manner they close.

In addition to aerodynamic forces, phonation depends on muscular forces that can modulate vocal fold thickness, tension, and shape, and that bring the vocal folds together to initiate and sustain voicing. The degree of force required depends not only on individual laryngeal configuration, but also on further properties of the vocal fold tissue, which may change during exercise, for example due to dehydration or increased blood flow. Taken together, these considerations suggest that even under the same external workload, speakers may experience markedly different combinations of aerodynamic conditions and tissue states. Additionally, individual production differences are not unique to speech during exercise. Even in speech at rest, speakers have been shown to differ in how they coordinate aerodynamic and laryngeal conditions to achieve voicing. For example, Koenig et al. (2005) examined voiced and voiceless /h/ in running speech and found that each of six female speakers used individual combinations of laryngeal, supralaryngeal, and aerodynamic conditions for devoicing. These findings underscore the many-to-one relationship that lies at the heart of phonetics: similar phonatory goals can be realized through different physiological means. Viewed in this light, the variability observed during exercise in the present study is not surprising: speakers who already differ at rest may continue to do so under the demands of physical activity.

Yet while speech production is inherently variable, exercise introduces physiological constraints that alter the conditions under which this variability unfolds. Speakers enter exercise with individual baseline production patterns, but increased ventilation progressively shifts speech production farther from those baselines. As ventilation increases, all speakers must accommodate larger volumes of air, higher airflow, and an increasing urge to breathe, making it more difficult to delay exhalation in order to complete utterances. Under modest ventilatory changes, speakers retain considerable flexibility in how they produce speech; as ventilation increases, however, the range of viable paths to phonation may narrow. In this context, the patterns observed in the present study could reflect groups of speakers converging on particular combinations of phonatory adjustments.

This type of parameter-based group was also observed by Patil et al. (2010), who examined links between breathing and acoustic–prosodic outcomes during exercise. The authors describe distinct response profiles that are characterized by alterations to a single parameter – f0, energy, or duration – or show combined changes across multiple parameters. For example, speakers adopting an "energy-dominant strategy" were observed to "shorten the duration of CV segments while pumping more energy into them" (Patil et al. 2010: 5148). The authors proposed that this strategy may help align speech and respiratory demands by allowing physiological needs for exhalation to be met during speech itself. This interpretation resonates with the present findings: Increases in SoE and shifts in harmonic energy suggest that some speakers may respond to increased ventilation by channeling more airflow through speech itself. Whether this reflects an efficient means of managing excess air, a response to respiratory discomfort, or a more general constraint on phonatory control remains an open question. Nevertheless, these observations point to the value of treating variability as structured and potentially informative.

5.4.3.1 Methodological considerations and future directions

Two methodological considerations should be noted. First, the prosodic environments of the vowels in the reading task were not controlled, and differences in segmental context, pitch accent, and phrase position may have influenced some of the acoustic measures. However, because these prosodic contexts were held constant across workload conditions, any observed exercise effects remain interpretable within each context. Future work might examine how the same vowel behaves across different prosodic contexts, especially since several participants reported difficulty with intonation under exercise.

Second, prior experience speaking during physical activity could have given rise to some of the observed individual variability. A few participants reported regularly talking during exercise, echoing observations by Sandage and colleagues that some of their participants may already have developed adaptive strategies. With practice, muscles become more efficient and precise, so speakers used to talking during exercise may be better able to control the intrinsic laryngeal muscles for stable voicing under increased respiratory drive. This suggests that variability is not purely physiological but may also be experiential. Future studies should therefore gather information about habitual speaking patterns during exercise.

5.4.4 Conclusion

This study examined phonation during physical activity using multiple acoustic measures. At the group level, exercise was associated with increased f0 alongside decreases in CPP and HNR, indicating faster but less regular vocal fold vibration. Increases in acoustic energy and SoE reflected greater vocal intensity and a greater push of energy into each glottal cycle. Changes in harmonic measures showed a redistribution of energy toward higher frequencies, suggesting that as ventilation increased, speakers altered how airflow is released and shaped during each glottal cycle. Crucially, however, visualizations of individual responses revealed differences in both the magnitude and direction of these effects, indicating that exercise does not elicit a single phonatory pattern but rather multiple, speaker-specific adjustments. While the source of this variability remains unclear, its prevalence underscores the importance of treating individual differences as informative for understanding speech production during physical activity.

6 Articulatory changes in speech during physical activity

6.1 Introduction

At the last level of speech production, coordinated movements of the articulators shape the glottal source signal into distinct speech sounds. While various studies have reported exercise-related adjustments at the respiratory and laryngeal levels, comparatively little work has examined whether articulatory adjustments occur during physical activity. Nevertheless, there are physiological reasons to expect such effects: the body's response to exercise alters the speech-production environment, including breath volume, subglottal pressure, blood flow to tissues, and muscle activation. Depending on their extent and nature, these changes could plausibly influence supralaryngeal configuration.

If such adjustments occur, they should be observable acoustically. The current study therefore investigates the effects of physical activity on articulation by assessing changes in formant frequencies as physical workload increases. While direct measurement of articulatory movements would be ideal, common methods to do so, such as electromagnetic articulography and magnetic resonance imaging, require the speaker to remain still during recording. The nature of the current study thus necessitates the use of indirect measures of articulatory behavior. Formant frequencies, which have a well-established relationship to articulatory configuration, provide a feasible alternative for capturing articulatory changes. The aim is to provide an empirical characterization of articulatory behavior during exercise and to compare it with speech produced at rest.

This chapter opens with a review of relevant literature, focusing on formant-frequency changes during physical activity and in contexts of psychological stress and loud speech, which serve as useful comparisons. It then describes the methodology used to obtain formant frequencies in the current study (Section 6.2), presents the results of the statistical analyses along with visual explorations of speaker differences (Section 6.3), and discusses the implications of these findings and directions for future research (Section 6.4).

6.1.1 An overview of articulatory–acoustic relations

What are known as formants today have intrigued scholars for centuries, including a young Isaac Newton, who observed that vowels possess characteristic "tones" (for a historical overview, see Vilain et al. 2015). This phenomenon is explained by the resonant properties of the vocal tract, which can be modeled as a tube comprising the pharynx and oral (and nasal) cavities. These structures act as interconnected resonators for the source signal, boosting some frequencies and attenuating others. The boosted frequencies are of greater relative amplitude, yielding concentrations of energy at certain frequency bands – these are the formants. The precise frequencies reflect articulatory configuration within the resonator and thus distinguish different vowels.

The link between articulator position and vowel identity was made explicit in 1948, when both Delattre (1948) and Joos (1948) showed that the articulatory positions represented by the IPA's vowel quadrilateral corresponded to the first two formant frequencies plotted on a two-dimensional plane. Later studies provided further empirical evidence for articulatory–acoustic relations by extending the tube model of the vocal tract to incorporate lip-rounding (Fant 1960) and by demonstrating that acoustic variability closely reflects articulatory variability (Whalen et al. 2018).

In the vocal tract, constrictions are formed along two dimensions: high to low and front to back. Phonetic height is related to F1, while phonetic backness is related to F2. When F1 is plotted on the y-axis and F2 on the x-axis, with both axes inverted (so higher values appear toward the origin), the F1–F2 coordinates of each vowel approximate its articulatory target in a mid-sagittal view. This F1–F2 vowel plane is the traditional way to represent vowels in the acoustic domain, following early perceptual work showing that the first two formants are the primary perceptual cues to vowel identity (Peterson & Barney 1952).

Formant-frequency data can be analyzed either at the level of individual vowel positions within this plane or through measures that summarize patterns across multiple vowels. Examining single vowels makes it possible to test whether particular vowel qualities are affected by the experimental manipulation, which is useful in exploratory studies where it is unclear where effects may appear. In contrast, metrics that combine information across vowels and/or formants can capture broader changes in the overall organization of the vowel space.

Some of the most common summary measures are reviewed by Kent & Vorperian (2018). Two measures are particularly relevant for the current study. The first is the centroid, a measure of central tendency calculated as the grand mean of the frequencies of the corner vowels for a given formant i:

$$\text{Centroid}(F_i) = \frac{F_i/\text{i}/ + F_i/\text{a}/ + F_i/\text{u}/}{3}$$

The centroid thus summarizes overall shifts in vowel height and/or backness, depending on the formant. The second measure is vowel space area (VSA), defined as the area enclosed by the corner vowels. These vowels occupy the extremes of the acoustic space and are thought to reflect articulatory working space. When corner vowels are hyperarticulated, the degree of acoustic separation between them grows, resulting in larger VSA values. Smaller VSA values result when the corner vowels are hypoarticulated and move toward the center of the vowel space. Because of this acoustic–articulatory relation, VSA has been used to investigate diverse speech contexts that may modulate intelligibility, like psychological stress (Scherer 2018) and registers such as infant-directed speech (Kuhl et al. 1997). Thus, VSA can serve as a quantitative index of intelligibility as well as a means to characterize global changes in vowel articulation.

6.1.2 Formant-frequency changes in speech during exercise

The systematic review presented in Chapter 2 identified three articulatory investigations of speech during exercise, all of which were conference papers assessing the acoustic signal. In the first study, Godin & Hansen (2011b), four speakers of American English (two females) exercised on an elliptical machine at a pace of 10 mph while repeating the vowels /i, ɑ/ 5 times each in both VCV and CV contexts with the consonants /t, d, m, n/. Changes in the first two formants (F1 and F2) were assessed using ANOVAs with physical workload, vowel, speaker, and preceding consonant as fixed effects. For F1, the expected significant main effects of speaker, vowel, and consonant were found but workload was just above the significance cutoff (p = .051), indicating that F1 frequency was different for each speaker, vowel, and consonant but did not change when speakers exercised. The interaction between speaker and workload was significant, indicating that speakers responded to exercise differently. Although the authors cite this as one of the main results of the study, it is not clear if the participants were responding differently to exercise or if they were responding to *different levels* of exercise – because the exercise task was defined as a fixed pace, participants with lower or higher levels of fitness would have to expend more or less physical effort, respectively, to maintain the same pace. The authors report rather small changes in F1 frequency (though it is not clear for which vowel): +20 Hz, +7 Hz, and +15 Hz for one male and both female speakers, respectively, and a decrease of 20 Hz for the second male speaker. For F2, workload was found to have a significant effect (p = .001). In sum, the study provided evidence that physical workload influences F1 differently for each speaker, but has a general influence on F2. Since speakers were probably exercising at different workloads, it may be the case that F1 is affected by the level of physical workload.

Marquard et al. (2017) investigated vowel production during exercise while wearing an oxygen mask, reporting results from four male volunteer firefighters. The participants cycled on a stationary bicycle with and without a mask while speaking freely and while producing target words containing the vowels /i, e, a, o, u/. The ANOVA results were vowel-specific, but generally suggest a fronting, centralizing tendency: F1 decreased for /a/ and increased for /u/, and F2 increased for /i, a/ and decreased for /e, o/. Adding the oxygen mask during exercise tended to change the direction of effect for F1 and increase the effect on F2.

Williamson et al. (2018) investigated the cumulative effects of exercise, heat, and altitude on articulatory precision. Thirteen participants (males and females, but no details) read a passage, produced sustained vowels, and spoke freely after a 45-minute bout of cycling in a hypobaric chamber at altitude (3000 m) and then at altitude plus heat. Articulatory precision was quantified using eigenspectra of correlation matrices constructed from time series data of the first three formants. Contrary to expectations, larger eigenvalues were found during physical activity. While the authors interpret this as evidence of *improved* articulatory coordination, they acknowledge that the observed changes in eigenvalues could also reflect greater articulatory variability (i.e., less precision).

Taken together, these studies provide some evidence that articulation is affected during physical activity, though the nature and extent of these effects remain unclear. The results obtained by Godin & Hansen (2011b) and Marquard et al. (2017) suggest that backness (F2) is affected across all vowels, while changes in height (F1) are small at best and may be speaker-specific. The latter study also suggests a tendency toward a smaller and more fronted vowel space in speech during physical activity. These findings are difficult to compare with those reported by Williamson et al. (2018) because those authors assessed articulatory precision rather than the size of the vowel space. Further, the eigenspectra measure used in that study supports two contradictory interpretations: greater exercise intensity may be associated with greater precision or with greater variability. Further investigation is needed to clarify whether, and in what way, physical activity affects articulation.

6.1.3 Related work: Speech under stress

Speech under stress refers to speech produced by an individual experiencing some form of stress, whether physical, psychological, environmental, or otherwise. Importantly, the body tends to respond to different stressors in similar ways, typically with increased heart rate, faster breathing, and heightened muscle tension (Sapolsky 2004, Selye 1950). Because these physiological responses are shared across stress contexts, findings on speech under cognitive or emotional stress may help inform expectations about speech during physical stress, including exercise.

Phonetic studies of speech under stress have not treated formants in much detail, and the findings of existing studies are not uniform. One group of studies suggests a centralizing tendency for vowels in speech under stress. An early experiment (Hecker et al. 1968) used math tasks under time pressure to induce cognitive stress in 10 male participants who uttered the phrase "generator current." The spectrograms were visually assessed, and the authors hypothesized that faster speech resulted in a centralization of the vowel space: "Some subjects, when they were under stress, had a tendency to generate certain consonants and vowels more rapidly, often with a less constricted vocal tract. In a sense, their articulatory movements for a given phonetic segment fell short of the intended target" (Hecker et al. 1968: 1000). Similar results were obtained by Karlsson et al. (2000: 124), who described a "de-peripheralisation" in the vowel space of 6 male Swedish speakers reading sequences of numbers during a divided-attention task with background noise. These authors also observed faster speech in the stressed condition.

In a more naturalistic stress situation, Huttunen et al. (2011) generated cognitive load in 13 Finnish male pilots using a flight simulator with helmet and oxygen mask. Formants were tracked automatically in Praat and hand-corrected as needed. The authors found that during flight simulations, F1 increased and F2 decreased for the front vowels /i, e, y/ (by an average of 47 Hz and 72 Hz, respectively) while both F1 and F2 significantly increased for back vowels /a, o/ (by an average of 11 and 51 Hz, respectively). No effect of articulation rate was found. Finally, in an analysis of speech under real-life stress – an in-flight emergency – Benson (1995) analyzed speech from a male F-16 pilot and his wingman. Formant values were obtained for the segments enclosed in angle brackets in the words "em<er>gency," "eng<in>e," "<new>t" with a peak-picking algorithm based on line spectral pairs, though only the latter two samples were reported in detail. For <in>, F1 was found to increase by 80 Hz and F2 by 100 Hz. For <new>, F1 increased by 20–220 Hz and F2 was not reported. Together, these results suggest a centralizing, fronting tendency, though the elevated F1 may also reflect nasalization.

A second group of studies concluded that stress effects were vowel-specific. Ruiz et al. (1996) investigated the vowels /u, ɛ, o, ø/ produced by one Belgian French speaker during a Stroop test[1] and the vowels /i, e, ɛ, a, u/ produced by a French-speaking pilot and copilot during an in-flight emergency. Results varied by participant: the speaker performing the Stroop task showed vowel- and formant-specific effects; the pilot showed only effects for F2 and the copilot only for F3. Yap et al. (2010) also used a Stroop test to generate cognitive load. Sixteen English speakers (7 males) produced the vowels /i, e, a, u/ (contained in color words). Formant estimates made using WaveSurfer (Snack algorithm; settings: 4 formants, 12th-order LPC) showed for F1 "a generally increasing trend as cognitive load increased" (*ibid.*, p. 2010) especially for /i/ and /u/. F2 also increased for /i/ but decreased for /a, u/. These changes imply expansion on the front–back dimension, but lowering on the high–low dimension, suggesting a *shift*, rather than a centralization, of the vowel space.

A third group of studies (to which Hecker et al. 1968 and Ruiz et al. 1996 also belong) concluded that the effects of cognitive load are *speaker* specific. In an analysis of real-life speech under stress – speech recorded during oral university exams – Sigmund (2012: 48) analyzed tokens of /i, a, u/ from 10 male Czech speakers, concluding that "speakers tend to follow an individual trend rather than a global trend valid for all speakers." However, data from individual speakers were not presented. In a laboratory study, Tolkmitt & Scherer (1986) tested the hypothesis that individual differences are related to personality traits, dividing speakers into groups based on stress-coping style. The 60 German-speaking participants (33 males) were given logic problems to generate cognitive load and shown gruesome images to generate emotional load. The vowels /i, e, a, o, u/ were elicited in random strings (e.g., A2E). Analysis of F1 and F2 estimates, which were combined into a single index, showed both individual differences and sex differences, with coping strategy playing a key role for females: "If we equate formant movement with articulatory effort, the stepping up of the stress in the cognitive situation causes articulation of the anxiety deniers to improve, whereas it deteriorates for the low- and high-anxious groups" (Tolkmitt & Scherer 1986: 307). These findings thus suggest that some speakers will exhibit an *expansion* of the vowel space under stressed conditions.

Finally, one study reported null effects. In an analysis of 10 vowels in a hVd context produced by 5 males performing a visual tracking test, Lively et al. (1993)

[1]The Stroop test assesses selective attention by asking participants to name the color of a word's font while ignoring the word itself (i.e., the word 'red' appears in blue font).

found no significant change in the frequencies or bandwidths of the first three formants. However, the authors also observed a large amount of intersubject variability, finding that often only one or two participants would show significant changes in formant frequency between baseline and stressed speech. The authors hypothesized that individuals react to stress in different ways, although another explanation could be that the participants felt *different levels* of stress in response to the lab task.

When the formant studies in speech under physical and psychological stress are viewed together, three themes emerge. First, while most studies reported changes in formant frequencies in speech under stress, the results are complex and vary across speakers, vowels, and formants. This variability may reflect genuine individual differences, but it could also arise from diverse methodological choices. For instance, some studies did not control for stress level (e.g., exercise intensity in Godin & Hansen 2011b). Furthermore, stress can be difficult to generate in a laboratory setting, and certain tasks, such as a Stroop test or timed calculations, might not have been perceived as equally stressful by all participants. Such uncertainty about perceived stress levels makes it unclear whether participants reacted differently to comparable levels of stress or whether they experienced different levels of stress altogether.

A second theme that emerged was a general tendency toward vowel-space centralization in stressed conditions. This was either described qualitatively (e.g., Karlsson et al. 2000) or inferred from the direction of formant-frequency changes for the corner vowels (e.g., Huttunen et al. 2011). However, one recurring pattern across several studies was an increase in F1 values for all vowels, which is more consistent with a lowering of the tongue or jaw position than with vowel-space centralization.

The third theme is that the magnitude of reported changes is relatively small – typically less than 50 Hz. It is possible that real-life stress might elicit larger shifts (Ruiz et al. 1996). Formant-frequency changes were also found to vary across the duration of speech samples, such as between more and less demanding phases of flight simulations (Huttunen et al. 2011) or at various points during real-life emergencies (Ruiz et al. 1996). This variation is thought to reflect fluctuations in stress arousal. In any case, reporting absolute formant values allows for a more comprehensive interpretation of these vowel-space changes.

Table 6.1 summarizes the methodological choices and findings for the corner vowels /i, a, u/, highlighting several knowledge gaps. For one, participants were predominantly male, raising questions about potential differences in formant-frequency changes for female speakers, who typically have larger vowel spaces. For another, studies focused on European, mostly Germanic languages with large vowel inventories, leaving open the question of whether effects are language-specific or if they apply more broadly across speech in general.

Overall, the existing literature on vowel formant-frequency changes in speech under stress is fragmented.[2] Many studies did not investigate all corner vowels or did not report formant values in Hz,[3] making it difficult to compare findings across studies. Although half of the studies reported individual differences, it is difficult to assess this variation due to small sample sizes. Furthermore, the research predominantly used binary comparisons between stressed and unstressed conditions, leaving the effects of varying stress levels unexplored. While the current body of work suggests that supralaryngeal modifications occur in speech under stress, there is a clear need for more systematic and controlled investigations to better understand these changes.

6.1.4 Related work: Loud speech

Loud speech shares several characteristics with speech produced during physical activity. In the respiratory domain, larger volumes of inspired air and increased subglottal pressure are found in both loud speech (Ladefoged & McKinney 1963) and speech during exercise (Ziegler 2014, Bailey & Hoit 2002). Acoustically, speech during exercise is louder than speech at rest (Dallaston & Rumbach 2016, Doust & Patrick 1981, Fuchs et al. 2015, McCaig 2013, Otis & Clark 1968, Patil et al. 2010, Trouvain & Truong 2015, Ziegler et al. 2019), a pattern also observed in the present dataset (Chapter 5). Given these similarities, loud speech can serve as a rough model for understanding speech production during physical activity.

Previous research on shouted and Lombard speech, which has investigated both acoustic and articulatory changes, may help interpret the increases in F1 reported in some studies of speech under stress. For the high corner vowels /i/ and /u/, higher F1 values are consistent with either vowel centralization or a general lowering of the jaw. For /a/, however, these two accounts make opposite predictions: centralization predicts a decrease in F1, whereas jaw-lowering predicts an increase. The behavior of /a/ during exercise is therefore particularly informative for interpreting articulatory changes during physical activity.

In studies of loud speech, the most widely obtained acoustic finding is an increase in F1 in all vowels (Bond et al. 1989, Garnier et al. 2006, Huber &

[2] This assessment is not unique to speech under stress. In their review of formant-measurement methods, Kent & Vorperian (2018) emphasize the general gaps in knowledge of vowel acoustics.

[3] Two studies made qualitative assessments (Hecker et al. 1968, Karlsson et al. 2000), two studies used a unit other than Hz (Tolkmitt & Scherer 1986, Williamson et al. 2018), and one study did not investigate formant frequencies *per se* but instead their usefulness as features for the automatic detection of speech under stress (Sigmund 2012).

Table 6.1: Overview of methods and results for corner vowels in formant-frequency studies under stress. ↑/↓: increase/decrease; ↕: mixed pattern; †: effect direction not specified; —: not investigated; blank cells: specific formant results not reported. **Bolded stressor types** reflect real-life stress conditions.

Study	Lang.	N	Stressor type	/i/		/a/		/u/		Speaker specific?
				F1	F2	F1	F2	F1	F2	
Benson (1995)	English	2 ♂	**Emergency**	—	—	↑				
Godin & Hansen (2011b)	English	4 (2 ♀)	**Exercise**	↕	†	↕	†	—		yes
Hecker et al. (1968)	English	10 ♂	Cognitive							yes
Huttunen et al. (2011)	Finnish	13 ♂	Cognitive	↑	↓	↑	↑	—		
Karlsson et al. (2000)	Swedish	6 ♂	Cognitive							
Lively et al. (1993)	English	5 ♂	Cognitive	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	yes
Marquard et al. (2017)	German	4 ♂	**Exercise**	n.s.	n.s.	↓	↑	↑	n.s.	
Ruiz et al. (1996)	French	1 ♂	Cognitive	—	—	↕				
Ruiz et al. (1996)	French	2 ♂	**Emergency**	↑	n.s.	n.s.	↑	n.s.	↓	yes
Sigmund (2012)	Czech	10 ♂	**Oral exam**							yes
Tolkmitt & Scherer (1986)	German	60 (27 ♀)	Cognitive & emotional							yes
Williamson et al. (2018)	English	13 ♂	**Exercise**							
Yap et al. (2010)	English	16 (9 ♀)	Cognitive	↑	↑	n.s.	↓	↑	↓	

Chandrasekaran 2006, Junqua 1993, Liénard & Di Benedetto 1999, Lu & Cooke 2008, Xue et al. 2021). However, in a study of German vowels in different types of speech samples, Koenig & Fuchs (2019) found that F1 increased in a *vowel-dependent* way. As for F2, results were less systematic and rarely significant, though general increases were reported (Garnier et al. 2006, Huber & Chandrasekaran 2006, Xue et al. 2021) in addition to increases for back vowels and decreases for front vowels (Bond et al. 1989, Garnier et al. 2006). However, not all studies found a consistent increase in formant frequency in loud speech. Several studies reported a decrease in formant values, either in both F1 and F2 (Huber et al. 1999) or only F2 (Lu & Cooke 2008). Methodological differences may account for these differences. Another explanation could lie in speaker differences – an early study reported considerable individual variation (Tartter et al. 1993).

Most of the acoustic studies saw a likely explanation for the observed effects on F1 in a lower jaw/tongue position and/or greater oral aperture in loud speech. Jaw height is inversely related to the frequency of the first formant, so a lower jaw would result in a higher F1. Why would speakers open their mouths wider during loud speech? For one, a larger mouth opening facilitates clear speech by avoiding the turbulent noise arising from increased subglottal pressure. For another, a larger mouth opening facilitates louder speech, much like the bell of a horn amplifies sounds, and may thus help speakers produce louder speech.

Articulatory studies consistently found that speakers increased oral aperture in loud speech compared with typical speech. The corresponding acoustic signal, as expected, showed a higher F1. Huber & Chandrasekaran (2006) were the first to demonstrate such a relationship, reporting an F1 increase of around 80 Hz[4] in a study of the low vowel /ɑ/ in loud speech and a significant increase in opening displacement for lower lip and jaw. Later studies investigated multiple vowels and found similar results. In a conference paper assessing six vowels in Australian English, Garnier et al. (2006) reported a significant increase in F1 (+118 Hz) in shouted speech and a significant increase in lip aperture and lip spreading across vowels. In German, Xue et al. (2021) investigated 8 tense and 7 lax vowels, finding a significantly higher F1 (+22%) and larger lip and jaw opening, and lower tongue position, across speakers.

These findings are relevant for the current study because they establish a link between increased mouth aperture and higher F1 values across vowels. During exercise, the mouth aperture is expected to widen to accommodate an expected

[4]Loud speech was generated in three conditions (shouting, double comfortable speech volume, and Lombard speech with babble noise of 70 dBA) but the mean increase in loudness was similar (10.23 dB, 9.35 dB, and 11.11 dB, respectively), so the average F1 increase across all three conditions is reported here.

increase in breath depth and flow rate. Consequently, it is reasonable to predict that speech produced during physical activity would show higher F1 values for all vowels. To interpret these changes in the context of hypo- or hyperarticulation, it is crucial to examine formant data for both high and low vowels. When considered in isolation, an increase in F1 for /a/ would be consistent with an expansion of the vowel space, suggesting hyperarticulation. In contrast, increases in F1 for /i/ and /u/ would be consistent with vowel centralization, pointing toward hypoarticulation. However, when the data from all vowels are considered together, an increase in F1 across all vowels would support the interpretation of an overall lower jaw or tongue position in speech during physical activity.

6.1.5 Objectives of the present articulatory study

Given the scarcity of articulatory studies on speech during physical activity, the primary objective of the present study is to systematically investigate how exercise affects vowel articulation. Specifically, the study examines changes in the first and second formant frequencies (F1 and F2) for the corner vowels /i, a, u/. Drawing on findings from research on speech under stress and loud speech, two broad hypotheses motivate the analysis. First, increased physical workload is expected to affect articulatory openness, reflected in higher F1 values, particularly for vowels produced with tighter vocal tract constrictions. Second, exercise may induce shifts in tongue position, leading to fronting or centralization patterns observable in F2 and in vowel space area (VSA). Beyond identifying group-level trends, a central goal of this study is to assess individual differences in articulatory adaptation. Previous work has reported mixed and sometimes contradictory findings, leaving open whether apparent effects reflect general articulatory tendencies, vowel- or task-specific constraints, or speaker-specific strategies. By examining individual trajectories alongside inferential statistics, the present study aims to contextualize group-level results and clarify their generalizability.

To achieve these goals, the study assesses articulation across two speech tasks: sustained vowels and read speech. Sustained vowels provide the most reliable formant estimates for a given vowel, while connected speech offers a more ecologically relevant context. The goal is not to directly compare tasks, but to evaluate whether articulatory patterns observed in sustained vowels extend to connected speech and thus plausibly generalize to real-world speaking conditions. Similarly, multiple vowels are assessed to better understand whether effects generalize across vowels. This approach thus aims to reveal articulatory adaptation during exercise as well as to clarify which methodologies are most appropriate for investigating the topic.

6.2 Methodology

6.2.1 Datasets

The corner vowels /i, a, u/ were selected to investigate whether vowel formant frequency and vowel space area change in response to varying levels of physical workload. Two speech tasks were selected for analysis: (1) production of sustained vowels and (2) reading a short passage (see Section 3.2 for details). The sustained-vowel task involves maintaining a single articulatory configuration and phonating over several seconds, which captures a speaker's prototypical vowel targets without the influence of neighboring segments. This makes any observed changes in formant frequencies more directly attributable to the experimental conditions. In contrast, the analysis of read speech data was intended to determine whether changes observed in sustained vowels could be generalized to more naturalistic speech contexts. Each speech task was produced in three physical workload conditions: at rest (control), low workload (35% of maximal HRR), and moderate workload (65% of maximal HRR).

Together, the two datasets contain 2,592 vowel tokens produced by 48 speakers. The sustained-vowel task elicited three repetitions of /i, a, u/ per condition per speaker (1,296 tokens). Although the reading task was not designed to elicit these vowels, three target words containing /i, a, u/ in stressed positions were identified, as shown in Table 6.2. These words were located in adjacent utterances at the beginning of the passage. There were three repetitions of each vowel per condition per speaker (1,296 tokens).

Table 6.2: Words used to extract corner vowel tokens from adjacent sentences in the read passage

Vowel	Word	Sentence
/i:/	*organis**ie**ren*	*Um ein perfektes Fest zu organisieren, braucht man eigentlich gar nicht so viel.* 'To organize a perfect party, you actually don't need all that much.'
/a:/	*spont**a**n*	*… man kann das auch relativ spontan machen, so lange Läden noch offen haben.* '… you can actually do it pretty spontaneously as long as the shops are still open.'
/u:/	*g**u**te*	*Wichtig ist einfach, dass man gute Laune hat…* 'The important thing is just being in a good mood…'

6.2.2 Vowel segmentation

The vowels of interest were delimited manually in each speech dataset using Praat (Boersma et al. 2026) at the onset and offset of F2, which provides a reliable landmark even in cases of gradual transition from aspiration to vowel (Fischer-Jørgensen & Hutters 1981). F2 was determined to be present at "the time of onset of the first vertical striation extending upward through the frequency regions of the first and second formants without interruption" (Francis et al. 2003). For /u/, the vertical striation extending to F3 was taken because F1 and F2 can be visually difficult to distinguish. To aid in identifying F2 onsets, the spectrogram was manipulated as follows: (a) the dynamic range was adjusted to help view vertical striations and distinguish between F1 and F2 (Derdemezis et al. 2016); and (b) the spectrogram view range was adjusted to help distinguish formants that may merge (e.g., F1 and F2 in /u/). The visual inspection of the spectrogram was supplemented by listening to the recording. The F2 criterion was applied to the end of the vowel to determine vowel offset. Figures 6.1 and 6.2 show example onsets for /i/ under moderate load and /u/ in the rest condition, respectively.

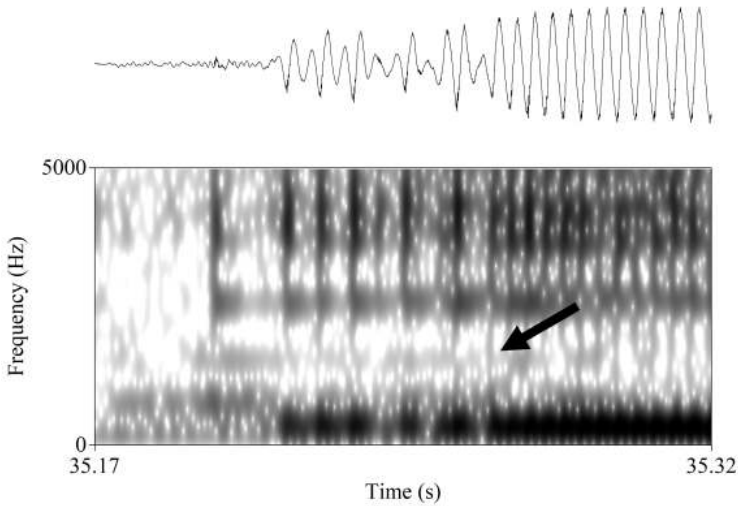

Figure 6.1: Sustained-vowel onset for /i/: The arrow indicates the first visible striation extending through the first and second formant frequencies (data from participant IRC, moderate-workload condition).

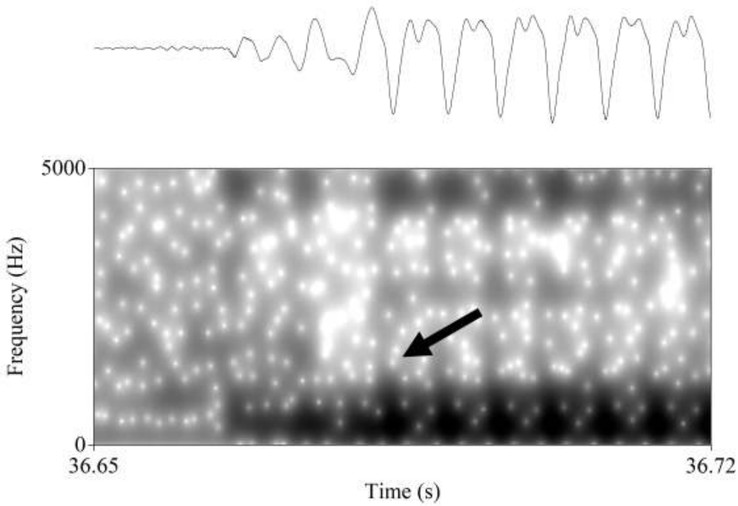

Figure 6.2: Sustained-vowel onset for /u/: The arrow highlights the first visible striation extending through the first to third formants (data from participant ITE, no-workload condition).

6.2.3 Selection of speech samples for analysis

To prepare the files needed for automatic formant extraction, the vowel tokens labeled in Praat were saved as separate WAV files using a Praat script. Each token was then checked for steady-state phonation (i.e., stable amplitude, f0, and formant structure). Different steps were required for the vowel tokens elicited as sustained vowels and in connected speech. For the sustained vowels, a 200 ms section of each token was selected for analysis (Burris et al. 2014, Huber et al. 1999). The section was extracted 1000 ms after vowel onset. A fixed point was chosen for reproducibility and consistency across speakers (Blomgren & Robb 1998), and 1000 ms was chosen because it was roughly mid-production for most speakers across conditions.[5] Importantly, both amplitude and f0 had stabilized by this point. After extracting the 200 ms section, the tokens were manually checked for steady-state phonation by visually inspecting the wideband spectrogram (view range: 0–5000 Hz). If the sample contained voicing breaks or sudden changes in amplitude, an adjacent 200 ms section was taken for analysis, using a starting point between 700–1100 ms after vowel onset. A total of 87 tokens (6.7% of tokens) were replaced in this manner. Nine tokens (0.7%) were excluded from

[5]The average duration across conditions and vowels was 2.5 seconds (IQR: 2.11–2.88 seconds).

analysis because formants could not be reliably distinguished – 5 /a/ tokens, 2 /i/ tokens, and 2 /u/ tokens. Figure 6.3 shows an example of a drop in amplitude that coincides with the sample taken at the fixed time-point 1000 ms post-onset. A typical sample is given in Figure 6.4 for comparison.

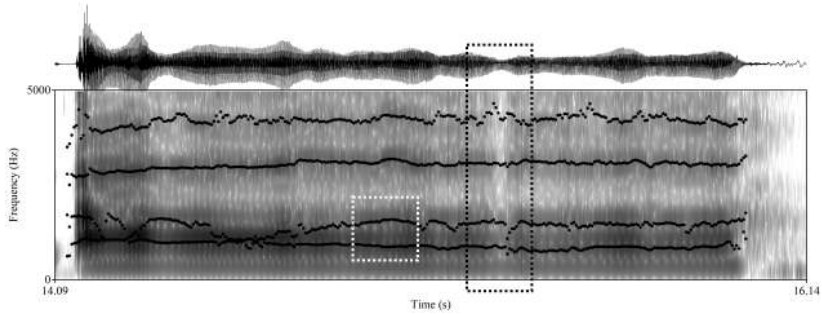

Figure 6.3: Example of sample replacement in a sustained /a/ token: A section containing a sharp drop in amplitude (*black dotted box*) was replaced with a nearby section (*white dashed box*) showing more stable phonation, amplitude, and formant structure (data from participant BRX, moderate workload).

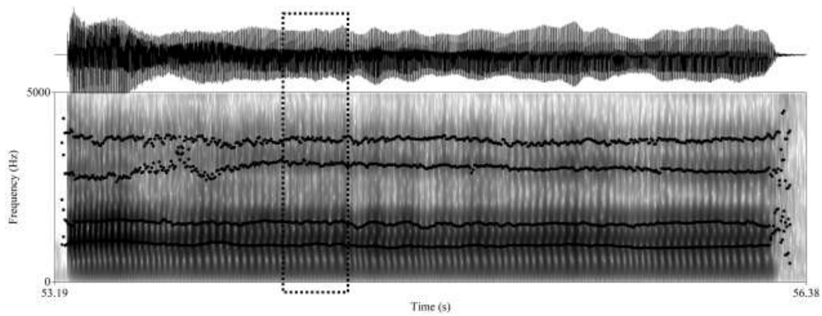

Figure 6.4: Example of a typical sample in a sustained /a/ token: The section at 1000 ms after onset shows stable phonation, amplitude, and formant structure (data from participant EGB, moderate workload).

A subset of speakers (n = 8, 16.7% of participants) did not consistently produce sustained vowels with steady pitch. For these speakers, the time-point for extraction was determined individually; for example, some speakers had steady sections initially and later modulated pitch, while others did not modulate pitch in all tokens. Steady-state phonation was assessed in these tokens using a fixed rate-of-change criterion for fundamental frequency of less than 20 Hz per 20 ms (Weismer et al. 1988). Tokens that showed greater change were excluded from analysis. Two participants (ACP, PZI; 4.2% of tokens) were excluded from analysis because no steady-state sections could be found.

For the vowel tokens in connected speech, the Praat script used to extract separate WAV files was modified to add a 25 ms buffer before and after the labeled vowel. This ensures that the entire produced vowel was later analyzed when using a 25 ms analysis window. This approach and the lines of code to implement it were taken from Katherine Crosswhite's Segmenter script.[6] The full vowel was used (rather than a middle section) because it is not clear whether physical activity affects formant trajectories and hence it is not clear where to extract the speech sample. As these speech samples are influenced by prosody and coarticulation, no steady-state criteria were applied. However, tokens containing sudden changes in voice quality were excluded – for example if the first half of the sample was produced with modal voice and the second half with creaky voice – because such fluctuations can interfere with formant tracking. One participant (HQN) produced highly reduced, breathy tokens of *gute* 'good', which were excluded from analysis. For one participant (RVH), the moderate-workload condition was not recorded due to a technical failure. Ten further tokens were excluded from analysis because formants could not be reliably distinguished: 3 /i/ tokens and 7 /u/ tokens.

6.2.4 Formant estimation and parameter extraction

Formants – resonances of the vocal tract – are not measured directly but inferred from other measurements. Hence, formant values should always be referred to as *estimates*. The most common method of obtaining formant estimates is via linear predictive coding (LPC), which uses an algorithm to predict future values based on past ones. Despite its wide usage, LPC has various pitfalls, which are well documented, for example, in the Praat manual (Boersma et al. 2026). For a critical overview of the problem, with examples and recommendations, readers are referred to "Formants are easy to measure; resonances not so much: Lessons from Klatt (1986)," (Whalen et al. 2022). Unfortunately, this paper was published after the current study had been conducted, but its recommendations have been followed where possible, starting with "(2) report[ing] analysis settings more fully" and "(3) justify[ing] choices made in those settings" (Whalen et al. 2022: 939). To this end, this section motivates the choice of software program and

[6] http://web.archive.org/web/20030620172734/ling.rochester.edu/people/cross/scripts.html

algorithm, lists the f0 settings and formant settings, and describes the manual validation process for each token.

VoiceSauce (Shue et al. 2011), implemented in MATLAB (version 2022a), was chosen to estimate formant values. VoiceSauce is an automated application that extracts multiple parameters from audio recordings, making it more powerful than alternatives such as Praat. An additional advantage is that it offers different LPC algorithms for tracking formants, including that used by Praat (Burg algorithm, as given by Childers 1978 and Press et al. 1992).

Although most phonetic studies of formants probably use Praat (Whalen et al. 2022: 934), the performance of LPC algorithms is known to vary. General opinion seems to favor the Snack Sound Toolkit (Sjölander 2021), which is used in WaveSurfer, over Praat. Yet a study comparing hand-labeled "ground-truth" formant values with the estimates generated by both programs found that "Snack was considerably worse in all instances" (Skarnitzl et al. 2015: 177). The authors speculated that since their results were obtained using high-quality audio samples, Snack might show superior performance with less ideal recordings, such as samples with background noise.

Given that the heavier breathing associated with exercise was predicted to add noise to the signal, the plotted outputs of the Praat and Snack algorithms were visually compared to select the best-performing algorithm for the current dataset. Formant estimates were obtained using the standard settings for each algorithm and one set of alternative settings for comparison. The alternative settings were those recommended in Skarnitzl et al. (2015) for Praat and their rough equivalent for Snack. Table 6.3 lists all settings.

Table 6.3: Settings for formant estimates obtained with Praat and Snack algorithms

Settings	Praat	Snack
Standard	5500 Hz ceiling 5 formants	5500 Hz ceiling 12th-order LPC (6 formants)
Alternative	5000 Hz ceiling 4.5 formants	5000 Hz ceiling 10th-order LPC (5 formants)

For each algorithm and setting, the respective F1 and F2 estimates were plotted on a two-dimensional vowel plane at the same scale, shown in Figure 6.5. Visual inspection of the plots showed that Praat outperformed Snack. In particular, /a/

was problematic for Snack, yielding numerous formant selection errors, as indicated by the cluster of points in the upper middle area of the plot. The high vowels also show lower F1 values, which could point to problems distinguishing the first formant from f0. On this basis, Praat was chosen over Snack to estimate formant values, and settings for ceiling and number of formants were determined individually (described below). Visual comparison of the standard and alternative settings for each algorithm showed little difference between settings.

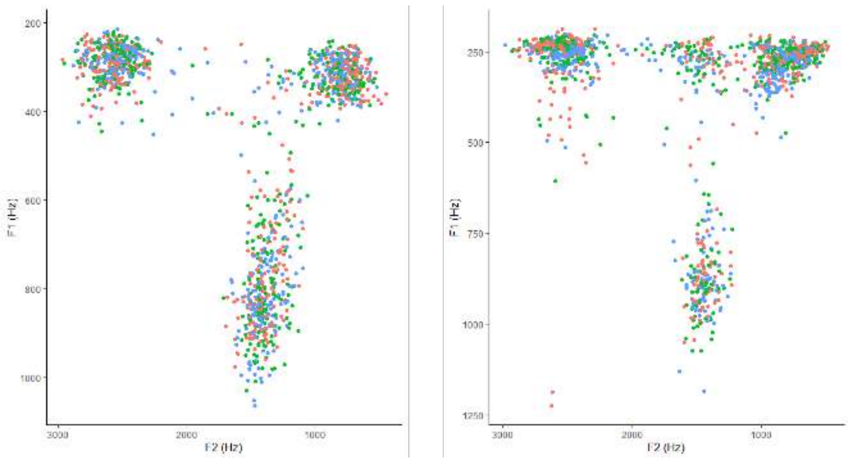

Figure 6.5: F1 and F2 outputs obtained with Praat standard settings (*left*) and Snack standard settings (*right*)

Formant estimation requires fundamental frequency (f0) to be tracked as well. This was done with Praat for consistency, using its autocorrelation method. Autocorrelation finds f0 by taking a small section of the waveform containing several pitch periods (the "window") and overlaying it with windows of different possible period lengths. Correlations are calculated between each set of windows, and the highest correlation is selected as the fundamental frequency of the waveform. The most important setting for determining how the resulting "pitch[7] track" is computed in Praat is the pitch range: the *pitch floor* determines the length of the analysis window and sets the lowest possible value for f0, and the *pitch ceiling* sets the highest value for f0. All candidates below the pitch floor and above the pitch ceiling are ignored by the algorithm.

Praat's default pitch range is 40–500 Hz, which covers a wide range of speakers, but the documentation recommends optimizing the floor and ceiling for each speaker individually (Boersma et al. 2026). The drawback of this approach is that

[7]This section follows Praat's use of the term "pitch" for settings related to f0.

it is time-consuming, especially in datasets with many speakers and experimental conditions that affect f0, such as those analyzed in the current study. A viable alternative, which balances pitch-tracking accuracy with efficiency, is to use a generic, but narrow, range for all speakers. In a comparison of pitch-tracking accuracy in Praat using nine pitch-range settings, Vogel et al. (2009) found that, for female speakers, a floor of 100 Hz and a ceiling of 250–300 Hz yielded f0 values that were not statistically different from those values obtained via hand-scoring, which is recognized as the gold standard. However, the authors cautioned that these settings may not be appropriate for pathological voice populations, given certain voice quality characteristics.

Because physical activity may affect voice quality in different ways, an appropriate generic pitch range for the current study was determined using a data-driven approach. First, f0 was determined using Praat's default pitch range to capture variation in the data. All sound files yielding mean values below 150 Hz were then checked manually. Most values of less than approximately 120 Hz turned out to be pitch-halving errors. The pitch floor was thus set to 130 Hz to avoid halving errors. The pitch ceiling was set at the highest value found in each dataset plus 10 Hz (350 Hz for the sustained-vowel and read-speech datasets). The remaining Praat settings were kept at the default values.[8]

The formant estimation settings were determined individually for each speaker and vowel. This approach is recommended to minimize formant-tracking errors (Harrison 2013, Shadle et al. 2016). There are two settings: the maximum formant frequency and the number of formants. The maximum formant frequency, which sets the range to look for bands of energy in the speech signal, is determined by the length of the speaker's vocal tract. The Praat manual states that "an average adult female speaker has a vocal tract length that requires an average ceiling of 5500 Hz" (Boersma et al. 2026). In contrast, a ceiling of 5000 Hz is recommended for males and 8000 Hz for a young child. Body size thus serves as a rough indication of formant ceiling. However, a complicating factor is that vocal tract length does not remain constant during speech but changes depending on vowel category. In a comparison of vowels in Brazilian and European Portuguese, (Escudero et al. 2009) found that vowel-related differences in vocal tract length were of comparable size to the average difference between male and female vocal tract lengths. The average vocal tract length was found to be longer for /u/ – due to a rounded, protruding lip posture – and shorter for

[8]Voice threshold: 0.45; silence threshold: 0.03; octave cost: 0.01; octave jump cost: 0.35; voiced unvoiced cost: 0.14; smoothing bandwidth: 5; boxes for kill octave jumps, smooth, and interpolate were unchecked.

/i/, which is generally produced with spread lips. The authors thus suggest that automatic formant estimates should be conducted with vowel-specific formant ceilings to reflect vowel-related differences in vocal tract length.

The second setting, the number of formants, specifies the algorithm's sensitivity to spectral variations. LPC predicts future formant values as a linear function of previous values. Changing the maximum number of formants thus tells the algorithm to assign formant values to more or fewer resonances in the spectrum. Thus, depending on the spectral properties of the speech sample, LPC may identify spurious formants or miss true formants. Selecting the appropriate number of formants is generally a matter of trial and error. Because few studies report the settings used, it is difficult to generalize between different types of data. For example, Harrison (2013) obtained the most accurate formant estimates in Praat using 6 formants for close English vowels like /i, u/ and 4 formants for open vowels like /a/. In contrast, Skarnitzl et al. (2015) found that for Czech, vowel-specific settings did not yield estimates that were closer to the ground-truth (hand-labeled) formant values, and thus recommended a single generic setting with a ceiling of 5000 Hz and 4.5 formants.[9] These disparate findings show that there is currently no one-size-fits-all set of parameter settings; rather, the most appropriate settings must be determined for each dataset.

The algorithm settings were determined for each speaker and vowel as follows. A generic vowel-specific setting was applied to the sound files across all conditions: for /i/ a 5500 Hz ceiling with 5 formants, and for /a/ and /u/ a 5000 Hz ceiling with 4.5 formants. The different formant ceilings reflect presumed differences in vocal tract length. Fewer formants were set for /a/ and /u/ because these vowels were generally breathier and a setting that was less sensitive to spectral energy yielded better results. To assess the settings for each vowel across workload conditions, Praat's formant estimates were overlaid on the wideband spectrograms and the trajectories for F1 and F2 were visually inspected to confirm that they were roughly parallel to each other and centered in the formant bar. If this was the case and no spurious outliers were observed, the setting was used for analysis.

If the generic settings produced spurious estimates, the ceiling and the number of formants were manually adjusted using the spectrograms as a guide. The new settings were then checked by visually inspecting the spectral slice of the entire vowel section (Boersma 2014: 388) and comparing the resulting formant

[9]The paper refers to "9th-order LPC," but this simply refers to the number of prediction coefficients – or "poles" – that this algorithm uses (here, 9). Since two poles are used to estimate each formant, any multiple of 0.5 is possible; the number of formants to be specified in Praat is thus 4.5. Harrison (2013) likewise specified "12th order" and "8th order."

estimates with reference values for female speakers for the German vowels /i, a, u/ as sustained vowels (Tolkmitt & Scherer 1986) and in spontaneous speech (Pätzold & Simpson 1997). This approach follows the recommendations of Kent & Vorperian (2018: 75), who note: "Formant estimation can be a process in which knowledge about likely formant locations leads to an efficient inspection of the acoustic data and can serve as a check on automatic analyses that may generate spurious formants or miss a formant altogether."

Spurious estimates were visually identified in the wideband spectrogram. These errors were generally unambiguous upon inspection, and two representative examples are provided below for illustration. Figure 6.6 displays poorly tracked ("wavy") formants with outlying estimates, while Figure 6.7 shows the appearance of a spurious formant that caused subsequent formants to be mislabeled. Both figures additionally include the corrected tracking following parameter adjustment for comparison. In some instances, no formant settings could be found to reliably track formants. In some cases (1.5% of tokens), a viable formant frequency value for F1 and/or F2 could not be obtained with LPC but the value could be taken manually from the spectral slice. For the vowel data, 12 tokens (0.9%) were excluded from analysis because no settings could be found to reliably track formants (7 /a/ tokens and 5 /u/ tokens).

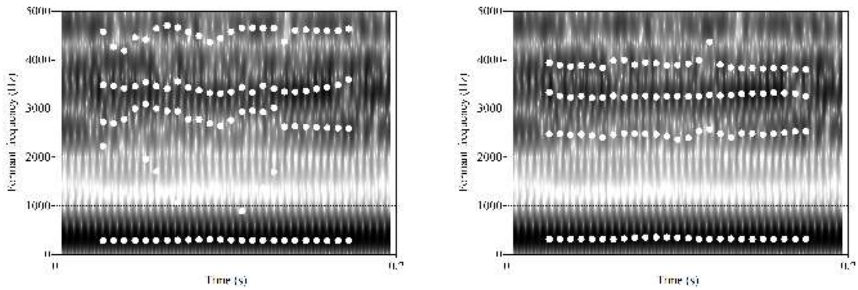

Figure 6.6: Example of poorly tracked higher formants in a sustained /i/ token (*left*) and corrected tracking after adjusting formant settings (*right*) (participant NPN, no-workload condition).

The process was repeated to determine the settings for the reading data. Initially, the sustained vowel settings were applied, but adjustments were often required. A total of 11 tokens (0.9%) were excluded from analysis because no settings could be found to reliably track formants (6 /a/ tokens, 2 /i/ tokens and 3 /u/ tokens).

Once the individual settings for each speaker–vowel combination had been identified, the sound files were processed in VoiceSauce to extract the F1 and F2

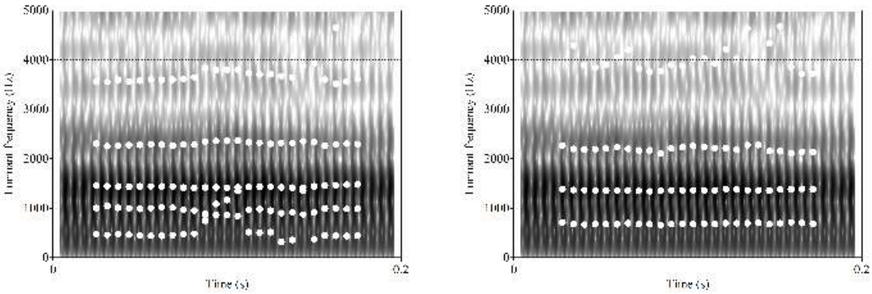

Figure 6.7: Example of a spurious F2 estimate in a sustained /a/ token (*left*) and corrected tracking after adjusting formant settings (*right*) (participant CSM, moderate workload).

values, which were plotted to assess outliers. Because of the challenges in estimating formants with LPC, outlying datapoints may be measurement errors. To account for this, upper and lower cutoffs were calculated as 1.5×IQR to exclude F1 and F2 outliers from analysis for each speech task, per vowel and workload condition. The cutoffs showed that for /a/ in the reading task, two participants (GMM, XFY) produced tokens of F1 and F2, respectively, that were consistently identified as outliers; auditory impressions of the recordings suggested that these speakers indeed produced a more close/back vowel than /a/ when reading; participant XFY also showed some inconsistency during sustained vowels. These /a/ tokens were thus excluded from analysis. A further 13 (1%) tokens were excluded as outliers from the reading task (7 /a/ tokens, 5 /i/ tokens, and 1 /u/ tokens) and 12 (0.9%) tokens from the sustained-vowel task (9 /a/ tokens, 2 /i/ tokens, and 1 /u/ tokens). Table 6.4 shows the total number of tokens analyzed for each speech task.

Table 6.4: Number of observations per condition, task, and vowel

	Read passage			**Sustained vowels**		
	/i/	/a/	/u/	/i/	/a/	/u/
rest	140	134	137	138	131	137
low	141	133	137	137	133	136
moderate	138	131	135	135	129	133
TOTAL	**419**	**398**	**409**	**410**	**393**	**406**

6.2.5 Formant measures

For each vowel token, mean values in Hz for the first two formants (F1, F2) were calculated over the entire section extracted from the vowel. Hertz was chosen over psychoacoustic scales like Bark or Mel because the object of inquiry was potential articulatory changes. The mean was chosen over the temporal midpoint or median because a single point may coincide with a sudden fluctuation in production and thus not be representative of the entire sample. The mean is a more stable measure, especially given that f0, amplitude, and formant structure became less steady during exercise.

To illuminate different aspects of the data, three measures were selected to assess formant changes in speech during exercise:

(1) Mean change in F1 and F2 for each vowel was used to assess changes in individual vowels to infer changes in articulatory targets.

(2) The centroid – the grand mean of all vowels for a given formant – was used as a vowel-extrinsic data reduction strategy to reveal general patterns in the formant data.

(3) The vowel space area (VSA) was used to assess global changes in the vowel space in speech during exercise. VSA was calculated in R using the package phonR (McCloy 2016) as a convex hull, which considers all observations (rather than grand means) and thus preserves speaker variation.

6.2.6 Statistical analyses

Statistical analyses were conducted using linear mixed-effects models[10] to account for multiple observations from the same participants across workload levels. Separate models were fitted for each speech task (reading and sustained vowels), with physical workload (rest, low, moderate) as the primary predictor and formant-based measures as outcome variables. This approach reflects the study's focus on the effects of exercise on articulatory targets, rather than differences between speech tasks themselves.

To capture between-speaker variability, two random-effects structures were compared for each outcome variable within each speech task. The more complex

[10]Analyses were implemented in R (Version 4.2.2) using the following packages: lme4 (Bates et al. 2015), lmerTest (Kuznetsova et al. 2017), and emmeans (Lenth et al. 2023). The BOBYQA optimizer (Powell 2009), with a set maximum of 200,000 iterations, was used to maximize the likelihood function.

structure allowed both participants' baseline formant levels (random intercepts) and their sensitivity to workload (random slopes) to vary across individuals. The simpler structure allowed only baseline differences between participants, assuming a similar response to workload. The best-fitting model was determined based on three criteria: lower Akaike information criterion (AIC) to indicate goodness of fit, lower Bayesian information criterion (BIC) to balance goodness of fit with model simplicity, and absence of singularity.

For the reading task, the best-fitting model for the vowel /u/ specified a random structure with speaker-specific intercepts only, as indicated by smaller AIC and BIC values. For vowels /i/ and /a/, models including random slopes showed smaller AIC but larger BIC values, suggesting an unclear contribution of the additional random-effects complexity. Here, the simplified random structure was selected as the best-fitting model. For the sustained-vowel task, models with both random intercepts and slopes provided the best fit for most measures. The exceptions were F2/i/ as well as the centroid and VSA measures, where the simpler random structure showed smaller AIC and BIC values. The selected random-effects structure for each measure is reported in Table 6.5.

6.3 Results

The effects of physical activity on supralaryngeal behavior were evaluated using three complementary measures to identify patterns at different levels of granularity: vowel-specific formants, a vowel-extrinsic summary measure, and vowel space area (VSA). The results are presented in this order, beginning with a visualization of the raw data. These quantitative analyses are followed by an exploratory, descriptive section on speaker differences. Formant data are given in hertz (Hz). Formants for specific vowels will be referred to using the symbolic notation in Kent & Vorperian (2018): the formant frequency for a given vowel is expressed as $Fi/x/$, where Fi indicates the ith formant and x represents a phonetic symbol (e.g., F2/a/ refers to the second formant of vowel /a/).

6.3.1 Descriptive statistics: Vowel formants

Before conducting statistical analyses, violin plots were created to examine the distribution of observations for each vowel. Violin plots depict the density of datapoints across the measurement range, revealing qualitative differences in distributional shape across workloads that are not captured by mean-based analyses. Reading and sustained-vowel tasks are shown as paired half-violins, enabling

Table 6.5: Random-effects structures for each measure and speech task

Measure	Speech task	Random-effects structure
F1 /a/	reading	(1 | participant)
	vowels	(1 + workload | participant)
F2 /a/	reading	(1 | participant)
	vowels	(1 + workload | participant)
F1 /i/	reading	(1 | participant)
	vowels	(1 + workload | participant)
F2 /i/	reading	(1 | participant)
	vowels	(1 | participant)
F1 /u/	reading	(1 | participant)
	vowels	(1 + workload | participant)
F2 /u/	reading	(1 | participant)
	vowels	(1 + workload | participant)
F1 centroid	reading	(1 | participant)
	vowels	(1 | participant)
F2 centroid	reading	(1 | participant)
	vowels	(1 | participant)
VSA	reading	(1 | participant)
	vowels	(1 | participant)

direct visual comparison of task-related differences in peaks, valleys, and distributional tails. Means and standard deviations calculated from the observed data are presented in Appendix C.

As seen in Figure 6.8, workload effects on distributional shape varied by vowel and formant, with the clearest changes observed for F1 in /i/ and /u/. For F1/i/, the distribution becomes increasingly bimodal with exercise in both speech tasks, suggesting a tendency for observations to cluster into two bands. In contrast, F2/i/ shows relatively little effect of workload. Distributions remain compact and largely symmetrical across conditions, although subtle task-specific changes are visible: the reading distributions become slightly flatter with increasing workload, whereas the sustained-vowel distributions show greater concentration around the mid-range.

For F1/a/, reading data exhibit relatively symmetrical distributions that become increasingly concentrated around the center of the measurement range as workload increases, indicating that extreme values become less common. Although the sustained-vowel distributions are less symmetrical overall, a similar tendency

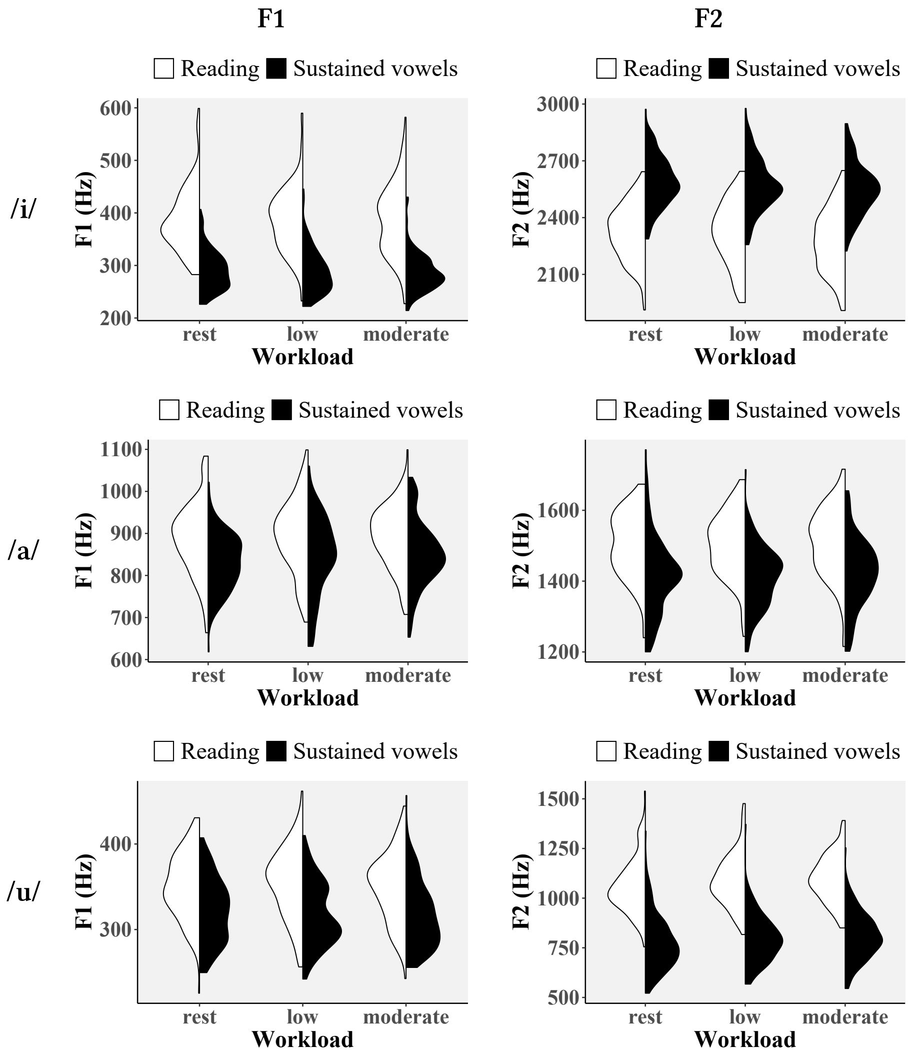

Figure 6.8: Half-violin plots showing distributions for F1 and F2 by workload and task for /i/, /a/, and /u/; white halves show reading, black halves show sustained vowels.

is visible. Effects for F2/a/ differ by task. Reading shows little systematic influence of workload, whereas sustained-vowel distributions become increasingly compact and symmetrical during exercise. This pattern suggests greater convergence across speakers, accompanied by a reduction in high F2 values.

For F1/u/, reading distributions widen and become more symmetrical with exercise, indicating a greater spread of observations across the measurement range. Sustained vowels, by contrast, display less symmetrical shapes, including a clear bimodality at low workload and a skew toward lower values at moderate workload. Together, these patterns suggest greater heterogeneity in the production of /u/ during exercise. Finally, F2/u/ shows compact, symmetrical distributions with minimal workload effects for both tasks, although upper outliers become less prominent as exercise intensity increases.

Across vowels, the clearest task-related effect is a systematic difference in the range and central tendency of formant values for reading versus sustained vowels, particularly for the high vowels. Similar patterns have been reported in reference data for German (Pätzold & Simpson 1997, Tolkmitt & Scherer 1986) and likely reflect a combination of phonetic context effects in reading – where surrounding articulatory targets impact vowel realization – and the greater articulatory freedom afforded by sustained vowels, which allow speakers more time to reach and maintain peripheral targets.

6.3.2 Inferential statistics: Formant measures

The statistical analyses were conducted separately for each vowel and formant in both speech tasks. Results are presented first for F1, with findings for /a/, /i/, and /u/ followed by the F1 centroid measure. Results for F2 are presented in the same order. Key results are summarized in the text, and pairwise comparisons are presented in Appendix C.

Speech during exercise was generally predicted to show higher F1 values based on the expectation that speakers would have greater mouth aperture to accommodate increased airflow. Figure 6.9 plots the model-predicted means for F1/a/ in each condition. In both speech tasks, F1/a/ tended to increase after exercise commenced but showed little further change with increasing exercise intensity. Pairwise comparisons revealed a significant difference between workload conditions only for the sustained-vowel task: F1/a/ increased significantly from rest to moderate workload ($p = .029$). No significant changes were found in the reading task, although confidence intervals were wide for both tasks.

F1/i/ (Figure 6.10) showed no clear effect of exercise in either speech task – pairwise comparisons did not show any significant differences between workload conditions. However, the wide confidence intervals indicate considerable uncertainty in the estimated means.

Similarly, no consistent increase was observed for F1/u/ (Figure 6.11). Instead, the plots suggest different trends for each speech task. In the reading task, F1/u/

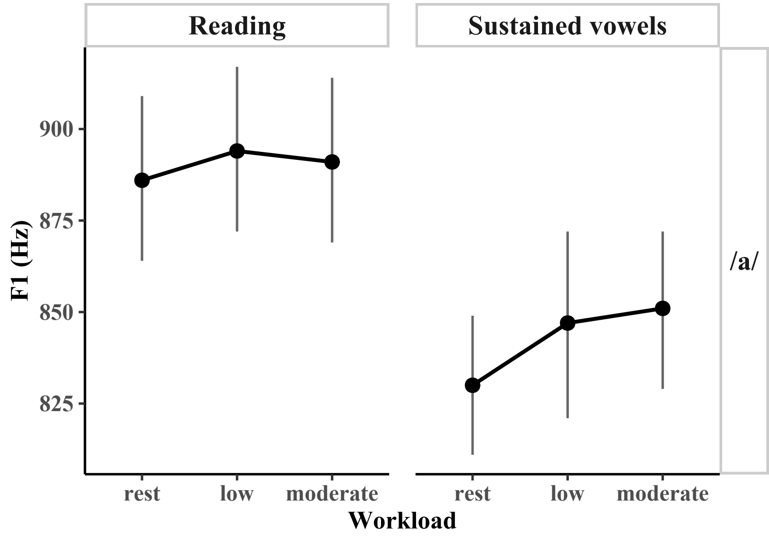

Figure 6.9: Estimated marginal means for F1/a/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

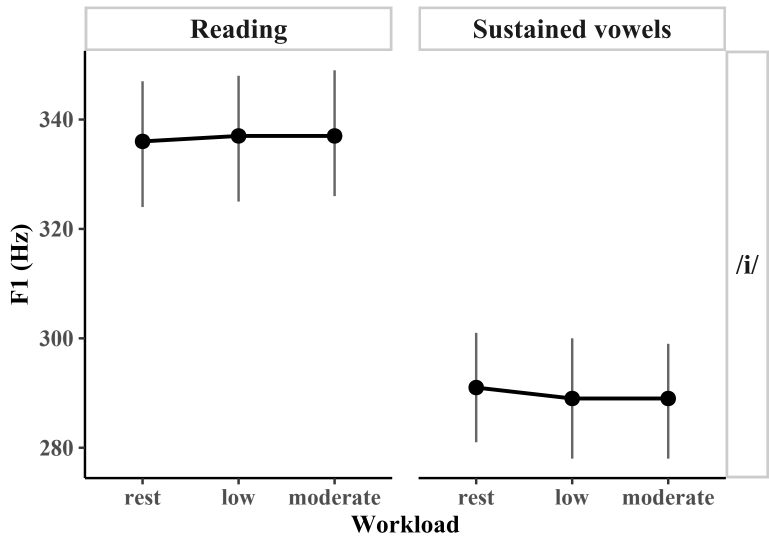

Figure 6.10: Estimated marginal means for F1/i/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

initially increased at low workload before returning to near-rest levels at moderate workload. In the sustained-vowel task, F1/u/ decreased somewhat at low workload and remained relatively stable as workload increased. However, the predicted differences between workloads remained small relative to the uncertainty surrounding the estimates: estimated marginal means differed by approximately 5 Hz, while confidence intervals spanned roughly 20 Hz.

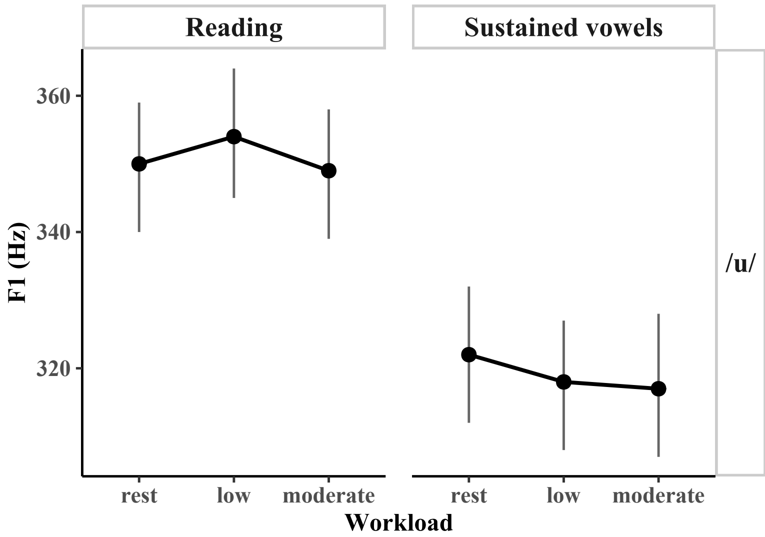

Figure 6.11: Estimated marginal means for F1/u/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

The F1 centroid revealed no clear effect of exercise (Figure 6.12), consistent with the largely non-significant F1 results for the individual vowels. While the estimated means trended upward with exercise intensity for both speech tasks, pairwise comparisons did not show any significant changes between conditions.

A change in F2/a/ was anticipated based on the mixed results reported in earlier studies (Godin & Hansen 2011b, Marquard et al. 2017). However, pairwise comparisons for both speech tasks in the current study found no significant differences between workload conditions. Although the plot (Figure 6.13) suggests a slight tendency toward higher F2/a/ values during exercise, particularly at moderate workload, the observed differences between the model-predicted means were minimal – less than 10 Hz in both speech tasks. Furthermore, the wide confidence intervals indicate low precision in the estimated effects of physical activity.

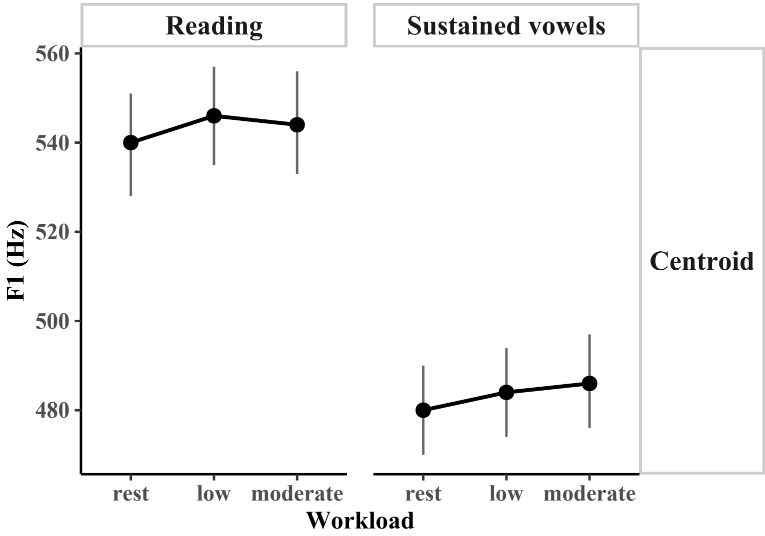

Figure 6.12: Estimated marginal means for F1 centroid values across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

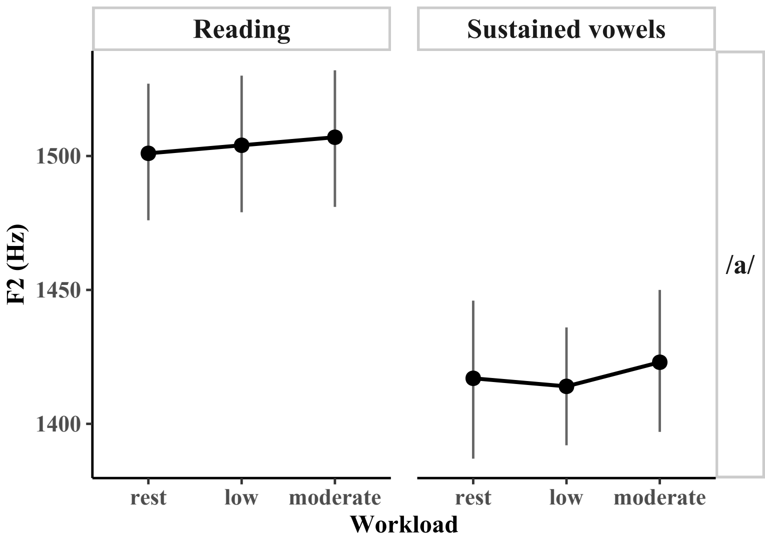

Figure 6.13: Estimated marginal means for F2/a/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

Previous studies on speech during exercise provided limited evidence for predictions regarding F2/i/. The estimated marginal means (Figure 6.14) showed a general downward trend for F2 with increasing exercise intensity. In the reading task, pairwise comparisons revealed significant decreases from rest to moderate workload ($p < .001$) and from low to moderate workload ($p = .001$). In the sustained-vowel task, significant decreases were observed for all workload comparisons: rest vs. low workload ($p = .015$), rest vs. moderate workload ($p < .001$), and low vs. moderate workload ($p = .002$). The estimated decrease between rest and moderate workload was 44 Hz, a change that would be compatible with a more posterior tongue position for /i/ during exercise.

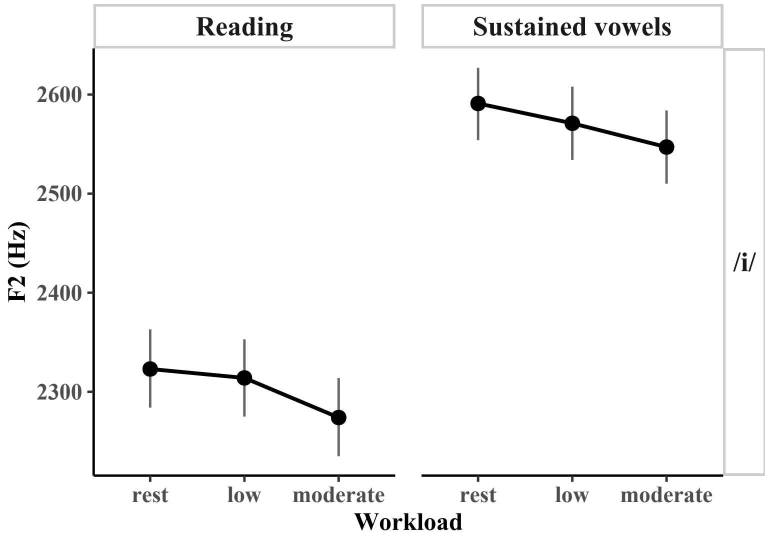

Figure 6.14: Estimated marginal means for F2/i/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

No prior studies of speech under exercise had reported on F2/u/. The plots show an upward trend with exercise across both speech tasks (Figure 6.15). In the reading task, pairwise comparisons revealed significant increases from rest to low workload ($p = .031$) and from rest to moderate workload ($p = .006$), corresponding to an increase of approximately 30 Hz from rest to moderate workload. No significant changes were found in the sustained-vowel task.

For the F2 centroid (Figure 6.16), a downward trend was observed with exercise for both read speech and sustained vowels, but no significant differences were

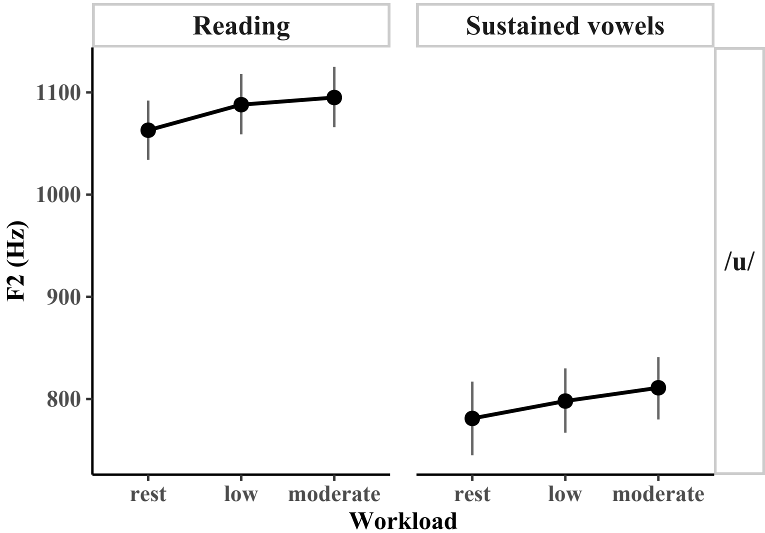

Figure 6.15: Estimated marginal means for F2/u/ across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

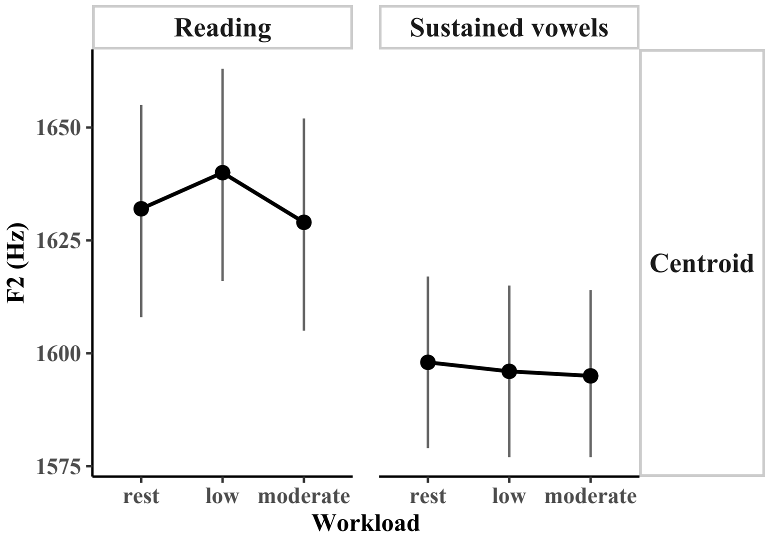

Figure 6.16: Estimated marginal means for F2 centroid values across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

identified. However, as a composite measure, the centroid reflects the combined behavior of the individual vowels and therefore largely mirrors the patterns observed in the separate vowel analyses. Consequently, the absence of significant effects in the centroid measure is consistent with the mixed findings observed for the individual F2 measures.

Tables 6.6 and 6.7 give the estimated marginal means for the formant-based measures. Pairwise comparison statistics are provided in Appendix C.

Table 6.6: Model statistics for individual formant measures (F1, F2) and centroids across workloads for read speech

Measure	Workload	EMM	SE	df	Lower CL	Upper CL
F1 /a/	rest	886	11.2	53.0	864	909
	low	894	11.2	53.1	872	917
	moderate	891	11.2	53.5	869	914
F2 /a/	rest	1501	12.7	52.1	1476	1527
	low	1504	12.7	52.1	1479	1530
	moderate	1507	12.7	52.5	1481	1532
F1 /i/	rest	336	5.67	87.9	324	347
	low	337	5.66	87.1	325	348
	moderate	337	5.68	88.5	326	349
F2 /i/	rest	2323	19.7	59.9	2284	2363
	low	2314	19.7	59.9	2275	2353
	moderate	2274	19.7	60.7	2235	2314
F1 /u/	rest	350	4.67	68.8	340	359
	low	354	4.66	68.6	345	364
	moderate	349	4.69	69.7	339	358
F2 /u/	rest	1063	14.7	72.9	1034	1092
	low	1088	14.7	72.9	1059	1118
	moderate	1095	14.8	73.9	1066	1125
F1 centroid	rest	540	5.65	57.4	528	551
	low	546	5.64	56.8	535	557
	moderate	544	5.67	58.1	533	556
F2 centroid	rest	1632	11.8	57.5	1608	1655
	low	1640	11.8	57.5	1616	1663
	moderate	1629	11.8	58.1	1605	1652

Table 6.7: Estimated marginal means (EMMs) for individual formant measures (F1, F2) and centroids across workloads for sustained vowels

Measure	Workload	EMM	SE	df	Lower CL	Upper CL
F1 /a/	rest	830	9.51	44.0	811	849
	low	847	12.74	44.0	821	872
	moderate	851	10.75	44.0	829	872
F2 /a/	rest	1417	14.70	44.0	1387	1446
	low	1414	11.10	44.0	1392	1436
	moderate	1423	13.10	44.0	1397	1450
F1 /i/	rest	291	4.94	45.0	281	301
	low	289	5.52	45.0	278	300
	moderate	289	5.18	45.0	278	299
F2 /i/	rest	2591	18.30	51.2	2554	2627
	low	2571	18.30	51.3	2534	2608
	moderate	2547	18.30	51.3	2510	2584
F1 /u/	rest	322	4.95	45.0	312	332
	low	318	4.80	45.0	308	327
	moderate	317	5.18	44.5	307	328
F2 /u/	rest	781	18.00	45.0	745	817
	low	798	15.60	45.0	767	830
	moderate	811	15.30	44.7	780	841
F1 centroid	rest	480	5.07	68.4	470	490
	low	484	5.07	68.4	474	494
	moderate	486	5.10	69.5	476	497
F2 centroid	rest	1598	9.32	62.0	1579	1617
	low	1596	9.32	62.0	1577	1615
	moderate	1595	9.36	62.8	1577	1614

6.3.3 Vowel space area

Vowel space area (VSA) was expected to contract during exercise based on previous findings for speech under cognitive and psychological stress. To quantify VSA, a convex hull – representing the smallest boundary enclosing all vowel observations – was calculated for each workload condition. While the results of the statistical analyses indicate a general decrease in VSA as workload increased, substantial differences were observed between the speech tasks (Figure 6.17). In the reading task, a mild downward trend is evident, with pairwise comparisons (Appendix C) showing a significant decrease in VSA from rest to moderate workload ($p = .024$). In the sustained-vowel task, VSA was markedly larger than in read speech at rest and low workload, and pairwise comparisons revealed a significant *increase* in VSA from rest to low workload ($p < .001$). This was followed by a sharp decrease at moderate workload ($p < .001$), resulting in VSA values similar to those observed for read speech. Table 6.8 presents the estimated marginal means across speech tasks and workloads.

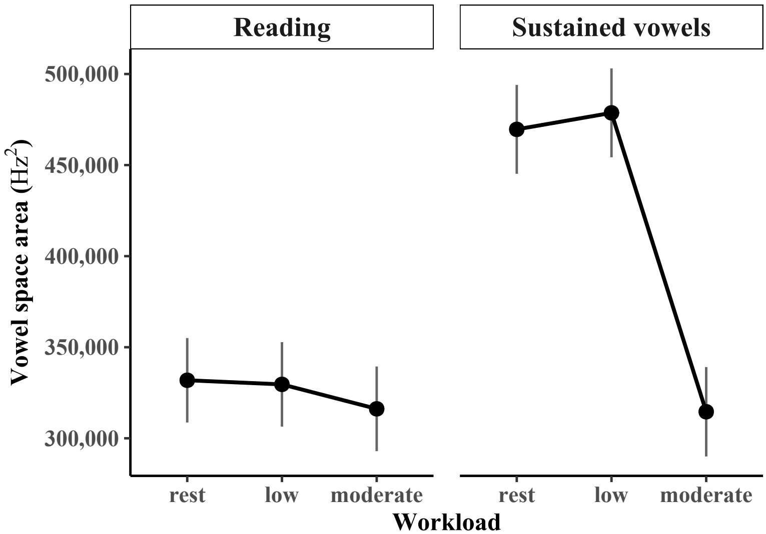

Figure 6.17: Estimated marginal means for vowel space area (Hz^2) across workload conditions in reading (*left*) and sustained vowels (*right*); error bars represent 95% confidence intervals.

Table 6.8: Model statistics for vowel space area (VSA) across speech tasks and workloads

Task	Workload	EMM	SE	df	Lower CL	Upper CL
Read passage	rest	331,857	11,568	55.6	308,679	355,035
	low	329,600	11,568	55.6	306,422	352,779
	moderate	316,166	11,601	56.1	292,927	339,405
Sustained vowels	rest	469,605	12,285	95.5	445,217	493,993
	low	478,662	12,285	95.5	454,274	503,050
	moderate	314,565	12,364	97.3	290,027	339,102

6.3.4 Individual differences by formant measure: Line plots

While previous work on speech during exercise has noted considerable speaker differences, few studies have provided detailed analyses of these variations. The current study aimed to explore the extent of between-speaker variation, addressing a knowledge gap highlighted by the mixed results in previous work and the wide confidence intervals observed in the current study. To explore individual results, this section presents two types of data visualizations: line plots, which display individual means across conditions compared to group means, and heatmaps, which show changes in all vowels and formants simultaneously, identifying groups of participants with similar responses to exercise. These visualizations address two main questions: Do speakers show opposite effects of physical workload on formant frequency? Do speakers exhibit different general tendencies (e.g., contraction vs. expansion of the vowel space) in response to exercise? The line plots will be presented first for each measure in turn, followed by the heatmaps.

The line plots display the experimental conditions on the x-axis and frequencies for a given formant (in Hz) on the y-axis. Individual speaker means for each condition are plotted in gray, while the group mean is plotted in bold black. In general, the plots look quite similar across measures, showing a collection of diverging lines around the mean. For this reason, only the first figure, for vowel /a/, is described in detail to illustrate different phenomena in the data. The descriptions of the plots for the vowels /i/ and /u/, the centroid, and vowel space area highlight selected phenomena and discuss the data with respect to the statistically significant results.

Figure 6.18 displays individual results for /a/. The group mean for F1 in the reading data shows minimal change across exercise intensities. However, individual patterns vary considerably. Some speakers align with the group mean – for example, two speakers with a mean of around 750 Hz at rest show relatively stable F1 across conditions. But other speakers show different tendencies, including opposite effects. For instance, at around 1050 Hz in the rest condition, a speaker shows an increase in F1 frequency at low workload and then a decrease at moderate workload – the "mountain" pattern that was also observed in the individual data in the respiratory and laryngeal studies (Chapters 4 and 5). At the bottom of the plot, the "valley" pattern is also visible, whereby speakers show an initial decrease from baseline with low workload and then a subsequent increase at moderate workload. Other speakers show steady decreases or increases with exercise. These diverse individual responses highlight the limitations of relying solely on group means to understand the effects of exercise.

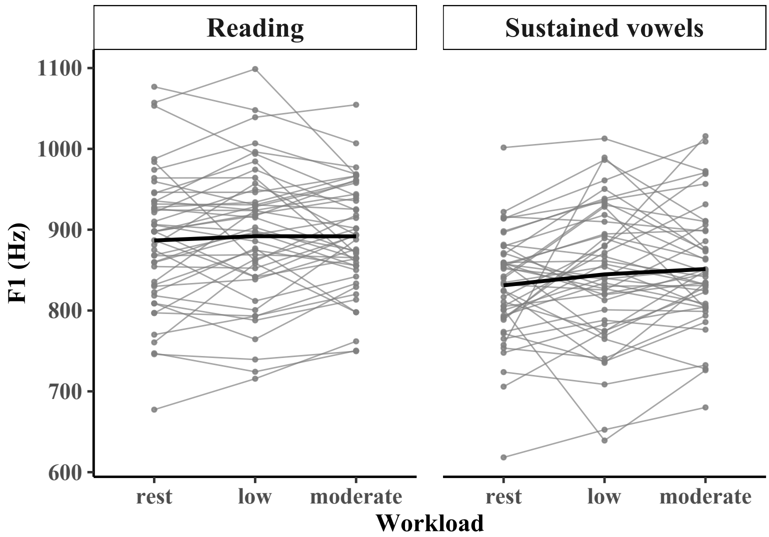

Figure 6.18: F1 values for /a/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

No obvious systematic differences between exercise effects on F1 and F2, or between read speech and sustained vowels, are apparent from the line plots. One exception was a significant increase in F1/a/ from rest to moderate in the sustained vowel data, though this was not consistent across all speakers – some speakers showed *decreases* with exercise (e.g., speaker at 1000 Hz). Such speaker differences likely contribute to the wide confidence intervals in the statistical analyses, and it is conceivable that future studies could obtain different results, depending in part on the random sampling of speakers. This variability could also account for some of the mixed results of previous studies. The substantial individual variation in speech responses to exercise observed here emphasizes the importance of considering speaker differences in future research.

Figure 6.20 presents the individual patterns for /i/, revealing a mix of responses similar to those observed for /a/. Notably, in the F1 plots, several speakers exhibit considerably lower or higher formant frequency values than the group average. For F2/i/, a significant decrease in formant frequency was observed in both speech tasks during exercise. In the reading task, some speakers showed steep decreases in formant frequency with low workload. However, these speakers tended to return toward their resting baselines at moderate exercise levels.

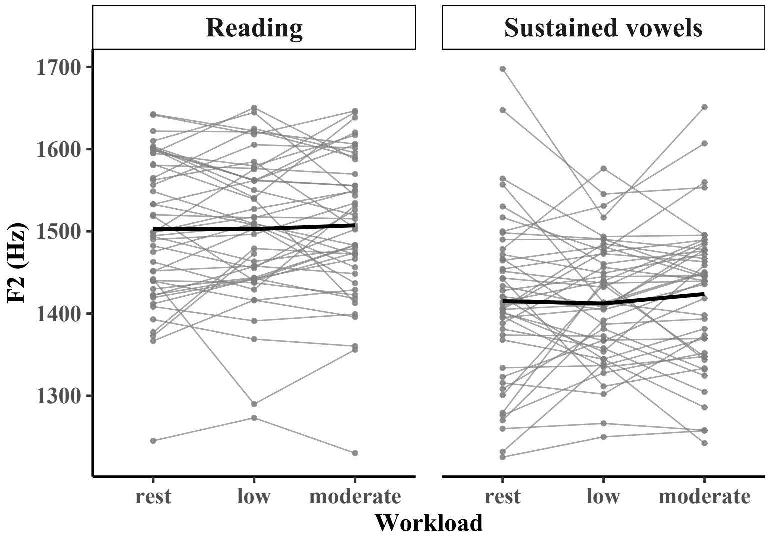

Figure 6.19: F2 values for /a/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

Other speakers showed marked increases in F2/i/ with low workload, followed by decreases as exercise intensity increased. The line plots suggest that light exercise may be associated with decreases or increases in F2/i/ (or little change), indicating that participants did not respond uniformly. At moderate workload, participants appeared more consistent, with most showing a decrease in F2.

The sustained vowel results for F2/i/ (Figure 6.21) showed significant decreases for all workload comparisons. Impressionistically, there appears to be less speaker variation in this task, with a more apparent downward trend across both exercise levels. The reasons for this apparent difference between speech tasks remain unclear. Coarticulatory effects could potentially influence responses in the reading task. Additionally, the F2 mean formant frequencies (and model-predicted means) were somewhat higher in the sustained vowel data, suggesting that /i/ was produced in a more fronted position.

For F1/u/, Figure 6.22 displays numerous mountain and valley patterns in both speech tasks, indicating opposite effects on formant frequency during light activity among speakers. This variation leads to diverse speaker responses when activity is increased to moderate intensity – an increase in exercise level will lead to a decrease in F1/u/ compared to low intensity for some speakers and an

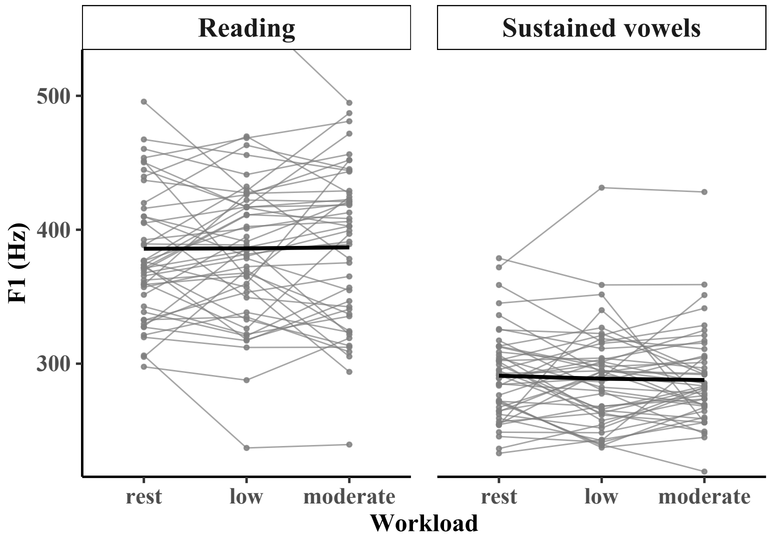

Figure 6.20: F1 values for /i/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black. *NOTE:* One participant exhibited higher values at rest (max = 600 Hz); the y-axis is truncated to improve readability.

increase for others. In addition to these patterns, some speakers showed apparently linear decreases or increases across conditions. Consequently, while individual plots show substantial changes in F1/u/ across workload conditions, any group-level effect may be obscured by the opposite directions of change observed across speakers.

F2/u/ (Figure 6.23) showed a significant increase in the reading task from rest to low workload and from low to moderate workload. This is consistent with a more fronted realization of /u/ relative to speech at rest and may reflect greater hypoarticulation. In the sustained vowels, some speakers showed rather sharp drops in F2 when exercise commenced, while others increased. Similarly, the transition from low to moderate workload was accompanied by an increase in F2 for many speakers but a decrease for others. One extreme speaker was noted, whose data were verified and retained. The reasons for this participant's divergence from others remain unclear, potentially due to a relatively fronted target or exaggerated lip-rounding at low workload. This participant's F2 observations align more closely with those in the read data. As with F1/u/, individual speakers show formant changes with exercise, but the variability is too great for the model to make precise estimates of the means.

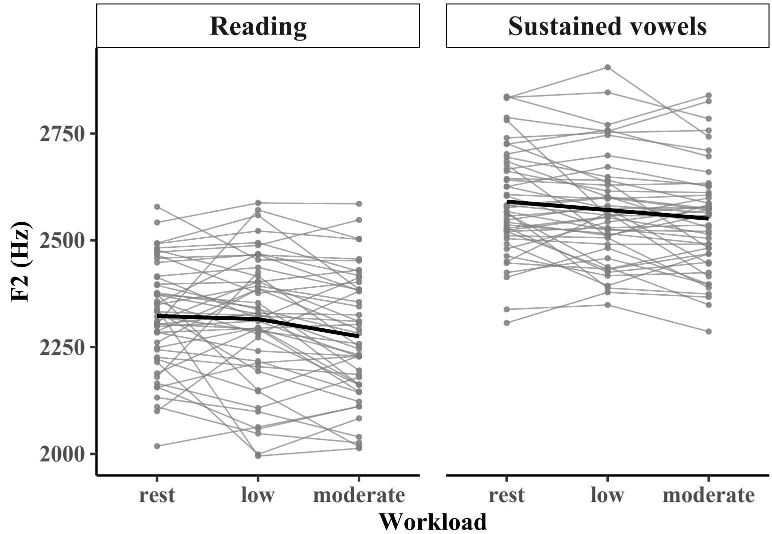

Figure 6.21: F2 values for /i/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

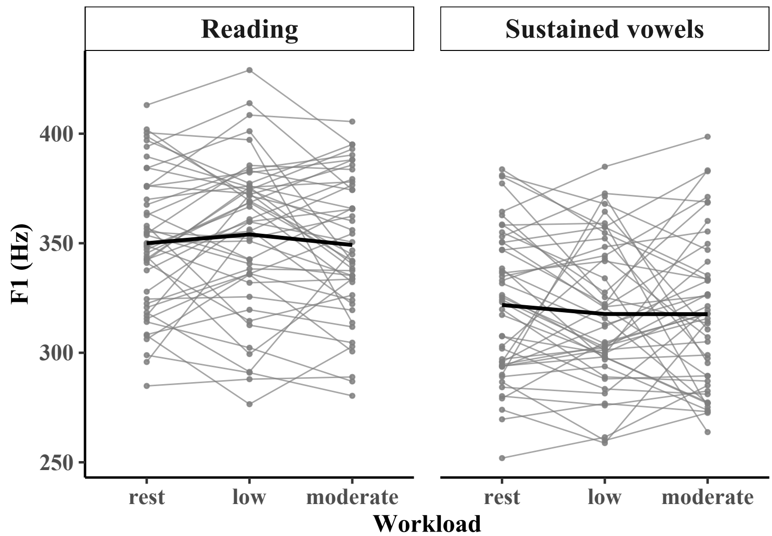

Figure 6.22: F1 values for /u/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

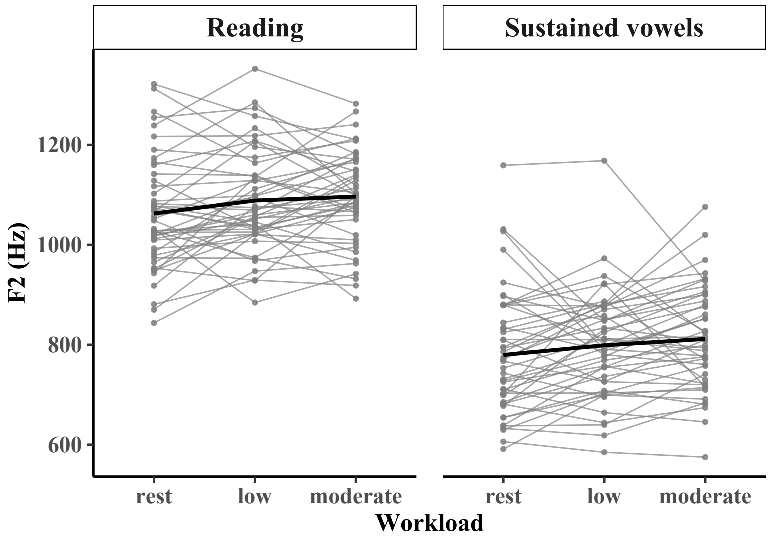

Figure 6.23: F2 values for /u/ in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

Figure 6.24 illustrates the individual results for the VSA measure, revealing a difference in speaker variability between speech tasks. In the sustained vowels, participants appeared more consistent in their response to exercise, showing a small uptick in total VSA at low workload followed by a sharp decrease at moderate workload. This pattern is consistent with an initial expansion of the vowel space followed by a larger contraction for many speakers, although some speakers showed a decrease in VSA even at low workload. In the reading task, participants displayed greater variability in their responses to exercise, with VSA increasing for some participants and decreasing for others at low workload. At moderate workload, some participants showed little apparent change from rest, while a general downward trend characterized the change from low to moderate workload for many speakers. These patterns help to contextualize the statistical analysis, which found a significant difference only between rest and moderate workload in the reading task.

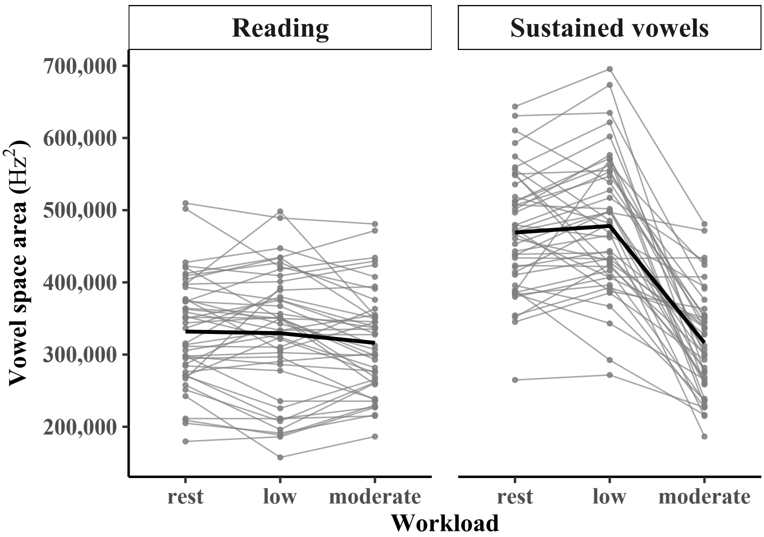

Figure 6.24: Vowel space area (Hz^2) in read speech (*left*) and sustained vowels (*right*); individual participant means (gray) are plotted to show variability, with the group mean overlaid in black.

6.3.5 Speaker groups across formant measures: Heatmaps

Heatmaps displaying changes across measures in read speech were generated to reveal shared response patterns across speakers and/or measures.[11] Figure 6.25 displays the changes between rest and low workload in read speech. The color of the cells represents changes in formant frequency (in Hz): red hues indicate increases and blue hues indicate decreases. Visually, the heatmap is dominated by scattered dark cells, indicating large changes for isolated speakers and measures. However, these speakers do not exhibit consistently large responses across measures, so there is no evident pattern of speaker-specific articulatory responses to exercise. The hierarchical clustering is largely driven by the magnitude of responses for a handful of individuals. Although the remaining participants cluster into two roughly equal groups, neither displays a clear or readily interpretable pattern of responses. Indeed, most cells show only negligible to modest increases, suggesting that exercise has little overall effect on formant-frequency changes.

[11]Guidance on interpreting heatmaps is given in Section 4.3.4.

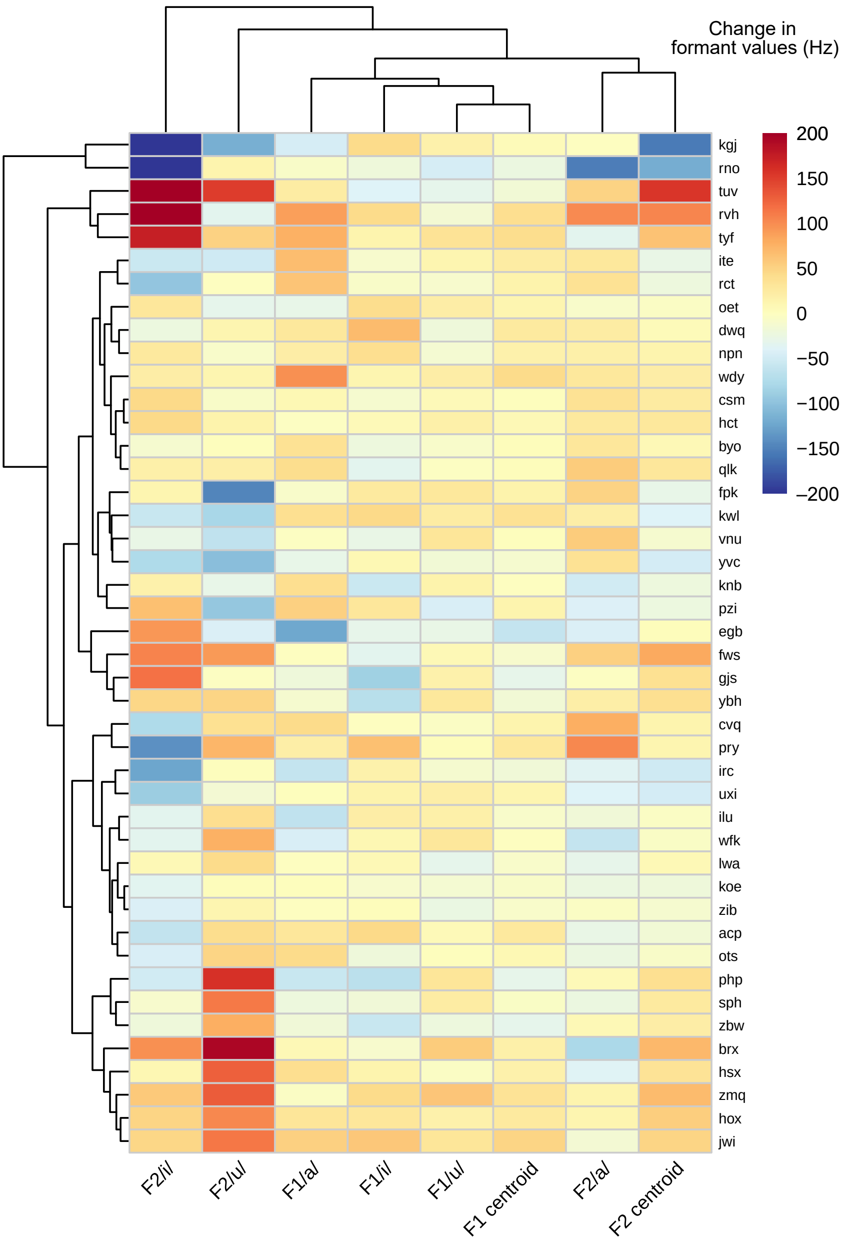

Figure 6.25: Heatmap showing change in formant frequency (Hz) from rest to low workload for all formant measures; hierarchical clustering is shown by the dendrograms.

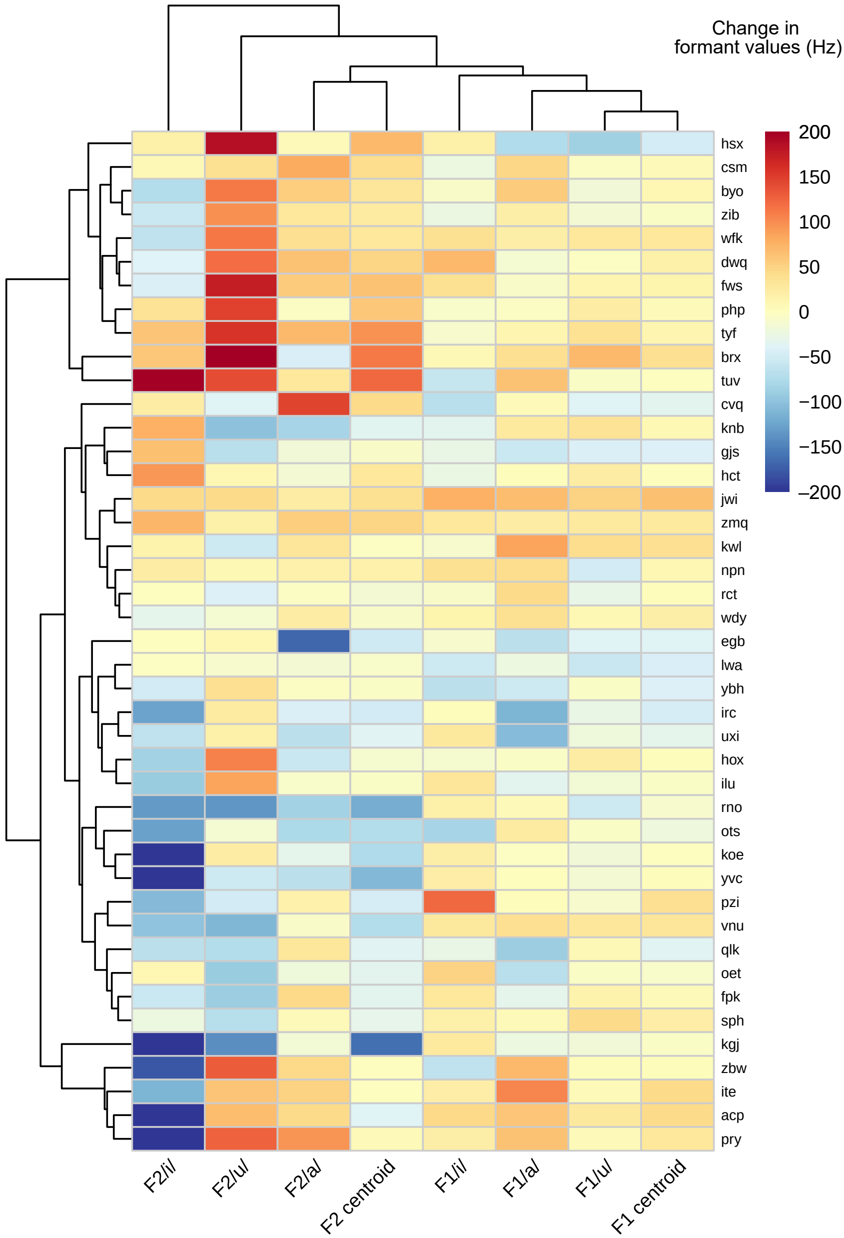

Figure 6.26: Heatmap showing change in formant frequency (Hz) from rest to moderate workload for all formant measures; hierarchical clustering is shown by the dendrograms.

The measures themselves also do not cluster into clear groups. If, as predicted, exercise systematically affected the front–back or high–low dimensions of vowel production, F1 or F2 measures would be expected to cluster together. Instead, the column dendrogram mixes these measures, providing no evidence of a single underlying articulatory adjustment. There is likewise no consistent direction of effect across measures: every column contains both increases and decreases, indicating that exercise does not induce a uniform directional change in formant values across speakers. Although the largest changes are observed for the high vowels, speakers adapt in both directions, so no clear effect of exercise emerges.

At moderate workload (Figure 6.26), no coherent multivariate response patterns are evident across speakers or measures. Although changes appear more pronounced, they remain largely confined to F2/i/ and F2/u/. F2/i/ shows a stronger overall tendency toward decreases, whereas F2/u/ exhibits mixed responses. The hierarchical clustering identifies a group of speakers, but this grouping appears to be driven largely by high F2/u/ values. Another group of speakers shows marked decreases in F2/i/, but, as at low workload, no systematic response patterns are evident across measures. Likewise, no clear clusters emerge for the measures themselves. Although some F1 and F2 measures cluster together at the lower levels of the dendrogram, this appears to reflect the extreme values for F2/i/ rather than a common underlying pattern. Finally, every column contains both increases and decreases, again providing no indication of a uniform effect of exercise.

6.4 Discussion

This study investigated whether increasing levels of physical activity affected articulatory position, as inferred from multiple formant-based measures. The discussion addresses systematic tendencies across articulatory dimensions (tongue height vs. tongue backness) as well as two combined measures, the centroid and vowel space area (VSA). Because vowel identity and speech task influenced the results, these factors are considered alongside group patterns. Speaker-related variability is also addressed to contextualize the statistical findings. The section concludes with methodological considerations and recommendations for future work.

The group results for F1, indexing tongue height, showed little change across vowels and tasks. Only moderate workload for /a/ reached significance, and even then the model-predicted increase was just 20 Hz (2.5%). Despite this lack of clear group-level effect, some speakers showed substantially larger changes (up to 100

Hz) in both directions. The question of tongue-height effects therefore remains rather open, although the individual data do not provide strong evidence for widespread effects.

The F2 results showed clearer patterns of anterior–posterior changes in tongue position for the high vowels. For /i/, F2 decreased with workload in both sustained vowels and read speech, suggesting a relatively robust exercise effect. In contrast, the increase for /u/ appeared only in read speech and may reflect its phonetic context rather than an exercise effect on the vowel itself. While the shifts in both high vowels could suggest a centralizing tendency, with /i/ moving back and /u/ moving forward, the heatmaps again reveal substantial speaker variability, with even greater differences in effect magnitude than for F1.

The F1 and F2 centroids were included to capture global shifts in vowel articulation, but neither measure showed a systematic effect of exercise. Given the modest group-level changes observed for individual vowels and the high degree of speaker variability, this outcome is unsurprising: aggregating inconsistent vowel-specific effects can dilute real differences rather than reveal them. The null centroid results therefore do not contradict the significant F2/i/ findings but rather suggest that this composite metric is not well suited to the present context. Because centroid measures have not, to the author's knowledge, been used to assess speech during exercise, the present findings offer a useful reference for future work. The only related study the author is aware of is Tolkmitt & Scherer (1986), who used a combined F1–F2 index in examining cognitive and emotional stress. Although they reported significant articulatory changes, their analyses were conducted within personality-based subgroups, reducing speaker variability. While more targeted applications of such metrics may prove informative, the heatmaps in the present study suggest that speaker differences are more clearly revealed by individual vowel formants than by centroids. Overall, these findings suggest that analyses focused on single vowels are likely to be more informative for detecting articulatory changes during exercise.

Like the centroids, vowel space area (VSA) was intended to capture broad articulatory changes. VSA decreased at moderate workload, with a particularly sharp drop observed in sustained vowels. This effect's magnitude stems from considerable task differences at baseline: without coarticulation, speakers can achieve more extreme vowel targets. Task-specific formant values have been previously observed in reference values for female German speakers. For example, Pätzold & Simpson (1997) reported a mean F2/u/ of 1048 Hz in read speech, while Tolkmitt & Scherer (1986) reported 672 Hz for sustained vowels.

While these baseline differences made the exercise effect more visible in sustained vowels, both tasks showed a decrease in VSA at moderate workload, suggesting an exercise-related shift. The results from the respiratory study (Chapter 4) confirmed substantially increased breath volume at this level, making it possible that speakers adopt a more biomechanically and aerodynamically economical tongue posture to accommodate higher airflow. Increased laryngeal tension at moderate workload, suggested by the phonatory study in Chapter 5, may also contribute. Because tongue and laryngeal muscles are often co-activated (Fregosi & Ludlow 2014), laryngeal adjustments made to support phonation could affect muscles involved in articulation, such as the genioglossus. This could manifest either as increased stiffness, resulting in target undershoot (VSA contraction), or greater force, leading to target overshoot. This account is consistent with the overall pattern observed here and may also help explain speaker differences.

Between-speaker differences were observed across all measures, as reflected in wide confidence intervals, opposing speaker trajectories in the line plots, and the heatmaps. However, there was little evidence of clear multivariate patterns. Indeed, in the heatmaps, the absence of robust response patterns is itself a result. This suggests that, in the present data, articulatory modifications may have been more speaker-specific than related to workload level, articulatory dimension, or vowel category. At the same time, the high vowels (/i/ and /u/) were generally the most sensitive to exercise, even though speakers adapted in different ways. Hierarchical clustering showed that the largest shifts across speakers occurred for F2/i/ and F2/u/. This pattern may plausibly reflect the greater precision required for high vowels: /i/ and /u/ involve tight palatal and velar constrictions that are more biomechanically constrained than the jaw-dominated articulation of /a/. If fine tongue-body control becomes more difficult during exercise, high vowels may therefore be more susceptible to change than low vowels.

The observed speaker heterogeneity may explain mixed findings in the literature. The group decrease in F2/i/ aligns with Huttunen et al. (2011) but contrasts with Yap et al. (2010), who reported an increase, and Godin & Hansen (2011b), who did not specify direction. The present group increase in F2/u/ also diverges from previous reports of decreases under stress (Ruiz et al. 1996, Yap et al. 2010). However, in smaller samples, results would depend heavily on which individuals were included. If, as demonstrated in the present study, speakers vary in direction, not just magnitude, then apparently contradictory results across studies may partly reflect who happened to be in the sample. At the same time, work on speech under stress has also demonstrated that non-linguistic factors can influence articulation. Tolkmitt & Scherer (1986)'s study on speech under cognitive stress revealed that different strategies for coping with anxiety correlated with either expansion or contraction of the vowel space. Their findings parallel some of the speaker differences observed in the current study's heatmap analysis. In particular, some speakers showed an apparent contraction of the front-back dimension for high vowels (higher F2/u/ and lower F2/i/), whereas others showed the opposite.

The theoretical takeaway from these findings is that exercise does not give rise to a single, uniform articulatory pattern but instead elicits diverse, speaker-specific changes. The results provide no evidence for the kind of hyperarticulation predicted from studies of loud or clear speech. Although speech during exercise shares characteristics with these speech modes, production conditions and speaker goals are fundamentally different: loud and clear speech involve *deliberately* altering subglottal pressure and articulatory targets for communicative goals on an utterance-by-utterance basis, whereas exercise continuously elevates subglottal pressure, requiring sustained vocal tract adjustments to allow phonation. Therefore, when articulatory alterations *do* occur during exercise, it is likely in response to other physiological adjustments at earlier speech production levels. For example, greater respiratory drive may prompt speakers to shift tongue posture to maintain airflow, while different levels of laryngeal tension could lead to co-activation effects in the tongue root, altering the precision of vowel targets in various ways. In this view, articulatory change reflects integration with respiratory–laryngeal adjustments, with high vowels acting as "pressure points" where disruptions to fine tongue-body control are most likely to surface.

When interpreting the magnitude of the observed effects, it is important to consider the limits of formant estimation. Previous studies of speech during exercise have reported modest changes (e.g., 20 Hz; Godin & Hansen 2011b). In the present study, the largest group-level effect occurred in F2/i/, with a model-predicted change of roughly 50 Hz from rest to moderate workload. Although this value is close to the roughly 50 Hz uncertainty margin associated with LPC-based formant estimation (Skarnitzl et al. 2015, Whalen et al. 2022), this margin of error applies to individual token estimates, not to differences between modelled means. Random tracking error is expected to average out across tokens, so a model-estimated difference of this magnitude is not inherently questionable, but it does warrant cautious interpretation. For context, shouted speech can show F2 increases exceeding 100 Hz (Garnier et al. 2006), and speech during real-life emergencies has shown changes of 100–200 Hz (Benson 1995). In the current dataset, the heatmap analysis showed speaker-specific shifts of similar magnitude, indicating that while group-level effects were modest, some individuals exhibited substantial articulatory changes during exercise.

Some limitations of the measures must also be considered. Although vowel formants reflect supralaryngeal articulatory configuration, they can be influenced by laryngeal adjustments as well. Consequently, the formant-frequency changes observed here may reflect a combination of tongue, jaw, and laryngeal modifications. Direct observation techniques would be required to disentangle these contributions. Recent technological developments make such approaches increasingly feasible in physically demanding contexts, including the MARRYS helmet for tracking vertical jaw displacement (Svensson Lundmark et al. 2023) and mobile endoscopic systems capable of capturing laryngeal behavior in strenuous exercise contexts such as martial arts (Naito & Niimi 2000). Further acoustic work could also examine dynamic measures, given the perceptual importance of formant dynamics in running speech (Kent & Vorperian 2018, McDougall 2006). Additionally, examining coarticulation rather than isolated vowel targets could yield valuable insights, particularly since some results in this analysis were specific to the reading task. It is plausible that coarticulatory effects are heightened during exercise, potentially impacting intelligibility and thus having real-world implications.

This study set out to determine how exercise affects articulatory position for the corner vowels /i, a, u/. The group results suggested that tongue height/jaw position remained largely stable, while the high vowels /i/ and /u/ were consistent with a centralizing pattern that was also reflected in the contraction of vowel space area. Although these trends point to a more centralized tongue position during exercise, substantial individual differences in effect direction and magnitude indicate that speakers make a range of articulatory adjustments. The findings thus suggest a theoretical view in which exercise-induced changes do not elicit a single, articulatory pattern but rather arise from interactions with speaker-specific respiratory and laryngeal adjustments. Methodologically, the work underscores the importance of multiple measures and speaker-level analysis for understanding speech during exercise. Overall, the study illustrates how varying physiological demands can shape speech production in diverse ways, offering new insight into the flexibility and constraints of the speech production system.

7 General discussion

Much of human communication occurs while people are moving, exerting themselves, or otherwise not at rest. Yet speech production research typically relies on data from seated speakers in recording booths, meaning that much of what we know about speech comes from a single, constrained context, leaving open fundamental questions about the system's range and adaptability. Because speech depends on the respiratory and motor systems, changes in physiological state – such as altered breathing, muscle activation, or metabolic demand – should influence how it is produced, though little is known about how or when these effects emerge. Speech is also characterized by substantial inter- and intra-speaker variability, which points to an inherent capacity for adjustment. Exercise provides a controlled physiological perturbation that reveals how speakers behave when the system is pushed away from rest: which aspects of production remain stable, which become constrained, and where speakers diverge. Studying speech during exercise therefore offers both a realistic communication setting and a way to probe the flexibility of the speech production system.

This study adopted a multi-level, data-driven approach to investigating speech production during physical activity, incorporating several methodological innovations to address its complexity. First, it compared moderate workload with low workload, capturing commonplace levels of physical activity that have received little attention in previous research. Second, it examined two distinct speech tasks – reading and sustained vowels – to determine how speech material shapes exercise effects, including the first systematic analysis of respiratory data for sustained vowels during exercise. Third, by integrating respiratory, phonatory, and articulatory measures, the design enabled cross-level comparisons that were rarely available in earlier work. In addition, the analysis treated individual differences as central rather than peripheral: speaker-level patterns were examined at each speech production level, revealing patterns of variability. Qualitative interview data further enriched these findings by providing direct insight into speakers' intentions and experiences. Together, these features allowed for a richer and more comprehensive dataset than earlier studies, yielding both generalizations about how the speech production system adapts during physical activity and insights into sources of individual variability.

This chapter synthesizes the findings across the respiratory, phonatory, and articulatory studies to identify the general patterns that characterize speech during physical activity. It integrates results across workload levels and speech tasks to determine which aspects of the system show consistent change and how these patterns relate to the underlying physiology. At the same time, the synthesis examines how these general trends coexist with marked individual differences, drawing on both quantitative results and interview data that highlight the multiple physical, emotional, and cognitive factors shaping speakers' responses. Building on this analysis, the chapter introduces a conceptual schematization to account for the observed variability. Finally, the chapter reflects on broader implications for understanding speech in altered bodily states, considers methodological limitations, and outlines directions for future research.

7.1 Summary of findings

The onset of physical activity produced more extensive changes than expected: significant adjustments emerged across all levels of speech production even in the low workload condition, which was functionally equivalent to brisk walking. At the respiratory level, participants immediately increased breath volume, reflecting the non-negotiable physiological requirement to bring in more oxygen and expel more CO_2. These volume-based adjustments were the most consistent patterns in the dataset, although speakers differed in how they distributed this volume over time – some took *deeper* breaths, while others took *more frequent* breaths. At the group level, however, expiratory duration remained largely unchanged at low workload, consistent with interview reports that speakers tried to preserve resting-level linguistic phrasing. At the phonatory level, acoustic energy, f0, and SoE all increased at low workload, reflecting a common physiological response: greater driving pressure prompted stiffer vocal fold behavior and more abrupt closure. High vowels, with their tighter vocal-tract constrictions, also showed the earliest signs of phonatory strain: reduced periodicity (HNR decreases) and shifts in spectral tilt in sustained vowels (H1*–H2*, H1*–A1*, H1*–A3*) emerged specifically for /i/ and /u/. These patterns suggest that high vowels function as "pressure points" when respiratory drive increases, prompting greater laryngeal tension to counteract subglottal pressure.

In contrast, articulatory patterns showed little convergence at low workload. However, while group means suggested minimal change – F1 remained stable, centroid measures did not shift, and VSA did not contract – these null effects masked variability across speakers. The heatmaps showed clear divergence between individuals, with some fronting or backing high vowels by meaningful amounts. Thus, while mild exertion elicited broadly aligned adjustments in respiratory depth and phonatory settings, articulatory responses remained heterogeneous. Low workload therefore reveals not a uniformly stable articulatory system, but one in which speakers diverge along different paths even before stronger physiological pressures emerge.

As physical activity increased to moderate intensity, effects that were subtle or inconsistent at low workload became stronger and more widespread. At the respiratory level, speakers not only continued to increase breath volume but now showed clear breakdowns in timing: rising respiratory drive accelerated breathing rate and shortened expiratory durations, reducing the time available for speech. Interview comments corroborated this shift, with speakers reporting that they began shortening utterances, either by intentionally breaking the text into smaller phrases or by running out of breath mid-utterance. Moderate workload thus marks the point at which early compensatory adjustments no longer suffice and speech breathing begins to reorganize. Phonatory adjustments also intensified, with further rises in f0, acoustic energy, and SoE. Group-level changes in spectral tilt and periodicity broadened from task- or vowel-specific effects into more system-wide patterns. High vowels /i/ and /u/ showed pronounced decreases in H1*–H2*, H1*–A1*, and H1*–A3*, reflecting greater laryngeal constriction in response to elevated subglottal pressure, while decreases in HNR and CPP indicated reduced periodicity. Articulatory behavior showed somewhat greater convergence than at low workload. Larger F2/i/ decreases pointed to more pronounced backing of /i/ across tasks, while F2/u/ fronting now emerged in reading. This apparent centralizing pattern was reinforced by vowel-space contraction in both tasks, suggesting that many speakers could no longer maintain peripheral articulations.

Comparison of the two speech tasks revealed aspects of speech breathing and its adaptation to exercise that would not have emerged from a single task. One key insight appeared already at baseline: sustained vowels showed considerably longer inspirations than reading despite similar breath volumes. Inspection of RIP traces showed initial tidal breathing, with the vowel incorporated mid-cycle, illustrating how speakers can alternate between tidal and speech breathing within a single breath. While this pattern was elicited by stimulus presentation, it demonstrates that short stretches of speech can be immediately incorporated into a tidal breath. This flexibility suggests that speakers can also produce brief, responsive utterances in everyday speech, such as backchannels, with minimal restructuring of the tidal breathing pattern. A second insight emerged during exercise, when sustained vowels elicited larger breath volumes than reading – an unexpected result given prior findings that breath volume is determined solely by workload. The key lies in the specific production conditions for sustained vowels: because they require continuous, unchecked airflow, they require greater initial volume to be sustained over several seconds when driving pressure increases. This demonstrates that certain types of speech do impose additional respiratory demands beyond those of exercise alone.

The sustained vowel task also clarified the interpretation of stable timing patterns observed in reading at low workload. If reading had been examined in isolation, the absence of timing changes might suggest minimal respiratory impact. However, the sustained vowel data show that respiratory rate already increased at low workload, showing that participants did respond to the increased drive to breathe. The maintenance of timing in reading therefore reflects active support of linguistic goals rather than the absence of respiratory drive. Overall, these task-specific findings demonstrate how speech breathing is tailored to utterance demands and flexibly adapts on a breath-by-breath basis. Consequently, exercise affects speech breathing differently according to an utterance's duration and breath support requirements.

Comparison of multiple vowels revealed vowel-specific effects, highlighting where the speech system becomes most vulnerable during exercise. Across the phonatory and articulatory measures, the clearest effects emerged for /i/ and /u/. This likely reflects vowel-specific production requirements that become more difficult to satisfy as breath volume increases. Specifically, these high vowels require a relatively constricted vocal tract, which amplifies the aerodynamic consequences of increased subglottal pressure and airflow, making it more difficult to maintain phonation using the speaker's baseline articulatory-phonatory configuration. In contrast, the low vowel /a/ is produced with a more open vocal tract, allowing airflow to be less constrained and potentially buffering the effects of heavier breathing. However, even though /i/ and /u/ were the most affected vowels, the heatmaps revealed substantial speaker differences in both effect direction and magnitude, particularly in articulatory measures. While some speakers showed centralizing tendencies, others exhibited forward shifts, and still others showed evidence of jaw lowering for /a/. Thus, although the group results indicate general vowel-specific phonatory and articulatory effects of exercise, the extent of individual variation points to a different perspective: vowels are better viewed not as acoustic objects that change during exercise, but as articulatory conditions that expose pressure on the speech production system under increased respiratory drive. And crucially, at these shared sites of vulnerability, speakers diverged in their responses.

The extent of individual differences is a central finding of the study. Despite a shared physiological manipulation imposed by exercise, speakers did not converge on a single speech response. Variability was evident across respiratory, phonatory, and articulatory levels for most measures, with individual line plots showing opposing trajectories for some speakers, particularly at low workload. While dominant patterns emerged at the group level, these patterns were at times absent or reversed in a substantial minority of participants (up to 40%). However, the multivariate heatmap visualizations suggested that this variability was not random. Instead, speakers frequently clustered into shared response patterns. At the respiratory level, for example, clusters revealed distinct volume–timing tradeoffs during exercise: some speakers took much deeper but less frequent breaths, others took moderately deeper but more frequent breaths, and still others made smaller changes across both dimensions. While such patterns were less sharply defined at the phonatory level, subsets of speakers showing parallel versus opposing responses were still evident (e.g., in HNR for /a/). At the articulatory level, fewer coherent clusters emerged, but opposite effects were observed across measures (e.g., fronting versus backing of /i/). The emergence of speaker groups across speech tasks and speech production levels suggests that variability is a characteristic feature of speech during exercise. Rather than reflecting random inter-speaker variability, these groupings suggest systematic modes of adaptation, with subsets of participants sharing preferred strategies for modifying speech production during exercise.

Such variability is not unique to the present study. Prior work on speech during physical activity and stress more broadly has reported substantial between-speaker differences at different levels of speech production. At the respiratory level, speakers adopt distinct breathing strategies during exercise, such as deeper but less frequent breaths versus a series of smaller "gulps" of air (Patil et al. 2010). At the phonatory level, individual participants within the same study have shown increases, no change, or decreases in key parameters during exercise (Godin & Hansen 2008, 2015) and under cognitive stress (Scherer et al. 2002). Articulatory studies likewise report that formant changes tend to follow individual rather than global trends, with effects driven by subsets of speakers (Sigmund 2012, Tolkmitt & Scherer 1986, Lively et al. 1993). These findings suggest that while significant group-level patterns may emerge, substantial individual variability is a recurring feature of speech under stress. Indeed, several authors explicitly caution that group means can obscure meaningful individual responses, with strong effects in some speakers coexisting with null or opposite effects in others (e.g., Scherer et al. 2002). Currently, the reasons for such variability are unclear, though potential factors are discussed in the next section. Despite this uncertainty, the results of the present study add further evidence of substantial individual differences and underscore the need for future studies of speech under stress to systematically examine speaker variability.

7.2 A branching account of speech adaptation during exercise

This commonly observed but unexplained variability calls for a concept that accounts for how a shared physiological stressor can give rise to divergent speech responses. Speech production involves multiple interacting subsystems, and exercise – and stress more generally – creates a body-wide perturbation rather than being limited to individual subsystems. Adaptation therefore unfolds across respiration, phonation, and articulation, with each adjustment reshaping the conditions under which subsequent adjustments occur. As a result, even a small number of early divergences – such as different ways speakers manage increased respiratory drive – place speakers in distinct production states from which further adaptations proceed. To illustrate this unfolding divergence, the present study proposes a branching schematization for speech adaptation during exercise, shown in Figure 7.1. This diagram shows early divergence at the respiratory level, depicted by three vertical bars representing distinct patterns of adaptation, with divergence increasing at the phonatory and articulatory levels. In this representation, all bars (adaptation patterns) appear to be equal. However, this is an abstract schematization for conceptual purposes. In the current study, the group-level results showed that some adaptive pathways appear to be more likely than others among the participants. For that reason, the diagram could also depict how such significant group-level effects can coexist with variability – for example, by using different line weights.

The core concept of the branching schematization is the progressive diversification of responses across levels of the speech production mechanism. This proposal emerged as a way to represent the groupings revealed by the hierarchical clustering in the heatmaps, which were clearest at the respiratory level and appeared increasingly heterogeneous at subsequent levels. However, the approach is also grounded in physiology. Given a certain physiological load, speakers must take in a certain amount of air. This already constrains the number of possible ways to modify breathing behavior. While speakers have some flexibility as to how often to take breaths, they are again constrained by utterance length and increasingly by the drive to breathe as exercise intensifies.

Another key proposal is that speakers are conscious of their breathing and make choices about how to control it. Such management of breath-taking is nat-

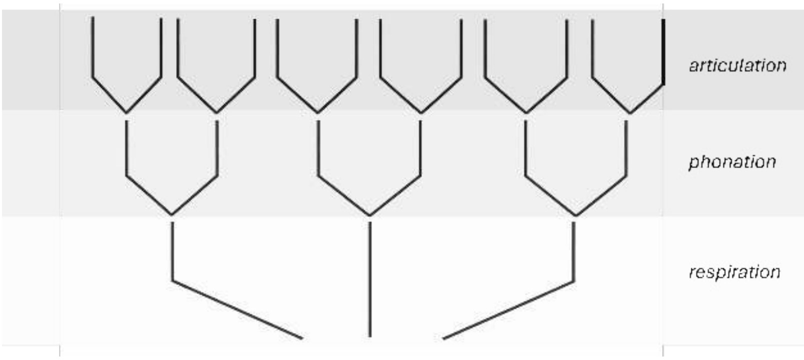

Figure 7.1: Schematic representation of how diverse speech outcomes may emerge across speakers during exercise. Each branch represents a distinct physical configuration at a given level of speech production. Speakers with similar configurations at one level may diverge at later levels, creating additional branches and ultimately yielding different speech profiles despite comparable physical workloads.

ural to speech – speakers must override tidal breathing to speak by actively controlling inspiration. But active control of breathing occurs frequently in many other contexts, such as sniffing, holding one's breath, or meditative breathing. Breathing is also salient in terms of proprioception. The feeling of an uncomfortably deep breath or of dyspnea is immediately perceptible to the individual. Consequently, a typical speaker is both sensitive to breath-related sensations and used to controlling breathing for different purposes. Together, this suggests that participants would have registered changes to their breathing during exercise and actively modified it to achieve their speech goals. Indeed, the qualitative data corroborate that speakers strategically adjusted their breathing, explicitly describing various changes such as taking deeper breaths to accommodate their utterances or breathing more often to avoid discomfort.

The presence of different breathing strategies opens the door to a greater number of possible outcomes at phonatory level for several reasons. First, two speakers that adopt the same respiratory strategy – say, fewer but deeper breaths – will experience the same consequence: increased breath volume, which likely raises subglottal pressure. However, the speakers need not respond in the same way. For example, one may choose to counteract the pressure by narrowing the posterior glottal gap, while the other may release some pressure by widening the gap. The complexity of laryngeal adjustments includes positioning but also differing amounts of force or tension, allowing for many different configurations of adjustments. After responding to the aerodynamic conditions in different ways, speakers now face different downstream constraints, which could then lead to disparate articulatory outcomes. In this way, each branch represents a response to a specific level of speech production but also reflects the state of the individual's body. This opens theoretical space for an account that includes both active adaptation – especially at respiratory level – and less conscious compensatory adjustments that may affect different aspects of speech production. The qualitative data in the present study did not indicate that speakers deliberately adjusted laryngeal settings in the same way as their respiratory behavior, but speakers did note difficulties maintaining intonation patterns. While it is not to be expected that a speaker would say "I increased medial compression," this does not mean that speakers were not targeting a specific vocal outcome. Speaker knowledge of phonatory/articulatory to acoustic mappings is likely more subconscious. Still, speakers were aware of changes in their speech and actively tried to reattain their baseline, which points toward an adaptive control in real time to changes arising from respiratory perturbation and other body-wide physiological responses to exercise.

While the branching schematization proposes an account of *how* divergent speech outcomes can emerge from a shared physiological perturbation, it does not explain *why* a given speaker follows one adaptive pathway rather than another. Several plausible sources of divergence are already well established in the literature. In the physical domain, breath patterns are known to be genetically predetermined (Benchetrit 2000) and also to vary widely in depth across speakers in both reading and spontaneous speech (Winkworth et al. 1995, 1994). Speakers are also known to have different habitual laryngeal settings (Laver 1980) and diverse articulatory strategies (e.g., Johnson et al. 1993). These differences are present in speech at rest and may persist under exercise, biasing speakers toward particular adaptive pathways or making some compensations more efficient than others when ventilation increases.

At the same time, other bodies of literature highlight cognitive, experiential, and intentional factors that can shape speech production under load. Studies of dual-task performance suggest that individuals differ in their ability to allocate attention when speaking while performing another task, with implications for speech production (Plummer-D'Amato et al. 2011, LeMonda et al. 2015). Subjective assessments of difficulty and vocal effort during exercise have also been shown to vary widely across speakers at the same workload (Baker et al. 2008, Rotstein et al. 2004, Sandage et al. 2013), and speaker-perceived effort is known to influence glottal source measures (McKenna & Stepp 2018). In addition, different affective states are known to give rise to distinct vocal profiles (e.g., Banse & Scherer 1996). Finally, speakers may have different vocal goals that give rise to varying adjustments. For example, speakers aiming to produce loud speech during exercise make additional laryngeal adjustments to those made for neutral speech during exercise (Ziegler et al. 2019).

Many of these factors were evident in the speaker reports from the present study,[1] which revealed markedly different experiences and adaptation strategies at identical workloads. In terms of physical speech production, participants varied widely in how difficult they perceived speaking during exercise to be, particularly at low workload and in the contrast between reading and sustained vowels. Crucially, speakers also described distinct approaches to breath planning as exercise intensity increased. Some consciously took deeper breaths and extended the expiratory phase to maintain baseline phrasing, while others spoke through discomfort until running out of breath. Still others allowed breathing demands to shape phrasing, strategically breaking up utterances at phrase boundaries to insert additional breaths. These reports align closely with the quantitative respiratory patterns in the heatmaps, where speakers exhibited systematic volume–timing tradeoffs and stable individual profiles across workloads. Together, these data provide evidence of active, speaker-specific decisions about breath management, providing empirical grounding for the first branching point in the proposed schematization.

Participants also described cognitive and emotional differences in how they experienced speaking during exercise. Some participants reported that needing to focus on breath planning left them with less attention for their reading performance. Others reported feeling that their intonation suffered, or described actively shifting attention between speaking, breathing, and cycling. Emotional responses likewise diverged, ranging from embarrassment about heavy breathing to feeling freer or more relaxed during exercise. Several speakers additionally described familiarization effects across the session, noting that they had figured out how to adjust their breathing in order to maintain their intended phrasing during exercise. Others reported getting used to the feeling of talking during exercise. While common themes emerge across participants in the analysis, the things each one reported as "affecting their speech" were often quite individual. Such different intentions and internal experiences may also influence which adaptations are made under the same physical workload. Thus, while the branching schematization proposes an account of how individual speakers can vary widely in the way they produce speech during exercise, the question of "who does what and why" remains open for future studies. New experiments may benefit from either a grouping approach that compares speakers with specific traits or in-depth

[1]Semi-structured questionnaires about perceived task difficulty and effects on speech were completed after the experiment (see Section 4.4).

case studies of individuals to explore links between physiological responses and psychological and otherwise internal factors.

7.3 Methodological considerations

The aim of this study was to investigate speech during exercise from a production perspective. While respiratory behavior could be assessed directly, laryngeal and articulatory changes were inferred from acoustic measures. These acoustic results provide rich evidence of speech adaptation under exercise, but it is important to bear in mind that individual acoustic measures do not map uniquely onto single physiological adjustments – similar acoustic outcomes can arise from different laryngeal or articulatory configurations. Accordingly, the interpretations offered here describe likely patterns of adaptation: while jointly considering results from multiple glottal source measures helps constrain plausible physiological accounts, the underlying physical adjustments may nevertheless vary across speakers. This reflects the inherent complexity of the speech production system rather than a limitation of the measures themselves, whose relationships to production are well established. Importantly, this interpretive stance aligns with the proposed branching schematization, which does not depend on identifying a single physical mechanism but instead accounts for how diverse adaptive pathways can emerge under shared physiological constraints.

The speech materials used in this study were selected to balance experimental control with ecological relevance, and their characteristics shape how the findings should be interpreted. The use of reading and sustained vowels is not a limitation per se, but it does constrain direct generalization to fully spontaneous dialogue, which remains an open question for future work. At the time of study design, controlled connected speech was necessary to ensure comparability across speakers and workloads, while sustained vowels were required to obtain reliable and interpretable acoustic measures of phonation. Unlike classic written texts used in previous studies, which typically consist of short, idealized sentences (e.g., the Rainbow Passage), the reading material in the present study was transcribed from spontaneous speech and intentionally contained longer sentences with lists and multiple clauses. Short sentences are easier to produce during exercise and tend to elicit highly uniform pausing behavior, leaving little room for individual variation. In contrast, the more demanding sentence structure used here required speakers to actively plan breath support and speech chunking, thereby revealing speaker-specific strategies that may have remained hidden in more constrained tasks. Likewise, the sustained vowel task elicited distinct breathing patterns that likely extend to other brief speech events that can be incorporated into an ongoing breath cycle. Together, the tasks show that exercise does not have a single effect on speech, but *different effects* depending on the type of speech being produced, thereby revealing the strategic flexibility of the speech production system. Future work should therefore consider the nature and duration of proposed speech materials.

7.4 Future directions

In keeping with the nested structure of this book – where level-specific findings are discussed in each chapter – ideas for future work have been detailed in the individual respiratory, phonatory, and articulatory studies. This chapter takes a broader, more theoretical perspective on future work, outlining two directions that both build on the findings of the current study and point toward greater theoretical implications for understanding speech production more broadly.

A key direction for future work is the investigation of spontaneous speech during physical activity, which is necessary to assess how the patterns observed here generalize beyond controlled tasks. Speech production requires anticipatory regulation of breath support: speakers gauge how much air is needed for different utterance types and tailor inspirations to the planned duration of the upcoming speech stretch (Huber 2008, Sperry & Klich 1992, Winkworth et al. 1995). When physical activity increases respiratory drive, this introduces constraints on breathing resources that shorten the amount of time available for speaking. An open question is then how such constraints affect speech planning. In the current study, speaker reports on unscripted speech indicate that respiratory constraints can shift planning priorities. Some speakers reported disruptions to content planning, such as reduced coherence across utterances or losing their train of thought after running out of breath, while others described opportunistically aligning thinking pauses with breathing. These observations point to speech planning as a general control problem in which communicative goals must be continuously mapped onto changing respiratory resources, relying not only on top-down intentions but also on bottom-up feedback about bodily state. Investigating spontaneous speech under increasing physical workload would therefore offer a valuable window into how speakers dynamically integrate linguistic planning with physiological constraints in real time.

A further direction for future research is the ability to become familiar with speaking under physical load. Both the interview data and potentially some quantitative patterns for individual participants in the present study suggest that some individuals adapted over the course of the experiment, reporting improved breath coordination or acclimatization to the task. Such observations raise the possibility that speech during exercise involves not only moment-to-moment compensation but learning over time. In everyday speech, the patterns observed at rest reflect long-established habits that speakers acquire early and rarely revisit consciously. Yet experience shows that speech production can become specialized: singers, actors, fitness instructors, and other professional voice users learn to manage breath support and vocal output under demanding physical conditions. The present findings suggest that a similar, if subtler, process may occur even over short timescales, as speakers learn how to consciously adjust their breathing "to speak as normally as possible," as one participant described it. From this perspective, adaptation to exercise may involve recalibrating mappings between available breathing resources, vocal fold behavior, articulation, and acoustic outcome. Future work using longitudinal or repeated-session designs could directly test this possibility by tracking how speech patterns evolve with sustained exposure to physical load. Such approaches would connect speech under exercise to broader work on motor learning and sensorimotor adaptation, positioning physical activity not only as a perturbation, but as a context in which new speech production modes can be learned.

7.5 Conclusion

This book set out to investigate how speech production adapts when the body shifts out of a resting state, using exercise as a controlled perturbation to investigate speech during different levels of physical activity. Across respiratory, phonatory, and articulatory levels, the results demonstrate that even mild exertion produces systematic changes in speech, while stronger workloads expose increasing constraints on breath timing, phonatory stability, and articulatory control. Speakers increased breath depth and shortened breath cycles; glottal source measures indicate a general increase in f0 and intensity, as well as increased tension in the intrinsic laryngeal muscles, suggesting a pressed voice quality; and centralizing articulatory patterns were observed for high vowels. At the same time, the data showed that speakers do not respond uniformly to exercise. Dominant group-level trends coexisted with substantial individual differences, including opposite response directions under identical physiological demands.

By integrating multi-level acoustic analyses with individual visualizations and qualitative reports, this study strongly suggests that variability in speech during exercise shows meaningful structure rather than being wholly random. Speakers use different strategies to manage increased respiratory drive, and those adjustments create distinct starting points that may give rise to different phonatory and articulatory adjustments. This pattern motivates a branching view of speech adaptation, in which speakers exposed to the same physical workload may arrive at different speech outcomes.

More broadly, the findings underscore that speech production cannot be fully understood as an activity confined to the resting body. Speech is produced by a system that must continuously negotiate competing physiological, cognitive, and communicative demands, and physical activity makes these negotiations visible. Studying speech during exercise therefore provides not only insight into a common yet understudied communicative context, but also a powerful lens for probing the flexibility and organization of the speech production system.

Appendix A: Chapter 4 supplement

Table A.1: Mean ± SD values for respiratory measures by speech task and workload level, calculated from the observed data

Measure	Task	Rest	Low workload	Moderate workload
Insp. duration (sec.)	reading	0.4 ± 0.2	0.5 ± 0.2	0.6 ± 0.2
	vowels	1.1 ± 0.6	1.0 ± 0.5	0.9 ± 0.4
Exp. duration (sec.)	reading	3.6 ± 1.9	3.5 ± 1.6	2.9 ± 1.2
	vowels	4.2 ± 1.3	4.1 ± 1.0	3.7 ± 0.9
Cycle duration (sec.)	reading	4.1 ± 2.0	4.1 ± 1.8	3.4 ± 1.3
	vowels	5.3 ± 1.5	5.1 ± 1.3	4.6 ± 1.0
Insp. fraction (%)	reading	0.13 ± 0.06	0.15 ± 0.06	0.18 ± 0.06
	vowels	0.21 ± 0.11	0.20 ± 0.08	0.20 ± 0.07
Resp. rate (breaths/min.)	reading	13.6 ± 4.0	13.9 ± 3.6	16.7 ± 4.0
	vowels	12.4 ± 2.8	14.1 ± 3.2	16.6 ± 3.4
Insp. amplitude (arbitrary units)	reading	0.12 ± 0.07	0.23 ± 0.11	0.34 ± 0.13
	vowels	0.13 ± 0.07	0.28 ± 0.13	0.38 ± 0.14
Exp. amplitude (arbitrary units)	reading	0.13 ± 0.08	0.22 ± 0.13	0.33 ± 0.13
	vowels	0.15 ± 0.07	0.30 ± 0.13	0.39 ± 0.14
% max. insp. displacement	reading	16.8 ± 10.0	30.4 ± 15.3	44.0 ± 17.9
	vowels	17.9 ± 11.1	38.3 ± 19.3	50.2 ± 19.9
% max. exp. displacement	reading	17.5 ± 11.3	29.5 ± 18.1	43.1 ± 19.0
	vowels	20.0 ± 10.4	39.8 ± 19.5	51.3 ± 20.8

Table A.2: Model statistics for temporal measures in read speech: Estimated marginal means, standard error, degrees of freedom and confidence level for all measures for normal and log models

Variable	Workload	Model Statistics				
		EMM	SE	df	Lower CL	Upper CL
Insp. duration (sec.)	rest	0.46	0.01	45.88	0.43	0.48
	low	0.54	0.02	45.98	0.51	0.57
	moderate	0.58	0.01	45.70	0.55	0.61
	log rest	0.43	0.01	45.93	0.41	0.45
	log low	0.51	0.01	45.98	0.48	0.54
	log moderate	0.55	0.01	45.70	0.52	0.58
Exp. duration (sec.)	rest	3.90	0.18	46.0	3.54	4.26
	low	3.74	0.15	45.99	3.44	4.04
	moderate	3.02	0.11	45.99	2.80	3.24
	log rest	3.36	0.14	45.97	3.10	3.65
	log low	3.35	0.13	45.97	3.10	3.62
	log moderate	2.78	0.10	45.71	2.59	2.98
Cycle duration (sec.)	rest	4.41	0.19	45.99	4.03	4.80
	low	4.32	0.16	45.99	3.99	4.64
	moderate	3.60	0.12	45.77	3.36	3.83
	log rest	3.88	0.15	45.97	3.59	4.20
	log low	3.93	0.14	45.98	3.65	4.23
	log moderate	3.37	0.11	45.74	3.16	3.59
Inspiratory fraction	rest	0.129	0.003	68.12	0.124	0.135
	low	0.144	0.003	67.52	0.138	0.149
	moderate	0.174	0.002	62.98	0.169	0.180
	log rest	0.117	0.002	64.86	0.113	0.122
	log low	0.133	0.003	64.36	0.128	0.138
	log moderate	0.165	0.003	60.55	0.158	0.171
Respiratory rate (breaths/min.)	rest	13.50	0.55	47.00	12.41	14.62
	low	13.89	0.50	47.00	12.89	14.90
	moderate	16.71	0.54	46.58	15.61	17.80

Table A.3: Model statistics for respiratory amplitude in read speech: Estimated marginal means, standard error, degrees of freedom and confidence level for all measures for normal and log models

Variable	Workload	Model Statistics				
		EMM	SE	df	Lower CL	Upper CL
Inspiratory amplitude (arbitrary units)	rest	0.13	0.01	45.92	0.12	0.14
	low	0.24	0.01	45.99	0.22	0.26
	moderate	0.34	0.01	45.75	0.31	0.36
	log rest	0.11	0.01	45.99	0.10	0.12
	log low	0.21	0.01	45.99	0.19	0.24
	log moderate	0.31	0.01	45.75	0.29	0.34
Expiratory amplitude (arbitrary units)	rest	0.13	0.01	45.84	0.12	0.14
	low	0.23	0.01	45.99	0.21	0.25
	moderate	0.33	0.01	45.75	0.30	0.35
	log rest	0.11	0.01	45.96	0.10	0.12
	log low	0.19	0.01	45.98	0.17	0.21
	log moderate	0.30	0.01	45.72	0.27	0.33

Table A.6: Pairwise workload contrasts for respiratory measures. Significant p-values ($p < .05$) are shown in bold.

Variable	Model	Contrast	Estimate	SE	df	t	p
Inspiratory duration (sec.)	Linear	rest–low	-0.09	0.01	45.4	-8.71	**< .001**
		rest–mod	-0.12	0.01	45.1	-10.59	**< .001**
		low–mod	-0.04	0.01	44.1	-3.93	**< .001**
	Log	rest–low	-0.08	0.01	45.9	-8.79	**< .001**
		rest–mod	-0.12	0.01	45.7	-11.19	**< .001**
		low–mod	-0.04	0.01	45.7	-4.76	**< .001**
Expiratory duration (sec.)	Linear	rest–low	0.16	0.11	45.8	1.46	.151
		rest–mod	0.88	0.14	45.8	6.48	**< .001**
		low–mod	0.72	0.08	45.0	8.67	**< .001**
	Log	rest–low	0.02	0.09	46.0	0.19	.854
		rest–mod	0.59	0.11	45.7	5.37	**< .001**
		low–mod	0.57	0.08	45.7	7.60	**< .001**
Cycle duration (sec.)	Linear	rest–low	0.10	0.12	45.8	0.84	.401

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Variable	Model	Contrast	Estimate	SE	df	*t*	*p*
		rest–mod	0.82	0.15	45.8	5.62	< **.001**
		low–mod	0.72	0.09	45.1	8.02	< **.001**
	Log	rest–low	-0.05	0.10	46.0	-0.46	.651
		rest–mod	0.52	0.12	45.7	4.32	< **.001**
		low–mod	0.56	0.08	45.7	6.92	< **.001**
Inspiratory fraction	Linear	rest–low	-0.015	0.002	4798	-7.12	< **.001**
		rest–mod	-0.045	0.002	4818	-22.80	< **.001**
		low–mod	-0.030	0.002	4812	-15.57	< **.001**
	Log	rest–low	-0.016	0.002	64.4	-9.18	< **.001**
		rest–mod	-0.047	0.002	60.5	-23.46	< **.001**
		low–mod	-0.031	0.002	60.5	-15.82	< **.001**
Respiratory rate (breaths/min.)	Linear	rest–low	-0.38	0.36	47.0	-1.05	.300
		rest–mod	-3.19	0.53	46.5	-6.01	< **.001**
		low–mod	-2.82	0.34	45.3	-7.51	< **.001**

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Variable	Model	Contrast	Estimate	SE	df	t	p
Inspiratory amplitude (arb. units)	Linear	rest–low	-0.109	0.01	45.9	-14.87	< **.001**
		rest–mod	-0.207	0.01	45.3	-20.35	< **.001**
		low–mod	-0.098	0.01	44.0	-13.16	< **.001**
	Log	rest–low	-0.102	0.01	46.0	-14.00	< **.001**
		rest–mod	-0.200	0.01	45.8	-20.35	< **.001**
		low–mod	-0.099	0.01	45.8	-13.82	< **.001**
Expiratory amplitude (arb. units)	Linear	rest–low	-0.099	0.007	45.8	-13.84	< **.001**
		rest–mod	-0.197	0.010	45.3	-19.57	< **.001**
		low–mod	-0.099	0.007	43.9	-13.44	< **.001**
	Log	rest–low	-0.078	0.007	46.0	-11.54	< **.001**
		rest–mod	-0.186	0.010	45.7	-17.73	< **.001**
		low–mod	-0.107	0.008	45.7	-13.62	< **.001**
% Maximal inspiratory displacement	Linear	rest–low	-14.4	1.081	45.9	-13.33	< **.001**
		rest–mod	-27.4	1.57	45.5	-17.50	< **.001**
		low–mod	-13.0	1.10	44.0	-11.84	< **.001**
	Log	rest–low	-13.2	1.03	46.0	-12.91	< **.001**
		rest–mod	-26.0	1.52	45.8	-17.08	< **.001**
		low–mod	-12.8	1.03	45.8	-12.44	< **.001**
% Maximal expiratory displacement	Linear	rest–low	-12.8	1.06	45.8	-12.14	< **.001**
		rest–mod	-25.8	1.53	45.5	-16.90	< **.001**
		low–mod	-13.0	1.01	43.9	-12.87	< **.001**
	Log	rest–low	-10.3	0.96	46.0	-10.71	< **.001**
		rest–mod	-24.1	1.53	45.8	-15.77	< **.001**
		low–mod	-13.9	1.99	45.8	-12.80	< **.001**

Table A.4: Model statistics for temporal measures in sustained vowels: Estimated marginal means, standard error, degrees of freedom and confidence level for all measures for normal and log models

Variable	Workload	Model statistics				
		EMM	SE	df	Lower CL	Upper CL
Insp. duration (sec.)	rest	1.11	0.05	45.9	1.01	1.21
	low	1.04	0.04	45.9	0.96	1.13
	moderate	0.90	0.03	45.3	0.84	0.95
	log rest	0.95	0.04	45.9	0.87	1.05
	log low	0.92	0.04	45.9	0.85	1.00
	log moderate	0.83	0.03	45.5	0.78	0.88
Exp. duration (sec.)	rest	4.30	0.10	62.7	4.11	4.49
	low	4.10	0.10	61.8	3.91	4.29
	moderate	3.70	0.10	62.3	3.51	3.89
Cycle duration (sec.)	rest	5.40	0.11	62.4	5.17	5.63
	low	5.13	0.11	61.7	4.90	5.36
	moderate	4.61	0.11	62.1	4.38	4.83
Inspiratory fraction	rest	0.21	0.01	122.7	0.20	0.22
	low	0.20	0.01	121.4	0.19	0.21
	moderate	0.20	0.01	121.9	0.19	0.21
	log rest	0.18	0.01	113.0	0.17	0.19
	log low	0.18	0.01	112.7	0.17	0.19
	log moderate	0.19	0.01	113.2	0.18	0.20
Respiratory rate (breaths/min.)	rest	12.35	0.45	86.9	11.45	13.25
	low	14.07	0.45	86.9	13.17	14.97
	moderate	16.64	0.45	86.9	15.74	17.54

Table A.5: Model statistics for respiratory amplitude in sustained vowels: Estimated marginal means, standard error, degrees of freedom and confidence level for all measures for normal and log models.

Variable	Workload	Model statistics				
		EMM	SE	df	Lower CL	Upper CL
Inspiratory amplitude (arbitrary units)	rest	0.13	0.01	45.2	0.12	0.14
	low	0.29	0.01	46.0	0.26	0.31
	moderate	0.38	0.01	46.0	0.36	0.42
	log rest	0.11	0.01	45.9	0.10	0.13
	log low	0.26	0.01	45.9	0.24	0.28
	log moderate	0.36	0.02	45.9	0.33	0.39
Expiratory amplitude (arbitrary units)	rest	0.15	0.01	45.7	0.13	0.16
	low	0.30	0.02	46.0	0.27	0.33
	moderate	0.39	0.02	46.0	0.36	0.42
	log rest	0.13	0.01	45.8	0.12	0.15
	log low	0.26	0.02	46.0	0.23	0.30
	log moderate	0.35	0.02	45.9	0.32	0.39

Appendix B: Chapter 5 supplement

Table B.1: Mean ± SD values for glottal source measures by vowel, speech task, and workload level, calculated from the observed data

Measure	Vowel	Task	Rest	Low workload	Moderate workload
CPP	/a/	reading	23.3 ± 2.6	23.2 ± 2.1	23.0 ± 2.1
		vowels	24.2 ± 2.6	23.6 ± 2.5	23.7 ± 2.3
	/i/	reading	22.1 ± 1.9	21.1 ± 2.0	20.9 ± 1.9
		vowels	23.4 ± 3.2	22.3 ± 3.0	22.5 ± 2.7
	/u/	reading	18.2 ± 1.5	17.7 ± 1.4	17.4 ± 1.2
		vowels	20.1 ± 2.7	19.5 ± 2.6	19.8 ± 2.6
Energy	/a/	reading	1.6 ± 1.6	2.2 ± 2.1	4.0 ± 3.9
		vowels	1.2 ± 1.0	2.1 ± 2.1	3.4 ± 2.4
	/i/	reading	1.5 ± 1.1	1.9 ± 1.6	3.1 ± 2.5
		vowels	0.9 ± 0.6	1.9 ± 2.1	3.2 ± 2.7
	/u/	reading	0.7 ± 0.7	1.1 ± 0.9	1.7 ± 2.2
		vowels	0.8 ± 0.6	1.5 ± 1.2	2.4 ± 1.6
f_0	/a/	reading	198.6 ± 20.5	204.9 ± 22.7	214.9 ± 23.3
		vowels	206.6 ± 20.5	219.5 ± 25.8	235.5 ± 27.1
	/i/	reading	217.7 ± 21.1	225.2 ± 24.9	235.3 ± 25.1
		vowels	213.2 ± 21.6	228.4 ± 26.4	243.2 ± 30.7
	/u/	reading	200.5 ± 18.6	205.9 ± 21.6	217.2 ± 23.1
		vowels	215.6 ± 20.8	229.3 ± 27.5	245.8 ± 30.4

Table continues on next page

Measure	Vowel	Task	Rest	Low workload	Moderate workload
H1*–A1*	/a/	reading	31.0 ± 4.0	30.5 ± 4.0	30.0 ± 4.0
		vowels	32.0 ± 7.0	31.0 ± 6.3	29.4 ± 5.9
	/i/	reading	17.9 ± 4.5	17.6 ± 5.1	17.0 ± 5.7
		vowels	11.8 ± 7.8	8.5 ± 7.9	6.1 ± 7.1
	/u/	reading	17.2 ± 3.3	17.3 ± 4.1	17.1 ± 4.9
		vowels	16.3 ± 6.3	13.8 ± 8.0	10.9 ± 8.4
H1*–A3*	/a/	reading	22.7 ± 5.0	22.1 ± 4.9	22.5 ± 4.8
		vowels	23.6 ± 6.2	21.9 ± 5.5	20.7 ± 5.6
	/i/	reading	18.1 ± 4.7	17.8 ± 5.4	16.3 ± 5.2
		vowels	11.9 ± 6.0	10.3 ± 6.7	8.6 ± 5.3
	/u/	reading	11.7 ± 3.8	11.0 ± 3.2	9.8 ± 4.5
		vowels	5.8 ± 6.6	3.5 ± 6.0	1.3 ± 6.9
H1*–H2*	/a/	reading	7.6 ± 3.1	7.5 ± 3.0	8.1 ± 3.3
		vowels	9.0 ± 3.2	9.4 ± 4.0	9.1 ± 3.6
	/i/	reading	10.3 ± 4.1	10.0 ± 5.0	8.8 ± 4.8
		vowels	4.6 ± 4.5	3.0 ± 4.7	1.1 ± 3.6
	/u/	reading	9.7 ± 3.6	9.3 ± 3.4	7.5 ± 3.7
		vowels	7.1 ± 4.1	5.4 ± 4.6	3.7 ± 4.4
HNR <500 Hz	/a/	reading	30.8 ± 9.2	33.2 ± 8.0	32.5 ± 5.7
		vowels	43.3 ± 6.4	41.3 ± 7.1	41.5 ± 5.6
	/i/	reading	29.1 ± 5.6	28.1 ± 6.3	27.3 ± 5.3
		vowels	38.1 ± 7.3	34.0 ± 6.5	33.8 ± 6.6
	/u/	reading	19.2 ± 4.9	18.4 ± 4.7	17.5 ± 4.3
		vowels	30.8 ± 7.4	27.2 ± 6.8	27.2 ± 6.6
HNR <1500 Hz	/a/	reading	29.2 ± 8.3	31.3 ± 7.2	30.1 ± 5.1
		vowels	41.6 ± 5.5	39.2 ± 6.6	38.1 ± 5.0
	/i/	reading	42.9 ± 4.7	41.0 ± 5.6	39.3 ± 4.8
		vowels	56.0 ± 6.3	50.3 ± 5.9	49.1 ± 5.9
	/u/	reading	29.9 ± 4.5	28.5 ± 4.6	26.7 ± 4.2
		vowels	43.1 ± 7.4	38.3 ± 6.7	37.1 ± 6.3

Table continues on next page

Measure	Vowel	Task	Rest	Low workload	Moderate workload
HNR <2500 Hz	/a/	reading	33.3 ± 7.6	34.7 ± 6.7	33.3 ± 4.8
		vowels	45.7 ± 5.4	43.3 ± 6.3	41.6 ± 4.8
	/i/	reading	45.4 ± 4.4	43.7 ± 5.4	41.6 ± 4.7
		vowels	60.2 ± 5.8	54.8 ± 5.5	52.8 ± 5.2
	/u/	reading	38.1 ± 4.4	36.2 ± 4.5	34.0 ± 4.2
		vowels	51.6 ± 7.3	46.8 ± 7.1	45.3 ± 6.3
HNR <3500 Hz	/a/	reading	35.5 ± 6.7	36.2 ± 6.1	34.8 ± 4.4
		vowels	46.8 ± 5.1	44.0 ± 5.9	42.1 ± 4.9
	/i/	reading	44.4 ± 4.2	42.7 ± 5.2	40.5 ± 4.5
		vowels	57.8 ± 5.4	52.4 ± 5.3	50.0 ± 5.0
	/u/	reading	41.8 ± 4.1	39.8 ± 4.3	37.4 ± 4.1
		vowels	55.6 ± 6.5	50.7 ± 6.7	48.7 ± 5.6
SoE	/a/	reading	0.08 ± 0.01	0.08 ± 0.01	0.09 ± 0.01
		vowels	0.11 ± 0.01	0.11 ± 0.02	0.12 ± 0.02
	/i/	reading	0.09 ± 0.01	0.09 ± 0.01	0.10 ± 0.01
		vowels	0.11 ± 0.01	0.12 ± 0.01	0.12 ± 0.02
	/u/	reading	0.07 ± 0.01	0.07 ± 0.01	0.08 ± 0.01
		vowels	0.11 ± 0.01	0.12 ± 0.02	0.12 ± 0.02

Appendix C: Chapter 6 supplement

Table C.1: Mean ± SD F1 and F2 values (Hz) by vowel, speech task, and workload level, calculated from the observed data

Vowel	Task	Rest		Low workload		Moderate workload	
		F1	F2	F1	F2	F1	F2
/a/	reading	887 ± 84	1503 ± 90	892 ± 86	1503 ± 89	892 ± 74	1507 ± 96
	vowels	831 ± 67	1415 ± 106	845 ± 93	1412 ± 86	851 ± 78	1424 ± 97
/i/	reading	386 ± 60	2323 ± 144	386 ± 59	2316 ± 166	387 ± 66	2275 ± 164
	vowels	291 ± 37	2591 ± 131	289 ± 42	2571 ± 136	288 ± 37	2551 ± 135
/u/	reading	350 ± 38	1063 ± 136	354 ± 41	1089 ± 130	349 ± 39	1097 ± 115
	vowels	322 ± 38	780 ± 139	318 ± 36	799 ± 120	318 ± 39	812 ± 118

Table C.2: Read speech: Pairwise contrasts for formant measures across workload levels

Vowel	Measure	Contrast	Estimate	SE	df	*t*	*p*
/a/	F1	rest–low	-7.98	5.49	352.13	-1.45	.147
		rest–mod	-4.75	5.53	352.29	-0.86	.391
		low–mod	3.22	5.55	352.39	0.58	.562
	F2	rest–low	-3.16	5.89	354.06	-0.54	.592
		rest–mod	-5.69	5.94	354.31	-0.96	.339
		low–mod	-2.53	5.94	354.31	-0.43	.670
/i/	F1	rest–low	-2.91	4.40	370.12	-0.66	.509
		rest–mod	-4.64	4.43	370.53	-1.05	.296
		low–mod	-1.73	4.42	370.38	-0.39	.695
	F2	rest–low	9.51	11.62	372.10	0.82	.414
		rest–mod	49.17	11.74	372.49	4.19	**< .001**
		low–mod	39.66	11.73	372.43	3.38	**.001**
/u/	F1	rest–low	-4.58	3.48	361.29	-1.32	.189
		rest–mod	1.26	3.51	361.87	0.36	.720
		low–mod	5.84	3.50	361.92	1.67	.096
	F2	rest–low	-25.34	11.69	360.39	-2.17	**.031**
		rest–mod	-32.41	11.76	361.01	-2.76	**006**
		low–mod	-7.07	11.77	361.11	-0.60	.548
Centroid	F1	rest–low	-6.48	3.50	85.10	-1.85	.068
		rest–mod	-4.62	3.56	85.30	-1.30	.198
		low–mod	1.86	3.53	85.20	0.53	.599
	F2	rest–low	-7.80	7.36	87.00	-1.06	.292
		rest–mod	3.38	7.42	87.10	0.46	.650
		low–mod	11.18	7.42	87.10	1.51	.135

Table C.3: Sustained vowels: Pairwise contrasts for formant measures across workload levels

Vowel	Measure	Contrast	Estimate	SE	df	*t*	*p*
/a/	F1	rest–low	-16.95	9.56	43.93	-1.77	.083
		rest–mod	-21.07	9.34	43.90	-2.26	**.029**
		low–mod	-4.11	8.17	43.95	-0.50	.617
	F2	rest–low	2.40	10.01	43.98	0.24	.811
		rest–mod	-6.95	9.16	43.96	-0.76	.452
		low–mod	-9.35	7.65	43.94	-1.22	.228
/i/	F1	rest–low	2.19	4.09	44.98	0.53	.596
		rest–mod	2.39	4.80	44.99	0.50	.620
		low–mod	0.21	4.06	44.96	0.05	.959
	F2	rest–low	19.44	7.98	363.03	2.44	**.015**
		rest–mod	43.96	7.99	363.02	5.50	**< .001**
		low–mod	24.51	8.01	363.06	3.06	**.002**
/u/	F1	rest–low	4.19	3.95	44.97	1.06	.294
		rest–mod	4.72	5.27	44.48	0.90	.375
		low–mod	0.52	5.05	44.39	0.10	.918
	F2	rest–low	-17.62	12.83	44.97	-1.37	.176
		rest–mod	-29.73	16.19	44.73	-1.84	.073
		low–mod	-12.11	12.54	44.28	-0.97	.340
Centroid	F1	rest–low	-3.78	4.06	87.00	-0.93	.355
		rest–mod	-6.42	4.09	87.18	-1.57	.120
		low–mod	-2.65	4.09	87.18	-0.65	.519
	F2	rest–low	1.87	6.59	87.00	0.28	.777
		rest–mod	2.47	6.65	87.13	0.37	.711
		low–mod	0.60	6.65	87.13	0.09	.928

Table C.4: Pairwise contrasts for vowel space area (VSA) across speech tasks and workloads.

Task	Contrast	Estimate	SE	df	*t*	*p*
Read passage	rest–low	2,257	6,781	87.0	0.33	.740
	rest–mod	15,691	6,837	87.1	2.30	**.024**
	low–mod	13,434	6,837	87.1	1.96	.053
Sustained vowels	rest–low	-9,057	12,780	85.3	-0.71	.480
	rest–mod	155,040	13,001	87.7	11.92	$< .001$
	low–mod	164,097	13,001	87.7	12.62	$< .001$

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Name index

Speech during exercise

Much of human communication occurs while people are moving, exerting themselves, or otherwise not at rest, yet speech production in such contexts has received little systematic attention. Previous studies have typically addressed respiration, phonation, or articulation separately, making it difficult to characterize changes in speech production as a whole. This book presents a multi-level investigation of speech during controlled physical activity. Combining respiratory data, acoustic analysis, and speaker reports, it examines how different exercise levels affect breathing patterns, phonation, and articulatory realization.

The analyses reveal adjustments across all production levels, but no uniform production mode: individual speakers show distinct patterns of response under the same physical workload. By integrating quantitative and qualitative evidence, the book highlights the role of between-speaker variability for understanding speech behavior under real-life conditions. The findings inform models of speech production and variability, and they support a broader view of speech as a dynamically coordinated system that adapts across changing bodily states.